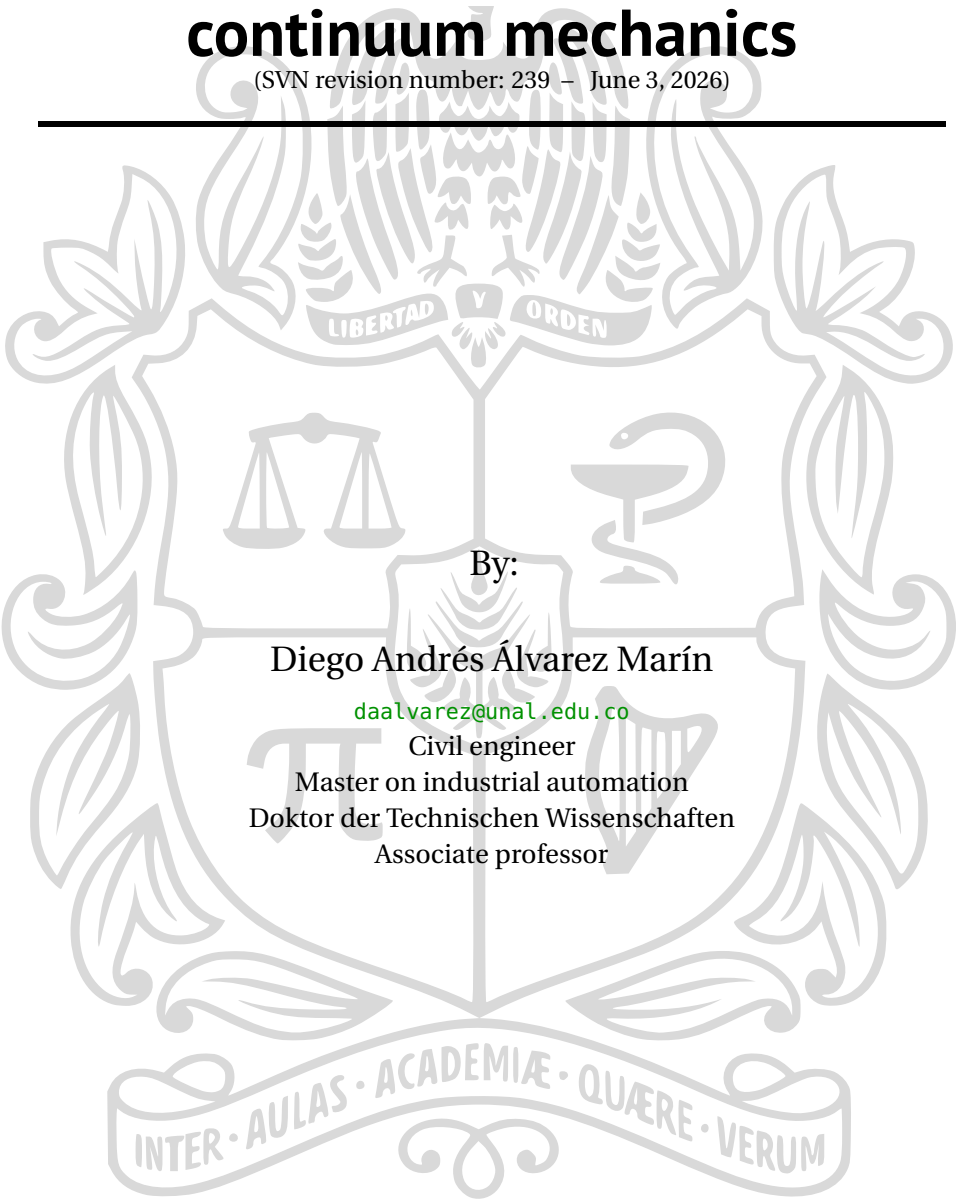
Notes for a course on continuum mechanics

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Chapter 1

Continuum mechanics

Let's start by thinking about the nature of matter itself. We know from modern physics that at the tiniest scales, matter is made up of discrete particles - molecules, atoms, and even smaller subatomic particles. However, in our everyday experiences and most practical applications in engineering, physics, and beyond, we deal with chunks of matter that are absolutely enormous compared to those tiny building blocks.

At these larger scales, we're not really concerned with the precise motions of each individual atom or molecule. What matters is their overall, averaged-out collective behavior as a bulk material. In principle, if we knew enough about how matter behaves at the molecular level, we could use statistical techniques to derive and predict the large-scale properties from the small-scale physics. But those calculations are really complicated - we can only pull it off for the simplest of systems, and even then, we have to make tons of approximations.

So in practice, pretty much our entire understanding of how materials deform, flow, and mechanically respond comes from direct observation and experimental testing at these larger length scales where matter appears continuous. This macroscopic perspective on material behavior is the domain of continuum mechanics.

Continuum mechanics[★] deals with the mechanical behaviour of solids and fluids on the macroscopic scale. In continuum mechanics, we treat matter as a seamless continuity filling up regions of space, rather than a discrete collection of particles. That allows us to define smoothly varying fields, like density, velocity, displacement, and so on as continuous functions throughout the material's volume. Of course, this continuum approximation only works if we're looking at chunk sizes much bigger than the microscopic scales where the discrete atomistic nature becomes apparent.

Those characteristic micro-lengths could be atomic spacings in a crystalline solid, the distance between molecules in a gas, or even the grains in soil. As long as the system dimensions dwarf those micro-scales, the continuous treatment is valid.

The core premise is that we associate a particle of the continuum material with every single point in the region of space it occupies. We can then attribute field quantities at that point, like its density or velocity, to that particle. While this seems like an obvious stretch from reality, it ends up being an incredibly successful framework, firmly grounded in solid and fluid mechanics and validated by experiments.

Mechanics itself studies the interplay between forces and motions. In continuum mechanics then, the key variables are functions describing forces and stresses on the one side, and kinematic fields like displacement, velocity, acceleration on the other side. For rigid bodies, the shape constraints mean the particle motions are highly coupled. But continuum mechanics really shines for deformable bodies, where particles can move much more freely to distort and flow.

A key feature of the continuum mechanics framework is that it contains two distinct classes of governing equations that apply to all materials, whether solids or fluids. The first type encapsulates fundamental physical laws that must be obeyed by any continuum model. These include *conservation laws** like conservation of mass, which states that mass cannot be created or destroyed, only transported from one region to another. Additionally, the conservation of momentum and conservation of energy laws are inherent constraints that any valid continuum mechanics description must satisfy. These general conservation equations essentially define the kinematic relationships and force equilibrium requirements that are universal across all materials.

The second type of governing equation arises from the *constitutive laws** that characterize and distinguish the behavior of different classes of materials. While the conservation equations are universal, the constitutive relations specify how a particular material deforms or flows in response to applied forces and motions. For example, the constitutive model for an elastic solid like rubber relates the stresses that develop to the strains induced by deformation. For a viscous liquid like honey, the constitutive equation describes the strain rates that arise from applied stresses. Formulating accurate constitutive models from experimental observations is one of the central challenges in continuum mechanics. But once determined, these constitutive equations can be combined with the general conservation laws to provide a complete predictive model for that material's mechanical response in any potential loading scenario.

Overall, continuum mechanics provides an impressively powerful framework for understanding the mechanics of common matter at human scales. By wisely blurring over the discrete atomistic details, we can develop relatively simple formulations that brilliantly capture the rich complexity of how bulk materials deform, flow and respond to forces in the situations we care about most.

1.1 Density in the continuum

In continuum mechanics, *mass density*★ (or simply *density*★) is defined as the limit of the ratio of mass to volume as the volume shrinks to a point, but with an important caveat: it is not a mathematical limit in the strict sense, but rather a *physical limit* where the volume must remain large enough to contain a statistically meaningful number of particles. Mathematically, we write:

$$\rho(\boldsymbol{x}, t) = \lim_{V \to V^*} \frac{m(V)}{V}$$

where $m(V)$ is the mass contained in a small volume V surrounding point $\boldsymbol{x}$ at time t, and V^* is the so-called *representative volume element*★, which is small enough to capture local properties without seeing molecular noise, yet large enough to be considered a point in the macroscopic field and to contain a sufficient number of particles to ensure that statistical fluctuations are negligible. It is important to stress that this expression should not be interpreted as the strict mathematical limit $V \to 0$, but rather a physical limit where V must be small enough to resolve macroscopic variations but large enough to contain many particles.

According to White (2015), a typical value of the representative volume element V^* is about $10^{-9}\,\mathrm{mm}^3$ for liquids and for gases at atmospheric pressure.



Chapter 2

A summary of elasticity theory

2.1 Infinitesimal deformations

$$\boldsymbol{\varepsilon} = \underbrace{\begin{pmatrix} \varepsilon_x & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_y & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_z \end{pmatrix} = \begin{pmatrix} \varepsilon_{11} & \varepsilon_{12} & \varepsilon_{13} \\ \varepsilon_{21} & \varepsilon_{22} & \varepsilon_{23} \\ \varepsilon_{31} & \varepsilon_{32} & \varepsilon_{33} \end{pmatrix}}_{\text{mathematical strains}} = \underbrace{\begin{pmatrix} \varepsilon_x & \frac{1}{2}\gamma_{xy} & \frac{1}{2}\gamma_{xz} \\ \frac{1}{2}\gamma_{yx} & \varepsilon_y & \frac{1}{2}\gamma_{yz} \\ \frac{1}{2}\gamma_{zx} & \frac{1}{2}\gamma_{zy} & \varepsilon_z \end{pmatrix}}_{\text{engineering strains}}. \tag{2.1}$$

Interpretation of Cauchy's infinitesimal strain tensor

2.1.1 Compatibility equations

Saint-Venant

2.2 Transformation of first- and second-order tensors

$$\underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_{\vec{\boldsymbol{v}}} = \underbrace{\begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{pmatrix}}_{\boldsymbol{T}} \underbrace{\begin{pmatrix} x' \\ y' \\ z' \end{pmatrix}}_{\vec{\boldsymbol{v}}'}.$$

$$\boldsymbol{T} = \left[\hat{\boldsymbol{e}}'_1, \quad \hat{\boldsymbol{e}}'_2, \quad \hat{\boldsymbol{e}}'_3\right].$$

$$\boxed{\underline{\underline{\boldsymbol{\sigma}}}' = \boldsymbol{T}^T \underline{\underline{\boldsymbol{\sigma}}} \boldsymbol{T},}$$

$$\underbrace{\begin{pmatrix} \sigma_{x'} & \tau_{x'y'} & \tau_{x'z'} \\ \tau_{x'y'} & \sigma_{y'} & \tau_{y'z'} \\ \tau_{x'z'} & \tau_{y'z'} & \sigma_{z'} \end{pmatrix}}_{\underline{\underline{\boldsymbol{\sigma}}}'} = \underbrace{\begin{pmatrix} \alpha_1 & \beta_1 & \gamma_1 \\ \alpha_2 & \beta_2 & \gamma_2 \\ \alpha_3 & \beta_3 & \gamma_3 \end{pmatrix}}_{\boldsymbol{T}^T} \underbrace{\begin{pmatrix} \sigma_x & \tau_{xy} & \tau_{xz} \\ \tau_{xy} & \sigma_y & \tau_{yz} \\ \tau_{xz} & \tau_{yz} & \sigma_z \end{pmatrix}}_{\underline{\underline{\boldsymbol{\sigma}}}} \underbrace{\begin{pmatrix} \alpha_1 & \alpha_2 & \alpha_3 \\ \beta_1 & \beta_2 & \beta_3 \\ \gamma_1 & \gamma_2 & \gamma_3 \end{pmatrix}}_{\boldsymbol{T}}.$$

Aquí $\hat{\boldsymbol{e}}'_1 := [\alpha_1,\ \beta_1,\ \gamma_1]^T$, $\hat{\boldsymbol{e}}'_2 := [\alpha_2,\ \beta_2,\ \gamma_2]^T$ y $\hat{\boldsymbol{e}}'_3 := [\alpha_3,\ \beta_3,\ \gamma_3]^T$ son vectores mutuamente ortogonales que representan los ejes del sistema de coordenadas para el cual se refieren las direcciones x', y' y z' (especificando estos en el sistema de coordenadas de la mano derecha).

$$\boxed{\underline{\underline{\boldsymbol{\sigma}}} = \boldsymbol{T} \underline{\underline{\boldsymbol{\sigma}}}' \boldsymbol{T}^T,}$$

ya que $\boldsymbol{T}\boldsymbol{T}^T = \boldsymbol{T}^T\boldsymbol{T} = \boldsymbol{I}$.

2.3 The Saint-Venant compatibility conditions

$$\begin{aligned} &\frac{\partial^2 \varepsilon_x}{\partial y^2} + \frac{\partial^2 \varepsilon_y}{\partial x^2} = \frac{\partial^2 \gamma_{xy}}{\partial x \partial y} \quad 2\frac{\partial^2 \varepsilon_x}{\partial y \partial z} = \frac{\partial}{\partial x}\left(-\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{xz}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z}\right) \\ &\frac{\partial^2 \varepsilon_y}{\partial z^2} + \frac{\partial^2 \varepsilon_z}{\partial y^2} = \frac{\partial^2 \gamma_{yz}}{\partial y \partial z} \quad 2\frac{\partial^2 \varepsilon_y}{\partial x \partial z} = \frac{\partial}{\partial y}\left(\frac{\partial \gamma_{yz}}{\partial x} - \frac{\partial \gamma_{xz}}{\partial y} + \frac{\partial \gamma_{xy}}{\partial z}\right) \\ &\frac{\partial^2 \varepsilon_z}{\partial x^2} + \frac{\partial^2 \varepsilon_x}{\partial z^2} = \frac{\partial^2 \gamma_{xz}}{\partial x \partial z} \quad 2\frac{\partial^2 \varepsilon_z}{\partial x \partial y} = \frac{\partial}{\partial z}\left(\frac{\partial \gamma_{yz}}{\partial x} + \frac{\partial \gamma_{xz}}{\partial y} - \frac{\partial \gamma_{xy}}{\partial z}\right); \end{aligned} \tag{2.2}$$

Chapter 3

Kinematics

This chapter deals with *kinematics**, that is, is the study of motion and deformation of a body without directly considering the forces influencing the motion. We will say that the body *deforms* whenever it moves and changes its shape. In particular, we shall analyze *large* displacements and deformations, also known as *finite* displacements and deformations, in contrast to the infinitesimal deformation approach employed in elasticity theory, that was reviewed in Section 2.1.This study will require a fresh start because, in contraposition to the infinitesimal deformations' viewpoint, which assumes that the displacements are small in relation to the initial dimension of the body, in finite deformation analysis we shall make no suppositions regarding the magnitude of the displacements and strains, the material that conforms the body or their internal structure; in fact, large displacements may even surpass the initial dimensions of the body, as is the case, for example in metal forming. However, in Section 3.15, we will reconcile both points of view.

Therefore, the concepts covered in this chapter apply to all kinds of materials, solids and fluids, independently of their internal physical structure or the degree of deformation or displacement that the body is experiencing.

3.1 Initial and current positions and configurations

Let us assume that a *body** $\mathscr{B}$, depicted in Figure 3.1, also referred to as a *continuum body** or simply as a *continuum**, is embedded into the three-dimensional Euclidean space $\mathbb{R}^3$; this body $\mathscr{B}$ is made up by an infinite set of *particles** also known as *material*

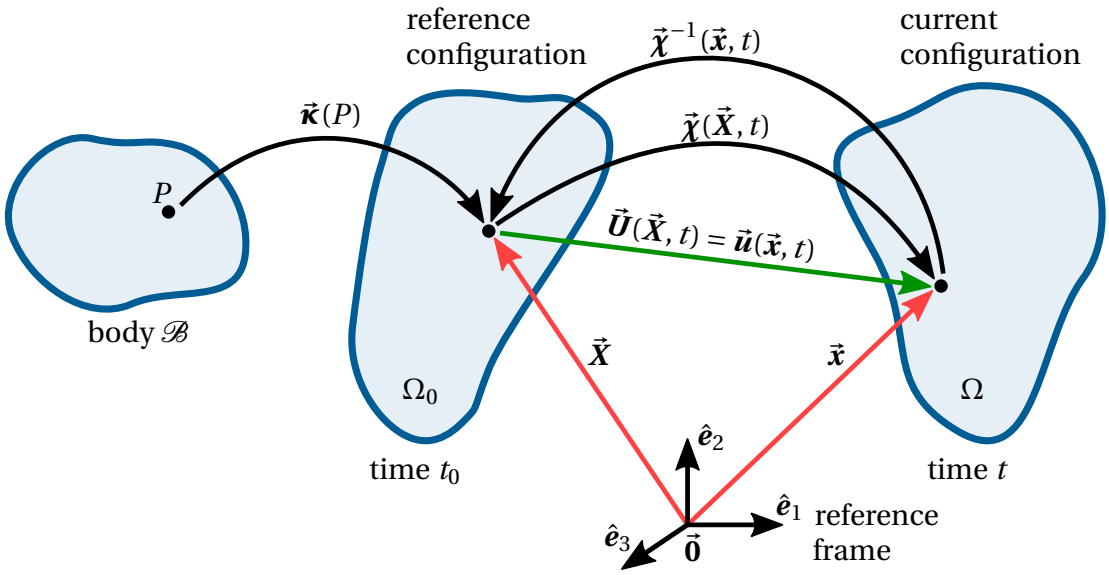


Figure 3.1: *When a body $\mathscr{B}$ moves, it changes its position, and possibly its shape, through time. In order to study the body, we specify at time t_0, the so-called reference time, where the body takes a position called the reference (undeformed or initial) configuration Ω_0; also, at time $t > t_0$ the body takes another position, the so-called current (or deformed) configuration Ω. One of the particles of $\mathscr{B}$, namely P, takes at times t_0 and t the positions $\vec{X}$ and $\vec{x}$, respectively. Both $\vec{X}$ and $\vec{x}$ are defined with respect to a reference frame denoted by the orthonormal vectors $\hat{e}_1$, $\hat{e}_2$, and $\hat{e}_3$. Finally, we denote the displacement of P from $\vec{X}$ to $\vec{x}$ as the green vector $\vec{U}(\vec{X},t)=\vec{u}(\vec{x},t)$.*

points* or *material elements**; even though each particle is an infinitesimal piece of continuous matter that is identifiable throughout its dynamic history, it is big enough to reflect averages over a length scale connected to molecules or atoms, so that it can be viewed as a continuum. Let us call one of those particles P.

In order to assign coordinates to each particle $P \in \mathscr{B}$, it is required to introduce a right-handed orthonormal coordinate system, that we will call the *reference frame**, and that is formed by the basis $\{\hat{\boldsymbol{\imath}}, \hat{\boldsymbol{\jmath}}, \hat{\boldsymbol{k}}\}$; it is customary in continuum mechanics to express this basis as $\{\hat{\boldsymbol{e}}_1, \hat{\boldsymbol{e}}_2, \hat{\boldsymbol{e}}_3\}$, in such a way that $\hat{\boldsymbol{e}}_1 \equiv \hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{e}}_2 \equiv \hat{\boldsymbol{\jmath}}$ and $\hat{\boldsymbol{e}}_3 \equiv \hat{\boldsymbol{k}}$; the origin of coordinates of this reference frame will be identified by vector $\vec{\boldsymbol{0}}$.

Since the body moves, we require to define an *initial** or *reference time**, which shall be identified as $t = t_0$ and that we usually set to zero, i.e. $t_0 = 0$; at that time, the body will adopt a position Ω_0 that will be referred to as the *initial**, *reference** or *undeformed configuration** of the body $\mathscr{B}$. We will identify the position of the particle P at time t_0 by the so-called *initial** or *referential position** $\vec{X} \in \Omega_0$ relative to the origin of coordinates $\vec{\boldsymbol{0}}$, as shown in Figure 3.1; the coordinate $\vec{X}$ will also be called the *material** or *Lagrangian coordinate**, after the Italian mathematician, physicist and astronomer Joseph-Louis Lagrange (1736 – 1813). In order to locate $\vec{X}$, we may define

a bijection (see Appendix A.1) $\vec{\boldsymbol{\kappa}} : \mathscr{B} \to \Omega_0$, such that $\vec{\boldsymbol{X}} = \vec{\boldsymbol{\kappa}}(P)$, that locates P in Ω_0 at time t_0; notice that the image (see Appendix A.2) of the body $\mathscr{B}$ through $\vec{\boldsymbol{\kappa}}$ is Ω_0, that is $\Omega_0 = \vec{\boldsymbol{\kappa}}(\mathscr{B})$. Since $\vec{\boldsymbol{\kappa}}$ is a bijection, we may do an abuse of notation and label particle P, at time t_0, by its initial position $\vec{\boldsymbol{X}}$.

As *time* $t \in T$ goes on, the continuum body $\mathscr{B}$ may move from one instant of time to another; here $T \subseteq \mathbb{R}$ is a real interval. The body $\mathscr{B}$ will occupy a continuous sequence of positions in space, which at time t will be Ω_t. In case subindex t is evident from the context, we will simply write it down as Ω; here each of the positions Ω is called a *configuration** of $\mathscr{B}$; when t is the present time, Ω_t will be called the *current** or *deformed configuration**; note that $\mathscr{B}$ may have infinitely many configurations that are distinguished by time $t \in T$.

Particle P will move from the referential position $\vec{\boldsymbol{X}}$ at time t_0 to the *current position** $\vec{\boldsymbol{x}} \in \Omega$ at time t relative to the origin of coordinates $\vec{\boldsymbol{0}}$. The coordinate $\vec{\boldsymbol{x}}$ is also called the *spatial** or *Eulerian coordinate**, after the Swiss mathematician, physicist, astronomer and engineer Leonhard Euler (1707 – 1783).

The Cartesian components of the material position vector $\vec{\boldsymbol{X}}$ according to the reference frame are usually denoted either as X, Y and Z or as X_1, X_2 and X_3, that is $\vec{\boldsymbol{X}} = [X, Y, Z]^T$ or $\vec{\boldsymbol{X}} = [X_1, X_2, X_3]^T$; similarly, the spatial position vector $\vec{\boldsymbol{x}}$ is usually denoted in terms of its components as $\vec{\boldsymbol{x}} = [x, y, z]^T$ or as $\vec{\boldsymbol{x}} = [x_1, x_2, x_3]^T$.

3.2 Motion and trajectory

Let us define a vector field $\vec{\boldsymbol{\chi}} : \Omega_0 \times T \to \Omega$ that maps the initial position $\vec{\boldsymbol{X}}$ of particle P to current position $\vec{\boldsymbol{x}}$ at time t, that is

$$\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t); \tag{3.1}$$

this function describes the position of each particle at time t. Since the image of Ω_0 through $\vec{\boldsymbol{\chi}}(\cdot, t)$ is Ω, that is $\Omega = \vec{\boldsymbol{\chi}}(\Omega_0, t)$, we will call the vector field $\vec{\boldsymbol{\chi}}$ the *motion of the body* $\mathscr{B}$ or simply the *equation of motion**. Take into account that at time t_0, $\vec{\boldsymbol{X}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t_0)$ and $\vec{\boldsymbol{x}} = \vec{\boldsymbol{X}}$.

Note that $\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, \cdot)$ acts as a parametric equation in t; in other words, the function $\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$ for a fixed $\vec{\boldsymbol{X}} \in \Omega_0$ and for all $t \in T$ describes the *trajectory** or *pathline** of the particle P in the three-dimensional Euclidean space $\mathbb{R}^3$.

We will assume that $\vec{\boldsymbol{\chi}}$ is a continuous function with respect to its first argument, the position, because otherwise, the particle P would be teleported in the sense that there is a transfer of matter from one point to another without traversing the physical space between them; in addition, this guarantees that there will not be cracks, holes or gaps in the current configuration when the body deforms. Moreover, $\vec{\boldsymbol{\chi}}$ is a continuous

function with respect to time (that is, its second argument) because if it didn't the particle P would be traveling in time from time $t = a$ to time $t = b$ without physically existing at those times $t \in (a, b)$.

In addition, the motion $\vec{\chi}(\cdot, t)$ must be a bijective function (see Appendix A.1) in t; thus, it must be both injective and surjective:

- Injectivity guarantees that two distinct particles from the initial configuration do not occupy the same position in the current configuration; physically, this prevents the interpenetration of matter; mathematically, if $\vec{X}_1 \neq \vec{X}_2$, then $\vec{\chi}(\vec{X}_1, t) \neq \vec{\chi}(\vec{X}_2, t)$ for all t.
- Surjectivity ensures that all particles in the current configuration have a corresponding particle from the initial configuration; this means the deformation covers the entire body without leaving "holes" or "gaps" and secures that no new material is spontaneously created during the deformation process); mathematically, for every $\vec{x}$ in the current configuration Ω, there exists a unique $\vec{X} \in \Omega_0$ such that $\vec{\chi}(\vec{X}, t) = \vec{x}$.

The bijectivity of $\vec{\chi}(\cdot, t)$ guarantees the existence of its inverse function (see Appendix A.3) $\vec{\chi}^{-1}(\cdot, t)$ for each fixed time t. This inverse function maps points in the current configuration back to their corresponding points in the initial configuration. We will call $\vec{\chi}^{-1}(\cdot, t)$ the *equation of inverse motion*.★ Note that $\vec{\chi}^{-1}(\cdot, t)$ maps the particle P from the position $\vec{x}$ at time t to the referential or initial position $\vec{X}$ at time t_0, that is

$$\vec{X} = \vec{\chi}^{-1}(\vec{x}, t). \tag{3.2}$$

In the following, we will assume that both the motion $\vec{\chi}$ and its inverse $\vec{\chi}^{-1}$ have continuous derivatives with respect to position and time; the rationale behind this assumption will be explained, in detail, in Sections 3.9 and 3.22.1.

3.3 Material and spatial descriptions

Any kinematic property of a body such as displacement, velocity or acceleration and any physical property of a body such as temperature, chemical concentration or density can be characterized, at time t, in terms of either the material coordinates $\vec{X}$ or the spatial coordinates $\vec{x}$ since equation (3.1) and its inverse relation (3.2) allow to transform one representation to the other.

On the one hand, every time that we employ the material coordinate $\vec{X}$ as the independent variable, as in equation (3.1), we will say that we are using a *material*,★ *referential*★ or *Lagrangian description*★ (Figure 3.2a). In this description, we follow single particles, distinguished by its initial position $\vec{X}$, as they move along a trajectory; let us remember that $\vec{X}$ acts as a label that identifies each particle; here each piece

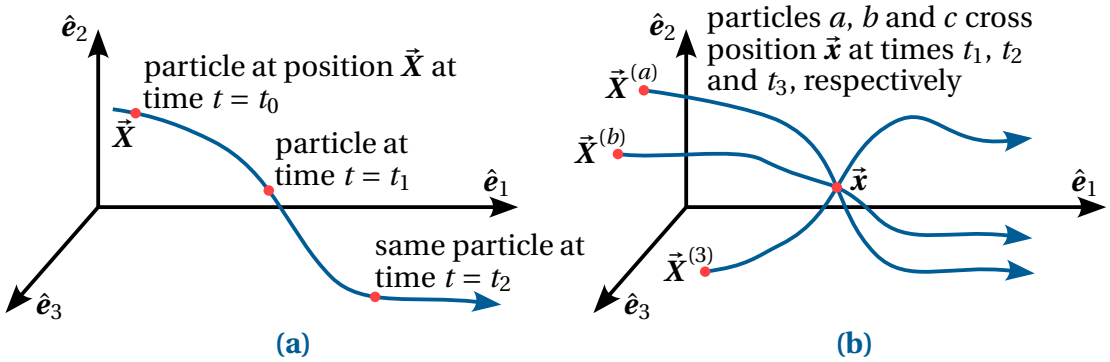


Figure 3.2: *The diagrams illustrate the Lagrangian and Eulerian perspectives in continuum mechanics. Figure* ***(a)*** *shows the Lagrangian description, where a specific particle's path is tracked over time; the basis* $\{\hat{e}_1, \hat{e}_2, \hat{e}_3\}$ *represents the reference frame; the point* $\vec{X}$ *indicates the particle's position at the initial time* t_0*. Figure* ***(b)*** *depicts the Eulerian description, focusing on a fixed spatial location* $\vec{x}$ *and the paths of different particles (a, b and c) passing through that point at different time instances (*t_1*,* t_2 *and* t_3*), as two particles cannot occupy the same spatial position simultaneously. In this case,* $\vec{x} = \vec{\chi}\left(\vec{X}^{(a)}, t_1\right) = \vec{\chi}\left(\vec{X}^{(b)}, t_2\right) = \vec{\chi}\left(\vec{X}^{(c)}, t_3\right)$.

of matter under study may change its shape, density or velocity but it is always the same piece of material.

On the other hand, every time that we employ the spatial coordinate $\vec{x}$ as the independent variable, as in equation (3.2), we will say that we are using a *spatial** or *Eulerian description** (Figure 3.2b). In this description, attention is focused on a specific location in space, namely $\vec{x}$; since the material may flow through this location while in motion, different particles are being studied at different times, and as a result, the spatial description cannot directly provide information about the properties of specific particles as they move.

For example, let us suppose that we are analyzing traffic flow. In the material description, we study the velocity of a single car as it moves along a highway; in the spatial description, we observe the velocity of cars at a fixed street intersection as time goes on. Also, in modern times the use of sensors is ubiquitous; their measurements can be collected in either an Eulerian framework, where the sensors are fixed spatially, like in the case of an anemometer[1] that is attached to the roof of a building, or in a Lagrangian framework, where the sensors move with the dynamics, like in the case of a sensor installed in a drifter to measure the ocean salinity, temperature, current direction and speed (see e.g. `https://gdp.ucsd.edu/ldl/`).

The spatial description is usually employed in fluid mechanics, since the position of the fluid particles at time t_0 is usually unknown, that is, it is difficult to know the

[1] In meteorology, an *anemometer* is a device that measures wind velocity; it is a common instrument used in weather stations.

position $\vec{\boldsymbol{X}}$ for each particle. However, solid mechanics employs both the material or the spatial description, although the former is usually preferred, inasmuch as it is natural to choose as initial configuration the one for which the material is unstressed, which is usually the case in the undeformed configuration.

It must bear in mind that, there are many practical problems in which bodies deformations are very small, like in elasticity theory; for example, beams, columns and slabs in a building have very small deformations during service conditions; usually, in this case, no distinction is made between material and spatial descriptions, and therefore the mathematical formulation simplifies notably.

In continuum mechanics, it is usual to identify a quantity in the material description by uppercase letters, like in X, $\vec{\boldsymbol{X}}$ or $\boldsymbol{X}$, while those in the spatial description are written down in lowercase letters, like in x, $\vec{\boldsymbol{x}}$ or $\boldsymbol{x}$; in addition, oftentimes a zero subindex is employed for variables in the material configuration, like in Ω_0 or ρ_0. Even though this convention is often used in continuum mechanics, it usually leads to confusions with some other popular convention that is used in finite element, structural analysis and elasticity theory: matrices (and second-order tensors) are denoted by upper cases and vectors (and first-order tensors) are denoted by lower cases. We will tackle this issue by usually *a)* writing first- and second-order tensors, vectors and matrices in bold; *b)* writing down zero-order tensors, i.e., scalars, without bold; *c)* drawing an upper arrow above each first-order tensor and vector. Thus, $\boldsymbol{A}$, $\boldsymbol{a}$ or $\boldsymbol{\alpha}$ stand for second-order tensors and matrices; $\vec{\boldsymbol{A}}$, $\vec{\boldsymbol{a}}$ or $\vec{\boldsymbol{\alpha}}$ represent first-order tensors and vectors; and A, a and α depict scalars. Also, a unit vector, that is one whose norm is one, will be denoted by an upper hat; thus $\hat{\vec{\boldsymbol{n}}} := \frac{\vec{\boldsymbol{n}}}{\|\vec{\boldsymbol{n}}\|}$ denotes a unit vector; however, it is redundant to write down both, the arrow and the hat, and in this case, we will simplify the notation of this unit vector to $\hat{\boldsymbol{n}}$.

To illustrate the notational difference between a material and a spatial description consider, for example, a scalar quantity such as the contaminant concentration C in a lake: in the material description the concentration might be denoted as $C(\vec{\boldsymbol{X}}, t)$ or C_0 while in the spatial description it is $c(\vec{\boldsymbol{x}}, t)$ or simply c. On the one side, in the material description $\vec{\boldsymbol{X}}$ serves to label particle P in the continuum at time t so that a change in time implies that the same particle $\vec{\boldsymbol{X}}$ has a different contaminant concentration; therefore the attention is focused on the particle itself, and the material description tracks the evolution of a property associated with a specific particle as it moves through space and time. On the other side, in the spatial description $\vec{\boldsymbol{x}}$ denotes a position in space at time t; here a change in time t implies that a different contaminant concentration is perceived at the same spatial position $\vec{\boldsymbol{x}}$, now probably occupied by another particle; hence, the attention focuses on a particular spatial location $\vec{\boldsymbol{x}}$, and the property of interest (in this case, contaminant concentration) is evaluated at that fixed position as different particles pass through it over time.

3.4 Displacement vector field

When a particle moves, it changes its location from its initial position $\vec{\boldsymbol{X}}$ at time t_0 to its current position $\vec{\boldsymbol{x}}$ at time t, as shown in Figure 3.1. This change of location is best described by what we will call the *displacement vector field* o simply *displacement field*[★]; this vector encodes the direction and magnitude of the movement and is described in the material configuration as

$$\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t) = \underbrace{\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}_{\text{current position}} - \underbrace{\vec{\boldsymbol{X}}}_{\text{initial position}} \tag{3.3}$$

or in the spatial configuration as

$$\vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t) = \underbrace{\vec{\boldsymbol{x}}}_{\text{current position}} - \underbrace{\vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}},t)}_{\text{initial position}} ; \tag{3.4}$$

both descriptions are associated to the same values since $\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t) = \vec{\boldsymbol{U}}(\vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}},t),t) = \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)$, even though $\vec{\boldsymbol{U}}$ and $\vec{\boldsymbol{u}}$ are functions of different arguments.

The Cartesian components of this displacement vector in the spatial frame are usually denoted either as u, v and w or as u_1, u_2 and u_3, that is $\vec{\boldsymbol{u}} = [u,v,w]^T$ or $\vec{\boldsymbol{u}} = [u_1,u_2,u_3]^T$; similarly, the displacement vector in the material frame is usually denoted component-wise either as $\vec{\boldsymbol{U}} = [U,V,W]^T$ or as $\vec{\boldsymbol{U}} = [U_1,U_2,U_3]^T$.

Inasmuch as the current and the initial positions coincide for t_0, the displacements vanish at that time, that is, $\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t_0) = \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t_0) = \vec{\boldsymbol{0}}$.

3.5 Rigid-body motion

When a body $\mathscr{B}$ moves without changing its shape, we shall say that it experiences a *rigid-body motion*[★]; in this situation, neither the distance between any two particles nor the angle between any two lines drawn on $\mathscr{B}$ change, and hence the body does not deform. In the next, we will study the two kinds of rigid-body motion: translation and rotation; also, a formula that describes the rigid-body motion of any body $\mathscr{B}$ will be shown.

3.5.1 Rigid-body translation

In a *rigid-body translation*[★], every particle sustains, at time $t \in T$, the same displacement $\vec{\boldsymbol{c}}(t)$; consequently the motion is described by the equation

$$\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t) = \vec{\boldsymbol{X}} + \vec{\boldsymbol{c}}(t), \tag{3.5}$$

where $\vec{\boldsymbol{c}}(t_0) = \vec{\mathbf{0}}$; in this case, $\vec{\boldsymbol{c}}$ is independent of the material position $\vec{\boldsymbol{X}}$ and only depends on time t.

Notice that in the case of rigid-body translation, all particles $\vec{\boldsymbol{X}} \in \Omega_0$ are subject to an identical displacement $\vec{\boldsymbol{c}}(t)$ at time t, with the same magnitude and direction, that is, $\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t) = \vec{\boldsymbol{c}}(t)$, according to equations (3.3) and (3.4); this is illustrated in Figure 3.3.

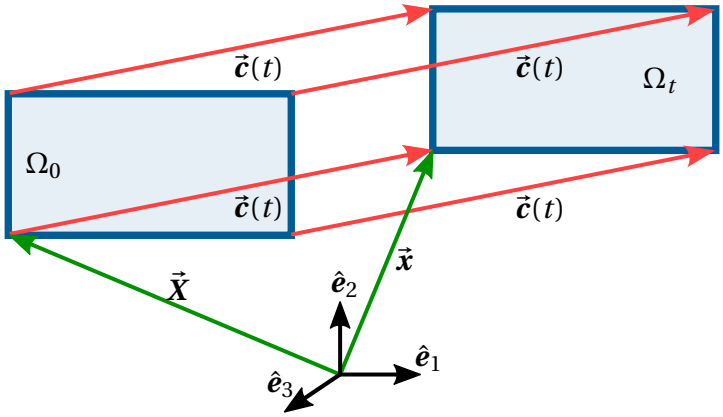


Figure 3.3: *Body that is subject to a rigid-body translation. Observe that all particles are subject to an identical displacement* $\vec{\boldsymbol{c}}(t)$.

3.5.2 Rigid-body rotation

★

Consider a particle located at material position $\vec{\boldsymbol{X}}$ that rotates an angle θ, according to the right-hand rule, about an axis of rotation, that passes through the origin of coordinates $O \equiv \vec{\mathbf{0}}$, described by the unit vector $\hat{\boldsymbol{n}} = \left[\hat{n}_x,\ \hat{n}_y,\ \hat{n}_z\right]^T$; here both, $\hat{\boldsymbol{n}}$ and θ may depend on time t. Let us calculate the spatial position $\vec{\boldsymbol{x}}$ for the rotated $\vec{\boldsymbol{X}}$.

With reference to Figure 3.4, let $\overrightarrow{NC}$ be a vector that is obtained by rotating $\overrightarrow{NA}$ an angle of 90° around $\hat{\boldsymbol{n}}$ according to the right-hand side rule, and since it is orthogonal to both $\hat{\boldsymbol{n}}$ and $\overrightarrow{NA}$, it follows that[2] $\overrightarrow{NC} = \hat{\boldsymbol{n}} \times \overrightarrow{NA}$ and that $\left\|\overrightarrow{NA}\right\| = \left\|\overrightarrow{NC}\right\|$; also let $\hat{\boldsymbol{e}}_{x'}$ and $\hat{\boldsymbol{e}}_{y'}$ be equal to the normalized $\overrightarrow{NA}$ and $\overrightarrow{NC}$ vectors, that is,

$$\hat{\boldsymbol{e}}_{x'} = \frac{\overrightarrow{NA}}{\left\|\overrightarrow{NA}\right\|} \qquad \hat{\boldsymbol{e}}_{y'} = \frac{\overrightarrow{NC}}{\left\|\overrightarrow{NC}\right\|}. \tag{3.6}$$

[2] To show this, use the fact that the cross product between vectors $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$ can be defined as $\vec{\boldsymbol{a}} \times \vec{\boldsymbol{b}} = \|\vec{\boldsymbol{a}}\|\|\vec{\boldsymbol{b}}\| \sin\theta\, \hat{\boldsymbol{m}}$, where $\hat{\boldsymbol{m}}$ is the unit vector orthogonal to vectors $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$, with its direction determined by the right-hand rule, and θ represents the smaller angle between vectors $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$.

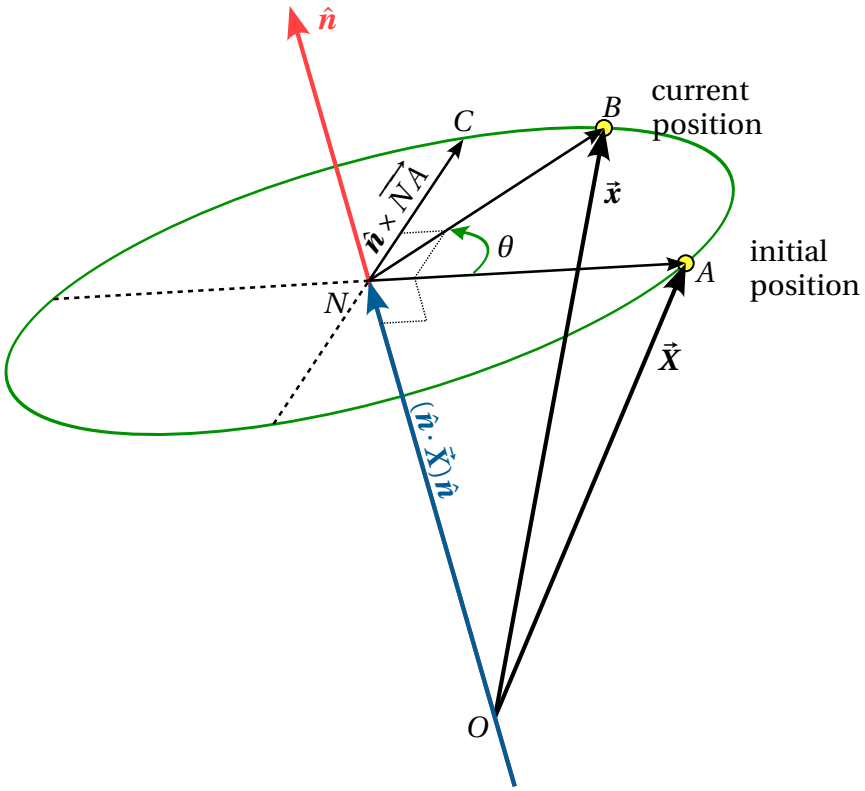


Figure 3.4: *Rodrigues' rotation formula is useful for rotating a vector $\vec{\boldsymbol{X}}$ around an axis $\hat{\boldsymbol{n}}$ an angle θ; the resulting vector $\vec{\boldsymbol{x}}$ is given by equation (3.10).*

Note that

$$\vec{\boldsymbol{x}} = \overrightarrow{ON} + \overrightarrow{NB}; \tag{3.7}$$

here, the vector $\overrightarrow{ON}$ is the projection[3] of $\vec{\boldsymbol{X}} = \overrightarrow{OA}$ over $\hat{\boldsymbol{n}}$, that is,

$$\overrightarrow{ON} = \text{proj}\,\vec{\boldsymbol{X}}/\hat{\boldsymbol{n}} = (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}}; \tag{3.8}$$

also,

$$\overrightarrow{NB} = \left\|\overrightarrow{NB}\right\| \cos\theta \hat{\boldsymbol{e}}_{x'} + \left\|\overrightarrow{NB}\right\| \sin\theta \hat{\boldsymbol{e}}_{y'};$$

since $\overrightarrow{NB}$ and $\overrightarrow{NC}$ are obtained by the rotation of $\overrightarrow{NA}$ around $\hat{\boldsymbol{n}}$, then the length of $\overrightarrow{NB}$ and $\overrightarrow{NC}$ are equal to that of $\overrightarrow{NA}$, that is, $\left\|\overrightarrow{NA}\right\| = \left\|\overrightarrow{NB}\right\| = \left\|\overrightarrow{NC}\right\|$; therefore, by using (3.6), $\overrightarrow{NB}$ can be expressed as

$$\overrightarrow{NB} = \left\|\overrightarrow{NA}\right\| \cos\theta \hat{\boldsymbol{e}}_{x'} + \left\|\overrightarrow{NC}\right\| \sin\theta \hat{\boldsymbol{e}}_{y'} = \overrightarrow{NA}\cos\theta + \overrightarrow{NC}\sin\theta;$$

now, $\overrightarrow{NA} = \overrightarrow{OA} - \overrightarrow{ON} = \vec{\boldsymbol{X}} - (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}}$ and $\overrightarrow{NC} = \hat{\boldsymbol{n}} \times \overrightarrow{NA} = \hat{\boldsymbol{n}} \times \left(\vec{\boldsymbol{X}} - (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}}\right) = \hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}}$ (since $\hat{\boldsymbol{n}} \times \alpha\hat{\boldsymbol{n}} = \vec{\boldsymbol{0}}$, for any scalar α); therefore,

$$\overrightarrow{NB} = \left(\vec{\boldsymbol{X}} - (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}}\right)\cos\theta + \left(\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}}\right)\sin\theta; \tag{3.9}$$

[3] See Appendix A.4, for a review of the concept of the projection of a vector over another vector.

by replacing (3.8) and (3.9) into (3.7), it follows that

$$\vec{\boldsymbol{x}} = (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}} + \big(\vec{\boldsymbol{X}} - (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}}\big)\cos\theta + \big(\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}}\big)\sin\theta$$

and after reorganizing terms, we obtain the so-called *Rodrigues' rotation formula*,[★] named after the french mathematician Benjamin Olinde Rodrigues (1795 – 1851):

$$\boxed{\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{X}}\cos\theta(t) + \big(\hat{\boldsymbol{n}}(t) \times \vec{\boldsymbol{X}}\big)\sin\theta(t) + \big(\hat{\boldsymbol{n}}(t) \cdot \vec{\boldsymbol{X}}\big)\hat{\boldsymbol{n}}(t)\big[1 - \cos\theta(t)\big];} \tag{3.10}$$

this equation allows to estimate the rotation of a point $\vec{\boldsymbol{X}}$ given an axis $\hat{\boldsymbol{n}}$ and an angle of rotation θ that may vary over time.

Let us remember that the cross product $\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}}$ can be expressed as the matrix product

$$\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}} = \boldsymbol{K}\vec{\boldsymbol{X}} \tag{3.11}$$

where $\boldsymbol{K}$ is the skew-symmetric matrix whose axial vector is $\hat{\boldsymbol{n}}$:

$$\boldsymbol{K} = \begin{pmatrix} 0 & -\hat{n}_z & \hat{n}_y \\ \hat{n}_z & 0 & -\hat{n}_x \\ -\hat{n}_y & \hat{n}_x & 0 \end{pmatrix}; \tag{3.12}$$

sometimes $\boldsymbol{K}$ is denoted as $[\hat{\boldsymbol{n}}]_\times$ in the literature.

A well documented identity is the vector triple product formula:[4] given three arbitrary vectors $\vec{\boldsymbol{a}}$, $\vec{\boldsymbol{b}}$, and $\vec{\boldsymbol{c}}$ in $\mathbb{R}^3$, this equation allows to expand $\vec{\boldsymbol{a}} \times (\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}})$ as an expression involving dot products:

$$\vec{\boldsymbol{a}} \times (\vec{\boldsymbol{b}} \times \vec{\boldsymbol{c}}) = (\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{c}})\vec{\boldsymbol{b}} - (\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}})\vec{\boldsymbol{c}};$$

making the substitutions $\vec{\boldsymbol{a}} = \vec{\boldsymbol{b}} = \hat{\boldsymbol{n}}$ and $\vec{\boldsymbol{c}} = \vec{\boldsymbol{X}}$, we arrive to

$$\hat{\boldsymbol{n}} \times (\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}}) = (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}} - \underbrace{(\hat{\boldsymbol{n}} \cdot \hat{\boldsymbol{n}})}_{=1}\vec{\boldsymbol{X}} = (\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}} - \vec{\boldsymbol{X}},$$

and using equation (3.11), it follows that

$$(\hat{\boldsymbol{n}} \cdot \vec{\boldsymbol{X}})\hat{\boldsymbol{n}} = \hat{\boldsymbol{n}} \times (\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}}) + \vec{\boldsymbol{X}} = \vec{\boldsymbol{K}}(\vec{\boldsymbol{K}}\vec{\boldsymbol{X}}) + \vec{\boldsymbol{X}} = \vec{\boldsymbol{K}}^2\vec{\boldsymbol{X}} + \vec{\boldsymbol{X}}.$$

In conclusion, the Rodrigues' rotation formula (3.10) can be written as the product

$$\vec{\boldsymbol{x}} = \vec{\boldsymbol{X}}\cos\theta + \underbrace{(\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}})}_{\boldsymbol{K}\vec{\boldsymbol{X}}}\sin\theta + \boldsymbol{K}^2\vec{\boldsymbol{X}}(1 - \cos\theta) + \vec{\boldsymbol{X}}(1 - \cos\theta)$$

[4] See, for example, https://en.wikipedia.org/wiki/Triple_product.

$$= \boldsymbol{I}\vec{\boldsymbol{X}} + \boldsymbol{K}\vec{\boldsymbol{X}}\sin\theta + \boldsymbol{K}^2\vec{\boldsymbol{X}}(1-\cos\theta)$$
$$= \boldsymbol{R}_\theta\vec{\boldsymbol{X}}.$$

where $\boldsymbol{I}$ is the 3×3 identity matrix and $\boldsymbol{R}_\theta$ is defined as:

$$\boldsymbol{R}_\theta(t) = \boldsymbol{I} + \big(\sin\theta(t)\big)\boldsymbol{K}(t) + \big(1-\cos\theta(t)\big)\boldsymbol{K}(t)^2 \tag{3.13}$$

where $\theta(t_0) = 0$, and consequently, $\boldsymbol{R}_\theta(t_0) = \boldsymbol{R}_0 = \boldsymbol{I}$. The matrix $\boldsymbol{R}_\theta$ (or simply $\boldsymbol{R}$ when the angle θ is clear from the context) is known as a *rotation matrix*[★] or a *proper orthogonal matrix*[★] (see Appendix A.9). It possesses several key properties:

- Rotation composition: $\boldsymbol{R}_\theta(t)\boldsymbol{R}_\phi(t) = \boldsymbol{R}_\phi(t)\boldsymbol{R}_\theta(t) = \boldsymbol{R}_{\theta+\phi}(t)$; this property means that if a body rotates θ degrees and then ϕ degrees, the total rotation of the body is $\theta+\phi$ degrees. It demonstrates that successive rotations can be combined into a single rotation.
- Inverse rotation: $\boldsymbol{R}_\theta\boldsymbol{R}_{-\theta} = \boldsymbol{R}_{\theta+(-\theta)} = \boldsymbol{R}_0 = \boldsymbol{I}$; this property illustrates that if a body rotates θ degrees and then $-\theta$ degrees, no net rotation occurs. In other words, rotating by an angle and then by its negative results in the original orientation.
- It is orthogonal, meaning $\boldsymbol{R}^T = \boldsymbol{R}^{-1}$; this implies that $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{R}\boldsymbol{R}^T = \boldsymbol{I}$. Orthogonality ensures that rotations preserve distances and angles between vectors, maintaining the rigidity of objects during transformation (this will be shown in Section 3.5.4).
- Its determinant is $+1$, i.e., $\det\boldsymbol{R} = +1$; this property distinguishes rotation matrices from *reflection matrices*[★], which are orthogonal matrices that have a determinant of -1.
- Its trace is $\operatorname{tr}\boldsymbol{R} = 2\cos\theta + 1$; this value encodes information about the angle of rotation θ.
- The eigenvalues of $\boldsymbol{R}$ are $+1$, and the complex conjugates $\exp(\pm i\theta) = \cos\theta \pm i\sin\theta$, where θ is the angle of rotation and i is the imaginary unit (i.e., $i^2 = -1$). The eigenvector corresponding to the eigenvalue $+1$ is the axis of rotation associated to $\boldsymbol{R}$.
- The axis of rotation $\hat{\boldsymbol{n}}$ of $\boldsymbol{R}$ can also be calculated from the relation:

$$2\sin\theta\,\hat{\boldsymbol{n}} = \begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix}.$$

  where, R_{ij} denotes the (i,j)-th element of $\boldsymbol{R}$.

Let us prove the last five properties. The proof of the first property is left as an exercise to the reader.

Example 3.1. *Determinant of the rotation matrix* $\boldsymbol{R}$

Demonstrate that the determinant of the rotation matrix $\boldsymbol{R}$ is equal to $+1$, that is $\det \boldsymbol{R} = +1$.

Solution:

To verify this property, we use Python and symbolic computation with SymPy:

```
from sympy import symbols, eye, Matrix, sin, cos, expand, det, simplify

# Initialize the variables and associate them the respective symbols
nx, ny, nz, t = symbols('n_x, n_y, n_z, theta')

# Construct the identity matrix
I = eye(3)                              # 3x3 identity matrix

# Construct the skew-symmetric matrix K
K = Matrix([[  0,  -nz,   ny],
            [ nz,    0,  -nx],          # equation (3.12)
            [-ny,   nx,    0]])

# Compute the rotation matrix R using Rodrigues' formula
R = I + sin(t)*K + (1 - cos(t))*K**2   # equation (3.13)

# Propose an equation for the determinant of R, eq. (3.14)
det_R = 1 + (nx**2 + ny**2 + nz**2)**2 * (1 - cos(t))**2 \
          + (nx**2 + ny**2 + nz**2) * (sin(t)**2 + 2*cos(t) - 2)

# Check if "det_R" is equivalent to det(R)
print(expand(det(R) - det_R))          # prints 0
```

This code demonstrates that:

$$\det \boldsymbol{R} = 1 + \left(\hat{n}_x^2 + \hat{n}_y^2 + \hat{n}_z^2\right)^2 (1-\cos\theta)^2 + \left(\hat{n}_x^2 + \hat{n}_y^2 + \hat{n}_z^2\right)\left(\sin^2\theta + 2\cos\theta - 2\right) \tag{3.14}$$

since $\hat{\boldsymbol{n}}$ is a unit vector, we know that $\hat{n}_x^2 + \hat{n}_y^2 + \hat{n}_z^2 = 1$; thus,

$$\det \boldsymbol{R} = 1 + (1-\cos\theta)^2 + \left(\sin^2\theta + 2\cos\theta - 2\right).$$

Using Python we can evaluate this expression:

```
# Verify the trigonometric identity
print(simplify(1 + (1 - cos(t))**2 + sin(t)**2 + 2*cos(t) - 2))  # prints 1
```

Consequently, the determinant of the rotation matrix $\boldsymbol{R}$ is equal to $+1$, i.e., $\det \boldsymbol{R} = +1$.

Now, let's examine the orthogonality property of rotation matrices.

Example 3.2. *Orthogonality of the rotation matrix* $\boldsymbol{R}$

Prove that the rotation matrix $\boldsymbol{R}$ is orthogonal.

Solution:

We can extend our previous Python code to demonstrate this property:

```
# Compute R(+t) and R(-t)
R_pt = I + sin(+t)*K + (1 - cos(+t))*K**2   # equation (3.13), for R(+t)
R_mt = I + sin(-t)*K + (1 - cos(-t))*K**2   # equation (3.13), for R(-t)

# Verify that R(+t).T == R(-t)
print((R_pt.T - R_mt).is_zero_matrix)       # prints "True"
```

This code verifies that $\boldsymbol{R}_\theta^T = \boldsymbol{R}_{-\theta}$. To prove orthogonality, from the inverse rotation property, $\boldsymbol{R}_\theta \boldsymbol{R}_{-\theta} = \boldsymbol{I}$, we can deduce that $\boldsymbol{R}_\theta^{-1} = \boldsymbol{R}_{-\theta}$. Combining this with our initial observation that $\boldsymbol{R}_\theta^T = \boldsymbol{R}_{-\theta}$, we conclude that $\boldsymbol{R}_\theta^T = \boldsymbol{R}_\theta^{-1}$, which is the defining property of an orthogonal matrix. Therefore, $\boldsymbol{R}_\theta$ is indeed orthogonal.

An important consequence of orthogonality is that rotations preserve distances and angles between vectors, which is important for maintaining the rigidity of objects during rotation; this will be illustrated in Section 3.5.4.

Example 3.3. *Trace of the rotation matrix* $\boldsymbol{R}$

Demonstrate that the trace of the rotation matrix $\boldsymbol{R}$ is equal to $2\cos\theta + 1$, where θ is the angle of rotation.

Solution:

Recall that the trace of a matrix is the sum of its diagonal elements, that is, $\operatorname{tr}\boldsymbol{R} = R_{11} + R_{22} + R_{33}$, where R_{ii} are the diagonal elements of $\boldsymbol{R}$. Running the following Python code (it continues from the last example):

```
from sympy import trace

# Propose an equation for the trace(R), eq. (3.15)
trace_R = 2*(nx**2 + ny**2 + nz**2)*(cos(t) - 1) + 3

# Check if "trace_R" is equivalent to trace(R)
print(expand(trace(R) - trace_R))   # prints 0
```

we can confirm that:

$$\operatorname{tr}\boldsymbol{R} = 2\left(\hat{n}_x^2 + \hat{n}_y^2 + \hat{n}_z^2\right)(\cos\theta - 1) + 3; \tag{3.15}$$

since $\hat{\boldsymbol{n}}$ is a unit vector, we know that $\hat{n}_x^2 + \hat{n}_y^2 + \hat{n}_z^2 = 1$; substituting this result back into the expression (3.15) for $\operatorname{tr}\boldsymbol{R}$, we find:

$$\operatorname{tr}\boldsymbol{R} = 2\cos\theta + 1.$$

This result provides us with a simple way to compute cosine of the rotation angle θ from any rotation matrix:

$$\cos\theta = \frac{\operatorname{tr}\boldsymbol{R} - 1}{2}. \tag{3.16}$$

However, note that this formula alone cannot determine the sign of θ or the axis of rotation $\hat{\boldsymbol{n}}$, as $\cos\theta$ is an even function, that is $\cos(\theta) = \cos(-\theta)$; as a result, formula (3.16) yields the same value for both θ and $-\theta$, leaving the direction of rotation (clockwise or counterclockwise) unresolved. Therefore, the sign of the rotation angle must be estimated using additional information about the rotation, as we will see in the following.

When we have a rotation matrix $\boldsymbol{R}$, but the axis of rotation $\hat{\boldsymbol{n}}$ is unknown, we can determine it through a property of rotation matrices that states that the axis of rotation remains unchanged under the rotation it defines. Mathematically, this means that $\boldsymbol{R}\hat{\boldsymbol{n}} = \hat{\boldsymbol{n}}$; in other words, the rotation axis $\hat{\boldsymbol{n}}$ is the eigenvector of $\boldsymbol{R}$ corresponding to the eigenvalue $+1$.

To illustrate this concept, let us continue with the previous Python code:

```
# Create the vector n
n = Matrix([nx, ny, nz])

# Verify that R*n == n
print(simplify(R*n))      # prints [nx, ny, nz]
```

This code will output $\left[\hat{n}_x,\ \hat{n}_y,\ \hat{n}_z\right]^T$, that is, the vector $\hat{\boldsymbol{n}}$. This result confirms that $\boldsymbol{R}\hat{\boldsymbol{n}} = \hat{\boldsymbol{n}}$, verifying our initial statement.

Now, using the following Python code (continuation):

```
from sympy import roots
lam = symbols('lambda')

# Calculate the characteristic polynomial (det(lambda*I - R))
char_poly = R.charpoly(lam).as_expr()

```

```
# Apply the unit vector constraint: n_x^2 + n_y^2 + n_z^2 = 1
# by substituting n_z^2 = 1 - n_x^2 - n_y^2
char_poly_simplified = simplify(char_poly.subs(nz**2, 1 - nx**2 - ny**2))
print(char_poly_simplified)

# Calculate the eigenvalues of R
eigenvalues = roots(char_poly_simplified, lam)
print(eigenvalues)
```

we are able to find that the characteristic polynomial of the rotation matrix $\boldsymbol{R}$ is:

$$\begin{aligned}\det(\lambda \boldsymbol{I} - \boldsymbol{R}) &= \lambda^3 - (2\cos\theta + 1)\lambda^2 + (2\cos\theta + 1)\lambda - 1 \\ &= (\lambda - 1)(\lambda^2 - 2\lambda\cos\theta + 1).\end{aligned}$$

We can extract the roots of this polynomial to conclude that: *a)* the eigenvalue $+1$ is always present for all rotation matrices (this eigenvalue corresponds to the rotation axis $\hat{\boldsymbol{n}}$); *b)* the other two eigenvalues of $\boldsymbol{R}$ are the complex conjugates, $\exp(\pm i\theta) = \cos\theta \pm i\sin\theta$, where θ is the angle of rotation and i is the imaginary unit (i.e., $i^2 = -1$).

While the eigendecomposition approach is theoretically elegant, it may not be the most computationally efficient method for determining the rotation axis. A more practical alternative is to calculate the axis of rotation using the method explained in the following example.

Example 3.4. *Axis of rotation associated to the the rotation matrix $\boldsymbol{R}$*

Show that the axis of rotation $\hat{\boldsymbol{n}}$ corresponding to the rotation matrix $\boldsymbol{R}$ can be calculated as:

$$\hat{\boldsymbol{n}} = \frac{1}{2\sin\theta}\begin{bmatrix} R_{32} - R_{23} \\ R_{13} - R_{31} \\ R_{21} - R_{12} \end{bmatrix}. \tag{3.17}$$

Here, R_{ij} denotes the (i, j)-th element of $\boldsymbol{R}$. This formula is valid as long as $\theta \neq 0°$ or $180°$, since $\sin\theta = 0$ in these cases, leading to division by zero.

Solution:

To verify this equation, consider the following Python code, which continues from our earlier example:

```
# Define the vector nhat using equation (3.17)
n_ = Matrix([R[2,1] - R[1,2],
             R[0,2] - R[2,0],
             R[1,0] - R[0,1]])/(2*sin(t))

# Check if n_ matches the original nhat defined in line 39
print(n - n_)   # prints [0, 0, 0]
```

The output confirms that the computed vector $\hat{\boldsymbol{n}}$ matches the original axis of rotation $\hat{\boldsymbol{n}}$, validating the formula.

Notice that the norm of vector $[R_{32} - R_{23}, R_{13} - R_{31}, R_{21} - R_{12}]^T$ is equal to $2|\sin\theta|$; this operation provides us with twice the sine of the angle of rotation θ, up to a sign.

It is important to note that equation (3.17) breaks down when $\sin\theta = 0$, i.e., when $\theta = 0°$ or $180°$. In these cases:

- For $\theta = 0°$, no rotation occurs, the rotation matrix reduces to the identity matrix ($\boldsymbol{R} = \boldsymbol{I}$), and no unique axis of rotation exists.
- For $\theta = 180°$, the rotation matrix becomes symmetric, with $\boldsymbol{R} = 2\hat{\boldsymbol{n}}\hat{\boldsymbol{n}}^T - \boldsymbol{I}$ (this demonstration is left as an exercise to the reader – use SymPy), and the axis of rotation must be determined using alternative methods, such as examining the eigenvectors of $\boldsymbol{R}$.

In summary, from equations (3.16) and (3.17) we observe that the diagonal elements of $\boldsymbol{R}$ encode the angle of rotation θ, while the off-diagonal elements provide information about the axis of rotation $\hat{\boldsymbol{n}}$. In the next, we will show an example that illustrates the procedure for calculating these variables from the rotation matrix $\boldsymbol{R}$.

Example 3.5. *Extracting $\hat{\boldsymbol{n}}$ and θ from the rotation matrix $\boldsymbol{R}$*

Create a Python program that illustrates how to extract the axial vector $\hat{\boldsymbol{n}}$ and the rotation angle θ from a given rotation matrix $\boldsymbol{R}$. The code should include handling of special cases where $\theta = 0$ and $\theta = \pi$.

Solution:

Consider a rotation matrix $\boldsymbol{R}$ defined by an axis of rotation $\hat{\boldsymbol{n}} = \left[\hat{n}_x, \hat{n}_y, \hat{n}_z\right]^T$ and a rotation angle θ. Our goal is to compute $\hat{\boldsymbol{n}}$ and θ directly from $\boldsymbol{R}$ using Python. Below is the complete implementation:

```
from numpy import array, eye, sin, cos, deg2rad, rad2deg, arctan2, allclose
from numpy import argmax, nan
from numpy.linalg import norm, det, eig

# Define the rotation angle (in degrees) and convert it to radians
t = deg2rad(-30)

# Define an unnormalized axial vector
n = array([-1, 2, 3])

# Normalize the axial vector to obtain nhat
nx, ny, nz = n/norm(n)
print(f"nhat = [nx, ny, nz] = [{nx}, {ny}, {nz}]")

```

```
15 # Construct the identity matrix
16 I = eye(3)
17
18 # Construct the skew-symmetric matrix K associated with nhat
19 K = array([[  0,  -nz,   ny],
20            [ nz,    0,  -nx],          # equation (3.12)
21            [-ny,   nx,    0]])
22
23 # Compute the rotation matrix R using Rodrigues' formula
24 R = I + sin(t)*K + (1 - cos(t))*(K@K)  # equation (3.13)
25 print("Rotation matrix = \n", R)
26
27 # Verify that R is a valid rotation matrix
28 print(allclose(R.T @ R, I))  # Should print "True"  ->  R is orthogonal
29 print(det(R))                # Should print "+1"    ->  det(R) = +1
30
31 # Extract the axis of rotation and the rotation angle, using eq. (3.17)
32 axis_n = array([R[2,1] - R[1,2],   R[0,2] - R[2,0],   R[1,0] - R[0,1]])
33 two_sin_t = norm(axis_n)
34
35 # Handle the case where theta is not 0° or 180°
36 if abs(two_sin_t) > 1e-10:                    # Avoid division by zero
37     n_ = axis_n/two_sin_t                     # Equation (3.17)
38     sin_t = two_sin_t/2
39     cos_t = (R[0,0] + R[1,1] + R[2,2] - 1)/2   # Equation (3.16)
40
41     # The angle is obtained using "arctan2" to resolve the sign ambiguity
42     t_ = rad2deg(arctan2(sin_t, cos_t))
43 else:
44     # Special case: theta = 0° or 180°
45     if allclose(R, I):
46         # Theta = 0°, there is not rotation; so, no unique axis exists
47         n_ = array([nan, nan, nan])
48         t_ = 0
49     else:
50         # Theta = 180°, use eigenvector corresponding to eigenvalue +1
51         eigenvalues, eigenvectors = eig(R)
52         idx = argmax(isclose(eigenvalues, +1.0))
53         n_  = eigenvectors[:, idx].real
54         t_  = 180
55
56 print(f"Calculated nhat  = {n_}")
57 print(f"Calculated theta = {t_}°")
```

Some additional comments are necessary to fully understand the code. In line 24, the command `K@K` calculates $\boldsymbol{K}^2$; this operation could also be performed using the command `numpy.linalg.matrix_power(K, 2)`. Also, lines 28 and 29 verify that $\boldsymbol{R}$ is a proper orthogonal (rotation) matrix by checking that $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}$ and $\det(\boldsymbol{R}) = +1$.

Running the code produces the following output. Line 13 prints the original $\hat{\boldsymbol{n}}$:

```
nhat = [nx, ny, nz] = [-0.26726124, 0.53452248, 0.80178373]
```

Line 25 outputs the associated rotation tensor $\boldsymbol{R}$:

```
Rotation matrix =
[[ 0.87559502  0.38175263 -0.29597008]
 [-0.42003109  0.90430386 -0.07621294]
 [ 0.2385524   0.19104831  0.95215193]]
```

Lines 28 and 29 confirm that $\boldsymbol{R}$ is indeed orthogonal:

```
True
1.0
```

Finally, lines 56 and 57 estimate $\hat{\boldsymbol{n}}$ and θ, respectively:

```
Calculated nhat  = [ 0.26726124 -0.53452248 -0.80178373]
Calculated theta = 30.0°
```

The computed values of $\hat{\boldsymbol{n}}$ and θ match the input parameters but with the opposite signs. Remember that the orientation of $\hat{\boldsymbol{n}}$ indicates how to perform rotation using the right-hand rule. Observe that a rotation of θ degrees around $\hat{\boldsymbol{n}}$ is equivalent to a rotation of $-\theta$ degrees around $-\hat{\boldsymbol{n}}$, so the computed results are consistent with this symmetry. This confirms the correctness of the procedure.



Finally, let's consider a more general case of rigid-body rotation. When the axis of rotation is defined by the unit vector $\hat{\boldsymbol{n}}$ and passes through the point $\tilde{\vec{\boldsymbol{X}}}_0(t)$, equation (3.13) changes to:

$$\vec{\boldsymbol{x}} = \boldsymbol{R}(t)\left(\vec{\boldsymbol{X}} - \tilde{\vec{\boldsymbol{X}}}_0(t)\right); \tag{3.18}$$

this equation describes the *rigid-body rotation* of every particle of the body $\mathscr{B}$ at time t. Bear in mind that rotating the body $\mathscr{B}$ about a given axis $\hat{\boldsymbol{n}}$, when $\tilde{\vec{\boldsymbol{X}}}_0 = \vec{\boldsymbol{0}}$, is equivalent to holding $\mathscr{B}$ fixed and rotating the referential coordinate system about the same axis through the same angle but in the opposite direction. Additionally, the position of $\tilde{\vec{\boldsymbol{X}}}_0(t)$ may vary with time.

3.5.3 General rigid-body motion

Using equations (3.5) and (3.18), any rigid-body motion can be specified as a combination of a rotation about an axis with direction $\hat{\boldsymbol{n}}$ that passes through a point $\tilde{\vec{\boldsymbol{X}}}_0(t)$ and a translation described by $\vec{\boldsymbol{c}}(t)$ as

$$\boxed{\vec{\boldsymbol{x}}(t) = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) = \boldsymbol{R}(t)\left(\vec{\boldsymbol{X}} - \tilde{\vec{\boldsymbol{X}}}_0(t)\right) + \vec{\boldsymbol{c}}(t);} \tag{3.19}$$

this equation determines the *rigid-body motion* that undergoes every material particle $\vec{\boldsymbol{X}} \in \Omega_0$ at time t. Take into account that, in this general case, $\vec{\boldsymbol{c}}(t_0) = \tilde{\vec{\boldsymbol{X}}}_0(t_0)$. Here

$\tilde{\vec{X}}_0(t)$ defines the position of the axis of rotation in the reference configuration, and $\vec{c}(t)$ is the new position of $\tilde{\vec{X}}_0(t)$ after the rigid-body motion; to show this, simply set $\vec{X} = \tilde{\vec{X}}_0$. Note that $\tilde{\vec{X}}_0(t)$ may change with time, as in the case of an rock with an irregular shape rolling down a hill: during flight, the rock rotates about its own center of mass; however, once it makes contact with the ground, the instantaneous center of rotation shifts to the point of contact on its surface.

3.5.4 Rigid-body motion does not change distances and angles

Let us mathematically verify that the distance between two points, namely $\vec{X}_1$ and $\vec{X}_2$ at t_0, does not change as the body rigidly moves. To do this, we will use equation (3.19):

$$\vec{x}_1(t) = \boldsymbol{R}(t)\left(\vec{X}_1 - \tilde{\vec{X}}_0(t)\right) + \vec{c}(t) \qquad \vec{x}_2(t) = \boldsymbol{R}(t)\left(\vec{X}_2 - \tilde{\vec{X}}_0(t)\right) + \vec{c}(t);$$

the vector between these points in the current configuration is:

$$\underbrace{\vec{x}_1(t) - \vec{x}_2(t)}_{\Delta\vec{x}(t)} = \boldsymbol{R}(t)\underbrace{\left(\vec{X}_1 - \vec{X}_2\right)}_{\Delta\vec{X}}.$$

Notice that

$$\|\Delta\vec{x}(t)\|^2 = \Delta\vec{x}(t)^T\Delta\vec{x}(t) = \Delta\vec{X}^T\underbrace{\boldsymbol{R}(t)^T\boldsymbol{R}(t)}_{=\boldsymbol{I}}\Delta\vec{X} = \Delta\vec{X}^T\Delta\vec{X} = \|\Delta\vec{X}\|^2;$$

in conclusion, since $\|\Delta\vec{x}(t)\| = \|\Delta\vec{X}\|$, the distance between any two points, for all t, remains unchanged during the rigid-body motion.

Now, let us show that in rigid-body motion the angle between two lines drawn on Ω_0 does not change in rigid-body motion. Consider two lines defined by points 1 and 2, and points 3 and 2, as shown in Figure 3.5. The angle between these lines is related to the dot product:

$$\begin{aligned}\left(\vec{x}_1(t) - \vec{x}_2(t)\right)\cdot\left(\vec{x}_3(t) - \vec{x}_2(t)\right) &= \left(\boldsymbol{R}(t)\left(\vec{X}_1 - \vec{X}_2\right)\right)^T\left(\boldsymbol{R}(t)\left(\vec{X}_3 - \vec{X}_2\right)\right)\\ &= \left(\vec{X}_1 - \vec{X}_2\right)^T\underbrace{\boldsymbol{R}(t)^T\boldsymbol{R}(t)}_{=\boldsymbol{I}}\left(\vec{X}_3 - \vec{X}_2\right)\\ &= \left(\vec{X}_1 - \vec{X}_2\right)\cdot\left(\vec{X}_3 - \vec{X}_2\right);\end{aligned}$$

since inner product satisfies the equality $\left(\vec{x}_1(t) - \vec{x}_2(t)\right)\cdot\left(\vec{x}_3(t) - \vec{x}_2(t)\right) = (\vec{X}_1 - \vec{X}_2)\cdot(\vec{X}_3 - \vec{X}_2)$, and the distances between points stay constant, that is $\left\|\vec{x}_1(t) - \vec{x}_2(t)\right\| = \left\|\vec{X}_1 - \vec{X}_2\right\|$ and $\left\|\vec{x}_2(t) - \vec{x}_3(t)\right\| = \left\|\vec{X}_2 - \vec{X}_3\right\|$, the angle between any two lines remains unchanged, for all t, during rigid-body motion.

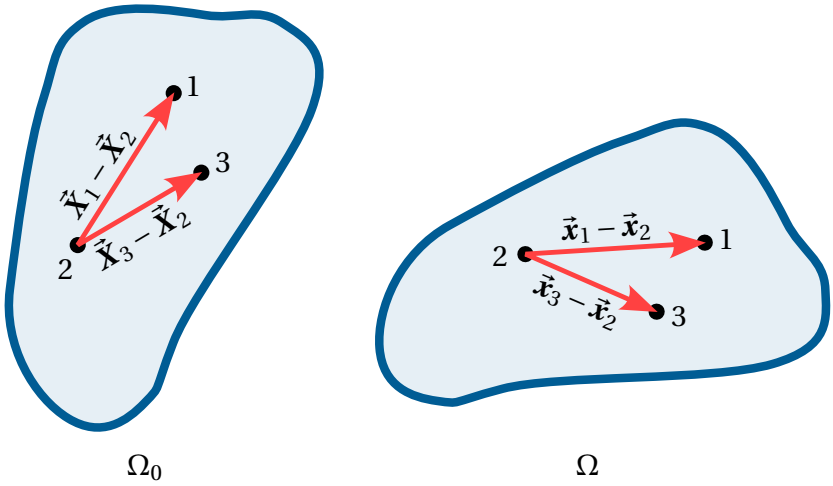


Figure 3.5: *Illustration of rigid body motion preserving distances and angles. The left figure shows the body in its reference configuration Ω_0 with points 1, 2, and 3. The right figure shows the same body after rigid body motion in the current configuration Ω. Note that the relative positions and angles between the points are preserved, demonstrating key properties of rigid body motion.*

3.5.5 Rigid-body motion in the two-dimensional case

In order to describe the rigid-body motion for the two-dimensional case, we will suppose that the body rotates counterclockwise over an angle θ about an axis with direction $\hat{\boldsymbol{k}}$, passing through a point $[X_0, Y_0, 0]^T$ and experiencing a translation defined by $[C_x, C_y, 0]^T$.

Using the following Python code:

```
from sympy import symbols, Matrix, eye, sin, cos

# Initialize the variables and associate them the respective symbols
t = symbols('theta')

# Define the axial vector nhat
nx, ny, nz = [0, 0, 1]                  # = \hat{k}

# Construct the identity matrix
I = eye(3)                              # 3x3 identity matrix

# Construct the skew-symmetric matrix K associated with nhat
K = Matrix([[  0,  -nz,   ny],
            [ nz,    0,  -nx],          # equation (3.12)
            [-ny,   nx,    0]])
print("K = ", K)

# Compute the rotation matrix R using Rodrigues' formula
R = I + sin(t)*K + (1 - cos(t))*K**2    # equation (3.13)
print("R = ", R)
```

it is possible to show that, in this case,

$$\boldsymbol{K} = \begin{pmatrix} 0 & -1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 0 \end{pmatrix}$$

and that

$$\boldsymbol{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix};$$

this matrix is the usual rotation matrix for the two-dimensional case. Therefore, equation (3.19) particularizes to:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} X - X_0 \\ Y - Y_0 \\ 0 \end{pmatrix} + \begin{pmatrix} C_x \\ C_y \\ 0 \end{pmatrix} = \begin{pmatrix} (X - X_0)\cos\theta - (Y - Y_0)\sin\theta + C_x \\ (X - X_0)\sin\theta + (Y - Y_0)\cos\theta + C_y \\ 0 \end{pmatrix}. \tag{3.20}$$

In the two-dimensional case, (X_0, Y_0) become the coordinates of the rotation center, and (C_x, C_y) are the new coordinates of the rotation center (X_0, Y_0) after the translation.

Let us illustrate this result with a programming example.

Example 3.6. *Rigid-body motion in the two-dimensional case*

Consider a triangle with vertices at $(3,1)$, $(5,1)$, $(5,2)$ in the reference configuration. This triangle undergoes a rigid-body motion consisting of a rotation of 30° counterclockwise around the point $(X_0, Y_0) = (3,1)$ and a translation that moves the point (X_0, Y_0) to the new position $(C_x, C_y) = (6,2)$. Depict this transformation using Python.

Solution:

We can implement this rigid-body motion using the following Python code:

```
from matplotlib.pyplot import figure, fill, axis, show
from numpy import array, deg2rad, sin, cos

# Coordinates of the triangle vertices in the reference configuration
XY  = array([[3, 1], [5, 1], [5, 2]])

# Separate the X and Y coordinates
X, Y = XY[:,0], XY[:,1]

# Coordinates of the rotation center in the reference configuration
X0, Y0 = [3, 1]

# New coordinates of the rotation center in the current configuration
```

```
Cx, Cy = [6, 2]

# Rotation angle in radians
t = deg2rad(30)    # convert 30° to radians

# Calculate the rigid body displacement in the current configuration
x = (X - X0)*cos(t) - (Y - Y0)*sin(t) + Cx  # equation (3.20)
y = (X - X0)*sin(t) + (Y - Y0)*cos(t) + Cy

# Plot the undeformed and deformed triangles
figure()
fill(X, Y, edgecolor='b', linestyle='-',  fill=False) # undeformed
fill(x, y, edgecolor='r', linestyle='--', fill=False) # deformed
axis('equal')
show()
```

This code implements the rigid-body motion using equation (3.20). The resulting visualization is shown in Figure 3.6.

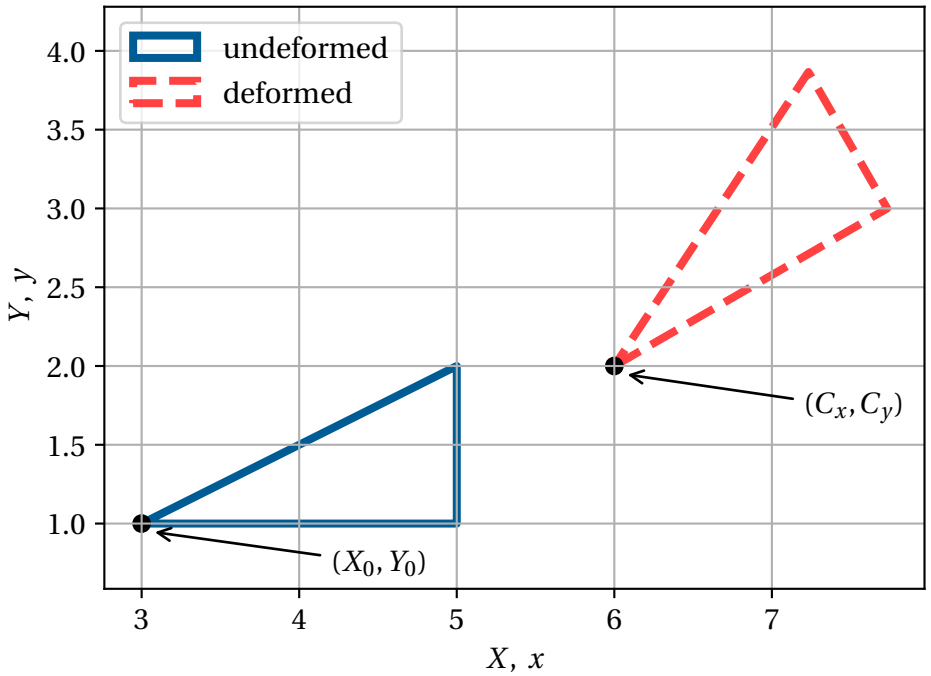


Figure 3.6: *Rigid body motion of a triangle. The blue solid line represents the triangle in its reference configuration, while the red dashed line shows the triangle after rotation and translation. The dot at (X_0, Y_0) indicates the center of rotation in the reference configuration, and the dot at (C_x, C_y) shows its new position after the transformation.*

3.6 Spatial and material fields and scalar, vector and tensor fields

Scalar, vector, and tensor fields play a very important role in describing and analyzing the behavior of continuous media, such as solids, fluids, and gases. These fields represent various physical quantities and properties associated with the material under consideration.

A *material field* is one in which the independent variables are the initial position $\vec{\boldsymbol{X}}$ and time t. In contrast, a *spatial field* is one in which the independent variables are the current position $\vec{\boldsymbol{x}}$ and time t.

Material fields can represent scalar, vector, or tensor quantities and are represented as:

Scalar material field: $H := H(\vec{\boldsymbol{X}}, t)$, where $H : \Omega_0 \times T \to \mathbb{R}$ maps the initial configuration Ω_0 to a real number.

Vector material field: $\vec{\boldsymbol{H}} := \vec{\boldsymbol{H}}(\vec{\boldsymbol{X}}, t)$, where $\vec{\boldsymbol{H}} : \Omega_0 \times T \to \mathbb{R}^3$ maps the initial configuration Ω_0 to a vector in $\mathbb{R}^3$.

Tensor material field: $\boldsymbol{H} := \boldsymbol{H}(\vec{\boldsymbol{X}}, t)$, where $\boldsymbol{H} : \Omega_0 \times T \to \mathbb{R}^{3\times 3}$ maps the initial configuration Ω_0 to a tensor in $\mathbb{R}^{3\times 3}$.

Similarly, spatial fields can represent scalar, vector, or tensor quantities and are represented as:

Scalar spatial field: $h := h(\vec{\boldsymbol{x}}, t)$, where $h : \Omega \times T \to \mathbb{R}$ maps the current configuration Ω to a real number.

Vector spatial field: $\vec{\boldsymbol{h}} := \vec{\boldsymbol{h}}(\vec{\boldsymbol{x}}, t)$, where $\vec{\boldsymbol{h}} : \Omega \times T \to \mathbb{R}^3$ maps the current configuration Ω to a vector in $\mathbb{R}^3$.

Tensor spatial field: $\boldsymbol{h} := \boldsymbol{h}(\vec{\boldsymbol{x}}, t)$, where $\boldsymbol{h} : \Omega \times T \to \mathbb{R}^{3\times 3}$ maps the current configuration Ω to a tensor in $\mathbb{R}^{3\times 3}$.

Examples of scalar fields include temperature, density, pressure, chemical concentration, and other quantities that can be represented by a single numerical value at each point in space and time. Examples of vector fields include force, displacement, velocity, acceleration, and other quantities that require both magnitude and direction for their representation. Finally, examples of tensor fields include stress and strain tensors.

The choice between using a material field or a spatial field depends on the specific problem and the desired approach to analyzing the behavior of the continuous medium. Material fields are useful when tracking the evolution of properties associated with specific material particles, while spatial fields are more suitable when the

focus is on how properties vary at fixed spatial locations as different material particles pass through them.

3.7 Material and spatial gradients

In the discussion that follows, it will be necessary to take the derivative of scalar and vector fields with respect to position ($\vec{\boldsymbol{X}}$ or $\vec{\boldsymbol{x}}$) or time t. In this section, we will analyze the former case, while the latter situation will be analyzed in Section 3.21.

Firstly, let us consider the scalar fields; in this case, the derivative of the scalar material field H with respect to the initial position $\vec{\boldsymbol{X}}$, is defined as

$$\text{Grad}\, H(\vec{\boldsymbol{X}},t) = \nabla_{\vec{\boldsymbol{X}}} H(\vec{\boldsymbol{X}},t) = \frac{\partial H(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}} = \begin{pmatrix} \frac{\partial H(\vec{\boldsymbol{X}},t)}{\partial X_1} \\ \frac{\partial H(\vec{\boldsymbol{X}},t)}{\partial X_2} \\ \frac{\partial H(\vec{\boldsymbol{X}},t)}{\partial X_3} \end{pmatrix} = \begin{pmatrix} \frac{\partial H(\vec{\boldsymbol{X}},t)}{\partial X} \\ \frac{\partial H(\vec{\boldsymbol{X}},t)}{\partial Y} \\ \frac{\partial H(\vec{\boldsymbol{X}},t)}{\partial Z} \end{pmatrix}$$

(Grad with uppercase G), and is referred to as the *material gradient of a scalar material field* H, at a fixed time t. Similarly, the derivative of a scalar spatial field h with respect to the current position $\vec{\boldsymbol{x}}$, that is,

$$\text{grad}\, h(\vec{\boldsymbol{x}},t) = \nabla_{\vec{\boldsymbol{x}}} h(\vec{\boldsymbol{x}},t) = \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}} = \begin{pmatrix} \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial x_1} \\ \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial x_2} \\ \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial x_3} \end{pmatrix} = \begin{pmatrix} \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial x} \\ \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial y} \\ \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial z} \end{pmatrix}$$

(grad with lowercase g), is referred to as the *spatial gradient of a scalar spatial field* h, at a fixed time t.

Now, let us consider a vector field; in this case, the derivative of the vector material field $\vec{\boldsymbol{H}} = [H_1, H_2, H_3]^T = [H_X, H_Y, H_Z]^T$ with respect to the initial position $\vec{\boldsymbol{X}}$, namely,

$$\begin{aligned} \text{Grad}\, \vec{\boldsymbol{H}}(\vec{\boldsymbol{X}},t) = \nabla_{\vec{\boldsymbol{X}}} \vec{\boldsymbol{H}}(\vec{\boldsymbol{X}},t) &= \frac{\partial \vec{\boldsymbol{H}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}} \\ &= \begin{pmatrix} \frac{\partial H_1(\vec{\boldsymbol{X}},t)}{\partial X_1} & \frac{\partial H_1(\vec{\boldsymbol{X}},t)}{\partial X_2} & \frac{\partial H_1(\vec{\boldsymbol{X}},t)}{\partial X_3} \\ \frac{\partial H_2(\vec{\boldsymbol{X}},t)}{\partial X_1} & \frac{\partial H_2(\vec{\boldsymbol{X}},t)}{\partial X_2} & \frac{\partial H_2(\vec{\boldsymbol{X}},t)}{\partial X_3} \\ \frac{\partial H_3(\vec{\boldsymbol{X}},t)}{\partial X_1} & \frac{\partial H_3(\vec{\boldsymbol{X}},t)}{\partial X_2} & \frac{\partial H_3(\vec{\boldsymbol{X}},t)}{\partial X_3} \end{pmatrix} = \begin{pmatrix} \frac{\partial H_X(\vec{\boldsymbol{X}},t)}{\partial X} & \frac{\partial H_X(\vec{\boldsymbol{X}},t)}{\partial Y} & \frac{\partial H_X(\vec{\boldsymbol{X}},t)}{\partial Z} \\ \frac{\partial H_Y(\vec{\boldsymbol{X}},t)}{\partial X} & \frac{\partial H_Y(\vec{\boldsymbol{X}},t)}{\partial Y} & \frac{\partial H_Y(\vec{\boldsymbol{X}},t)}{\partial Z} \\ \frac{\partial H_Z(\vec{\boldsymbol{X}},t)}{\partial X} & \frac{\partial H_Z(\vec{\boldsymbol{X}},t)}{\partial Y} & \frac{\partial H_Z(\vec{\boldsymbol{X}},t)}{\partial Z} \end{pmatrix} \end{aligned} \tag{3.21}$$

(Grad with uppercase G), is referred to as the *material gradient of a vector material field* $\vec{\boldsymbol{H}}$, at a fixed time t. Similarly, the derivative of a vector spatial field $\vec{\boldsymbol{h}} = [h_1, h_2, h_3]^T = \left[h_x, h_y, h_z\right]^T$ with respect to the current position $\vec{\boldsymbol{x}}$, defined as,

$$\begin{aligned}\operatorname{grad}\vec{\boldsymbol{h}}(\vec{\boldsymbol{x}},t) = \nabla_{\vec{\boldsymbol{x}}}\vec{\boldsymbol{h}}(\vec{\boldsymbol{x}},t) &= \frac{\partial\vec{\boldsymbol{h}}(\vec{\boldsymbol{x}},t)}{\partial\vec{\boldsymbol{x}}} \\ &= \begin{pmatrix} \frac{\partial h_1(\vec{\boldsymbol{x}},t)}{\partial x_1} & \frac{\partial h_1(\vec{\boldsymbol{x}},t)}{\partial x_2} & \frac{\partial h_1(\vec{\boldsymbol{x}},t)}{\partial x_3} \\ \frac{\partial h_2(\vec{\boldsymbol{x}},t)}{\partial x_1} & \frac{\partial h_2(\vec{\boldsymbol{x}},t)}{\partial x_2} & \frac{\partial h_2(\vec{\boldsymbol{x}},t)}{\partial x_3} \\ \frac{\partial h_3(\vec{\boldsymbol{x}},t)}{\partial x_1} & \frac{\partial h_3(\vec{\boldsymbol{x}},t)}{\partial x_2} & \frac{\partial h_3(\vec{\boldsymbol{x}},t)}{\partial x_3} \end{pmatrix} = \begin{pmatrix} \frac{\partial h_x(\vec{\boldsymbol{x}},t)}{\partial x} & \frac{\partial h_x(\vec{\boldsymbol{x}},t)}{\partial y} & \frac{\partial h_x(\vec{\boldsymbol{x}},t)}{\partial z} \\ \frac{\partial h_y(\vec{\boldsymbol{x}},t)}{\partial x} & \frac{\partial h_y(\vec{\boldsymbol{x}},t)}{\partial y} & \frac{\partial h_y(\vec{\boldsymbol{x}},t)}{\partial z} \\ \frac{\partial h_z(\vec{\boldsymbol{x}},t)}{\partial x} & \frac{\partial h_z(\vec{\boldsymbol{x}},t)}{\partial y} & \frac{\partial h_z(\vec{\boldsymbol{x}},t)}{\partial z} \end{pmatrix}\end{aligned} \tag{3.22}$$

(grad with lowercase g), is referred to as the *spatial gradient of a vector spatial field* $\vec{\boldsymbol{h}}$, at a fixed time t.

Note that the gradients of the vector fields (3.21) and (3.22) are Jacobian matrices (see Appendix A.7).

Finally, it is possible to define material and spatial gradients of tensor fields, which requires the use of third-order tensors. However, we will not pursue this topic in this text.

3.8 Deformation gradient

When a body undergoes deformation, it experiences changes in shape and/or size, as well as changes in the angles between lines and distances between points within the body. In this section, we will focus on studying the change of distances between points using a quantity called the deformation gradient. This is a key concept in the study of finite deformation, which deals with large deformations where the underlying geometry is significantly altered.

The deformation gradient provides a mathematical description of how an infinitesimal line segment within the body is transformed by the deformation process. By analyzing the deformation gradient, we can quantify the stretching, rotation, and distortion of line, area and volume elements within the deforming body. This information is essential for understanding the kinematics of finite deformation and developing constitutive models that relate deformation to the internal stresses and strains.

With reference to Figure 3.7, let us consider a *material* or *underformed curve*, painted as a red dotted curve, in the initial configuration Ω_0. We may model it as the parametric function (see Appendix A.5) $\vec{\boldsymbol{\Gamma}} : \Xi \to \Omega_0$, $\xi \mapsto \vec{\boldsymbol{\Gamma}}(\xi)$; here ξ denotes a point on the interval $\Xi \subset \mathbb{R}$. This curve deforms into a *spatial* or *deformed curve*, painted in red, in the current configuration Ω, which is given by the image of $\vec{\boldsymbol{\Gamma}}$ through $\vec{\boldsymbol{\chi}}(\cdot, t)$, that is

$$\vec{\boldsymbol{\gamma}}(\xi, t) := \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{\Gamma}}(\xi), t); \tag{3.23}$$

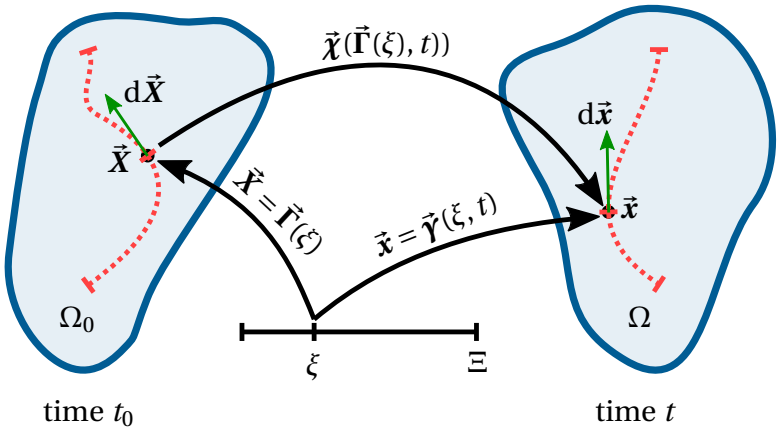


Figure 3.7: *The material curve $\vec{\Gamma}$ deforms into the spatial curve $\vec{\gamma}$. At a specific point $\vec{X}$ and its corresponding $\vec{x}$, two infinitesimal tangent vectors to those curves are $d\vec{X}$ and $d\vec{x}$. The deformation gradient $\boldsymbol{F}(\vec{X}, t)$ provides a mean to map vector $d\vec{X}$ to its deformed counterpart $d\vec{x}$, through equation (3.25). The deformation gradient $\boldsymbol{F}(\vec{X}, t)$ is a second-order tensor that captures the local deformation behavior of the material near the point $\vec{X}$ at time t; it describes how infinitesimal line elements in the material are transformed by the deformation process.*

note that $\vec{\gamma}(\xi, t)$ is also a parametric curve in ξ and that $\vec{\gamma} : \Xi \times T \to \Omega$. Let us assume that for a given ξ, $\vec{X} = \vec{\Gamma}(\xi)$ and $\vec{x} = \vec{\gamma}(\xi, t)$. The derivative of (3.23) can be estimated using the chain rule (see Appendix A.10) as:

$$\frac{\partial \vec{\gamma}(\xi, t)}{\partial \xi} = \frac{\partial \vec{\chi}(\vec{\Gamma}(\xi), t)}{\partial \vec{\Gamma}(\xi)} \frac{d\vec{\Gamma}(\xi)}{d\xi}; \tag{3.24}$$

this equation can be expressed compactly using our notation for material ($\vec{X}$) and spatial ($\vec{x}$) points as:

$$\frac{\partial \vec{x}}{\partial \xi} = \frac{\partial \vec{x}}{\partial \vec{X}} \frac{d\vec{X}}{d\xi}.$$

According to the explanation given in Appendix A.6, $\frac{d\vec{\Gamma}(\xi)}{d\xi}$ and $\frac{\partial \vec{\gamma}(\xi,t)}{\partial \xi}$ are tangent vectors to the parametric curves $\vec{\Gamma}(\xi)$ and $\vec{\gamma}(\xi, t)$ at the point ξ. Since we would like to study the deformation locally to $\vec{X}$ and $\vec{x}$, we will define a pair of differential vectors $d\vec{X}$ and $d\vec{x}$ in the neighborhoods of $\vec{X}$ and $\vec{x}$ that point in the direction of $\frac{d\vec{\Gamma}(\xi)}{d\xi}$ and $\frac{\partial \vec{\gamma}(\xi,t)}{\partial \xi}$, respectively; for this reason, both tangent vectors are scaled by the differential $d\xi$, so that:

$$\begin{aligned} d\vec{X} &:= \frac{d\vec{\Gamma}(\xi)}{d\xi} d\xi \quad = \frac{d\vec{X}}{d\xi} d\xi \\ d\vec{x} &:= \frac{\partial \vec{\gamma}(\xi, t)}{d\xi} \partial\xi = \frac{\partial \vec{x}}{\partial \xi} d\xi. \end{aligned}$$

These equations define $\mathrm{d}\vec{\boldsymbol{X}}$ and $\mathrm{d}\vec{\boldsymbol{x}}$, which are referred to as the *material** and the *spatial tangent vectors** or *material** and *spatial line elements**, respectively. These terms are shorthands for referring to infinitesimal line elements in their respective configurations.

Important note: In these equations, $\frac{\mathrm{d}\vec{\boldsymbol{\Gamma}}(\xi)}{\mathrm{d}\xi}$ and $\frac{\partial\vec{\boldsymbol{\gamma}}(\xi,t)}{\partial\xi}$ are derivatives expressed in Leibniz notation. It is important to understand that $\frac{\mathrm{d}\cdot}{\mathrm{d}\cdot}$ is the notation of an operator, that it should not be interpreted as a fraction, and in consequence, one should not attempt to cancel the $\mathrm{d}\xi$ terms. For clarity, these tangent vectors can be expressed using Euler's notation for derivatives: $\mathrm{d}\vec{\boldsymbol{X}} := \mathrm{D}_{\xi}\vec{\boldsymbol{\Gamma}}(\xi)\,\mathrm{d}\xi$ and $\mathrm{d}\vec{\boldsymbol{x}} := \mathrm{D}_{\xi}\vec{\boldsymbol{\gamma}}(\xi,t)\,\mathrm{d}\xi$. This formulation helps avoid the temptation to perform inappropriate simplifications and emphasizes that we are dealing with scaled tangent vectors along the parametric curves.

After multiplying both sides of equation (3.24) by $\mathrm{d}\xi$ and taking into account that $\vec{\boldsymbol{X}} = \vec{\boldsymbol{\Gamma}}(\xi)$ we arrive to the linear transformation (see Appendix A.11):

$$\underbrace{\frac{\partial\vec{\boldsymbol{\gamma}}(\xi,t)}{\partial\xi}\,\mathrm{d}\xi}_{\mathrm{d}\vec{\boldsymbol{x}}} = \underbrace{\frac{\partial\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}}_{\boldsymbol{F}(\vec{\boldsymbol{X}},t)}\underbrace{\frac{\mathrm{d}\vec{\boldsymbol{\Gamma}}(\xi)}{\mathrm{d}\xi}\,\mathrm{d}\xi}_{\mathrm{d}\vec{\boldsymbol{X}}},$$

that is,

$$\boxed{\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\vec{\boldsymbol{X}};} \tag{3.25}$$

we will call the Jacobian matrix

$$\begin{aligned}\boldsymbol{F}(\vec{\boldsymbol{X}},t) = \operatorname{Grad}\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t) = \nabla_{\vec{\boldsymbol{X}}}\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t) = \frac{\partial\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}} = \frac{\partial\vec{\boldsymbol{x}}}{\partial\vec{\boldsymbol{X}}}\\ = \begin{pmatrix}\frac{\partial x_1}{\partial X_1} & \frac{\partial x_1}{\partial X_2} & \frac{\partial x_1}{\partial X_3}\\ \frac{\partial x_2}{\partial X_1} & \frac{\partial x_2}{\partial X_2} & \frac{\partial x_2}{\partial X_3}\\ \frac{\partial x_3}{\partial X_1} & \frac{\partial x_3}{\partial X_2} & \frac{\partial x_3}{\partial X_3}\end{pmatrix} = \begin{pmatrix}\frac{\partial x}{\partial X} & \frac{\partial x}{\partial Y} & \frac{\partial x}{\partial Z}\\ \frac{\partial y}{\partial X} & \frac{\partial y}{\partial Y} & \frac{\partial y}{\partial Z}\\ \frac{\partial z}{\partial X} & \frac{\partial z}{\partial Y} & \frac{\partial z}{\partial Z}\end{pmatrix}\end{aligned} \tag{3.26}$$

the *material deformation gradient** or simply the *deformation gradient** tensor; we will see that $\boldsymbol{F}$ is a fundamental tensor in continuum mechanics since it is the basis for defining many measures of deformation; it is a 3×3 unsymmetric second order tensor that characterizes the change of the material tangent vector $\mathrm{d}\vec{\boldsymbol{X}}$ during motion in the neighborhood of point $\vec{\boldsymbol{X}}$ at time t; in other words, $\boldsymbol{F}$ maps the material tangent vector $\mathrm{d}\vec{\boldsymbol{X}}$ to the spatial tangent vector $\mathrm{d}\vec{\boldsymbol{x}}$; notice that $\boldsymbol{F}(\vec{\boldsymbol{X}},t)$ embodies whatever deformation that occurs at a point $\vec{\boldsymbol{X}}$ for any time $t\in T$. It must be emphasized that, in general, $\boldsymbol{F}$ is an unsymmetric tensor. In Section 3.11.3, we will see that a requisite that $\boldsymbol{F}$ must fulfill, to be physically viable, is that $\det\boldsymbol{F} > 0$.

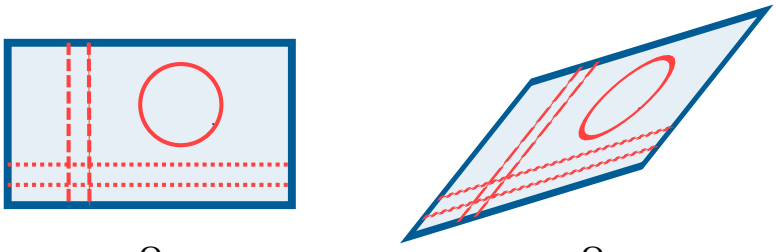


Figure 3.8: *In a homogeneous deformation, parallel lines in the reference configuration Ω_0 deform into parallel lines in the current configuration Ω; also a spherical surface in Ω_0 deforms into an ellipsoidal surface in Ω.*

It is said that matrix $\boldsymbol{F}$ makes a *linear transformation** over $\mathrm{d}\vec{\boldsymbol{X}}$; this means that $\boldsymbol{F}$ rotates and scales $\mathrm{d}\vec{\boldsymbol{X}}$ in order to convert it to $\mathrm{d}\vec{\boldsymbol{x}}$, as in (3.25); since $\boldsymbol{F}$ relates points on both configurations Ω_0 and Ω, we will say that $\boldsymbol{F}$ is a *two-point tensor**; in other words, two-point tensors are used to transform vectors between the reference and current configurations.

Alternatively, the deformation gradient can be calculated from the displacement vector. From equation (3.3), it follows that $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) + \vec{\boldsymbol{X}}$; taking the material gradient of this equation yields

$$\boldsymbol{F}(\vec{\boldsymbol{X}}, t) = \frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)}{\partial \vec{\boldsymbol{X}}} = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)}{\partial \vec{\boldsymbol{X}}} + \frac{\partial \vec{\boldsymbol{X}}}{\partial \vec{\boldsymbol{X}}} = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)}{\partial \vec{\boldsymbol{X}}} + \boldsymbol{I}, \tag{3.27}$$

where the second-order tensor

$$\mathrm{Grad}\,\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)}{\partial \vec{\boldsymbol{X}}} = \begin{pmatrix} \frac{\partial U_1}{\partial X_1} & \frac{\partial U_1}{\partial X_2} & \frac{\partial U_1}{\partial X_3} \\ \frac{\partial U_2}{\partial X_1} & \frac{\partial U_2}{\partial X_2} & \frac{\partial U_2}{\partial X_3} \\ \frac{\partial U_3}{\partial X_1} & \frac{\partial U_3}{\partial X_2} & \frac{\partial U_3}{\partial X_3} \end{pmatrix} = \begin{pmatrix} \frac{\partial U}{\partial X} & \frac{\partial U}{\partial Y} & \frac{\partial U}{\partial Z} \\ \frac{\partial V}{\partial X} & \frac{\partial V}{\partial Y} & \frac{\partial V}{\partial Z} \\ \frac{\partial W}{\partial X} & \frac{\partial W}{\partial Y} & \frac{\partial W}{\partial Z} \end{pmatrix}$$

is the so-called *displacement gradient in the material description**. Observe that if there is no motion the displacement is null, that is $\vec{\boldsymbol{U}} = \vec{\boldsymbol{0}}$, and in consequence, $\boldsymbol{F} = \boldsymbol{I}$ and $\vec{\boldsymbol{x}} = \vec{\boldsymbol{X}}$.

Finally, a deformation is said to be *homogeneous** when the deformation gradient is independent of the material coordinates $\vec{\boldsymbol{X}}$, that is, $\boldsymbol{F}(\vec{\boldsymbol{X}}, t) = \boldsymbol{F}(t)$; in other words, the deformation gradient is constant throughout the body and varies only with time. If this condition is not satisfied, the deformation is called *inhomogeneous**.

Homogeneous deformations possess several important properties, including (see Figure 3.8):

- Material surfaces that form planes in the reference configuration deform into planes in the deformed configuration. Additionally, two parallel planes in the reference configuration deform into two parallel planes.

- Material curves that form straight lines in the reference configuration deform into straight lines in the deformed configuration. Furthermore, two parallel straight lines in the reference configuration deform into two parallel straight lines.
- A material surface that forms a spherical surface in the reference configuration deforms into an ellipsoidal surface in the deformed configuration.

These properties arise from the fact that, in a homogeneous deformation, the deformation gradient is constant throughout the body, implying that the deformation behavior is uniform and consistent across all material points.

It is important to note that homogeneous deformations are idealized cases and are often used as approximations or simplifications in analyzing deformation processes. In reality, many deformations exhibit some degree of inhomogeneity due to factors such as boundary conditions, material inhomogeneities, or the presence of localized deformation mechanisms.

3.9 Inverse of the deformation gradient

A *singular matrix* is a square matrix that has a determinant of zero and cannot be inverted. Linear algebra says that the product of a square matrix $\boldsymbol{A}$ and a vector $\vec{\boldsymbol{x}}$ is zero, that is[5] $\boldsymbol{A}\vec{\boldsymbol{x}} = \vec{\boldsymbol{0}}$, if and only if $\boldsymbol{A}$ is singular or if $\vec{\boldsymbol{x}} = \vec{\boldsymbol{0}}$. Physics requires from $\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\vec{\boldsymbol{X}}$ that $\mathrm{d}\vec{\boldsymbol{x}} = \vec{\boldsymbol{0}}$ if and only if $\mathrm{d}\vec{\boldsymbol{X}} = \vec{\boldsymbol{0}}$; thus, $\boldsymbol{F}$ cannot be singular, $\det \boldsymbol{F} \neq 0$, the inverse $\boldsymbol{F}^{-1}$ exists for all $\vec{\boldsymbol{x}} \in \Omega$ and

$$\boxed{\mathrm{d}\vec{\boldsymbol{X}} = \boldsymbol{F}^{-1}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\vec{\boldsymbol{x}};} \tag{3.28}$$

in other words, the *inverse of the deformation gradient**, also known as the *spatial deformation gradient** tensor, $\boldsymbol{F}^{-1}$ maps the spatial tangent vector $\mathrm{d}\vec{\boldsymbol{x}}$ to the material tangent vector $\mathrm{d}\vec{\boldsymbol{X}}$; in short, the condition $\det \boldsymbol{F} \neq 0$ guarantees that the deformation process maintains a one-to-one correspondence between material and spatial points, which is a fundamental requirement for continuum mechanics.

Take into account that according to the inverse function theorem (see Appendix A.8), the matrix inverse of the Jacobian matrix of an invertible function is the Jacobian matrix of the inverse function; thus, we can write:

$$\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}}, t) = \operatorname{grad} \vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t) = \nabla_{\vec{\boldsymbol{x}}} \vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t) = \frac{\partial \vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}} = \frac{\partial \vec{\boldsymbol{X}}}{\partial \vec{\boldsymbol{x}}}$$

[5] In linear algebra, it is said that a linear system of equations $\boldsymbol{A}\vec{\boldsymbol{x}} = \vec{\boldsymbol{b}}$ is *homogeneous* if $\vec{\boldsymbol{b}} = \vec{\boldsymbol{0}}$. This system of equations has a solution $\vec{\boldsymbol{x}} \neq \vec{\boldsymbol{0}}$ if and only if $\det(\boldsymbol{A}) = 0$.

$$= \begin{pmatrix} \frac{\partial X_1}{\partial x_1} & \frac{\partial X_1}{\partial x_2} & \frac{\partial X_1}{\partial x_3} \\ \frac{\partial X_2}{\partial x_1} & \frac{\partial X_2}{\partial x_2} & \frac{\partial X_2}{\partial x_3} \\ \frac{\partial X_3}{\partial x_1} & \frac{\partial X_3}{\partial x_2} & \frac{\partial X_3}{\partial x_3} \end{pmatrix} = \begin{pmatrix} \frac{\partial x_1}{\partial X_1} & \frac{\partial x_1}{\partial X_2} & \frac{\partial x_1}{\partial X_3} \\ \frac{\partial x_2}{\partial X_1} & \frac{\partial x_2}{\partial X_2} & \frac{\partial x_2}{\partial X_3} \\ \frac{\partial x_3}{\partial X_1} & \frac{\partial x_3}{\partial X_2} & \frac{\partial x_3}{\partial X_3} \end{pmatrix}^{-1}$$
$$= \begin{pmatrix} \frac{\partial X}{\partial x} & \frac{\partial X}{\partial y} & \frac{\partial X}{\partial z} \\ \frac{\partial Y}{\partial x} & \frac{\partial Y}{\partial y} & \frac{\partial Y}{\partial z} \\ \frac{\partial Z}{\partial x} & \frac{\partial Z}{\partial y} & \frac{\partial Z}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial X} & \frac{\partial x}{\partial Y} & \frac{\partial x}{\partial Z} \\ \frac{\partial y}{\partial X} & \frac{\partial y}{\partial Y} & \frac{\partial y}{\partial Z} \\ \frac{\partial z}{\partial X} & \frac{\partial z}{\partial Y} & \frac{\partial z}{\partial Z} \end{pmatrix}^{-1} ; \tag{3.29}$$

in other words, the inverse of the deformation gradient is the spatial gradient of the inverse motion $\vec{\chi}^{-1}$.

Furthermore, from equation (3.4), it follows that $\vec{\chi}^{-1}(\vec{x}, t) = \vec{x} - \vec{u}(\vec{x}, t)$; taking the spatial gradient of this equation, we obtain:

$$\boldsymbol{F}^{-1}(\vec{x}, t) = \frac{\partial \vec{\chi}^{-1}(\vec{x}, t)}{\partial \vec{x}} = \frac{\partial \vec{x}}{\partial \vec{x}} - \frac{\partial \vec{u}(\vec{x}, t)}{\partial \vec{x}} = \boldsymbol{I} - \frac{\partial \vec{u}(\vec{x}, t)}{\partial \vec{x}}, \tag{3.30}$$

where the second-order tensor

$$\operatorname{grad} \vec{u}(\vec{x}, t) = \frac{\partial \vec{u}(\vec{x}, t)}{\partial \vec{x}} = \begin{pmatrix} \frac{\partial u_1}{\partial x_1} & \frac{\partial u_1}{\partial x_2} & \frac{\partial u_1}{\partial x_3} \\ \frac{\partial u_2}{\partial x_1} & \frac{\partial u_2}{\partial x_2} & \frac{\partial u_2}{\partial x_3} \\ \frac{\partial u_3}{\partial x_1} & \frac{\partial u_3}{\partial x_2} & \frac{\partial u_3}{\partial x_3} \end{pmatrix} = \begin{pmatrix} \frac{\partial u}{\partial x} & \frac{\partial u}{\partial y} & \frac{\partial u}{\partial z} \\ \frac{\partial v}{\partial x} & \frac{\partial v}{\partial y} & \frac{\partial v}{\partial z} \\ \frac{\partial w}{\partial x} & \frac{\partial w}{\partial y} & \frac{\partial w}{\partial z} \end{pmatrix}$$

is the so-called *displacement gradient in the spatial description*.★

Finally, it is important to note that for the deformation gradient tensor $\boldsymbol{F}$ and its inverse $\boldsymbol{F}^{-1}$ to exist, it is required that the motion function $\vec{\chi}$ and its inverse $\vec{\chi}^{-1}$ are continuous and differentiable with respect to position, as is evident from equations (3.26) and (3.29). This continuity requirement implies that no cracks or voids can appear during the deformation process.

3.10 Special cases of the deformation gradient

In this section, we will introduce several idealized deformation modes that are fundamental in the study of finite deformation kinematics. These modes are often used as building blocks for more complex deformation fields or as simplified models for various applications. Additionally, we will calculate the deformation gradient $\boldsymbol{F}$ for each of these modes.

3.10.1 Identity deformation gradient

★According to equation (3.5), in the case of a rigid-body translation, all particles move according to $\vec{x} = \vec{\chi}(t) = \vec{X} + \vec{c}(t)$. In this case, there is no deformation and the

deformation gradient is $\boldsymbol{F} = \frac{\partial \vec{\boldsymbol{\chi}}}{\partial \vec{\boldsymbol{X}}} = \boldsymbol{I}$, where $\boldsymbol{I}$ is the second-order identity tensor. This situation also occurs when there is no motion, since in such a case $\vec{\boldsymbol{c}}(t) = \vec{\boldsymbol{0}}$.

3.10.2 Orthogonal deformation gradient

★According to equation (3.19), which represents a rigid-body motion as $\vec{\boldsymbol{x}}(t) = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) = \boldsymbol{R}(t)\left(\vec{\boldsymbol{X}} - \tilde{\vec{\boldsymbol{X}}}_0(t)\right) + \vec{\boldsymbol{c}}(t)$, it follows that for a rigid-body motion, the deformation gradient is $\boldsymbol{F} = \frac{\partial \vec{\boldsymbol{\chi}}}{\partial \vec{\boldsymbol{X}}} = \boldsymbol{R}$. In this case, there is no deformation at $\vec{\boldsymbol{X}}$, but only a pure rotation, and possibly, a translation. Since $\boldsymbol{R}$ is a proper orthogonal tensor (see Appendix A.9), the determinant of the deformation gradient is $\det \boldsymbol{F} = +1$, and its inverse is $\boldsymbol{F}^{-1} = \boldsymbol{R}^T$.

3.10.3 Uniform deformation and dilation

Imagine stretching a rubber band uniformly in one direction, say x_i; then the entire band elongates uniformly λ_i times its original length where λ_i is the so-called *stretch ratio*★ in the x_i direction. Consequently, a material particle originally located at position X_i in the reference configuration maps to a new position x_i in the current configuration via the linear relation $x_i = \lambda_i X_i$.

This mapping describes a *uniform deformation*★. We distinguish two primary cases based on the value of the stretch ratio λ_i:

- **Elongation:** Occurs when $\lambda_i > 1$, indicating the material has been stretched.
- **Contraction (Shortening):** Occurs when $0 < \lambda_i < 1$, indicating the material has been compressed.

It is a fundamental requirement in continuum mechanics that the mapping between configurations be orientation-preserving. Therefore, the stretch ratio must satisfy $\lambda_i > 0$. A value of $\lambda_i \leq 0$ is physically inadmissible: $\lambda_i = 0$ would imply that a finite volume of matter has been crushed into a single point (violating the impenetrability of matter), while $\lambda_i < 0$ would imply a *reflection*★ (effectively turning the material inside out like a mirror image) which cannot occur in a continuous deformation of a physical body.

If the body undergoes uniform deformations, without rotation, in all three coordinate directions, the deformation is described by the equations

$$x_1 = \lambda_1 X_1 \qquad x_2 = \lambda_2 X_2 \qquad x_3 = \lambda_3 X_3$$

where $\lambda_1 > 0$, $\lambda_2 > 0$, and $\lambda_3 > 0$ are either constants or functions of time t. This is illustrated in Figure 3.9.

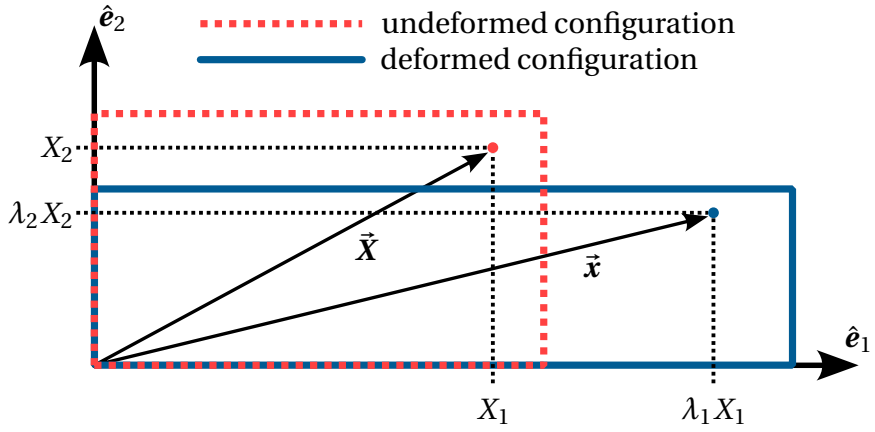


Figure 3.9: *Uniform deformation of a rectangle, where each edge is elongated or shortened by different stretch ratios λ_1 and λ_2. The dashed rectangle represents the undeformed configuration, while the one outlined in solid lines depicts the same rectangle after undergoing uniform deformation; in this example, $\lambda_1 > 1$ and $0 < \lambda_2 < 1$.*

In this case, the deformation gradient $\boldsymbol{F}$ is equal to a diagonal symmetric and positive definite tensor (see Section 3.10.4):

$$\boldsymbol{F}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} \lambda_1 & 0 & 0 \\ 0 & \lambda_2 & 0 \\ 0 & 0 & \lambda_3 \end{pmatrix};$$

observe that in this case, the deformation gradient $\boldsymbol{F}$ takes a diagonal form with the diagonal elements representing the stretch ratios along the respective coordinate directions.

If $\lambda_2 = \lambda_3$ then the body undergoes a uniform expansion or contraction in all directions transverse to the x_1 direction.

When $\lambda_1 = \lambda_2 = \lambda_3$ the body undergoes a uniform dilation or contraction in all directions. This type of deformation is also called *uniform dilation**, *isotropic expansion or contraction**; an example of this case is the inflating or deflating of a spherical balloon uniformly in all directions.

3.10.4 Positive definite deformation gradient

Consider the case where the deformation gradient $\boldsymbol{F}$ is equal to a positive-definite tensor $\boldsymbol{U}$ (see Appendix A.12); in this case, we will say that $\boldsymbol{U}$ is a *stretch tensor**, for reasons that will be understood later; remember that, by definition, all positive definite matrices are symmetric. In this scenario, equation (3.25) particularizes to:

$$\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{U}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\vec{\boldsymbol{X}};$$

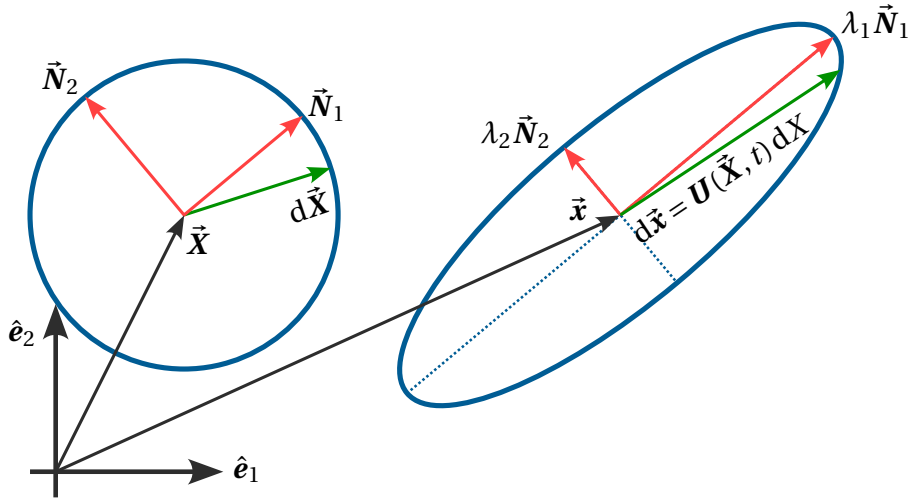


Figure 3.10: *Effect of a positive definite (and symmetric) deformation gradient $\boldsymbol{U}$ on a cylinder of constant radius $\|\mathrm{d}\vec{X}\|$; this cylinder is stretched along its principal material directions into an ellipse. The left side depicts the original undeformed circular cross-section in the reference configuration. It shows vectors $\hat{\boldsymbol{N}}_1$ and $\hat{\boldsymbol{N}}_2$ representing principal material directions, and $\mathrm{d}\vec{X}$ representing an infinitesimal material line element. The right side shows the deformed elliptical cross-section in the current configuration. The principal stretches λ_1 and λ_2 are applied to $\hat{\boldsymbol{N}}_1$ and $\hat{\boldsymbol{N}}_2$ respectively, resulting in the stretched $\lambda_1\hat{\boldsymbol{N}}_1$ and $\lambda_2\hat{\boldsymbol{N}}_2$. The infinitesimal element $\mathrm{d}\vec{X}$ is transformed to the stretched and rotated spatial line element $\mathrm{d}\vec{x}$ according to equation $\mathrm{d}\vec{x} = \boldsymbol{U}(\vec{X}, t)\,\mathrm{d}\vec{X}$.*

since $\boldsymbol{U}$ is symmetric, its eigenvalues λ_i are all real numbers, and since $\boldsymbol{U}$ is positive definite, they are also positive, i.e., $\lambda_i > 0$. Additionally, the corresponding eigenvectors $\hat{\boldsymbol{N}}_i$ are mutually orthogonal, and $\boldsymbol{U}\hat{\boldsymbol{N}}_i = \lambda_i\hat{\boldsymbol{N}}_i$.

We can perform an analysis similar to the one done for the Cauchy stress tensor $\underline{\underline{\boldsymbol{\sigma}}}$ and the equation $\vec{\boldsymbol{q}} = \underline{\underline{\boldsymbol{\sigma}}}\,\hat{\boldsymbol{n}}$ in elasticity theory (see Section 4.2). In that case, the eigenvectors of $\underline{\underline{\boldsymbol{\sigma}}}$ define the so-called *principal directions*, and their corresponding eigenvalues are the *principal stresses*. In our case, the eigenvectors of $\boldsymbol{U}$ define the *principal material directions** or *principal referential directions**, while their corresponding eigenvalues λ_i are referred to as the *principal stretches**.

The principal stretches quantify the stretching ($\lambda_i > 1$) or contraction ($0 < \lambda_i < 1$) of material line elements $\mathrm{d}\vec{X}$ along the principal material directions defined by the corresponding eigenvectors $\hat{\boldsymbol{N}}_i$. In fact, along each of these eigenvectors $\hat{\boldsymbol{N}}_i$, the deformed element $\mathrm{d}\vec{x}$ is in the same direction as the undeformed element $\mathrm{d}\vec{X}$, since $\mathrm{d}\vec{x} = \boldsymbol{U}\hat{\boldsymbol{N}}_i = \lambda_i\hat{\boldsymbol{N}}_i$; in other words, in each of the directions given by the three eigenvectors $\hat{\boldsymbol{N}}_i$, the body only stretches λ_i times but does not rotate.

Figure 3.10 illustrates the effect of a symmetric and positive definite $\boldsymbol{U}$ on a cylinder of constant radius $\|\mathrm{d}\vec{X}\|$.

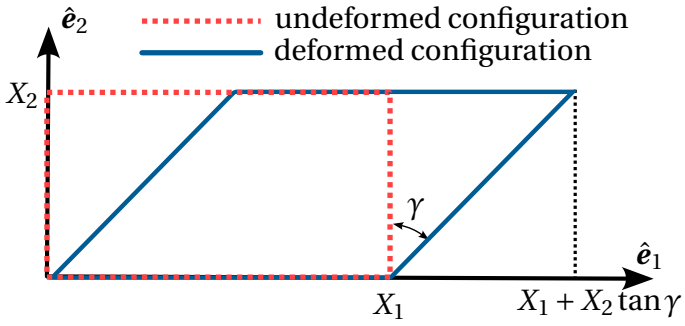


Figure 3.11: *The simple shear deformation represents a situation where layers of material slide over each other while preserving their original volume. The dashed rectangle represents the undeformed configuration, while the parallelogram outlined in solid lines depicts the same rectangle after undergoing shear deformation.*

3.10.5 Simple shear

In a *simple shear** deformation, parallel planes are displaced relative to each other by an amount proportional to the distance between the planes and in a direction parallel to the planes. For example, the simple shear deformation illustrated in Figure 3.11 is described by the equations:

$$x_1 = X_1 + X_2 \tan\gamma \qquad x_2 = X_2 \qquad x_3 = X_3$$

Here the *shear plane** is the one spanned by X_1 and X_2, where X_1 corresponds to the *shear direction**, and the angle γ is a measure of the amount of shear. Note that a simple shear involves no change in volume of any portion of the body (why?); in this case, the deformation gradient corresponds to

$$\boldsymbol{F}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} 1 & \tan\gamma & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Observe that shear is associated to the non-zero off-diagonal term $\tan\gamma$; also, Figure 3.11 shows that the rectangle has a tendency to rotate; the non-symmetric nature of the $\boldsymbol{F}$ reflects this.

3.10.6 Pure shear

Pure shear is a type of deformation in which a material body undergoes simultaneous elongation in one direction and shortening in the perpendicular direction. It is distinguished from simple shear by the fact that pure shear involves no rigid body rotation of material elements, despite the presence of shear strains.

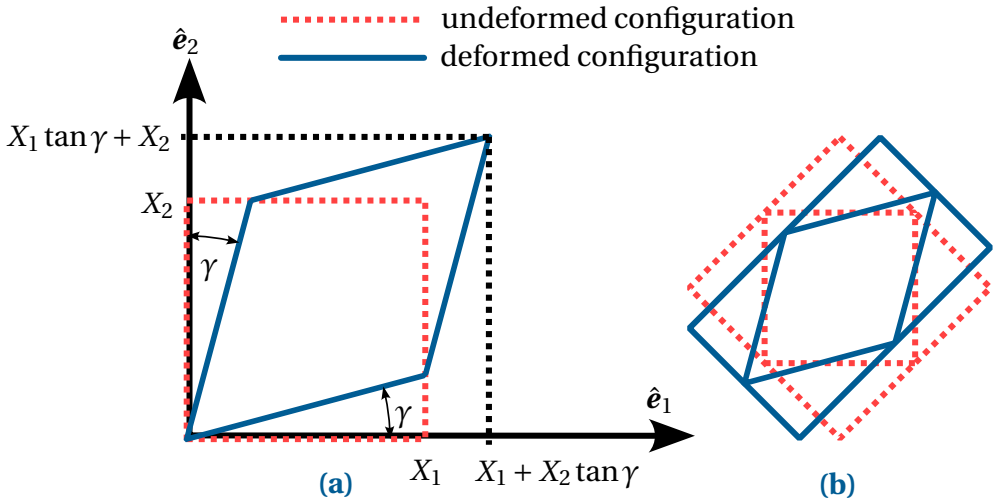


Figure 3.12: *Here a square element **(a)** is subject to pure shear deformation; this square is elongated along one diagonal while compressing it along the perpendicular diagonal as seen in **(b)**, resulting in zero net rotation.*

For the pure shear deformation illustrated in Figure 3.12, the equations relating the current position $\vec{\boldsymbol{x}}$ to the reference position $\vec{\boldsymbol{X}}$ are given by:

$$x_1 = X_1 + X_2 \tan\gamma \qquad x_2 = X_1 \tan\gamma + X_2 \qquad x_3 = X_3$$

where γ represents the shear angle parameter that controls the intensity of the deformation. Note that this deformation is assumed to occur in the X_1-X_2 plane, with no deformation in the X_3 direction.

Taking the derivatives of these transformation equations with respect to the reference coordinates, as indicated by equation (3.26), we obtain the deformation gradient tensor:

$$\boldsymbol{F}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} 1 & \tan\gamma & 0 \\ \tan\gamma & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

Note that the pure shear deformation gradient is symmetric as during pure shear, while the lower right region of the square tends to rotate counter-clockwise due to the shearing action, the upper left region simultaneously tends to rotate clockwise by exactly the same amount. These opposing rotational tendencies cancel each other out, resulting in zero net rotation for the material element as a whole.

It can be shown that the eigenvalues of $\boldsymbol{F}$ are $1 + \tan\gamma$, $1 - \tan\gamma$, and 1. Thus, pure shear represents a special case of a positive definite deformation gradient (see Section 3.10.4), as long as all eigenvalues are positive; this happens when $\gamma \in (-45°, 45°)$. This guarantees that the deformation remains physically realizable.

plain strain
$x_1 = \chi_1(X_1, X_2)$
$x_2 = \chi_2(X_1, X_2)$
$x_3 = X_3$

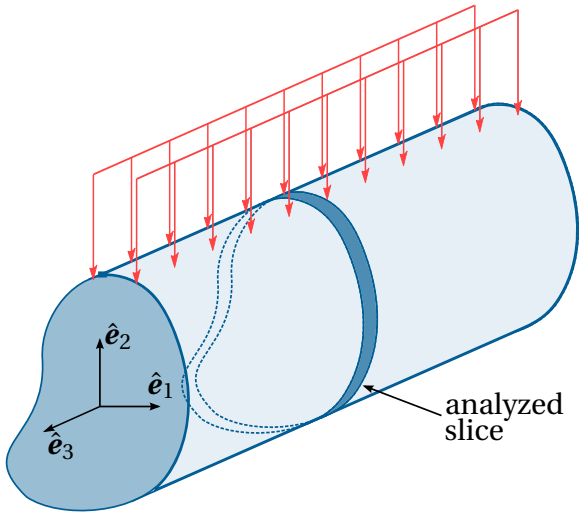


Figure 3.13: *In plain strain, the deformation is confined to a plane perpendicular to vector $\hat{\boldsymbol{e}}_3$, and there is no deformation along the $\hat{\boldsymbol{e}}_3$ axis. This is the situation of the thin slice that is unable to deform in the $\hat{\boldsymbol{e}}_3$ direction due to the restraint provided by the neighboring slices. Plain strain deformation is commonly encountered in various engineering applications involving long, prismatic structures, such as pipes, retaining walls, and shafts.*

3.10.7 Plain strain

The *plane strain*★ deformation is an idealized deformation mode that occurs in situations where the deformation is confined to a plane, and there is no deformation perpendicular to that plane. This type of deformation is commonly encountered in various applications, such as in the analysis of long prismatic structural members such as pipes, embankments, strip foundations or retaining walls, where the deformation is assumed to be constant along its length. Also, it occurs in situations where the deformation is restricted in one direction due to boundary conditions or geometric constraints.

Mathematically, this type of deformation is defined by

$$x_1 = \chi_1(X_1, X_2) \qquad x_2 = \chi_2(X_1, X_2) \qquad x_3 = X_3.$$

Here, the planes $x_3 = c$, where c is a constant, are called the *deformation planes*.★ There is no deformation in the x_3 direction, i.e., the deformation is confined to a plane perpendicular to the x_3 axis, as illustrated in Figure 3.13. Note that the deformation can vary within the deformation planes (x_1-x_2 planes) but is constant along the x_3 direction.

In this case, the deformation gradient corresponds to

$$\boldsymbol{F}(\vec{\boldsymbol{X}},t)=\begin{pmatrix}\frac{\partial\chi_1(X_1,X_2)}{\partial X_1} & \frac{\partial\chi_1(X_1,X_2)}{\partial X_2} & 0\\ \frac{\partial\chi_2(X_1,X_2)}{\partial X_1} & \frac{\partial\chi_2(X_1,X_2)}{\partial X_2} & 0\\ 0 & 0 & 1\end{pmatrix}.$$

Note that since the deformation gradient components F_{13}, F_{23}, F_{31} and F_{32} involving X_3 are zero, the deformation is effectively two-dimensional.

Example 3.7. *Deformation gradient*

Consider a body that undergoes motion given by:

$$\vec{\boldsymbol{x}}=\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)=\begin{pmatrix}(1+at^2)X_1\\ X_2\\ X_3\end{pmatrix}$$

where a is a constant. The motion of the body represents a uniaxial stretching in the $\hat{\boldsymbol{e}}_1$-direction that increases quadratically with time, while X_2 and X_3 remain unchanged. Calculate its corresponding deformation gradient and the inverse of the deformation gradient.

Solution:

To compute the deformation gradient $\boldsymbol{F}$, we take the Jacobian matrix of the motion with respect to the reference coordinates $\vec{\boldsymbol{X}}$, that is, $\boldsymbol{F}(\vec{\boldsymbol{X}},t)=\frac{\partial\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}$, according to equation (3.26). The components of $\boldsymbol{F}$ are:

$$F_{11}=\frac{\partial x_1}{\partial X_1}=1+at^2, \qquad F_{12}=\frac{\partial x_1}{\partial X_2}=0, \qquad F_{13}=\frac{\partial x_1}{\partial X_3}=0,$$
$$F_{21}=\frac{\partial x_2}{\partial X_1}=0, \qquad F_{22}=\frac{\partial x_2}{\partial X_2}=1, \qquad F_{23}=\frac{\partial x_2}{\partial X_3}=0,$$
$$F_{31}=\frac{\partial x_3}{\partial X_1}=0, \qquad F_{32}=\frac{\partial x_3}{\partial X_2}=0, \qquad F_{33}=\frac{\partial x_3}{\partial X_3}=1.$$

Thus,

$$\boldsymbol{F}(\vec{\boldsymbol{X}},t)=\begin{pmatrix}1+at^2 & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}.$$

This is a diagonal matrix, indicating pure stretching along the first axis with no shear or rotation. This deformation is homogeneous as $\boldsymbol{F}$ does not depend on the material coordinates $\vec{\boldsymbol{X}}$

The inverse deformation gradient is given by equation (3.29). Since $\boldsymbol{F}$ is diagonal, its inverse is simply

$$\boldsymbol{F}^{-1}(t)=\begin{pmatrix}\frac{1}{1+at^2} & 0 & 0\\ 0 & 1 & 0\\ 0 & 0 & 1\end{pmatrix}.$$

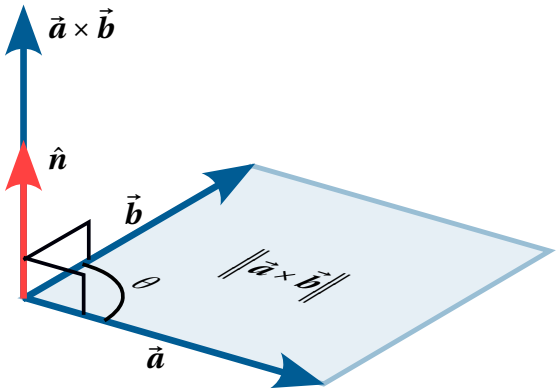


Figure 3.14: *The area of the parallelogram shown, whose sides are formed by vectors $\vec{a}$ and $\vec{b}$, is equal to $\|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin\theta$. This equation is valid as long as $\theta \in [0°, 180°]$.*

Include and example similar to the one I have on my GITHUB.

3.11 Change of length, area and volume due to deformation

When a continuum body undergoes deformation, the geometry of the body changes. The lengths, areas, and volumes of material elements within the body are altered from their initial configurations. In the following, we will study how to quantify these geometric changes in continuum bodies under deformation; these concepts are necessary for analyzing deformation processes and the resulting stresses and strains.

3.11.1 How to measure areas and volumes using vector calculus

Areas and volumes in $\mathbb{R}^3$ can be calculated using both the inner- and the cross-product. With reference to Figure 3.14, let us first remember that the cross product can be defined as

$$\vec{a} \times \vec{b} = \left(\|\vec{a}\| \|\vec{b}\| \sin\theta \right) \hat{n}$$

where $\theta \in [0°, 180°]$ is the angle between vectors $\vec{a}$ and $\vec{b}$, and $\hat{n}$ is the unit vector that is perpendicular to the containing plane of $\vec{a}$ and $\vec{b}$ and whose direction follows the right-hand rule. Here

$$A = \|\vec{a} \times \vec{b}\| = \|\vec{a}\| \|\vec{b}\| \sin\theta$$

denotes the signed area of the parallelogram with sides $\vec{a}$ and $\vec{b}$, as shown in Figure 3.14. When $A < 0$, it means that $\theta \notin [0°, 180°]$.

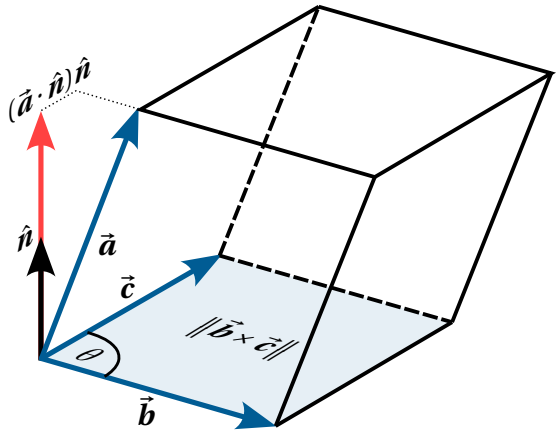


Figure 3.15: *Geometrical interpretation of the scalar triple product. The scalar triple product is important because its absolute value* $\left|\vec{a}\cdot(\vec{b}\times\vec{c})\right|$ *is the volume of the parallelepiped spanned by* $\vec{a}$, $\vec{b}$, *and* $\vec{c}$. *The absolute value is necessary in case* $\vec{a}$, $\vec{b}$, *and* $\vec{c}$ *are not positively oriented.*

Now, the volume $\mathscr{V}$ of a parallelepiped whose edges are $\vec{a}$, $\vec{b}$ and $\vec{c}$ is given by the so-called *scalar triple product*[★], so that (see Figure 3.15):

$$\mathscr{V} = \vec{a}\cdot(\vec{b}\times\vec{c});$$

when $\mathscr{V} > 0$ it is said that vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$ are *positively oriented*[★]. This formula applies because the volume of the parallelepiped is the area of the base times its height; here $\vec{b}\times\vec{c}$ acts as the area A of the base of the parallelogram with sides $\vec{b}$ and $\vec{c}$ multiplied by a unit normal vector $\vec{n}$, that is $\vec{b}\times\vec{c} = A\vec{n}$; the height of the parallelepiped is the component of $\vec{a}$ in the direction normal to the base, hence the height is the length of the projection of $\vec{a}$ over $\hat{n}$, that is $\vec{a}\cdot\hat{n}$; remember that, $\text{Proj}\,\vec{a}/\hat{n} = (\vec{a}\cdot\hat{n})\,\hat{n}$. Thus the volume $\mathscr{V}$ can also be estimated as:

$$\mathscr{V} = A(\vec{a}\cdot\vec{n}) = \vec{a}\cdot(A\vec{n}) = \vec{a}\cdot(\vec{b}\times\vec{c}). \tag{3.31}$$

Since the selection of $\vec{a}$, $\vec{b}$, and $\vec{c}$ is arbitrary, as long as they are positively oriented,

$$\mathscr{V} = \vec{a}\cdot(\vec{b}\times\vec{c}) = \vec{b}\cdot(\vec{c}\times\vec{a}) = \vec{c}\cdot(\vec{a}\times\vec{b}) = \det\Big(\big[\vec{a}\;\vec{b}\;\vec{c}\big]\Big), \tag{3.32}$$

that is, the scalar triple product can also be computed as the determinant of a 3×3 matrix whose columns are the vectors $\vec{a}$, $\vec{b}$ and $\vec{c}$.

Finally, notice that the scalar triple product vanishes if vectors $\vec{a}$, $\vec{b}$, and $\vec{c}$ are coplanar. In this case, the columns of matrix $\big[\vec{a}\;\vec{b}\;\vec{c}\big]$ are linearly dependent and thus its determinant is null.

3.11.2 Change of length due to deformation: the stretch ratio λ

The change of length during deformation of an infinitesimal material element $\mathrm{d}\vec{\boldsymbol{X}}$ can be measured by means of the so-called *stretch ratio*,★ the *extension ratio*★ or simply the *stretch* λ, which is defined as the ratio between its deformed and initial lengths:

$$\lambda = \frac{\|\mathrm{d}\vec{\boldsymbol{x}}\|}{\|\mathrm{d}\vec{\boldsymbol{X}}\|} \tag{3.33}$$

the material elongates when $\lambda > 1$, shortens when $0 < \lambda < 1$ and it remains unstretched when $\lambda = 1$; it is important to point out that $\lambda \leq 0$ is physically unfeasible. In short, the stretch ratio is a measure of how much $\mathrm{d}\vec{\boldsymbol{X}}$ has stretched.

Note that

$$\frac{\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|}{\|\mathrm{d}\vec{\boldsymbol{X}}\|} = \lambda - 1$$

can be understood as the "percentaje" of stretch or contraction of the material element $\mathrm{d}\vec{\boldsymbol{X}}$; thus, $\lambda - 1$ is directly comparable with the longitudinal strain ε that is usually employed in elasticity theory to measure infinitesimal deformations, and that was reviewed in Section 2.1; in this case, the material elongates when $\lambda - 1 > 0$, shortens when $\lambda - 1 < 0$ and it remains unstretched when $\lambda - 1 = 0$.

3.11.3 Change of volume due to deformation: the role of $\det \boldsymbol{F}$

★

In large deformation analysis, the change of volume cannot be disregarded. Furthermore, a variety of materials exhibit distinct behaviors between volume-preserving deformation (*distortion*★) and volume-changing deformation (*dilation*★). Therefore, it is necessary to describe the volume change in terms of deformation.

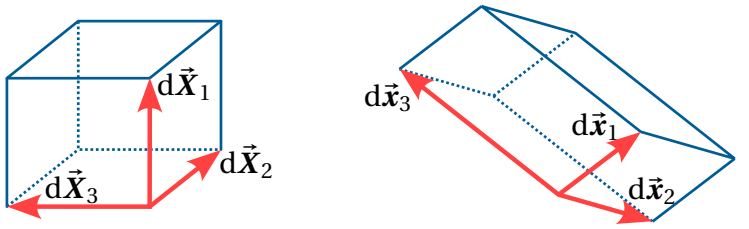
Let us consider a *material volume element*,★ $\mathrm{d}\mathscr{V}_0$ that is one defined in the initial configuration, which is specified by the positively oriented, non-coplanar vectors $\mathrm{d}\vec{\boldsymbol{X}}_1$, $\mathrm{d}\vec{\boldsymbol{X}}_2$ and $\mathrm{d}\vec{\boldsymbol{X}}_3$, as shown in Figure 3.16. The volume[6] $\mathrm{d}\mathscr{V}_0$ of that material volume element is given by the scalar triple product

$$\mathrm{d}\mathscr{V}_0 = \mathrm{d}\vec{\boldsymbol{X}}_1 \cdot (\mathrm{d}\vec{\boldsymbol{X}}_2 \times \mathrm{d}\vec{\boldsymbol{X}}_3) = \det\left(\left[\mathrm{d}\vec{\boldsymbol{X}}_1 \ \mathrm{d}\vec{\boldsymbol{X}}_2 \ \mathrm{d}\vec{\boldsymbol{X}}_3\right]\right); \tag{3.34}$$

since the triad of vectors $\mathrm{d}\vec{\boldsymbol{X}}_1$, $\mathrm{d}\vec{\boldsymbol{X}}_2$ and $\mathrm{d}\vec{\boldsymbol{X}}_3$ is positively oriented, $\mathrm{d}\mathscr{V}_0 > 0$.

Using the deformation gradient, it is possible to map the vectors $\mathrm{d}\vec{\boldsymbol{X}}_1$, $\mathrm{d}\vec{\boldsymbol{X}}_2$ and $\mathrm{d}\vec{\boldsymbol{X}}_3$ to the current configuration; in this way, the vectors $\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_1$, $\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_2$ and $\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_3$ form

[6] Throughout this text, the symbol $\mathrm{d}\mathscr{V}_0$ will denote either the volume element or the volume itself, with the interpretation determined by the context in which it is used.



material volume element $d\mathcal{V}_0$ **(a)**

spatial volume element dv_t **(b)**

Figure 3.16: *Figure* ***(a)*** *depicts a material volume element* $d\mathcal{V}_0$ *in the initial undeformed configuration* Ω_0*, represented by the parallelepiped formed by the vectors* $d\vec{\boldsymbol{X}}_1$, $d\vec{\boldsymbol{X}}_2$ *and* $d\vec{\boldsymbol{X}}_3$*. Figure* ***(b)*** *shows the same volume element after deformation, now mapped to the current configuration* Ω*. The deformation gradient* $\boldsymbol{F}$ *transforms the original vectors* $d\vec{\boldsymbol{X}}_1$, $d\vec{\boldsymbol{X}}_2$ *and* $d\vec{\boldsymbol{X}}_3$ *into the new vectors* $d\vec{\boldsymbol{x}}_1$, $d\vec{\boldsymbol{x}}_2$ *and* $d\vec{\boldsymbol{x}}_3$*, respectively. These transformed vectors form a new parallelepiped representing the spatial volume element* dv_t*. The determinant of the deformation gradient,* $\det(\boldsymbol{F})$*, relates the original material volume* $d\mathcal{V}_0$ *to the deformed spatial volume* dv_t *through the equation* $dv_t = \det(\boldsymbol{F})\, d\mathcal{V}_0$.

a so-called *spatial volume element** dv_t whose volume is given by the scalar triple product:

$$
\begin{aligned}
dv_t &= (\boldsymbol{F}\, d\vec{\boldsymbol{X}}_1) \cdot \big((\boldsymbol{F}\, d\vec{\boldsymbol{X}}_2) \times (\boldsymbol{F}\, d\vec{\boldsymbol{X}}_3)\big) \\
&= \det\big(\big[\boldsymbol{F}\, d\vec{\boldsymbol{X}}_1\ \boldsymbol{F}\, d\vec{\boldsymbol{X}}_2\ \boldsymbol{F}\, d\vec{\boldsymbol{X}}_3\big]\big) && \text{(using equation [3.32])} \\
&= \det\big(\boldsymbol{F}\big[d\vec{\boldsymbol{X}}_1\ d\vec{\boldsymbol{X}}_2\ d\vec{\boldsymbol{X}}_3\big]\big) && \text{(factorize } \boldsymbol{F}\text{)} \\
&= \det(\boldsymbol{F}) \underbrace{\det\big(\big[d\vec{\boldsymbol{X}}_1\ d\vec{\boldsymbol{X}}_2\ d\vec{\boldsymbol{X}}_3\big]\big)}_{= d\mathcal{V}_0 \text{ (using [3.34])}} && \text{(since } \det(\boldsymbol{AB}) = \det \boldsymbol{A} \det \boldsymbol{B}\text{);}
\end{aligned}
$$

in short, we arrive to the result,

$$
\boxed{dv_t = \det\big(\boldsymbol{F}(\vec{\boldsymbol{X}}, t)\big)\, d\mathcal{V}_0;} \tag{3.35}
$$

since the volumes of the material and spatial elements must be positive, that is $d\mathcal{V}_0 > 0$ and $dv_t > 0$, respectively, it follows that $\det \vec{\boldsymbol{F}}(\vec{\boldsymbol{X}}, t) > 0$ for all $\vec{\boldsymbol{X}} \in \Omega_0$ and for all times $t \in T$. This is a physical requisite for every deformation gradient $\boldsymbol{F}$. Note that an infinitesimal volume in the undeformed geometry cannot shrink to a point, i.e., a zero volume, hence $\det \vec{\boldsymbol{F}}(\vec{\boldsymbol{X}}, t) \neq 0$.

Let us remember that the deformation gradient $\boldsymbol{F}$ is a Jacobian matrix (Appendix A.7), and the determinant of a Jacobian matrix is the so-called Jacobian determinant.

From the demonstration it can be deduced that the determinant of the deformation gradient, that we will refer to as the *volume ratio** $J(\vec{\boldsymbol{X}}, t)$:

$$\boxed{J(\vec{\boldsymbol{X}}, t) = \det \vec{\boldsymbol{F}}(\vec{\boldsymbol{X}}, t) = \mathrm{d}v_t \Big/ \mathrm{d}\mathscr{V}_0,}$$

(do not understand this as a derivative but as a division of quantities) is a scale factor which specifies how many times the material volume element $\mathrm{d}\mathscr{V}_0$ expands (or contracts) in order to form the spatial volume element $\mathrm{d}v_t$; in this way, $\det \boldsymbol{F} > 1$ indicates volumetric expansion and $0 < \det \boldsymbol{F} < 1$ shows volumetric contraction. Notice that $\det \boldsymbol{F} < 0$ is not physically viable as it implies reflection* or inversion of the material, like in a mirror; this condition on the determinant of the deformation gradient tensor ensures that the mapping between the material and spatial configurations preserves the orientation and does not allow for self-intersections or folding of the deformed body.

Using the volume ratio, it is possible to define the *volumetric strain**, which is a measure of the relative change in volume due to deformation, as (we denote it as e in elasticity theory):

$$e := \frac{\mathrm{d}v_t - \mathrm{d}\mathscr{V}_0}{\mathrm{d}\mathscr{V}_0} = J - 1.$$

In general terms, it is said that a deformation is *isochoric** or *volume preserving** at $\vec{\boldsymbol{X}} \in \Omega_0$ if $\det \boldsymbol{F}(\vec{\boldsymbol{X}}, t) = 1$, that is, the volume does not change locally around $\vec{\boldsymbol{X}}$. If this is the fact for all $\vec{\boldsymbol{X}} \in \Omega_0$, then the deformation of the body is said to be *isochoric*; this is the kind of deformation that happens in *incompressible materials**. For example, according to Section 3.10.2, in rigid-body motion $\boldsymbol{F} = \boldsymbol{R}$, where $\boldsymbol{R}$ is a rotation matrix, that is an orthogonal matrix whose determinant is 1; since $\det(\boldsymbol{F}) = 1$, there is no change of volume in rigid-body motion. Also, since the determinant of a triangular matrix is the product of the elements of its main diagonal, the simple shear deformation, studied in Section 3.10.5, is also an example of an isochoric deformation.

The determinant of the deformation gradient tensor, $\det \boldsymbol{F}$, can also be employed to transform integrals of a function f over the deformed domain Ω_t to integrals over the undeformed domain Ω_0 (see Appendix A.15):

$$\boxed{\iiint_{\Omega_t} f(\vec{\boldsymbol{x}})\, \mathrm{d}v = \iiint_{\Omega_0} f(\vec{\boldsymbol{X}}) \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V};} \tag{3.36}$$

in this case we will denote $\mathrm{d}v \equiv \mathrm{d}v_t \equiv \mathrm{d}v(\vec{\boldsymbol{x}}, t)$ and $\mathrm{d}\mathscr{V} \equiv \mathrm{d}\mathscr{V}_0 \equiv \mathrm{d}\mathscr{V}(\vec{\boldsymbol{X}}, t_0)$; this transformation is particularly useful when the problem is initially defined in the material (undeformed) configuration Ω_0, as it enables the evaluation of integrals over the deformed domain Ω_t without explicitly tracking the deformed geometry. In other words,

by using the determinant of the deformation gradient tensor, one can effectively map the deformed domain back to the undeformed domain, where the integration is performed.

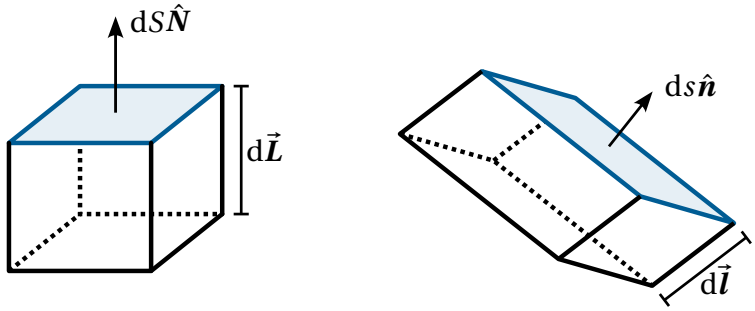
It's important to note that when $f = 1$, the integral represents the volume of the deformed configuration Ω_t at time t:

$$\iiint_{\Omega_t} \mathrm{d}v = \iiint_{\Omega_0} \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V}.$$

This equation shows that the volume of the deformed configuration can be obtained by integrating the determinant of the deformation gradient tensor over the undeformed domain. The determinant of the deformation gradient tensor acts as a scaling factor of $\mathrm{d}\mathscr{V}$ that accounts for the local volume change due to deformation.

3.11.4 Change of area due to deformation: Nanson's formula

We already know how lines and volumes deform. It is the turn of real and imaginary surfaces and areas. By a *real differential surface** we express a tiny portion of surface that is part of the body's contour, while an *imaginary differential surface** refers to one extracted from a plane that fictitiously cuts the body. Let us suppose that $\hat{\boldsymbol{N}}$ and $\hat{\boldsymbol{n}}$ are unit normal vectors to a real or imaginary differential surfaces $\mathrm{d}S \subset \partial\Omega_0$ and $\mathrm{d}s \subset \partial\Omega_t$, as shown in Figure 3.17; we will see in the following that $\hat{\boldsymbol{N}}$ does not map, through $\boldsymbol{F}$, to a unit normal vector $\hat{\boldsymbol{n}}$ related to a spatial differential surface $\mathrm{d}s$, that is, $\hat{\boldsymbol{n}} \neq \boldsymbol{F}\hat{\boldsymbol{N}}$.



material volume element $\mathrm{d}\mathscr{V}_0$ **(a)**

spatial volume element $\mathrm{d}v_t$ **(b)**

Figure 3.17: *Figure* ***(a)*** *represents a material volume element* $\mathrm{d}\mathscr{V}_0$ *in the reference configuration which is bounded by a differential surface with an area* $\mathrm{d}S$ *and an outward-pointing unit normal vector* $\hat{\boldsymbol{N}}$. *Figure* ***(b)*** *shows the corresponding deformed spatial volume element* $\mathrm{d}v_t$ *in the current configuration. The deformed differential surface has an area* $\mathrm{d}s$, *and the outward-pointing unit normal vector is denoted by* $\hat{\boldsymbol{n}}$. *Nanson's formula establishes the relationship between the undeformed area* $\mathrm{d}S$ *and the normal vector* $\hat{\boldsymbol{N}}$, *and their deformed counterparts* $\mathrm{d}s$ *and* $\hat{\boldsymbol{n}}$, *taking into account the deformation gradient tensor* $\boldsymbol{F}$.

We will characterize those material and spatial differential surfaces by vector elements that are normal to the infinitesimally small areas defined on the initial and current configurations, namely, $\mathrm{d}\vec{\boldsymbol{S}} = \mathrm{d}S\hat{\boldsymbol{N}}$ and $\mathrm{d}\vec{\boldsymbol{s}} = \mathrm{d}s\hat{\boldsymbol{n}}$; take into account that $\mathrm{d}s$ stands for a scalar differential area, while $\mathrm{d}\vec{\boldsymbol{s}}$ refers to a vector differential area, that is a vector normal to an infinitesimal surface whose area is $\mathrm{d}s$. According to equation (3.31), and using an arbitrary material element $\mathrm{d}\vec{\boldsymbol{L}} \neq \vec{\boldsymbol{0}}$ we may define an infinitesimal material volume element $\mathrm{d}\mathscr{V}_0 = \mathrm{d}\vec{\boldsymbol{S}} \cdot \mathrm{d}\hat{\boldsymbol{L}}$ which deforms into a spatial volume element $\mathrm{d}v_t = \mathrm{d}\vec{\boldsymbol{s}} \cdot \mathrm{d}\hat{\boldsymbol{l}}$ where $\mathrm{d}\vec{\boldsymbol{l}} = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{L}}$. Replacing these quantities into equation (3.35), it follows that:

$$\begin{aligned} \mathrm{d}v_t &= J\,\mathrm{d}\mathscr{V}_0 \\ \mathrm{d}\vec{\boldsymbol{s}} \cdot \mathrm{d}\hat{\boldsymbol{l}} &= J\,\mathrm{d}\vec{\boldsymbol{S}} \cdot \mathrm{d}\hat{\boldsymbol{L}} \\ \mathrm{d}\vec{\boldsymbol{s}} \cdot \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{L}} &= J\,\mathrm{d}\vec{\boldsymbol{S}} \cdot \mathrm{d}\hat{\boldsymbol{L}}; \end{aligned}$$

where $J = \det \boldsymbol{F}$ is the volume ratio. The product $\vec{\boldsymbol{a}} \cdot \boldsymbol{B}\vec{\boldsymbol{c}}$ is a scalar; therefore $\vec{\boldsymbol{a}} \cdot \boldsymbol{B}\vec{\boldsymbol{c}} = \left(\vec{\boldsymbol{a}}^T \boldsymbol{B}\vec{\boldsymbol{c}}\right)^T = \vec{\boldsymbol{c}}^T \boldsymbol{B}^T \vec{\boldsymbol{a}} = \left(\boldsymbol{B}^T \vec{\boldsymbol{a}}\right) \cdot \vec{\boldsymbol{c}}$ and $\mathrm{d}\vec{\boldsymbol{s}} \cdot \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{L}} = \left(\boldsymbol{F}^T\,\mathrm{d}\vec{\boldsymbol{s}}\right) \cdot \mathrm{d}\vec{\boldsymbol{L}}$; now,

$$\begin{aligned} \left(\boldsymbol{F}^T\,\mathrm{d}\vec{\boldsymbol{s}}\right) \cdot \mathrm{d}\vec{\boldsymbol{L}} - J\,\mathrm{d}\vec{\boldsymbol{S}} \cdot \mathrm{d}\hat{\boldsymbol{L}} = 0 \\ \underbrace{\left(\boldsymbol{F}^T\,\mathrm{d}\vec{\boldsymbol{s}} - J\,\mathrm{d}\vec{\boldsymbol{S}}\right)}_{=\vec{\boldsymbol{0}}} \cdot \mathrm{d}\vec{\boldsymbol{L}} = 0 \end{aligned}$$

since this equation holds true for an arbitrary $\mathrm{d}\vec{\boldsymbol{L}} \neq \vec{\boldsymbol{0}}$ then, $\boldsymbol{F}^T\,\mathrm{d}\vec{\boldsymbol{s}} = J\,\mathrm{d}\vec{\boldsymbol{S}}$ and $\mathrm{d}\vec{\boldsymbol{s}} = J\boldsymbol{F}^{-T}\,\mathrm{d}\vec{\boldsymbol{S}}$; taking into account that $\mathrm{d}\vec{\boldsymbol{S}} = \mathrm{d}S\hat{\boldsymbol{N}}$ and $\mathrm{d}\vec{\boldsymbol{s}} = \mathrm{d}s\hat{\boldsymbol{n}}$, we arrive to the so-called *Nanson's formula*:[7]

$$\boxed{\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}s = J(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S.} \tag{3.37}$$

This equation describes how the the unit normal vector $\hat{\boldsymbol{N}}$ transforms to $\hat{\boldsymbol{n}}$ under deformation and states that the area of the initial differential surface $\mathrm{d}S$ changes to $\left\| J\,\mathrm{d}S\boldsymbol{F}^{-T}\hat{\boldsymbol{N}} \right\|$. This equation will be used in Section 4.4 for the calculation of the "true" and the nominal "stress", when a deforming body is subject to a certain amount of force. This equation is important when analyzing surface phenomena, such as pressure loads, in deformable bodies: as the surface area changes during deformation, the applied pressure load must be adjusted accordingly to account for the change in the surface area.

3.12 Polar decomposition theorem

The *polar decomposition theorem* plays a pivotal role in continuum mechanics. It is useful to decompose a real invertible square matrix into a unique product of two

[7] After the British mathematician Edward John Nanson (1850–1936) who devised this formula in 1874: Nanson , E. J. (1874). *Note on hydrodynamics.* Mess. Math. 3, p. 120–121.

other matrices: one which is a positive-definite symmetric matrix and another that is an orthogonal matrix; In the case of the deformation gradient $\boldsymbol{F}$, these matrices represent the stretch and rotation components of $\boldsymbol{F}$, respectively. This decomposition is important because rotation does not contribute to stress while stretch does; also, this decomposition will allow us to provide a geometrical insight into the deformation of a body. In the next, we will delve into a mathematical explanation of the polar decomposition; then, we will apply it to the deformation gradient $\boldsymbol{F}$.

The polar decomposition of a square matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$ factorizes $\boldsymbol{A}$ into a product of the form $\boldsymbol{A} = \boldsymbol{RU} = \boldsymbol{vR}$, where $\boldsymbol{R}$ is an orthogonal matrix and $\boldsymbol{U}$ and $\boldsymbol{v}$ are positive semi-definite symmetric matrices, both real, square and of the same size as $\boldsymbol{A}$. We will say that $\boldsymbol{A} = \boldsymbol{RU}$ is the *right* and $\boldsymbol{A} = \boldsymbol{vR}$ is the *left polar decomposition* of $\boldsymbol{A}$. Since $\boldsymbol{R}$ is orthogonal, that is $\boldsymbol{R}^{-1} = \boldsymbol{R}^T$, matrices $\boldsymbol{U}$ and $\boldsymbol{v}$ are related by $\boldsymbol{U} = \boldsymbol{R}^T\boldsymbol{vR}$ and $\boldsymbol{v} = \boldsymbol{RUR}^T$.

The polar decomposition always exists; also, it is unique as long as $\boldsymbol{A}$ is invertible, which is the case of the deformation gradient $\boldsymbol{F}$ (see Section 3.9), inasmuch as $\det \boldsymbol{F} \neq 0$; in this latter case, $\boldsymbol{U}$ and $\boldsymbol{v}$ are positive definite matrices; this factorization is unique in the sense that there is only one orthogonal matrix $\boldsymbol{R}$ and unique positive definite matrices $\boldsymbol{U}$ and $\boldsymbol{v}$ that fulfill this decomposition; this will be shown in Section 3.19.4.

Let us apply the polar decomposition to the deformation gradient $\boldsymbol{F}$. In this case,

$$\boldsymbol{F} = \boldsymbol{RU} \qquad\qquad \boldsymbol{F} = \boldsymbol{vR};$$

at each point $\vec{\boldsymbol{X}} \in \Omega_0$ and at each time $t \in T$; on the one side, $\boldsymbol{R}$ describes pure rigid-body rotation and is known as the *rotation tensor** and is the same matrix on both equations; on the other side, $\boldsymbol{U}$ and $\boldsymbol{v}$ describe pure stretching or contraction around point $\vec{\boldsymbol{X}}$; in this case, $\boldsymbol{U}$ is called the *right* or *material stretch tensor* while $\boldsymbol{v}$ is referred to as the *left* or *spatial stretch tensor**. In the next, we will see that both $\boldsymbol{U}$ and $\boldsymbol{v}$ measure local stretching or contraction along its corresponding eigenvectors while $\boldsymbol{R}$ performs local rotation, that is a change of local orientation.

Following Figure 3.18, and the physical interpretation of the deformation gradient as an orthogonal tensor (see Section 3.10.2) or as a symmetric positive-definite tensor (see Section 3.10.4), we can provide a physical interpretation of the polar decomposition. The deformation gradient $\boldsymbol{F}$ transforms the material element $\mathrm{d}\boldsymbol{X}$ into the spatial element $\mathrm{d}\vec{\boldsymbol{x}}$ by means of equation (3.25), that is, $\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\vec{\boldsymbol{X}}$. This equation can be written as:

$$\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}} = \boldsymbol{R}(\boldsymbol{U}\,\mathrm{d}\vec{\boldsymbol{X}}) \tag{3.38}$$

or alternatively as:

$$\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}} = \boldsymbol{v}(\boldsymbol{R}\,\mathrm{d}\vec{\boldsymbol{X}}). \tag{3.39}$$

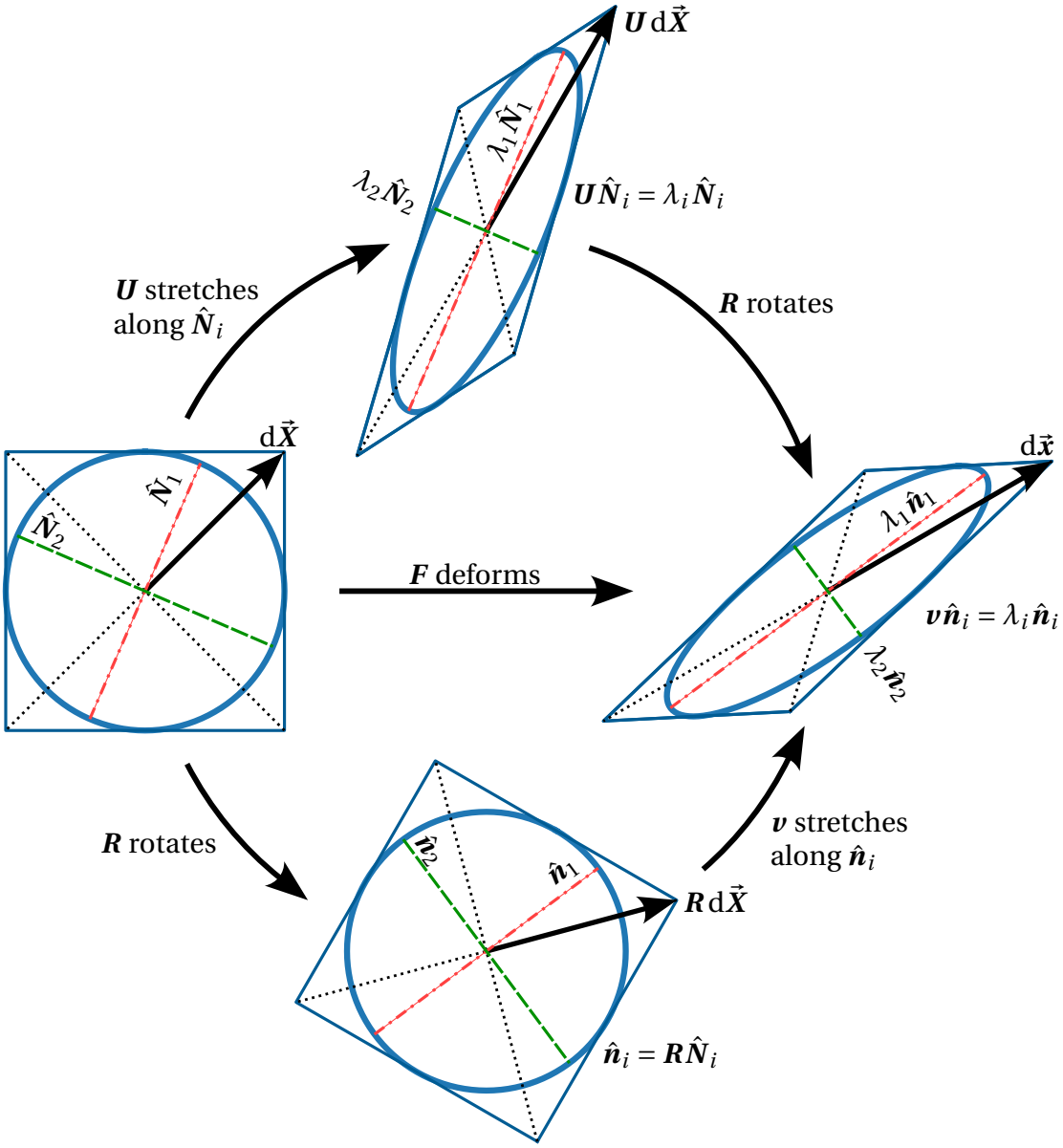


Figure 3.18: *The polar decomposition of the deformation tensor $\boldsymbol{F}$ decomposes the local motion into a pure stretch and a pure rotation. On the one hand, equation $\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}} = \boldsymbol{R}(\boldsymbol{U}\,\mathrm{d}\vec{\boldsymbol{X}})$ describes a pure stretch of $\mathrm{d}\vec{\boldsymbol{X}}$ by the material stretch tensor $\boldsymbol{U}$, followed by a pure rotation of $\boldsymbol{U}\,\mathrm{d}\vec{\boldsymbol{X}}$ performed by the rotation tensor $\boldsymbol{R}$; on the other hand, equation $\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}} = \boldsymbol{v}(\boldsymbol{R}\,\mathrm{d}\vec{\boldsymbol{X}})$ describes a pure rotation of the material element $\mathrm{d}\vec{\boldsymbol{X}}$ carried out by $\boldsymbol{R}$ followed by a pure stretch by the spatial stretch tensor $\boldsymbol{v}$ of the rotated tensor $\boldsymbol{R}\,\mathrm{d}\vec{\boldsymbol{X}}$ in order to form the spatial element $\mathrm{d}\vec{\boldsymbol{x}}$. It is clear that both processes yield the same deformation. Here, the dashed lines inside the circumferences and ellipses show the directions of the eigenvectors of $\boldsymbol{U}$ and $\boldsymbol{v}$.*

On the one hand, equation (3.38) describes a pure stretch of the material line element $d\vec{\boldsymbol{X}}$ by the material stretch tensor $\boldsymbol{U}$, followed by a pure rotation of $\boldsymbol{U}\,d\vec{\boldsymbol{X}}$ performed by the rotation tensor $\boldsymbol{R}$; on the other hand, equation (3.39) describes a pure rotation of the material line element $d\vec{\boldsymbol{X}}$ carried out by the rotation tensor $\boldsymbol{R}$ followed by a pure stretch of the rotated element $\boldsymbol{R}\,d\vec{\boldsymbol{X}}$ by the spatial stretch tensor $\boldsymbol{v}$; from a geometric point of view, it makes no difference which of these two paths is followed towards deformation, as the final result, the spatial line element $d\vec{\boldsymbol{x}}$, is the same in both cases.

Since $\boldsymbol{R}$, $\boldsymbol{U}$ and $\boldsymbol{v}$ can be understood as deformation gradients, they can be treated as two-point tensors. As discussed in Section 3.11.3, the determinants of two-point tensors are positive. Therefore, the determinants of $\boldsymbol{R}$, $\boldsymbol{U}$, and $\boldsymbol{v}$ are positive. Due to the fact that $\boldsymbol{R}$ is an orthogonal matrix, its determinant is either $+1$ or -1; however, due to the positive determinant requirement, we have $\det\boldsymbol{R} = +1$, which means that $\boldsymbol{R}$ is a proper rotation matrix representing a rigid body rotation without any change in volume. Furthermore, since the determinant of a product of matrices is equal to the product of their determinants, i.e., $\det(\boldsymbol{AB}) = \det\boldsymbol{A}\det\boldsymbol{B}$, it follows that $J = \det\boldsymbol{F} = \det\boldsymbol{U} = \det\boldsymbol{v} > 0$, where J is the Jacobian determinant. This means that either $\boldsymbol{U}$ or $\boldsymbol{v}$ are responsible for the change of volume in the material during deformation.

Observe that $\boldsymbol{R} = \boldsymbol{I}$ denotes *pure stretch*,* and in this case $\boldsymbol{F} = \boldsymbol{U} = \boldsymbol{v}$; on the other hand, $\boldsymbol{U} = \boldsymbol{v} = \boldsymbol{I}$ represents *pure rotation** as in the case of rigid-body rotation, and in this case $\boldsymbol{F} = \boldsymbol{R}$.

In Section 3.19, we will continue our discussion on the polar decomposition; in particular, we will show that $\boldsymbol{R}$ is an orthogonal tensor, that $\boldsymbol{U}$ and $\boldsymbol{v}$ are positive definite tensors and that this decomposition is unique. Also, we will present four different methods for the numerical calculation of the polar decomposition.

Example 3.8. *Polar decomposition*

Calculate the polar decomposition of the deformation analyzed in Example 3.7.

Solution:

Plot circumferences and ellipses with the corresponding eigenvalues and eigenvectors like in Kim example 3.4, page 151

3.13 Cauchy-Green deformation tensors

The deformation gradient $\boldsymbol{F}$, despite its importance in the analysis of deformation, is not a suitable measure of deformation. A suitable measure of deformation should satisfy the *invariability* property, which states that the deformation measure must not change in the absence of distortion, like in the case of rigid-body motion.

In order to verify that the deformation gradient is not invariant remember, from Section 3.10.2, that in case of rigid-body motion $\boldsymbol{F} = \boldsymbol{R}_\theta$, where $\boldsymbol{R}_\theta$ is a rotation matrix. Notice, $\boldsymbol{F}$ depends on the angle of rotation θ, even though in this situation there is no deformation or distortion.

Therefore, we are compelled to define another measure of deformation. In the following, we will define the Cauchy-Green deformation tensors, which will allow us to estimate the changes of length of material and spatial elements.

3.13.1 The right Cauchy-Green deformation tensor

*Consider two material elements $\mathrm{d}\vec{\boldsymbol{X}}_1$ and $\mathrm{d}\vec{\boldsymbol{X}}_2$, which deform into

$$\mathrm{d}\vec{\boldsymbol{x}}_1 = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_1 \quad \text{and} \quad \mathrm{d}\vec{\boldsymbol{x}}_2 = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_2. \tag{3.40}$$

We might indirectly measure the angle between $\mathrm{d}\vec{\boldsymbol{x}}_1$ and $\mathrm{d}\vec{\boldsymbol{x}}_2$ using the inner product, which serves as a measure of the parallelism between two vectors, so that:

$$\mathrm{d}\vec{\boldsymbol{x}}_1 \cdot \mathrm{d}\vec{\boldsymbol{x}}_2 = (\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_1)\cdot(\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_2) = \mathrm{d}\vec{\boldsymbol{X}}_1^{\,T}\underbrace{\boldsymbol{F}^T\boldsymbol{F}}_{\boldsymbol{C}}\mathrm{d}\vec{\boldsymbol{X}}_2 \tag{3.41}$$

where

$$\boxed{\boldsymbol{C}(\boldsymbol{X},t) := \boldsymbol{F}(\boldsymbol{X},t)^T\boldsymbol{F}(\boldsymbol{X},t)} \tag{3.42}$$

is a second-order tensor known as the *right Cauchy-Green deformation tensor*, named after the British miller and mathematical physicist George Green (1793–1841), who proposed it in 1839. Notice that $\det\boldsymbol{C} = (\det\boldsymbol{F})^2 = J^2 > 0$. For readers unfamiliar with the concept, the use of the dot product as a measure of parallelism may seem unconventional; however, it is one of the most popular tools in machine learning for measuring the similarity between two samples, in algorithms like recommendation systems (see e.g. https://en.wikipedia.org/wiki/Cosine_similarity).

The right Cauchy-Green tensor $\boldsymbol{C}$ is symmetric and positive definite[8]. In order to show symmetry, take into account that $\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F} = (\boldsymbol{F}^T\boldsymbol{F})^T = \boldsymbol{C}^T$, inasmuch as

[8] See Appendix A.12, for the definition of positive definiteness.

$\boldsymbol{B}^T\boldsymbol{A}^T = (\boldsymbol{AB})^T$. Now, when $\mathrm{d}\vec{\boldsymbol{X}} = \mathrm{d}\vec{\boldsymbol{X}}_1 = \mathrm{d}\vec{\boldsymbol{X}}_2$, it follows that $\mathrm{d}\vec{\boldsymbol{x}} = \mathrm{d}\vec{\boldsymbol{x}}_1 = \mathrm{d}\vec{\boldsymbol{x}}_2$ and equation (3.41) reduces to

$$\mathrm{d}\vec{\boldsymbol{x}}\cdot\mathrm{d}\vec{\boldsymbol{x}} = \|\mathrm{d}\vec{\boldsymbol{x}}\|^2 = \mathrm{d}\vec{\boldsymbol{X}}^T\boldsymbol{C}\,\mathrm{d}\vec{\boldsymbol{X}} > 0; \tag{3.43}$$

this shows that $\boldsymbol{C}(\boldsymbol{X}, t)$ is a positive definite tensor at each $\vec{\boldsymbol{X}} \in \Omega_0$, since this inequality holds for all $\mathrm{d}\vec{\boldsymbol{X}} \neq \vec{\boldsymbol{0}}$. Equation (3.43) also tells us that with the help of $\boldsymbol{C}$ we can compute the new squared length of $\mathrm{d}\vec{\boldsymbol{X}}$, that is $\|\mathrm{d}\vec{\boldsymbol{x}}\|^2$.

Alternatively, if we divide both sides of the equation by $\|\mathrm{d}\vec{\boldsymbol{X}}\|$, using the definition of the stretch ratio λ, in equation (3.33), and making $\hat{\boldsymbol{N}} = \frac{\mathrm{d}\vec{\boldsymbol{X}}}{\|\mathrm{d}\vec{\boldsymbol{X}}\|}$ we arrive to

$$\lambda^2 = \frac{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2}{\|\mathrm{d}\vec{\boldsymbol{X}}\|^2} = \frac{\mathrm{d}\vec{\boldsymbol{X}}}{\|\mathrm{d}\vec{\boldsymbol{X}}\|}\cdot\boldsymbol{C}\frac{\mathrm{d}\vec{\boldsymbol{X}}}{\|\mathrm{d}\vec{\boldsymbol{X}}\|} = \hat{\boldsymbol{N}}^T\boldsymbol{C}\hat{\boldsymbol{N}}; \tag{3.44}$$

this formula allows us to estimate the squared stretch ratio λ^2 of any material element in the direction $\hat{\boldsymbol{N}}$; therefore, the right Cauchy-Green deformation tensor $\boldsymbol{C}(\vec{\boldsymbol{X}}, t)$ determines the local deformation in the neighborhood of the material coordinate $\vec{\boldsymbol{X}}$ at time t. Take into account that $\lambda^2 = \hat{\boldsymbol{N}}^T\boldsymbol{C}\hat{\boldsymbol{N}} > 0$ inasmuch as $\boldsymbol{C}$ is a positive definite matrix.

According to the polar decomposition $\boldsymbol{F} = \boldsymbol{RU}$, and therefore,

$$\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F} = (\boldsymbol{RU})^T(\boldsymbol{RU}) = \boldsymbol{U}^T\underbrace{\boldsymbol{R}^T\boldsymbol{R}}_{=\boldsymbol{I}}\boldsymbol{U} = \underbrace{\boldsymbol{U}^T}_{=\boldsymbol{U}}\boldsymbol{U} = \boldsymbol{UU} = \boldsymbol{U}^2; \tag{3.45}$$

Therefore, using $\boldsymbol{F}$ we can calculate the 6 components of $\boldsymbol{C}$, but using $\boldsymbol{C}$ we cannot retrieve the 9 components of $\boldsymbol{F}$, inasmuch as $\boldsymbol{C}$ does not include information regarding the rotation that a particle undergoes during motion. Notice that $\boldsymbol{U}$, in contraposition to $\boldsymbol{C}$, has the advantage of the geometrical interpretation explained in Figures 3.10 and 3.18; however, it is easier to calculate $\boldsymbol{C}$ than $\boldsymbol{U}$; therefore, in applications $\boldsymbol{C}$ is usually employed.

Also, for a rigid-body motion, $\boldsymbol{F} = \boldsymbol{R}$; thus, $\boldsymbol{C} = \boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{I}$; this fact proofs that in rigid-body motion there is no deformation of the material.

Since $\boldsymbol{C} = \boldsymbol{U}^2$, we may write equation (3.45) as:

$$\lambda^2 = \hat{\boldsymbol{N}}^T\boldsymbol{C}\hat{\boldsymbol{N}} = \hat{\boldsymbol{N}}^T\boldsymbol{U}^T\boldsymbol{U}\hat{\boldsymbol{N}} = (\boldsymbol{U}\hat{\boldsymbol{N}})\cdot(\boldsymbol{U}\hat{\boldsymbol{N}}) = \left\|\boldsymbol{U}\hat{\boldsymbol{N}}\right\|^2;$$

that is, $|\lambda| = \left\|\boldsymbol{U}\hat{\boldsymbol{N}}\right\|$, and this justifies the fact that $\boldsymbol{U}$ is called an stretch tensor.

Finally, this is a good point to mention that the inverse of $\boldsymbol{C}$, namely $\boldsymbol{C}^{-1}(\vec{\boldsymbol{x}}, t) = (\boldsymbol{F}^T(\vec{\boldsymbol{X}}, t)\boldsymbol{F}(\vec{\boldsymbol{X}}, t))^{-1} = \boldsymbol{F}^{-1}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)$, is sometimes called the *Piola deformation tensor*.*

3.13.2 The left Cauchy-Green deformation tensor

★The two material elements $d\vec{\boldsymbol{X}}_1$ and $d\vec{\boldsymbol{X}}_2$ can be estimated from their spatial counterparts using equations (3.40) and (3.28), as

$$d\vec{\boldsymbol{X}}_1 = \boldsymbol{F}^{-1}\, d\vec{\boldsymbol{x}}_1 \quad \text{and} \quad d\vec{\boldsymbol{X}}_2 = \boldsymbol{F}^{-1}\, d\vec{\boldsymbol{x}}_2.$$

We wish to measure the similarity between $d\vec{\boldsymbol{X}}_1$ and $d\vec{\boldsymbol{X}}_2$ by using the inner product; in this case:

$$d\vec{\boldsymbol{X}}_1 \cdot d\vec{\boldsymbol{X}}_2 = (\boldsymbol{F}^{-1}\, d\vec{\boldsymbol{x}}_1)\cdot(\boldsymbol{F}^{-1}\, d\vec{\boldsymbol{x}}_2) = d\vec{\boldsymbol{x}}_1^T \underbrace{\boldsymbol{F}^{-T}\boldsymbol{F}^{-1}}_{\boldsymbol{b}^{-1}} d\vec{\boldsymbol{x}}_2 \tag{3.46}$$

where

$$\boldsymbol{b}^{-1}(\vec{\boldsymbol{x}}, t) = \boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}}, t) \tag{3.47}$$

and therefore

$$\boxed{\boldsymbol{b}(\boldsymbol{X}, t) := \boldsymbol{F}(\boldsymbol{X}, t)\boldsymbol{F}(\boldsymbol{X}, t)^T;}$$

this is a second-order symmetric tensor that is called the *left Cauchy-Green deformation tensor*[9] or the *Finger deformation tensor*, ★ after the Austrian physicist and mathematician Josef Finger (1841–1925). Symmetry follows from the fact that $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^T = (\boldsymbol{F}\boldsymbol{F}^T)^T = \boldsymbol{b}^T$.

When $d\vec{\boldsymbol{X}} = d\vec{\boldsymbol{X}}_1 = d\vec{\boldsymbol{X}}_2$, it follows that $d\vec{\boldsymbol{x}} = d\vec{\boldsymbol{x}}_1 = d\vec{\boldsymbol{x}}_2$ and equation (3.46) transforms into

$$d\vec{\boldsymbol{X}} \cdot d\vec{\boldsymbol{X}} = \|d\vec{\boldsymbol{X}}\|^2 = d\vec{\boldsymbol{x}}^T \boldsymbol{b}^{-1}\, d\vec{\boldsymbol{x}} > 0; \tag{3.48}$$

this shows that $\boldsymbol{b}^{-1}(\boldsymbol{x}, t)$ is a positive definite tensor at each $\vec{\boldsymbol{x}} \in \Omega_t$, since this inequality holds for all $d\vec{\boldsymbol{x}} \neq \vec{\boldsymbol{0}}$; the fact that every positive definite matrix is invertible guarantees that $\boldsymbol{b}$ exists; also since every positive definite matrix has positive definite inverse, then $\boldsymbol{b}(\boldsymbol{X}, t)$ is also positive definite at each $\vec{\boldsymbol{X}} \in \Omega_0$; equation (3.48) tells us that with the help of $\boldsymbol{b}$ we can compute the initial squared length of $d\vec{\boldsymbol{x}}$, that is $\|d\vec{\boldsymbol{X}}\|^2$; take into account that $\det \boldsymbol{b} = (\det \boldsymbol{F})^2 = J^2 > 0$.

Alternatively, if we divide both sides of the equation by $\|d\vec{\boldsymbol{x}}\|$, using the definition of the stretch λ, in equation (3.33), and making $\hat{\boldsymbol{n}} = \frac{d\vec{\boldsymbol{x}}}{\|d\vec{\boldsymbol{x}}\|}$, we arrive to

$$\lambda^{-2} = \frac{\|d\vec{\boldsymbol{X}}\|^2}{\|d\vec{\boldsymbol{x}}\|^2} = \frac{d\vec{\boldsymbol{x}}}{\|d\vec{\boldsymbol{x}}\|}\cdot \boldsymbol{b}^{-1}\frac{d\vec{\boldsymbol{x}}}{\|d\vec{\boldsymbol{x}}\|} = \hat{\boldsymbol{n}}^T \boldsymbol{b}^{-1}\hat{\boldsymbol{n}};$$

here λ^{-1} is known as the *inverse stretch ratio* ★ or simply the *inverse stretch*; this formula allows us to estimate the inverse of the squared stretch ratio of any spatial element in the direction $\hat{\boldsymbol{n}}$; therefore, the left Cauchy-Green deformation tensor $\boldsymbol{b}^{-1}(\vec{\boldsymbol{x}}, t)$

[9] The untransposed gradient deformation tensor $\boldsymbol{F}$ is on the left in $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^T$, while it is on the right in $\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F}$.

determines the local deformation in the neighborhood of the spatial coordinate $\vec{\boldsymbol{x}}$ at time t.

The polar decomposition states that $\boldsymbol{F} = \boldsymbol{v}\boldsymbol{R}$; therefore, we may write

$$\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^T = (\boldsymbol{v}\boldsymbol{R})(\boldsymbol{v}\boldsymbol{R})^T = \boldsymbol{v}\underbrace{\boldsymbol{R}^T\boldsymbol{R}}_{=\boldsymbol{I}}\boldsymbol{v}^T = \boldsymbol{v}\underbrace{\boldsymbol{v}^T}_{=\boldsymbol{v}} = \boldsymbol{v}\boldsymbol{v} = \boldsymbol{v}^2, \tag{3.49}$$

that is,

$$\boldsymbol{b} = \boldsymbol{v}^2; \tag{3.50}$$

this means $\boldsymbol{b}$ does not include information regarding the rotation that a particle undergoes during motion.

Note that, for a rigid-body motion, $\boldsymbol{F} = \boldsymbol{R}$; thus, $\boldsymbol{b} = \boldsymbol{R}\boldsymbol{R}^T = \boldsymbol{I}$; this fact shows that in rigid-body motion there is no deformation of the material.

3.13.3 Relationship between the left and the right Cauchy-Green tensors

In this section we will find the relationship between the left and the right Cauchy-Green tensors. Let us remember that by means of the polar decomposition $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{v}\boldsymbol{R}$; therefore, using the fact that $\boldsymbol{R}$ is an orthogonal matrix, i.e. $\boldsymbol{R}^{-1} = \boldsymbol{R}^T$,

$$\begin{aligned} \boldsymbol{U} &= \boldsymbol{R}^T\boldsymbol{v}\boldsymbol{R} \\ \boldsymbol{v} &= \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^T \end{aligned} \tag{3.51}$$

now,

$$\boldsymbol{C} = \boldsymbol{U}^2 = \left(\boldsymbol{R}^T\boldsymbol{v}\boldsymbol{R}\right)\left(\boldsymbol{R}^T\boldsymbol{v}\boldsymbol{R}\right) = \boldsymbol{R}^T\boldsymbol{v}\underbrace{\boldsymbol{R}\boldsymbol{R}^T}_{=\boldsymbol{I}}\boldsymbol{v}\boldsymbol{R} = \boldsymbol{R}^T\boldsymbol{v}^2\boldsymbol{R} = \boldsymbol{R}^T\boldsymbol{b}\boldsymbol{R}$$

in consequence,

$$\boxed{\boldsymbol{C} = \boldsymbol{R}^T\boldsymbol{b}\boldsymbol{R}.}$$

Now, following a similar procedure, it is left as an exercise to the reader to show that

$$\boxed{\boldsymbol{b} = \boldsymbol{R}\boldsymbol{C}\boldsymbol{R}^T.}$$

These equations enable us to transform the Cauchy-Green tensors between the material and spatial configurations using the rotation tensor $\boldsymbol{R}$; this transformation is very important when dealing with large deformations, where the material and spatial configurations can differ significantly.

Example 3.9. *Cauchy-Green tensors*

Calculate the left and right Cauchy-Green tensors of the deformation analyzed in Examples 3.7 and 3.8.

Solution:

To do. In Example 3.10, we will interpret what do these tensors mean.

3.14 Finite strain tensors

In continuum mechanics, the concept of strain is fundamental to quantifying deformation. While *deformation* describes any change in shape or size of a body, *strain* specifically quantifies that change relative to the original dimensions of the body. More precisely, strain tensors characterize how a given displacement field deviates locally from a rigid-body motion, capturing changes in shape, size, angles between material lines, and distances between material points.

In classical elasticity theory, the infinitesimal strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$, defined by equation (2.1), is commonly used to measure deformation. Its diagonal components quantify longitudinal strains (changes in length), while its off-diagonal components represent shear strains (changes in angles between material lines), as detailed in Section 2.1. However, $\underline{\underline{\boldsymbol{\varepsilon}}}$ is limited to scenarios involving small deformations where changes in length and angles are minimal.

To overcome this limitation, we need to develop strain measures that accurately quantify strain for large deformations, converge to the infinitesimal strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$ for small deformations, and remain invariant under rigid-body motions.

In this section, we will introduce two such finite strain tensors that fulfill these criteria. These tensors provide different perspectives on deformation, depending on whether they are expressed in terms of the body's initial (reference) or deformed (current) configuration.

As a starting point, recall from Section 3.13 that the Cauchy-Green deformation tensors ($\boldsymbol{C}$ and $\boldsymbol{b}$) effectively measure changes in length and angles. However, they fall short of our third criterion: they are not null during rigid-body motion. Specifically, we demonstrated that for rigid-body motions $\boldsymbol{C}(\boldsymbol{X}, t) = \boldsymbol{b}^{-1}(\boldsymbol{x}, t) = \boldsymbol{I}$.

While this equality confirms the absence of deformation around the material coordinate $\vec{\boldsymbol{X}}$ or the spatial coordinate $\vec{\boldsymbol{x}}$ during rigid-body motion, it does not align with our conventional use of $\mathbf{0}$ to denote the absence of a quantity.

To address these requirements, we will now introduce two finite strain tensors: the Cauchy-Lagrange strain tensor (also known as the Green-Lagrange strain tensor), and the Euler-Almansi strain tensor. They possess the following key properties:

- They are null during rigid-body motion, aligning with our intuitive understanding of zero strain.
- They generalize the concept of the infinitesimal strain tensor to large deformations.
- They converge to the infinitesimal strain tensor $\boldsymbol{\varepsilon}$ for small deformations.
- They remain invariant under rigid-body motions.

The Cauchy-Lagrange strain tensor is formulated with respect to the reference configuration, making it particularly useful for Lagrangian formulations. Conversely, the Euler-Almansi strain tensor is expressed in terms of the current configuration, making it suitable for Eulerian formulations.

In the following subsections, we will derive these strain tensors, explore their properties, and discuss their relationships to other deformation measures. We will also examine their physical interpretations.

3.14.1 Cauchy-Lagrange strain tensor

★If we subtract from the right Cauchy-Green tensor $\boldsymbol{C}$ the identity matrix $\boldsymbol{I}$, we will obtain a deformation measure that is zero in case of no deformation. In consequence, we will define a new measure of deformation by

$$\boldsymbol{E}(\vec{\boldsymbol{X}},t)=\frac{\boldsymbol{C}(\vec{\boldsymbol{X}},t)-\boldsymbol{I}}{2} \tag{3.52}$$

which will be referred to as the *Cauchy-Lagrange (material) strain tensor*. We shall see, in Section 3.15.1, that the $1/2$ factor is required to make tensor $\boldsymbol{E}$ identical to the deformation tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$, employed within elasticity theory, in the case of infinitesimal strains. This tensor is a symmetric one since both $\boldsymbol{C}$ and $\boldsymbol{I}$ are symmetric tensors.

Using equations (3.27) ($\boldsymbol{F}=\frac{\partial\vec{\boldsymbol{U}}}{\partial\vec{\boldsymbol{X}}}+\boldsymbol{I}$) and (3.42) ($\boldsymbol{C}=\boldsymbol{F}^T\boldsymbol{F}$), it yields

$$\begin{aligned}\boldsymbol{E}(\vec{\boldsymbol{X}},t)&=\frac{1}{2}\left[\left(\frac{\partial\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}+\boldsymbol{I}\right)^T\left(\frac{\partial\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}+\boldsymbol{I}\right)-\boldsymbol{I}\right]\\&=\frac{1}{2}\left[\frac{\partial\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}+\frac{\partial\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}^T+\frac{\partial\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}^T\frac{\partial\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial\vec{\boldsymbol{X}}}\right]\end{aligned}$$

or in indicial notation:

$$E_{ij}=\frac{C_{ij}-\delta_{ij}}{2}=\frac{1}{2}\left[\frac{\partial U_i}{\partial X_j}+\frac{\partial U_j}{\partial X_i}+\frac{\partial U_k}{\partial X_i}\frac{\partial U_k}{\partial X_j}\right],$$

where δ_{ij} is the Kronecker delta.

This equation can be expanded with the help of SYMPY, using the following code:

```
from sympy import expand, eye, Function, Matrix, symbols

# define the material coordinates
X, Y, Z = symbols('X, Y, Z')
vec_X = Matrix([X, Y, Z])

# define the displacement vector field in material coordinates
U = Function('U')(X, Y, Z)
V = Function('V')(X, Y, Z)
W = Function('W')(X, Y, Z)
vec_U = Matrix([U, V, W])

# calculate the deformation gradient, equation (3.27)
I = eye(3)
F = I + vec_U.jacobian(vec_X)

# calculate the right Cauchy-Green deformation tensor, equation (3.42)
C = F.T*F

# calculate the Cauchy-Lagrange strain tensor, equation (3.52)
E = (C - I)/2

# extract the terms from the Cauchy-Lagrange strain tensor, equations (3.53)
EX  = expand(E[0,0])
EY  = expand(E[1,1])
EZ  = expand(E[2,2])
EXY = expand(E[0,1])
EXZ = expand(E[0,2])
EYZ = expand(E[1,2])
```

so that,

$$\boldsymbol{E} = \begin{pmatrix} E_x & E_{xy} & E_{xz} \\ E_{xy} & E_y & E_{yz} \\ E_{xz} & E_{yz} & E_z \end{pmatrix}$$

with

$$
\begin{aligned}
E_x &= \frac{\partial U}{\partial X} + \frac{1}{2}\left[\left(\frac{\partial U}{\partial X}\right)^2 + \left(\frac{\partial V}{\partial X}\right)^2 + \left(\frac{\partial W}{\partial X}\right)^2\right] \\
E_y &= \frac{\partial V}{\partial Y} + \frac{1}{2}\left[\left(\frac{\partial U}{\partial Y}\right)^2 + \left(\frac{\partial V}{\partial Y}\right)^2 + \left(\frac{\partial W}{\partial Y}\right)^2\right] \\
E_z &= \frac{\partial W}{\partial Z} + \frac{1}{2}\left[\left(\frac{\partial U}{\partial Z}\right)^2 + \left(\frac{\partial V}{\partial Z}\right)^2 + \left(\frac{\partial W}{\partial Z}\right)^2\right] \\
E_{xy} &= \frac{1}{2}\left(\frac{\partial U}{\partial Y} + \frac{\partial V}{\partial X}\right) + \frac{1}{2}\left[\frac{\partial U}{\partial X}\frac{\partial U}{\partial Y} + \frac{\partial V}{\partial X}\frac{\partial V}{\partial Y} + \frac{\partial W}{\partial X}\frac{\partial W}{\partial Y}\right] \\
E_{xz} &= \frac{1}{2}\left(\frac{\partial U}{\partial Z} + \frac{\partial W}{\partial X}\right) + \frac{1}{2}\left[\frac{\partial U}{\partial X}\frac{\partial U}{\partial Z} + \frac{\partial V}{\partial X}\frac{\partial V}{\partial Z} + \frac{\partial W}{\partial X}\frac{\partial W}{\partial Z}\right] \\
E_{yz} &= \frac{1}{2}\left(\frac{\partial V}{\partial Z} + \frac{\partial W}{\partial Y}\right) + \frac{1}{2}\left[\frac{\partial U}{\partial Y}\frac{\partial U}{\partial Z} + \frac{\partial V}{\partial Y}\frac{\partial V}{\partial Z} + \frac{\partial W}{\partial Y}\frac{\partial W}{\partial Z}\right]
\end{aligned}
\tag{3.53}
$$

Take into account that in rigid-body motion $\boldsymbol{C} = \boldsymbol{I}$ and thus, $\boldsymbol{E} = \boldsymbol{0}$; this means that in rigid body motion there are neither strains not stresses.

Note that since $\boldsymbol{E}$ is calculated using the right Cauchy-Green tensor $\boldsymbol{C} = \boldsymbol{U}^2$, it does not include information about the rotation that is experienced by the particle at $\vec{\boldsymbol{X}}$ during motion, therefore this strain measure is invariant under rigid-body motions.

3.14.2 Euler-Almansi strain tensor

★If we subtract the inverse of the left Cauchy-Green tensor $\boldsymbol{b}^{-1}$ from the identity matrix $\boldsymbol{I}$, we will obtain a deformation measure that is zero in case of no deformation. Consequently, we define a new measure of deformation by

$$
\boldsymbol{e}(\vec{\boldsymbol{x}}, t) = \frac{\boldsymbol{I} - \boldsymbol{b}^{-1}(\vec{\boldsymbol{x}}, t)}{2} \tag{3.54}
$$

which will be referred to as the *Euler-Almansi (spatial) strain tensor*. The reason for the division by two will be understood later, in Section 3.15.1, when we compare this tensor to the infinitesimal deformation tensor $\boldsymbol{\varepsilon}$ employed within elasticity theory. This tensor is symmetric since both $\boldsymbol{b}^{-1}$ and $\boldsymbol{I}$ are symmetric tensors.

Using equations (3.30) ($\boldsymbol{F}^{-1} = \boldsymbol{I} - \frac{\partial \vec{\boldsymbol{u}}}{\partial \vec{\boldsymbol{x}}}$) and (3.47) ($\boldsymbol{b}^{-1} = \boldsymbol{F}^{-T}\boldsymbol{F}^{-1}$), it yields

$$
\begin{aligned}
\boldsymbol{e}(\vec{\boldsymbol{x}}, t) &= \frac{1}{2}\left[\boldsymbol{I} - \left(\boldsymbol{I} - \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}}\right)^T \left(\boldsymbol{I} - \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}}\right)\right] \\
&= \frac{1}{2}\left[\frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}} + \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}}^T - \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{X}}, t)}{\partial \vec{\boldsymbol{x}}}^T \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}}\right]
\end{aligned}
$$

or in indicial notation:

$$
e_{ij} = \frac{\delta_{ij} - \left[\boldsymbol{b}^{-1}\right]_{ij}}{2} = \frac{1}{2}\left[\frac{\partial u_i}{\partial x_j} + \frac{\partial u_j}{\partial x_i} - \frac{\partial u_k}{\partial x_i}\frac{\partial u_k}{\partial x_j}\right].
$$

This equation can be expanded with the help of SYMPY, using the following code:

```
from sympy import expand, eye, Function, Matrix, symbols

# define the spatial coordinates
x, x, z = symbols('x, x, z')
vec_x = Matrix([x, x, z])

# define the displacement vector field in spatial coordinates
u = Function('u')(x, x, z)
v = Function('v')(x, x, z)
w = Function('w')(x, x, z)
vec_u = Matrix([u, v, w])

# calculate the inverse of the deformation gradient, equation (3.30)
I = eye(3)
inv_F = I - vec_u.jacobian(vec_x)

# calculate the inverse of the left Cauchy-Green deformat. tensor, eq. (3.47)
inv_b = inv_F.T*inv_F

# calculate the Euler-Almansi strain tensor, equation (3.54)
e = (I - inv_b)/2

# extract the terms from the Euler-Almansi strain tensor, equations (3.55)
ex  = expand(e[0,0])
ey  = expand(e[1,1])
ez  = expand(e[2,2])
exy = expand(e[0,1])
exz = expand(e[0,2])
eyz = expand(e[1,2])
```

so that,

$$\boldsymbol{e} = \begin{pmatrix} e_x & e_{xy} & e_{xz} \\ e_{xy} & e_y & e_{yz} \\ e_{xz} & e_{yz} & e_z \end{pmatrix}$$

with

$$
\begin{aligned}
e_x &= \frac{\partial u}{\partial x} - \frac{1}{2}\left[\left(\frac{\partial u}{\partial x}\right)^2 + \left(\frac{\partial v}{\partial x}\right)^2 + \left(\frac{\partial w}{\partial x}\right)^2\right] \\
e_y &= \frac{\partial v}{\partial y} - \frac{1}{2}\left[\left(\frac{\partial u}{\partial y}\right)^2 + \left(\frac{\partial v}{\partial y}\right)^2 + \left(\frac{\partial w}{\partial y}\right)^2\right] \\
e_z &= \frac{\partial w}{\partial z} - \frac{1}{2}\left[\left(\frac{\partial u}{\partial z}\right)^2 + \left(\frac{\partial v}{\partial z}\right)^2 + \left(\frac{\partial w}{\partial z}\right)^2\right] \\
e_{xy} &= \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) - \frac{1}{2}\left[\frac{\partial u}{\partial x}\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\frac{\partial v}{\partial y} + \frac{\partial w}{\partial x}\frac{\partial w}{\partial y}\right] \\
e_{xz} &= \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) - \frac{1}{2}\left[\frac{\partial u}{\partial x}\frac{\partial u}{\partial z} + \frac{\partial v}{\partial x}\frac{\partial v}{\partial z} + \frac{\partial w}{\partial x}\frac{\partial w}{\partial z}\right] \\
e_{yz} &= \frac{1}{2}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right) - \frac{1}{2}\left[\frac{\partial u}{\partial y}\frac{\partial u}{\partial z} + \frac{\partial v}{\partial y}\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\frac{\partial w}{\partial z}\right].
\end{aligned}
\tag{3.55}
$$

Take into account that in rigid-body motion $\boldsymbol{b}^{-1} = \boldsymbol{I}$ and thus, $\boldsymbol{e} = \boldsymbol{0}$; this means that rigid body motion induces no strains, consequently the body remains unstressed.

Note that since $\boldsymbol{e}$ is calculated using the left Cauchy-Green tensor $\boldsymbol{b} = \boldsymbol{v}^2$, it does not include information about the rotation experienced by the particle at $\vec{\boldsymbol{x}}$ during motion, and thus this strain tensor is invariant under rigid-body motions.

In summary, both the Cauchy-Lagrange and Euler-Almansi strain tensors are measures of deformation that are zero in rigid-body motion; we will see in Section 3.15 that they converge to the infinitesimal deformation tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$ for small deformations. However, they differ in their formulations and the reference configurations they are based on (material and spatial, respectively). Additionally, neither tensor accounts for the rotation experienced by the particle during motion.

3.14.3 Relationship between the Cauchy-Lagrange and the Euler-Almansi strain tensors

Both the Cauchy-Lagrange and the Euler-Almansi strain tensors can be expressed as the change in the scalar product $\frac{1}{2}\left[\mathrm{d}\vec{\boldsymbol{x}}_i \cdot \mathrm{d}\vec{\boldsymbol{x}}_j - \vec{\boldsymbol{X}}_i \cdot \mathrm{d}\vec{\boldsymbol{X}}_j\right]$, where $\mathrm{d}\vec{\boldsymbol{x}}_i = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_i$ and $\mathrm{d}\vec{\boldsymbol{x}}_j = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_j$.

To illustrate this relationship, let's first express this quantity in terms of the material elements $\mathrm{d}\vec{\boldsymbol{X}}_i$ and $\mathrm{d}\vec{\boldsymbol{X}}_j$:

$$
\begin{aligned}
\frac{1}{2}\left[\mathrm{d}\vec{\boldsymbol{x}}_i \cdot \mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i \cdot \mathrm{d}\vec{\boldsymbol{X}}_j\right] &= \frac{1}{2}\left[\left(\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_i\right) \cdot \left(\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_j\right) - \mathrm{d}\vec{\boldsymbol{X}}_i \cdot \boldsymbol{I}\,\mathrm{d}\vec{\boldsymbol{X}}_j\right] \\
&= \frac{1}{2}\left[\left(\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_i\right)^T \left(\boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_j\right) - \mathrm{d}\vec{\boldsymbol{X}}_i^T \boldsymbol{I}\,\mathrm{d}\vec{\boldsymbol{X}}_j\right] \\
&= \frac{1}{2}\left[\mathrm{d}\vec{\boldsymbol{X}}_i^T \boldsymbol{F}^T \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i^T \boldsymbol{I}\,\mathrm{d}\vec{\boldsymbol{X}}_j\right]
\end{aligned}
$$

$$
\begin{aligned}
&= \frac{1}{2} \mathrm{d}\vec{\boldsymbol{X}}_i^T \left[\boldsymbol{F}^T \boldsymbol{F} - \boldsymbol{I} \right] \mathrm{d}\vec{\boldsymbol{X}}_j \\
&= \mathrm{d}\vec{\boldsymbol{X}}_i^T \frac{\boldsymbol{C} - \boldsymbol{I}}{2} \mathrm{d}\vec{\boldsymbol{X}}_j \\
&= \mathrm{d}\vec{\boldsymbol{X}}_i^T \boldsymbol{E} \, \mathrm{d}\vec{\boldsymbol{X}}_j ;
\end{aligned}
$$

this expression represents the change in scalar product in terms of the material elements and the Cauchy-Lagrange strain tensor.

Next, let's express the same quantity in terms of the spatial elements $\mathrm{d}\vec{\boldsymbol{x}}_i$ and $\mathrm{d}\vec{\boldsymbol{x}}_j$:

$$
\begin{aligned}
\frac{1}{2} \left[\mathrm{d}\vec{\boldsymbol{x}}_i \cdot \mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i \cdot \mathrm{d}\vec{\boldsymbol{X}}_j \right] &= \frac{1}{2} \left[\mathrm{d}\vec{\boldsymbol{x}}_i \cdot \mathrm{d}\vec{\boldsymbol{x}}_j - \left(\boldsymbol{F}^{-1} \mathrm{d}\vec{\boldsymbol{x}}_i \right) \cdot \left(\boldsymbol{F}^{-1} \mathrm{d}\vec{\boldsymbol{x}}_j \right) \right] \\
&= \frac{1}{2} \left[\mathrm{d}\vec{\boldsymbol{x}}_i^T \boldsymbol{I} \, \mathrm{d}\vec{\boldsymbol{x}}_j - \left(\boldsymbol{F}^{-1} \mathrm{d}\vec{\boldsymbol{x}}_i \right)^T \left(\boldsymbol{F}^{-1} \mathrm{d}\vec{\boldsymbol{x}}_j \right) \right] \\
&= \frac{1}{2} \left[\mathrm{d}\vec{\boldsymbol{x}}_i^T \boldsymbol{I} \, \mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{x}}_i^T \boldsymbol{F}^{-T} \boldsymbol{F}^{-1} \mathrm{d}\vec{\boldsymbol{x}}_j \right] \\
&= \frac{1}{2} \mathrm{d}\vec{\boldsymbol{x}}_i^T \left[\boldsymbol{I} - \boldsymbol{F}^{-T} \boldsymbol{F}^{-1} \right] \mathrm{d}\vec{\boldsymbol{x}}_j \\
&= \frac{1}{2} \mathrm{d}\vec{\boldsymbol{x}}_i^T \left[\boldsymbol{I} - \left(\boldsymbol{F} \boldsymbol{F}^T \right)^{-1} \right] \mathrm{d}\vec{\boldsymbol{x}}_j \\
&= \mathrm{d}\vec{\boldsymbol{x}}_i^T \frac{\boldsymbol{I} - \boldsymbol{b}^{-1}}{2} \mathrm{d}\vec{\boldsymbol{x}}_j \\
&= \mathrm{d}\vec{\boldsymbol{x}}_i^T \boldsymbol{e} \, \mathrm{d}\vec{\boldsymbol{x}}_j ;
\end{aligned}
$$

this expression represents the change in scalar product in terms of the spatial elements and the Euler-Almansi strain tensor.

In summary, we have the following relationship:

$$
\boxed{\frac{\mathrm{d}\vec{\boldsymbol{x}}_i \cdot \mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i \cdot \mathrm{d}\vec{\boldsymbol{X}}_j}{2} = \mathrm{d}\vec{\boldsymbol{X}}_i^T \boldsymbol{E} \, \mathrm{d}\vec{\boldsymbol{X}}_j = \mathrm{d}\vec{\boldsymbol{x}}_i^T \boldsymbol{e} \, \mathrm{d}\vec{\boldsymbol{x}}_j .} \tag{3.56}
$$

In the special case where $\mathrm{d}\vec{\boldsymbol{X}}_i = \mathrm{d}\vec{\boldsymbol{X}}_j = \mathrm{d}\vec{\boldsymbol{X}}$, and consequently $\mathrm{d}\vec{\boldsymbol{x}}_i = \mathrm{d}\vec{\boldsymbol{x}}_j = \mathrm{d}\vec{\boldsymbol{x}}$, the equations simplify to:

$$
\frac{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2 - \|\mathrm{d}\vec{\boldsymbol{X}}\|^2}{2} = \mathrm{d}\vec{\boldsymbol{X}}^T \boldsymbol{E} \, \mathrm{d}\vec{\boldsymbol{X}} = \mathrm{d}\vec{\boldsymbol{x}}^T \boldsymbol{e} \, \mathrm{d}\vec{\boldsymbol{x}} .
$$

This equation represents the change in square length that occurs as the body deforms from the initial to the current configuration.

3.14.4 Physical interpretation of the Cauchy-Lagrange strain tensor and the right Cauchy-Green tensor

To understand the physical interpretation of each term in $\boldsymbol{E}$, we employ equation (3.56), which is rewritten here for the sake of convenience:

$$\mathrm{d}\vec{\boldsymbol{X}}_i^T\,\boldsymbol{E}\,\mathrm{d}\vec{\boldsymbol{X}}_j = \frac{\mathrm{d}\vec{\boldsymbol{x}}_i\cdot\mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i\cdot\mathrm{d}\vec{\boldsymbol{X}}_j}{2}. \tag{3.57}$$

If we divide this equation by $\|\mathrm{d}\vec{\boldsymbol{X}}_i\|\,\|\mathrm{d}\vec{\boldsymbol{X}}_j\|$ and define $\hat{\boldsymbol{N}}_i = \frac{\mathrm{d}\vec{\boldsymbol{X}}_i}{\|\mathrm{d}\vec{\boldsymbol{X}}_i\|}$ and $\hat{\boldsymbol{N}}_j = \frac{\mathrm{d}\vec{\boldsymbol{X}}_j}{\|\mathrm{d}\vec{\boldsymbol{X}}_j\|}$ as the unit vectors in the directions of $\mathrm{d}\vec{\boldsymbol{X}}_i$ and $\mathrm{d}\vec{\boldsymbol{X}}_j$, respectively, it follows that

$$\frac{\mathrm{d}\vec{\boldsymbol{X}}_i^T}{\|\mathrm{d}\vec{\boldsymbol{X}}_i\|}\boldsymbol{E}\frac{\mathrm{d}\vec{\boldsymbol{X}}_j}{\|\mathrm{d}\vec{\boldsymbol{X}}_j\|} = \frac{1}{2}\frac{\mathrm{d}\vec{\boldsymbol{x}}_i\cdot\mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i\cdot\mathrm{d}\vec{\boldsymbol{X}}_j}{\|\mathrm{d}\vec{\boldsymbol{X}}_i\|\,\|\mathrm{d}\vec{\boldsymbol{X}}_j\|} = \hat{\boldsymbol{N}}_i^T\boldsymbol{E}\hat{\boldsymbol{N}}_j,$$

and in the special case where $\mathrm{d}\vec{\boldsymbol{X}}_i = \mathrm{d}\vec{\boldsymbol{X}}_j = \mathrm{d}\vec{\boldsymbol{X}}$ and $\hat{\boldsymbol{N}}_i = \hat{\boldsymbol{N}}_j = \hat{\boldsymbol{N}} = \frac{\mathrm{d}\vec{\boldsymbol{X}}}{\|\mathrm{d}\vec{\boldsymbol{X}}\|}$, we obtain the so-called *Green's scalar strain*[★] in the direction $\hat{\boldsymbol{N}}$:

$$\boxed{\hat{\boldsymbol{N}}^T\boldsymbol{E}\hat{\boldsymbol{N}} = \frac{1}{2}\frac{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2 - \|\mathrm{d}\vec{\boldsymbol{X}}\|^2}{\|\mathrm{d}\vec{\boldsymbol{X}}\|^2} = \frac{\lambda^2-1}{2};} \tag{3.58}$$

it represents the change in squared length of line element that was in the direction $\hat{\boldsymbol{N}}$ before deformation, relative to its undeformed squared length; here λ stands for the corresponding stretch ratio.

With reference to Figure 3.19, let us define the material elements $\mathrm{d}\vec{\boldsymbol{X}}_1 = \mathrm{d}S_1\hat{\boldsymbol{e}}_1$, $\mathrm{d}\vec{\boldsymbol{X}}_2 = \mathrm{d}S_2\hat{\boldsymbol{e}}_2$ and $\mathrm{d}\vec{\boldsymbol{X}}_3 = \mathrm{d}S_3\hat{\boldsymbol{e}}_3$, where $\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{\jmath}}$, and $\hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{k}}$; also let us define the spatial elements $\mathrm{d}\vec{\boldsymbol{x}}_1 = \mathrm{d}s_1\hat{\boldsymbol{n}}_1$, $\mathrm{d}\vec{\boldsymbol{x}}_2 = \mathrm{d}s_2\hat{\boldsymbol{n}}_2$ and $\mathrm{d}\vec{\boldsymbol{x}}_3 = \mathrm{d}s_3\hat{\boldsymbol{n}}_3$, where $\mathrm{d}\vec{\boldsymbol{x}}_i = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}_i$; here $\hat{\boldsymbol{n}}_i$ should not be understood as an eigenvector but as the normalized spatial element $\mathrm{d}\vec{\boldsymbol{x}}_i$, i.e., $\hat{\boldsymbol{n}}_i = \frac{\mathrm{d}\vec{\boldsymbol{x}}_i}{\|\mathrm{d}\vec{\boldsymbol{x}}_i\|}$; then, from (3.57):

$$\mathrm{d}S_i\,\mathrm{d}S_j\hat{\boldsymbol{e}}_i^T\boldsymbol{E}\hat{\boldsymbol{e}}_j = \frac{\mathrm{d}s_i\,\mathrm{d}s_j\hat{\boldsymbol{n}}_i\cdot\hat{\boldsymbol{n}}_j - \mathrm{d}S_i\,\mathrm{d}S_j\hat{\boldsymbol{e}}_i\cdot\hat{\boldsymbol{e}}_j}{2};$$

taking into account that $E_{ij} = \hat{\boldsymbol{e}}_i^T\boldsymbol{E}\hat{\boldsymbol{e}}_j$, that $\hat{\boldsymbol{n}}_i\cdot\hat{\boldsymbol{n}}_j = \cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{x}}_i,\mathrm{d}\vec{\boldsymbol{x}}_j\right)$ represents the cosine of the angle between $\mathrm{d}\vec{\boldsymbol{x}}_i$, and that $\hat{\boldsymbol{e}}_i\cdot\hat{\boldsymbol{e}}_j = \delta_{ij}$ is the Kronecker delta, we obtain:

$$\begin{aligned} E_{ij} &= \frac{C_{ij}-\delta_{ij}}{2} \\ &= \frac{\mathrm{d}s_i\,\mathrm{d}s_j\cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{x}}_i,\mathrm{d}\vec{\boldsymbol{x}}_j\right) - \mathrm{d}S_i\,\mathrm{d}S_j\delta_{ij}}{2\,\mathrm{d}S_i\,\mathrm{d}S_j} \end{aligned}$$

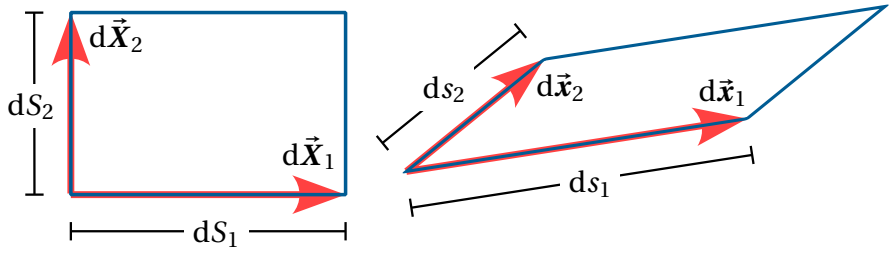


Figure 3.19: *Figure (a) shows an undeformed rectangular area element with sides* dS_1 *and* dS_2*, defined by material vectors* $d\vec{\boldsymbol{X}}_1 = dS_1\hat{\boldsymbol{i}}$ *and* $d\vec{\boldsymbol{X}}_2 = dS_2\hat{\boldsymbol{j}}$*. Figure (b) illustrates the corresponding deformed area element, showing how the original rectangular shape transforms into a parallelogram. The deformed element is characterized by spatial vectors* $d\vec{\boldsymbol{x}}_1 = \boldsymbol{F}\,d\vec{\boldsymbol{x}}_1 = ds_1\hat{\boldsymbol{n}}_1$ *and* $d\vec{\boldsymbol{x}}_2 = \boldsymbol{F}\,d\vec{\boldsymbol{x}}_2 = ds_2\hat{\boldsymbol{n}}_2$*, with magnitudes* ds_1 *and* ds_2 *respectively. The corresponding stretches are given by* $\lambda_i = ds_i / dS_i$*.*

$$= \frac{\lambda_i \lambda_j \cos\measuredangle\left(d\vec{\boldsymbol{x}}_i, d\vec{\boldsymbol{x}}_j\right) - \delta_{ij}}{2}$$

where λ_1, λ_2, and λ_3 are, respectively, the stretches of the material line elements that were originally in the $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$, and $\hat{\boldsymbol{k}}$ direction, that is $\lambda_i = \frac{ds_i}{dS_i}$.

Consequently, the diagonal terms of the Cauchy-Lagrange strain tensor correspond to:

$$E_x = E_{11} = \frac{C_{11} - 1}{2} = \frac{ds_1^2 - dS_1^2}{2\,dS_1^2} = \frac{\lambda_1^2 - 1}{2}$$
$$E_y = E_{22} = \frac{C_{22} - 1}{2} = \frac{ds_2^2 - dS_2^2}{2\,dS_2^2} = \frac{\lambda_2^2 - 1}{2}$$
$$E_z = E_{33} = \frac{C_{33} - 1}{2} = \frac{ds_3^2 - dS_3^2}{2\,dS_3^2} = \frac{\lambda_3^2 - 1}{2};$$

these are, according to equation (3.58), the Green's scalar strains in the $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$, and $\hat{\boldsymbol{k}}$ directions, respectively, representing the change in squared lengths of the material line elements along those directions.

On the other hand, the off-diagonal elements are:

$$E_{xy} = E_{yx} = E_{12} = E_{21} = \frac{C_{12}}{2} = \frac{C_{21}}{2} = \frac{\lambda_1\lambda_2}{2}\cos\measuredangle\left(d\vec{\boldsymbol{x}}_1, d\vec{\boldsymbol{x}}_2\right)$$
$$E_{xz} = E_{zx} = E_{13} = E_{31} = \frac{C_{13}}{2} = \frac{C_{31}}{2} = \frac{\lambda_1\lambda_3}{2}\cos\measuredangle\left(d\vec{\boldsymbol{x}}_1, d\vec{\boldsymbol{x}}_3\right)$$
$$E_{yz} = E_{zy} = E_{23} = E_{32} = \frac{C_{23}}{2} = \frac{C_{32}}{2} = \frac{\lambda_2\lambda_3}{2}\cos\measuredangle\left(d\vec{\boldsymbol{x}}_2, d\vec{\boldsymbol{x}}_3\right);$$

these off-diagonal terms are proportional to the product of the stretches along the corresponding coordinate directions, multiplied by the cosine of the angle between the deformed line elements that were originally in the coordinate directions; for instance, $E_{xy} = E_{12}$ corresponds to $\frac{\lambda_1\lambda_2}{2}$ times the cosine of the deformed angle between two spatial line elements that were originally in the $\hat{\boldsymbol{i}}$ and $\hat{\boldsymbol{j}}$ directions, an angle that was initially 90° before deformation. A non-zero value of these off-diagonal terms indicates a change in the angle between the material line elements.

Finally, we can express the right Cauchy-Green tensor components as:

$$\begin{aligned} C_x = C_{11} &= \lambda_1^2 & C_{xy} = C_{yx} = C_{12} = C_{21} &= \lambda_1\lambda_2 \cos\measuredangle(\mathrm{d}\vec{\boldsymbol{x}}_1, \mathrm{d}\vec{\boldsymbol{x}}_2) \\ C_y = C_{22} &= \lambda_2^2 & C_{xz} = C_{zx} = C_{13} = C_{31} &= \lambda_1\lambda_3 \cos\measuredangle(\mathrm{d}\vec{\boldsymbol{x}}_1, \mathrm{d}\vec{\boldsymbol{x}}_3) \\ C_z = C_{33} &= \lambda_3^2 & C_{yz} = C_{zy} = C_{23} = C_{32} &= \lambda_2\lambda_3 \cos\measuredangle(\mathrm{d}\vec{\boldsymbol{x}}_2, \mathrm{d}\vec{\boldsymbol{x}}_3)\,; \end{aligned}$$

that, is, the diagonal terms C_{11}, C_{22} and C_{33} are the squared stretch ratios in the material (undeformed) directions $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$, and $\hat{\boldsymbol{k}}$, respectively; it is important to note that $U_{ii} \neq \sqrt{C_{ii}}$, except when $\boldsymbol{C}$ is a diagonal tensor; the off-diagonal terms C_{ij} (with $i \neq j$) capture the cosine of the angle between the deformed spatial elements $\mathrm{d}\vec{\boldsymbol{x}}_i$ and $\mathrm{d}\vec{\boldsymbol{x}}_j$, scaled by their respective stretch ratios. These off-diagonal terms characterize the distortion or shearing of the material element during deformation, from the perspective of the material configuration.

3.14.5 Physical interpretation of the Euler-Almansi strain tensor and the inverse of the left Cauchy-Green tensor

To understand the physical interpretation of each term in $\boldsymbol{e}$, we employ equation (3.56), which is rewritten for the sake of convenience:

$$\mathrm{d}\vec{\boldsymbol{x}}_i^T\, \boldsymbol{e}\, \mathrm{d}\vec{\boldsymbol{x}}_j = \frac{\mathrm{d}\vec{\boldsymbol{x}}_i \cdot \mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i \cdot \mathrm{d}\vec{\boldsymbol{X}}_j}{2}. \tag{3.59}$$

If we divide this equation by $\|\mathrm{d}\vec{\boldsymbol{x}}_i\|\,\|\mathrm{d}\vec{\boldsymbol{x}}_j\|$ and define $\hat{\boldsymbol{n}}_i = \frac{\mathrm{d}\vec{\boldsymbol{x}}_i}{\|\mathrm{d}\vec{\boldsymbol{x}}_i\|}$ and $\hat{\boldsymbol{n}}_j = \frac{\mathrm{d}\vec{\boldsymbol{x}}_j}{\|\mathrm{d}\vec{\boldsymbol{x}}_j\|}$ as the unit vectors in the directions of $\mathrm{d}\vec{\boldsymbol{x}}_i$ and $\mathrm{d}\vec{\boldsymbol{x}}_j$, respectively, it follows that

$$\frac{\mathrm{d}\vec{\boldsymbol{x}}_i^T}{\|\mathrm{d}\vec{\boldsymbol{x}}_i\|}\, \boldsymbol{e}\, \frac{\mathrm{d}\vec{\boldsymbol{x}}_j}{\|\mathrm{d}\vec{\boldsymbol{x}}_j\|} = \frac{1}{2}\frac{\mathrm{d}\vec{\boldsymbol{x}}_i \cdot \mathrm{d}\vec{\boldsymbol{x}}_j - \mathrm{d}\vec{\boldsymbol{X}}_i \cdot \mathrm{d}\vec{\boldsymbol{X}}_j}{\|\mathrm{d}\vec{\boldsymbol{x}}_i\|\,\|\mathrm{d}\vec{\boldsymbol{x}}_j\|} = \hat{\boldsymbol{n}}_i^T\, \boldsymbol{e}\, \hat{\boldsymbol{n}}_j,$$

and in the special case where $\mathrm{d}\vec{\boldsymbol{x}}_i = \mathrm{d}\vec{\boldsymbol{x}}_j = \mathrm{d}\vec{\boldsymbol{x}}$ and $\hat{\boldsymbol{n}}_i = \hat{\boldsymbol{n}}_j = \hat{\boldsymbol{n}} = \frac{\mathrm{d}\vec{\boldsymbol{x}}}{\|\mathrm{d}\vec{\boldsymbol{x}}\|}$, we obtain the so-called *Almansi's scalar strain*[★] in the direction $\hat{\boldsymbol{n}}$:

$$\boxed{\hat{\boldsymbol{n}}^T\, \boldsymbol{e}\, \hat{\boldsymbol{n}} = \frac{1}{2}\frac{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2 - \|\mathrm{d}\vec{\boldsymbol{X}}\|^2}{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2} = \frac{1-\lambda^{-2}}{2};} \tag{3.60}$$

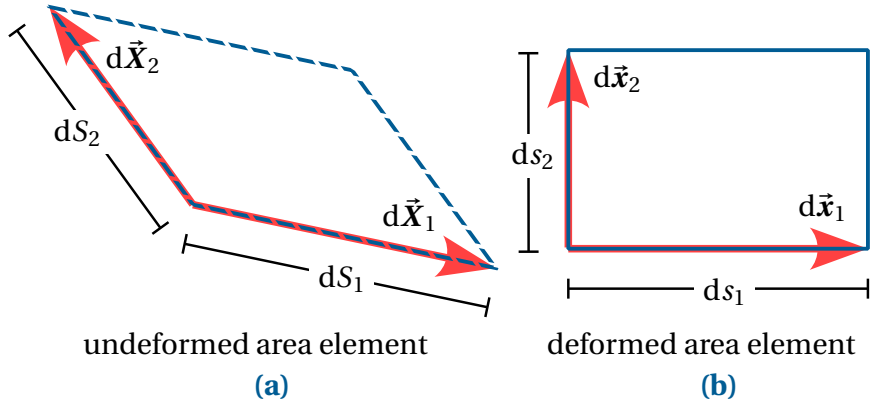


Figure 3.20: *Figure (a) shows how a parallelogram with sides* dS_1 *and* dS_2*, defined by material vectors* $d\vec{\boldsymbol{X}}_1 = \boldsymbol{F}^{-1}\, d\vec{\boldsymbol{x}}_1 = dS_1 \hat{\boldsymbol{N}}_1$ *and* $d\vec{\boldsymbol{X}}_2 = \boldsymbol{F}^{-1}\, d\vec{\boldsymbol{x}}_2 = dS_2 \hat{\boldsymbol{N}}_2$ *deforms into the rectangular area element shown in (b). The deformed element is characterized by spatial vectors* $d\vec{\boldsymbol{x}}_1 = ds_1 \hat{\boldsymbol{\imath}}$ *and* $d\vec{\boldsymbol{x}}_2 = ds_2 \hat{\boldsymbol{\jmath}}$*, with magnitudes* ds_1 *and* ds_2 *respectively. The corresponding stretches are given by* $\lambda_i = ds_i / dS_i$*.*

it represents the change in squared length of a spatial line element in a specific direction $\hat{\boldsymbol{n}}$ after deformation, relative to its deformed squared length; here λ stands for the corresponding stretch ratio.

With reference to Figure 3.19, let us define the spatial elements $d\vec{\boldsymbol{x}}_1 = ds_1 \hat{\boldsymbol{e}}_1$, $d\vec{\boldsymbol{x}}_2 = ds_2 \hat{\boldsymbol{e}}_2$, and $d\vec{\boldsymbol{x}}_3 = ds_3 \hat{\boldsymbol{e}}_3$, where $\hat{\boldsymbol{e}}_1 = \hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{e}}_2 = \hat{\boldsymbol{\jmath}}$, and $\hat{\boldsymbol{e}}_3 = \hat{\boldsymbol{k}}$; also, let us define the material elements $d\vec{\boldsymbol{X}}_1 = dS_1 \hat{\boldsymbol{N}}_1$, $d\vec{\boldsymbol{X}}_2 = dS_2 \hat{\boldsymbol{N}}_2$, and $d\vec{\boldsymbol{X}}_3 = dS_3 \hat{\boldsymbol{N}}_3$, where $d\vec{\boldsymbol{x}}_i = \boldsymbol{F}\, d\vec{\boldsymbol{X}}_i$; here $\hat{\boldsymbol{N}}_i = \frac{d\vec{\boldsymbol{X}}_i}{\|d\vec{\boldsymbol{X}}_i\|}$ are the unit vectors in the directions of the undeformed line elements; then, from (3.59):

$$ds_i\, ds_j \hat{\boldsymbol{e}}_i^T \boldsymbol{e} \hat{\boldsymbol{e}}_j = \frac{ds_i\, ds_j \hat{\boldsymbol{e}}_i \cdot \hat{\boldsymbol{e}}_j - dS_i\, dS_j \hat{\boldsymbol{N}}_i \cdot \hat{\boldsymbol{N}}_j}{2};$$

taking into account that $e_{ij} = \hat{\boldsymbol{e}}_i^T \boldsymbol{e} \hat{\boldsymbol{e}}_j$, that $\hat{\boldsymbol{e}}_i \cdot \hat{\boldsymbol{e}}_j = \delta_{ij}$ is the Kronecker delta, and that $\hat{\boldsymbol{N}}_i \cdot \hat{\boldsymbol{N}}_j = \cos \measuredangle \left(d\vec{\boldsymbol{X}}_i, d\vec{\boldsymbol{X}}_j\right)$ represents the cosine of the angle between $d\vec{\boldsymbol{X}}_i$ and $d\vec{\boldsymbol{X}}_j$, we obtain,

$$\begin{aligned} e_{ij} &= \frac{\delta_{ij} - \left[\boldsymbol{b}^{-1}\right]_{ij}}{2} \\ &= \frac{ds_i\, ds_j \delta_{ij} - dS_i\, dS_j \cos \measuredangle \left(d\vec{\boldsymbol{X}}_i, d\vec{\boldsymbol{X}}_j\right)}{2\, ds_i\, ds_j} \\ &= \frac{\delta_{ij} - \frac{1}{\lambda_i \lambda_j} \cos \measuredangle \left(d\vec{\boldsymbol{X}}_i, d\vec{\boldsymbol{X}}_j\right)}{2} \end{aligned}$$

where λ_1, λ_2, and λ_3 are, respectively, the stretches of the spatial line elements that are currently in the $\hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{\jmath}}$, and $\hat{\boldsymbol{k}}$ directions, that is $\lambda_i = \frac{ds_i}{dS_i}$.

Consequently, the diagonal terms of the Euler-Almansi strain tensor correspond to:

$$e_x = e_{11} = \frac{1-\left[\boldsymbol{b}^{-1}\right]_{11}}{2} = \frac{\mathrm{d}s_1^2 - \mathrm{d}S_1^2}{2\,\mathrm{d}s_1^2} = \frac{1-\lambda_1^{-2}}{2}$$
$$e_y = e_{22} = \frac{1-\left[\boldsymbol{b}^{-1}\right]_{22}}{2} = \frac{\mathrm{d}s_2^2 - \mathrm{d}S_2^2}{2\,\mathrm{d}s_2^2} = \frac{1-\lambda_2^{-2}}{2}$$
$$e_z = e_{33} = \frac{1-\left[\boldsymbol{b}^{-1}\right]_{33}}{2} = \frac{\mathrm{d}s_3^2 - \mathrm{d}S_3^2}{2\,\mathrm{d}s_3^2} = \frac{1-\lambda_3^{-2}}{2};$$

these are, according to equation (3.60), the Almansi's scalar strains in the $\hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{\jmath}}$, and $\hat{\boldsymbol{k}}$ directions, respectively, representing the changes in squared lengths of the spatial line elements along those directions.

On the other hand, the off-diagonal elements are:

$$e_{xy} = e_{yx} = e_{12} = e_{21} = -\frac{\left[\boldsymbol{b}^{-1}\right]_{12}}{2} = -\frac{\left[\boldsymbol{b}^{-1}\right]_{21}}{2} = -\frac{1}{2\lambda_1\lambda_2}\cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{X}}_1, \mathrm{d}\vec{\boldsymbol{X}}_2\right)$$
$$e_{xz} = e_{zx} = e_{13} = e_{31} = -\frac{\left[\boldsymbol{b}^{-1}\right]_{13}}{2} = -\frac{\left[\boldsymbol{b}^{-1}\right]_{31}}{2} = -\frac{1}{2\lambda_1\lambda_3}\cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{X}}_1, \mathrm{d}\vec{\boldsymbol{X}}_3\right)$$
$$e_{yz} = e_{zy} = e_{23} = e_{32} = -\frac{\left[\boldsymbol{b}^{-1}\right]_{23}}{2} = -\frac{\left[\boldsymbol{b}^{-1}\right]_{32}}{2} = -\frac{1}{2\lambda_2\lambda_3}\cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{X}}_2, \mathrm{d}\vec{\boldsymbol{X}}_3\right);$$

these off-diagonal terms are proportional to the product of the inverse stretches along the corresponding coordinate directions, multiplied by the cosine of the angle between the undeformed line elements that are, after deformation, in the coordinate directions; for instance, $e_{xy} = e_{12}$ corresponds to $\frac{1}{2\lambda_1\lambda_2}$ times the cosine of the undeformed angle between two material line elements that are currently in the $\hat{\boldsymbol{\imath}}$ and $\hat{\boldsymbol{\jmath}}$ directions, an angle that after deformation is 90°. A non-zero value of these off-diagonal terms indicates a change in the angle between the spatial line elements.

Using these results, we can express the components of the inverse of the left Cauchy-Green tensor as:

$$\left[\boldsymbol{b}^{-1}\right]_{11} = \lambda_1^{-2} \qquad \left[\boldsymbol{b}^{-1}\right]_{12} = \left[\boldsymbol{b}^{-1}\right]_{21} = \frac{\cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{x}}_1, \mathrm{d}\vec{\boldsymbol{x}}_2\right)}{\lambda_1\lambda_2}$$
$$\left[\boldsymbol{b}^{-1}\right]_{22} = \lambda_2^{-2} \qquad \left[\boldsymbol{b}^{-1}\right]_{13} = \left[\boldsymbol{b}^{-1}\right]_{31} = \frac{\cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{x}}_1, \mathrm{d}\vec{\boldsymbol{x}}_3\right)}{\lambda_1\lambda_3}$$
$$\left[\boldsymbol{b}^{-1}\right]_{33} = \lambda_3^{-2} \qquad \left[\boldsymbol{b}^{-1}\right]_{23} = \left[\boldsymbol{b}^{-1}\right]_{32} = \frac{\cos\measuredangle\left(\mathrm{d}\vec{\boldsymbol{x}}_2, \mathrm{d}\vec{\boldsymbol{x}}_3\right)}{\lambda_2\lambda_3};$$

that, is, the diagonal terms $\left[\boldsymbol{b}^{-1}\right]_{11}$, $\left[\boldsymbol{b}^{-1}\right]_{22}$ and $\left[\boldsymbol{b}^{-1}\right]_{33}$ are the inverse squared stretch ratios in the spatial (deformed) directions $\hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{\jmath}}$, and $\hat{\boldsymbol{k}}$, respectively; the off-diagonal terms $\left[\boldsymbol{b}^{-1}\right]_{ij}$ (with $i \neq j$) capture the cosine of the angle between the undeformed

material elements $d\vec{\boldsymbol{X}}_i$ and $d\vec{\boldsymbol{X}}_j$, scaled by the inverse of their respective stretch ratios. These off-diagonal terms characterize the distortion or shearing of the material element during deformation, from the perspective of the spatial configuration.

In summary, the Cauchy-Lagrange strain tensor $\boldsymbol{E}$ measures the deformation of a material line element from the reference (undeformed) configuration, while the Euler-Almansi strain tensor $\boldsymbol{e}$ measures the deformation of a spatial line element from the current (deformed) configuration. The diagonal terms of $\boldsymbol{E}$ represent the changes in squared lengths of the material line elements, while the off-diagonal terms represent the changes in angles between these line elements after deformation. Conversely, the diagonal terms of $\boldsymbol{e}$ represent the changes in squared lengths of the spatial line elements, and the off-diagonal terms represent the changes in angles between the undeformed material line elements.

Example 3.10. *Cauchy-Green tensors interpretation*

Interpret the meaning of the Cauchy-Green tensors calculated in Example 3.9.

Solution:

Missing. Make the physical interpretation, showing changes of angles and changes of length

3.14.6 Which deformation or strain tensor to use

When working with deformation and strain, the choice of deformation or strain tensor depends on whether a material description (where the material coordinate $\vec{\boldsymbol{X}}$ is used) or a spatial description (where the spatial coordinate $\vec{\boldsymbol{x}}$ is used) is employed.

In the material description, where $\vec{\boldsymbol{X}}$ is the independent variable it is straightforward to calculate the deformation gradient $\boldsymbol{F}(\vec{\boldsymbol{X}}, t)$ and therefore, it is natural to use the right $\boldsymbol{C}(\vec{\boldsymbol{X}}, t)$ or the left $\boldsymbol{b}(\vec{\boldsymbol{X}}, t)$ Cauchy-Green deformation tensors or the Cauchy-Lagrange strain tensor $\boldsymbol{E}(\vec{\boldsymbol{X}}, t)$. These deformation or strain measures are evaluated at the material point $\vec{\boldsymbol{X}}$ and describe the deformation in the neighborhood of that material point from the undeformed configuration.

In the spatial description, where $\vec{\boldsymbol{x}}$ is the independent variable and it is natural to calculate the inverse of the deformation gradient $\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}}, t)$; in this case, it is straightforward to use the inverse of the right $\boldsymbol{C}^{-1}(\vec{\boldsymbol{x}}, t)$ or the left $\boldsymbol{b}^{-1}(\vec{\boldsymbol{x}}, t)$ Cauchy-Green deformation tensors or the Euler-Almansi strain tensor $\boldsymbol{e}(\vec{\boldsymbol{x}}, t)$, which are based on the inverse of the deformation gradient. These deformation or strain measures are evaluated at the spatial point $\vec{\boldsymbol{x}}$ and describe the deformation in the neighborhood of that spatial point from the current (deformed) configuration.

3.15 Infinitesimal strain and rotation tensors

In many practical applications involving moderate forces, displacements that are small in comparison to the dimension of the body (to the point that geometrical changes are ignored) and small deformations, the finite strain tensors introduced in Section 3.14 can be simplified to the so-called infinitesimal strain tensor, which forms the basis of the classical theory of elasticity, and that was reviewed in Section 2.1. In the following, we will explore how the finite strain tensors $\boldsymbol{E}$ and $\boldsymbol{e}$ reduce to the infinitesimal strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$ in the small deformation setting.

3.15.1 Cauchy's infinitesimal strain tensor

When moderate forces are applied to bodies, minor changes in shape and size occur. The exact finite strain tensors of Cauchy-Lagrange $\boldsymbol{E}$ and Euler-Almansi $\boldsymbol{e}$, introduced in Section 3.14, are useful for estimating the deformation of the body; however, in the case of small deformations, these tensors can be simplified to the so-called *infinitesimal strain tensor*[★], that is the tensor studied in the theory of elasticity and which was reviewed in Section 2.1:

$$\underline{\underline{\boldsymbol{\varepsilon}}} = \begin{pmatrix} \varepsilon_x & \varepsilon_{xy} & \varepsilon_{xz} \\ \varepsilon_{yx} & \varepsilon_y & \varepsilon_{yz} \\ \varepsilon_{zx} & \varepsilon_{zy} & \varepsilon_z \end{pmatrix}$$

where

$$\varepsilon_x := \frac{\partial U}{\partial X} \qquad \varepsilon_y := \frac{\partial V}{\partial Y} \qquad \varepsilon_z := \frac{\partial W}{\partial Z}$$

are the so-called *longitudinal strains*[★] and

$$\varepsilon_{xy} := \frac{1}{2}\left(\frac{\partial U}{\partial Y} + \frac{\partial V}{\partial X}\right) \qquad \varepsilon_{xz} := \frac{1}{2}\left(\frac{\partial U}{\partial Z} + \frac{\partial W}{\partial X}\right) \qquad \varepsilon_{yz} := \frac{1}{2}\left(\frac{\partial V}{\partial Z} + \frac{\partial W}{\partial Y}\right)$$

stand for the *mathematical shear strains*[★].

In practice, for materials such as concrete, wood, or steel, deformations are typically on the order of 0.001 or even smaller, meaning that the components and angles of the strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$ are insignificant compared to one. In such cases, we can safely assume that $|\varepsilon_{ij}| \ll 1$ and that $\left|\frac{\partial U_i}{\partial X_j}\right| \ll 1$, for $i, j = 1, 2, 3$; this fact allows us to neglect the squares and products in the equations for the finite strain tensors (3.53) and (3.55).

Therefore, the components of the displacement gradients $\frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}$ and $\frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}$ are very small in comparison to unity, and the higher-order terms involving products of

displacement gradients become negligible, leading to these simplified expressions for the strain tensor components:

$$\boldsymbol{E}(\vec{\boldsymbol{X}},t) \approx \frac{1}{2}\left[\frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}} + \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}^T + \cancel{\frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}^T \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}}\right]$$
$$\boldsymbol{e}(\vec{\boldsymbol{x}},t) \approx \frac{1}{2}\left[\frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}} + \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}^T - \cancel{\frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{x}}}^T \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}}\right].$$

This means that, in the case of infinitesimal deformations, components of the Cauchy-Lagrange strain tensor (3.53) become:

$$E_{xx} \approx \frac{\partial U}{\partial X} \qquad E_{yy} \approx \frac{\partial V}{\partial Y} \qquad E_{zz} \approx \frac{\partial W}{\partial Z}$$
$$E_{xy} \approx \frac{1}{2}\left(\frac{\partial U}{\partial Y} + \frac{\partial V}{\partial X}\right) \quad E_{xz} \approx \frac{1}{2}\left(\frac{\partial U}{\partial Z} + \frac{\partial W}{\partial X}\right) \quad E_{yz} \approx \frac{1}{2}\left(\frac{\partial V}{\partial Z} + \frac{\partial W}{\partial Y}\right),$$

and those of the Euler-Almansi strain tensor (3.55) simplify to:

$$e_{xx} \approx \frac{\partial u}{\partial x} \qquad e_{yy} \approx \frac{\partial v}{\partial y} \qquad e_{zz} \approx \frac{\partial w}{\partial z}$$
$$e_{xy} \approx \frac{1}{2}\left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}\right) \quad e_{xz} \approx \frac{1}{2}\left(\frac{\partial u}{\partial z} + \frac{\partial w}{\partial x}\right) \quad e_{yz} \approx \frac{1}{2}\left(\frac{\partial v}{\partial z} + \frac{\partial w}{\partial y}\right).$$

In the case of infinitesimal deformations, the displacements of the body are very small in comparison to its dimensions, that is, $\|\vec{\boldsymbol{U}}\| \ll \|\vec{\boldsymbol{X}}\|$. Therefore, $\vec{\boldsymbol{x}} = \vec{\boldsymbol{X}} + \vec{\boldsymbol{U}} \approx \vec{\boldsymbol{X}}$ and the initial and current configurations are almost indistinguishable, i.e., $\Omega_0 \approx \Omega_t$; in addition, $\vec{\boldsymbol{u}} = \vec{\boldsymbol{U}}$. In this situation, it is irrelevant whether the displacement gradients are created by differentiation with respect to the material coordinates $\vec{\boldsymbol{X}}$ or the spatial coordinates $\vec{\boldsymbol{x}}$, that is, $\frac{\partial \vec{\boldsymbol{u}}}{\partial \vec{\boldsymbol{x}}} \approx \frac{\partial \vec{\boldsymbol{U}}}{\partial \vec{\boldsymbol{X}}}$. Therefore, in the case of infinitesimal deformations, since there is no sense of using the spatial or the material description:

$$\underline{\underline{\boldsymbol{\varepsilon}}} \approx \boldsymbol{E} \approx \boldsymbol{e}.$$

or alternatively,

$$\underline{\underline{\boldsymbol{\varepsilon}}} \approx \frac{1}{2}\left[\frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}} + \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}^T\right] \approx \frac{1}{2}\left[\frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}} + \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}^T\right].$$

Now, using equation (3.27), that is $\boldsymbol{F}(\vec{\boldsymbol{X}},t) = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}} + \boldsymbol{I}$, it follows that

$$\boxed{\underline{\underline{\boldsymbol{\varepsilon}}} = \frac{\boldsymbol{F} + \boldsymbol{F}^T}{2} - \boldsymbol{I}} \tag{3.61}$$

this equation is exact and it does not involve any kind of approximation; note that $\underline{\underline{\boldsymbol{\varepsilon}}}$ is symmetric second-order tensor.

However, it should be recognized that $\boldsymbol{F} = \boldsymbol{R}$ in the case of a rigid-body rotation, and in this scenario, $\underline{\underline{\boldsymbol{\varepsilon}}} = \frac{\boldsymbol{R}+\boldsymbol{R}^T}{2} - \boldsymbol{I} \neq \boldsymbol{0}$; this implies that the infinitesimal strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$ is not an exact representation of deformation for rigid-body rotations. Despite this limitation, for small deformations, the infinitesimal strain tensor provides an excellent approximation for measuring deformation, to the extent that linear elasticity theory and strength of materials are built upon such approximation. Additionally, the components of the infinitesimal strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$ have a clear geometric interpretation, as reviewed in Section 2.1.

The advantage of $\underline{\underline{\boldsymbol{\varepsilon}}}$ over the Cauchy-Lagrange $\boldsymbol{E}$ and the Euler-Almansi $\boldsymbol{e}$ strain tensors is that the components of $\underline{\underline{\boldsymbol{\varepsilon}}}$ are linear in the displacement components u_i; this linearity is of extreme importance in the theory of elasticity, as it permits the simplification of the governing equations, allows the use the superposition principle and facilitates analytical solutions for various problems in elasticity, which would be much more challenging or even impossible with nonlinear strain measures.

In conclusion, the choice of strain tensor depends on the problem at hand and the assumptions made regarding the magnitude of the deformations. For small deformations, the infinitesimal strain tensor may be a suitable approximation, while for large deformations, the Green-Lagrange or Euler-Almansi strain tensors may provide a more accurate representation of the deformation state.

Example 3.11. *Strains in rigid-body motion*

In Section 3.5.5, it was shown that the rigid-body motion that rotates counter-clockwise over an angle θ about an axis with direction $\hat{\boldsymbol{k}}$, passing through a point $[X_0, Y_0, 0]^T$ and experiencing a translation defined by $[C_x, C_y, 0]^T$ is given by:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} (X - X_0)\cos\theta - (Y - Y_0)\sin\theta + C_x \\ (X - X_0)\sin\theta + (Y - Y_0)\cos\theta + C_y \\ 0 \end{pmatrix}.$$

Using this description of motion, compare the Euler-Lagrange strain $\boldsymbol{E}$, the Euler-Almansi strain $\boldsymbol{E}$ and the infinitesimal strain $\boldsymbol{\varepsilon}$.

Solution:

Using symbolic computation, by means of the following Python code:

```
from sympy import symbols, Eq, cos, sin, solve, trigsimp, Matrix, eye

x, y, z, X, Y, Z  = symbols('x y z X Y Z')
X0, Y0, Cx, Cy, t = symbols('X0 Y0 Cx Cy theta')

vec_X = Matrix([X,Y,Z])    # material coordinates
```

```
vec_x = Matrix([x,y,z])    # spatial coordinates

# motion
chi = Matrix([ (X - X0)*cos(t) - (Y - Y0)*sin(t) + Cx,
               (X - X0)*sin(t) + (Y - Y0)*cos(t) + Cy,
                                                   Z ])

# inverse motion
sol = solve(Eq(vec_x,chi), vec_X)
inv_chi = trigsimp(Matrix([ sol[X].collect([sin(t), cos(t)]),
                            sol[Y].collect([sin(t), cos(t)]),
                            sol[Z].collect([sin(t), cos(t)]) ]))

# deformation gradient and its inverse
F     =     chi.jacobian(vec_X)
inv_F = inv_chi.jacobian(vec_x)

# 3x3 identity matrix
I = eye(3)

# calculation of the deformation and strain tensors
C = trigsimp(F.T*F)              # right Cauchy-Green deformation tensor
E = (C - I)/2                    # Cauchy-Lagrange strain tensor

inv_b = trigsimp(inv_F.T*inv_F)  # inverse of left Cauchy-Green def. tensor

e     = (I - inv_b)/2            # Euler-Almansi strain tensor

# infinitesimal strain tensor
eps = (F + F.T)/2 - I
```

it is possible to show that the associated inverse motion $\vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t)$ is given by:

$$\begin{pmatrix} X \\ Y \\ Z \end{pmatrix} = \vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t) = \begin{pmatrix} X_0 + (x - C_x)\cos\theta + (y - C_y)\sin\theta \\ Y_0 - (x - C_x)\sin\theta + (y - C_y)\cos\theta \\ 0 \end{pmatrix}.$$

Additionally, the deformation gradient $\boldsymbol{F}$ and its inverse $\boldsymbol{F}^{-1}$ are:

$$\boldsymbol{F} = \boldsymbol{R} = \begin{pmatrix} \cos\theta & -\sin\theta & 0 \\ \sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix} \qquad \boldsymbol{F}^{-1} = \boldsymbol{R}^T = \begin{pmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{pmatrix}.$$

The code also shows that the right Cauchy-Green tensor $\boldsymbol{C}$ and the inverse of the left Cauchy-Green tensor $\boldsymbol{b}^{-1}$ are equal to the identity matrix: $\boldsymbol{C} = \boldsymbol{b}^{-1} = \boldsymbol{I}$, indicating that there is no deformation.

Consequently, the Green-Lagrange $\boldsymbol{E}$ and Euler-Almansi $\boldsymbol{e}$ strain tensors are null: $\boldsymbol{E} = \boldsymbol{e} = \boldsymbol{0}$, as they are derived from $\boldsymbol{C}$ and $\boldsymbol{b}^{-1}$, respectively.

However, the infinitesimal strain tensor is:

$$\boldsymbol{\varepsilon} = \begin{pmatrix} \cos\theta - 1 & 0 & 0 \\ 0 & \cos\theta - 1 & 0 \\ 0 & 0 & 0 \end{pmatrix}.$$

Note that neither the Green-Lagrange $\boldsymbol{E}$ nor the Euler-Almansi $\boldsymbol{e}$ strain tensors are affected by the rigid-body rotation. However, the infinitesimal strain tensor $\boldsymbol{\varepsilon}$ is not null and varies with the angle of rotation θ. If θ is small, then $\cos\theta - 1 \approx 0$, and $\boldsymbol{\varepsilon} \approx \mathbf{0}$, indicating that the infinitesimal strain tensor becomes negligible for small rotations.

This example demonstrates that for rigid-body motions, the Green-Lagrange and Euler-Almansi strain tensors are zero, as they are derived from deformation measures that are invariant under rigid-body motions. However, the infinitesimal strain tensor, which is a linearized approximation, is not necessarily zero for rigid-body rotations. It becomes negligible only when the rotation angle is small.

Example 3.12. *Strains in rotation and deformation*

Example, Kim, page 147

Solution:

Using symbolic computation, by means of the following Python code:

```
from numpy import array, c_, ones
from numpy.linalg import solve as linsolve
from sympy import Matrix, nsimplify, symbols, Eq, solve, eye

# variables to make more readable the code:
X, Y = 0, 1
x, y = 0, 1

# coordinates in the reference configuration
cX = array([[  0, 0],
            [1.5, 0],
            [1.5, 1],
            [  0, 1]])

# coordinates in the current configuration
cx = array([[0.7, 0],
            [0.7, 2],
            [  0, 2],
            [  0, 0]])

# find the coefficients of the interpolation
```

```
coef_x = linsolve(c_[ones(4), cX[:,X], cX[:,Y], cX[:,X]*cX[:,Y]], cx[:, x])
coef_y = linsolve(c_[ones(4), cX[:,X], cX[:,Y], cX[:,X]*cX[:,Y]], cx[:, y])

# find the rational numbers associated to the interpolation
coef_x  = nsimplify(Matrix(coef_x), rational=True)
coef_y  = nsimplify(Matrix(coef_y), rational=True)

# redefinition of variables
x, y, z, X, Y, Z  = symbols('x y z X Y Z')

# material and spatial coordinate vectors
vec_X = Matrix([X,Y,Z])    # material coordinates
vec_x = Matrix([x,y,z])    # spatial coordinates

# motion
chi = Matrix([ coef_x[0] + coef_x[1]*X + coef_x[2]*Y + coef_x[3]*X*Y,
               coef_y[0] + coef_y[1]*X + coef_y[2]*Y + coef_y[3]*X*Y,
                                                                  Z ])
# inverse motion
sol     = solve(Eq(vec_x,chi), vec_X)
inv_chi = Matrix([ sol[X], sol[Y], sol[Z] ])

# deformation gradient and its inverse
F     =     chi.jacobian(vec_X)
inv_F = inv_chi.jacobian(vec_x)

# 3x3 identity matrix
I = eye(3)

# calculation of the deformation and strain tensors
C = F.T*F              # right Cauchy-Green deformation tensor
E = (C - I)/2          # Cauchy-Lagrange strain tensor

inv_b = inv_F.T*inv_F  # inverse of left Cauchy-Green deformation tensor
e     = (I - inv_b)/2  # Euler-Almansi strain tensor

# infinitesimal strain tensor
eps = (F + F.T)/2 - I
```

Bla bla bla

3.15.2 Comparison of the Green and the Almansi scalar strains in the infinitesimal case

The Cauchy-Lagrange and Euler-Almansi strain tensors provide measures of deformation that are valid for both small and large strains. However, their physical interpretations and mathematical formulations differ slightly. In the next, we will

compare these strain measures and their relationship to the infinitesimal strain tensor in the context of small deformations.

In Section 3.14.4, we explored how Green's scalar strain in the direction of $\hat{\boldsymbol{N}}$ is expressed as:

$$\hat{\boldsymbol{N}}^T \boldsymbol{E} \hat{\boldsymbol{N}} = \frac{1}{2} \frac{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2 - \|\mathrm{d}\vec{\boldsymbol{X}}\|^2}{\|\mathrm{d}\vec{\boldsymbol{X}}\|^2}; \tag{3.62}$$

this quantity signifies the change in the squared length of a line element oriented along $\hat{\boldsymbol{N}}$ before deformation, relative to its undeformed squared length. Thus, the diagonal terms of the Cauchy-Lagrange strain tensor:

$$\begin{aligned} E_x = E_{11} &= \frac{\mathrm{d}s_1^2 - \mathrm{d}S_1^2}{2\,\mathrm{d}S_1^2} = \frac{\lambda_1^2 - 1}{2} \\ E_y = E_{22} &= \frac{\mathrm{d}s_2^2 - \mathrm{d}S_2^2}{2\,\mathrm{d}S_2^2} = \frac{\lambda_2^2 - 1}{2} \\ E_z = E_{33} &= \frac{\mathrm{d}s_3^2 - \mathrm{d}S_3^2}{2\,\mathrm{d}S_3^2} = \frac{\lambda_3^2 - 1}{2}; \end{aligned} \tag{3.63}$$

correspond to the Green's scalar strains in the $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$, and $\hat{\boldsymbol{k}}$ directions, respectively, representing the change in squared lengths of the material line elements along those axes.

In contrast, in Section 3.14.5, we studied how Almansi's scalar strain in the direction $\hat{\boldsymbol{n}}$ is expressed by:

$$\hat{\boldsymbol{n}}^T \boldsymbol{e} \hat{\boldsymbol{n}} = \frac{1}{2} \frac{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2 - \|\mathrm{d}\vec{\boldsymbol{X}}\|^2}{\|\mathrm{d}\vec{\boldsymbol{x}}\|^2}; \tag{3.64}$$

this denotes the change in squared length of a spatial line element along the specific direction $\hat{\boldsymbol{n}}$ after deformation, relative to its deformed squared length. Consequently, the diagonal terms of the Euler-Almansi strain tensor

$$\begin{aligned} e_x = e_{11} &= \frac{\mathrm{d}s_1^2 - \mathrm{d}S_1^2}{2\,\mathrm{d}s_1^2} = \frac{1 - \lambda_1^{-2}}{2} \\ e_y = e_{22} &= \frac{\mathrm{d}s_2^2 - \mathrm{d}S_2^2}{2\,\mathrm{d}s_2^2} = \frac{1 - \lambda_2^{-2}}{2} \\ e_z = e_{33} &= \frac{\mathrm{d}s_3^2 - \mathrm{d}S_3^2}{2\,\mathrm{d}s_3^2} = \frac{1 - \lambda_3^{-2}}{2} \end{aligned} \tag{3.65}$$

depict Almansi's scalar strains along the $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$, and $\hat{\boldsymbol{k}}$ directions, elucidating the changes in squared lengths of spatial line elements along these axes.

Now, using the infinitesimal strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$, we can estimate the longitudinal strain in a given direction $\hat{\boldsymbol{N}}$ as:

$$\hat{\boldsymbol{N}}^T \underline{\underline{\boldsymbol{\varepsilon}}} \hat{\boldsymbol{N}} = \frac{\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|}{\|\mathrm{d}\vec{\boldsymbol{X}}\|}; \tag{3.66}$$

this equation denotes the change in length of a line element in the direction $\hat{\boldsymbol{N}}$, relative to its undeformed length. Consequently, the diagonal terms of the infinitesimal strain tensor correspond to:

$$\begin{aligned}\varepsilon_x = \varepsilon_{11} &= \frac{\mathrm{d}x_1 - \mathrm{d}X_1}{\mathrm{d}X_1} = \lambda_1 - 1 \\ \varepsilon_y = \varepsilon_{22} &= \frac{\mathrm{d}x_2 - \mathrm{d}X_2}{\mathrm{d}X_2} = \lambda_2 - 1 \\ \varepsilon_z = \varepsilon_{33} &= \frac{\mathrm{d}x_3 - \mathrm{d}X_3}{\mathrm{d}X_3} = \lambda_3 - 1\end{aligned} \tag{3.67}$$

these are called, in elasticity theory, the longitudinal strains along the $\hat{\boldsymbol{\imath}}$, $\hat{\boldsymbol{\jmath}}$, and $\hat{\boldsymbol{k}}$ directions, respectively.

Next, we will demonstrate that equations (3.62), (3.64), and (3.66) yield similar results in the small deformation setting, where $\|\mathrm{d}\vec{\boldsymbol{x}}\| \approx \|\mathrm{d}\vec{\boldsymbol{X}}\|$. For Lagrange's scalar strain, we have:

$$\mathrm{d}\hat{\boldsymbol{N}}^T \boldsymbol{E}\, \mathrm{d}\hat{\boldsymbol{N}} = \frac{(\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|)(\|\mathrm{d}\vec{\boldsymbol{x}}\| + \|\mathrm{d}\vec{\boldsymbol{X}}\|)}{2\left\|\mathrm{d}\vec{\boldsymbol{X}}\right\|^2} \approx \frac{(\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|)\|\mathrm{d}\vec{\boldsymbol{X}}\|}{\left\|\mathrm{d}\vec{\boldsymbol{X}}\right\|^2} = \frac{\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|}{\left\|\mathrm{d}\vec{\boldsymbol{X}}\right\|};$$

similarly, for Almansi's scalar strain:

$$\mathrm{d}\hat{\boldsymbol{n}}^T \boldsymbol{e}\, \mathrm{d}\hat{\boldsymbol{n}} = \frac{(\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|)(\|\mathrm{d}\vec{\boldsymbol{x}}\| + \|\mathrm{d}\vec{\boldsymbol{X}}\|)}{2\left\|\mathrm{d}\vec{\boldsymbol{x}}\right\|^2} \approx \frac{(\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|)\|\mathrm{d}\vec{\boldsymbol{X}}\|}{\left\|\mathrm{d}\vec{\boldsymbol{X}}\right\|^2} = \frac{\|\mathrm{d}\vec{\boldsymbol{x}}\| - \|\mathrm{d}\vec{\boldsymbol{X}}\|}{\left\|\mathrm{d}\vec{\boldsymbol{X}}\right\|}.$$

This demonstrates that for small deformations, where $\|\mathrm{d}\vec{\boldsymbol{x}}\| \approx \|\mathrm{d}\vec{\boldsymbol{X}}\|$, both Green's and Almansi's scalar strains converge to the infinitesimal strain measure.

Now, let us compare equations (3.63), (3.65) and (3.67). Note that the diagonal terms of the strain tensors can be expressed as:

$$E_{ii} = \frac{\lambda_i^2 - 1}{2} \qquad e_{ii} = \frac{1 - \lambda_i^{-2}}{2} \qquad \varepsilon_{ii} = \lambda_i - 1.$$

When plotting these functions against $\varepsilon_{ii} = \lambda - 1$, we obtain Figure 3.21; this illustration reveals the following observations:

- For small deformations ($|\lambda - 1| < 0.01$), all three scalar strain measures are nearly equivalent; this corresponds to strains of less than 1% and justifies the use of the simpler infinitesimal strain tensor in linear elasticity theory.
- As deformations become larger, the measures diverge. Green's scalar strain tends to overestimate the deformation compared to the infinitesimal measure, while Almansi's scalar strain underestimates it.

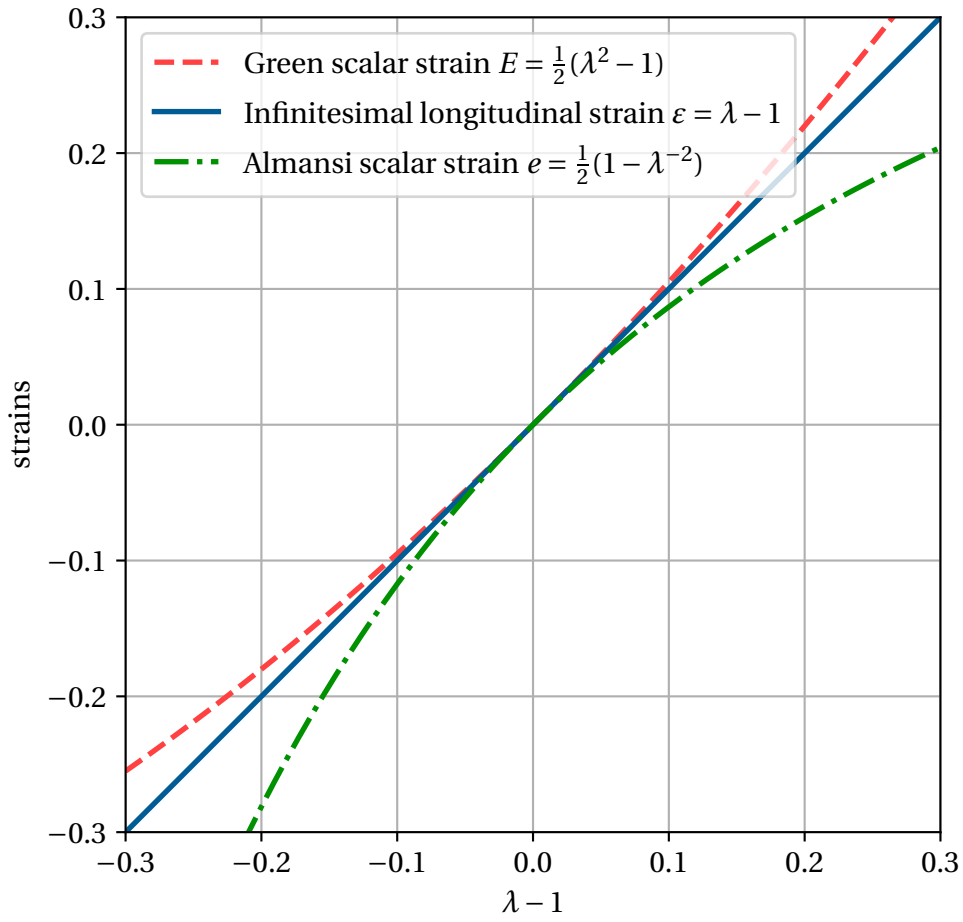


Figure 3.21: *Comparison of the different values of the longitudinal strain, using the formulas provided by the Cauchy-Lagrange, Euler-Almansi and infinitesimal strain tensors.*

- The Green's scalar strain is always larger than the Almansi's scalar strain for the same deformation, reflecting their different reference configurations (undeformed vs. deformed).

- Both finite strain measures remain valid for large deformations, where the infinitesimal strain approximation breaks down.

In conclusion, while the Green's and Almansi's scalar strain provide accurate measures for large deformations, they converge to the infinitesimal strain tensor for small deformations. This convergence justifies the use of the simpler infinitesimal strain

tensor in linear elasticity and small-strain analyses, while emphasizing the importance of finite strain measures in scenarios involving large deformations, such as in the study of elastomers, biological tissues, or severe plastic deformations in metals.

3.15.3 Comparison of the Cauchy-Lagrange the Euler-Almansi shear strains in the infinitesimal case

In Section 3.14.4, we examined the shear strains estimated using the Cauchy-Lagrange tensor $\boldsymbol{E}$, which are expressed as:

$$
\begin{aligned}
E_{xy} = E_{yx} = E_{12} = E_{21} &= \frac{\lambda_1 \lambda_2}{2} \cos \measuredangle (\mathrm{d}\vec{\boldsymbol{x}}_1, \mathrm{d}\vec{\boldsymbol{x}}_2) \\
E_{xz} = E_{zx} = E_{13} = E_{31} &= \frac{\lambda_1 \lambda_3}{2} \cos \measuredangle (\mathrm{d}\vec{\boldsymbol{x}}_1, \mathrm{d}\vec{\boldsymbol{x}}_3) \\
E_{yz} = E_{zy} = E_{23} = E_{32} &= \frac{\lambda_2 \lambda_3}{2} \cos \measuredangle (\mathrm{d}\vec{\boldsymbol{x}}_2, \mathrm{d}\vec{\boldsymbol{x}}_3) ;
\end{aligned}
$$

Similarly, in Section 3.14.5, we explored the shear strains estimated using the Euler-Almansi tensor $\boldsymbol{e}$, given by:

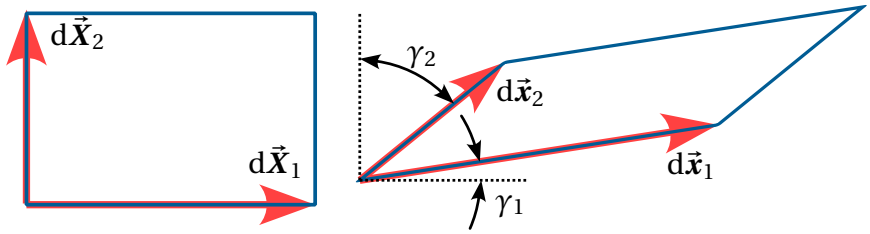
$$
\begin{aligned}
e_{xy} = e_{yx} = e_{12} = e_{21} &= -\frac{1}{2\lambda_1 \lambda_2} \cos \measuredangle \left(\mathrm{d}\vec{\boldsymbol{X}}_1, \mathrm{d}\vec{\boldsymbol{X}}_2\right) \\
e_{xz} = e_{zx} = e_{13} = e_{31} &= -\frac{1}{2\lambda_1 \lambda_3} \cos \measuredangle \left(\mathrm{d}\vec{\boldsymbol{X}}_1, \mathrm{d}\vec{\boldsymbol{X}}_3\right) \\
e_{yz} = e_{zy} = e_{23} = e_{32} &= -\frac{1}{2\lambda_2 \lambda_3} \cos \measuredangle \left(\mathrm{d}\vec{\boldsymbol{X}}_2, \mathrm{d}\vec{\boldsymbol{X}}_3\right) ;
\end{aligned}
$$

In the small strain framework, where $\|\mathrm{d}\vec{\boldsymbol{x}}\| \approx \|\mathrm{d}\vec{\boldsymbol{X}}\|$, we can approximate the stretches to be approximately equal to 1, i.e., $\lambda_1 \approx \lambda_2 \approx \lambda_3 \approx 1$. Consequently, these equations simplify to:

$$
\begin{aligned}
&\text{Cauchy-Lagrange:} \quad \Longrightarrow \quad 2E_{ij} \approx \cos \measuredangle (\mathrm{d}\vec{\boldsymbol{x}}_i, \mathrm{d}\vec{\boldsymbol{x}}_j). \\
&\text{Euler-Almansi:} \quad \Longrightarrow \quad 2e_{ij} \approx -\cos \measuredangle (\mathrm{d}\vec{\boldsymbol{X}}_i, \mathrm{d}\vec{\boldsymbol{X}}_j).
\end{aligned}
$$

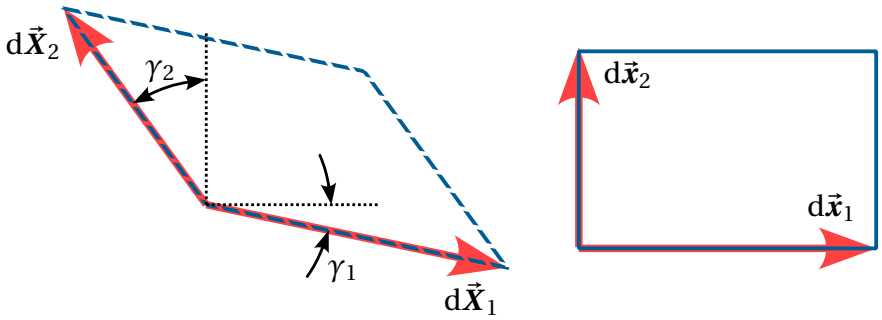
With reference to Figure 3.22, using the trigonometrical identity $\cos\left(\frac{\pi}{2} - \gamma\right) = \sin\gamma$, we can rewrite this equation in terms of the angle $\gamma_1 + \gamma_2$. This means that $2E_{ij} \approx \cos \measuredangle (\mathrm{d}\vec{\boldsymbol{x}}_1, \mathrm{d}\vec{\boldsymbol{x}}_2) = \sin(\gamma_1 + \gamma_2)$ and in the case of small angles, since $\sin\gamma \approx \gamma$, we have that $2E_{ij} \approx \gamma_1 + \gamma_2 = \gamma_{ij}$, that the engineering angular strain that is employed in elasticity theory, which measures the small decrease in the angle between $\mathrm{d}\vec{\boldsymbol{x}}_1$ and $\mathrm{d}\vec{\boldsymbol{x}}_2$.

With reference to Figure 3.23, using the trigonometrical identity $-\cos\left(\frac{\pi}{2} + \gamma\right) = \sin\gamma$, we can rewrite this equation in terms of the angle $\gamma_1 + \gamma_2$. This means that $2e_{ij} =$



undeformed area element **(a)** deformed area element **(b)**

Figure 3.22: *Understanding the shear strains in the Cauchy-Lagrange strain tensor. Using the trigonometric identity* $\sin(\gamma_1+\gamma_2) = \cos\left(\frac{\pi}{2} - (\gamma_1+\gamma_2)\right) = \cos\measuredangle(d\vec{x}_1, d\vec{x}_2)$*, it follows that* $E_{12} = E_{21} = \frac{\lambda_1\lambda_2}{2}\sin(\gamma_1+\gamma_2)$.



undeformed area element **(a)** deformed area element **(b)**

Figure 3.23: *Understanding the shear strains in the Euler-Almansi strain tensor. Using the trigonometric identity* $\sin(\gamma_1+\gamma_2) = -\cos\left(\frac{\pi}{2} + (\gamma_1+\gamma_2)\right) = -\cos\measuredangle\left(d\vec{X}_1, d\vec{X}_2\right)$*, it follows that* $e_{12} = e_{21} = \frac{1}{2\lambda_1\lambda_2}\sin(\gamma_1+\gamma_2)$.

$-\cos\measuredangle\left(d\vec{X}_1, d\vec{X}_2\right) = \sin(\gamma_1+\gamma_2)$ and in the case of small angles, since $\sin\gamma \approx \gamma$, we have that $2e_{ij} \approx \gamma_1+\gamma_2 = \gamma_{ij}$, the engineering angular strain that is employed in elasticity theory, which measures the small increase in the angle between $d\vec{X}_1$ and $d\vec{X}_2$.

These comparisons reveal the following observations:

- Both Cauchy-Lagrange and Euler-Almansi shear strains converge to half the engineering shear strain for small deformations.
- The Cauchy-Lagrange strain measures the decrease in angle in the deformed configuration, while the Euler-Almansi strain measures the increase in angle in the undeformed configuration. In fact, the opposite signs in the definitions of these strain tensors (cos vs. $-$cos) reflect this difference in perspective.

- For small deformations, both measures provide essentially the same numerical value, justifying the use of the simpler engineering shear strain in linear elasticity. Hence, any of these strain measures can be used interchangeably in the small deformation setting.

This comparison highlights the consistency between these finite strain measures and the infinitesimal strain theory for small deformations. It also emphasizes the importance of choosing the appropriate strain measure based on the reference configuration (material or spatial) and the magnitude of deformation being studied.

In practice, this means that for small strain analyses, engineers can confidently use the simpler engineering shear strain concept. However, for large deformation analyses, such as in plasticity, the choice between Cauchy-Lagrange and Euler-Almansi formulations becomes more significant and should be made based on the problem formulation and the desired reference configuration.

3.15.4 Infinitesimal rotation

In Section 3.5.2, we discussed the formulation associated to the rigid-body rotation through an angle θ about an axis $\hat{\boldsymbol{n}}$. For infinitesimal rotations, we can safely approximate $\sin\theta \approx \theta$ and $\cos\theta \approx 1$, simplifying Rodrigues' rotation formula (3.10) to:

$$\vec{\boldsymbol{x}} = \vec{\boldsymbol{X}} + (\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}})\theta,$$

where $\vec{\boldsymbol{x}}$ and $\vec{\boldsymbol{X}}$ represent the deformed and undeformed positions, respectively. Utilizing the fact that $\hat{\boldsymbol{n}} \times \vec{\boldsymbol{X}} = \boldsymbol{K}\vec{\boldsymbol{X}}$, where $\boldsymbol{K}$ is defined by equation (3.12), we find the displacement at point $\vec{\boldsymbol{X}}$:

$$\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{x}} - \vec{\boldsymbol{X}} = \boldsymbol{K}\vec{\boldsymbol{X}}\theta.$$

The material gradient of this displacement yields the skew-symmetric tensor $\boldsymbol{\Omega}$:

$$\boldsymbol{\Omega} := \frac{\partial \vec{\boldsymbol{U}}}{\partial \vec{\boldsymbol{X}}} = \theta\boldsymbol{K} = \begin{pmatrix} 0 & -\theta\hat{n}_z & \theta\hat{n}_y \\ \theta\hat{n}_z & 0 & -\theta\hat{n}_x \\ -\theta\hat{n}_y & \theta\hat{n}_z & 0 \end{pmatrix}.$$

This rotation is also characterized by the magnitude θ and direction $\hat{\boldsymbol{n}}$ of the vector $\vec{\boldsymbol{\omega}} = \theta\hat{\boldsymbol{n}} = \left[\theta\hat{n}_x, \theta\hat{n}_y, \theta\hat{n}_z\right]^T$, known as the *dual vector** or *axial vector** of $\boldsymbol{\Omega} := \theta\boldsymbol{K}$; this vector describes the angle θ and the axis $\hat{\boldsymbol{n}}$ of rotation of $\vec{\boldsymbol{X}}$; its components $\omega_i = \theta\hat{n}_i$ for $i = x,\ y,\ z$, represent the rotation angles around the $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$ and $\hat{\boldsymbol{k}}$ axes, respectively; this can be observed by decomposing the rotation into the canonical basis as $\vec{\boldsymbol{\omega}} = \omega_x\hat{\boldsymbol{i}} + \omega_y\hat{\boldsymbol{j}} + \omega_z\hat{\boldsymbol{k}}$.

Using $\boldsymbol{F}(\vec{\boldsymbol{X}},t) = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}} + \boldsymbol{I}$, as in equation (3.27), the deformation gradient for infinitesimal deformations becomes

$$\boldsymbol{F} = \begin{pmatrix} 1+\frac{\partial U}{\partial X} & \frac{\partial U}{\partial Y} & \frac{\partial U}{\partial Z} \\ \frac{\partial V}{\partial X} & 1+\frac{\partial V}{\partial Y} & \frac{\partial V}{\partial Z} \\ \frac{\partial W}{\partial X} & \frac{\partial W}{\partial Y} & 1+\frac{\partial W}{\partial Z} \end{pmatrix}.$$

Furthermore, the tensor $\boldsymbol{\Omega}$ can be extracted from $\boldsymbol{F}$ using:

$$\boldsymbol{\Omega} = \frac{\boldsymbol{F} - \boldsymbol{F}^T}{2} \tag{3.68}$$

and this shows that

$$\boldsymbol{\Omega} = \begin{pmatrix} 0 & \frac{1}{2}\left(\frac{\partial U}{\partial Y} - \frac{\partial V}{\partial X}\right) & \frac{1}{2}\left(\frac{\partial U}{\partial Z} - \frac{\partial W}{\partial X}\right) \\ \frac{1}{2}\left(\frac{\partial V}{\partial X} - \frac{\partial U}{\partial Y}\right) & 0 & \frac{1}{2}\left(\frac{\partial V}{\partial Z} - \frac{\partial W}{\partial Y}\right) \\ \frac{1}{2}\left(\frac{\partial W}{\partial X} - \frac{\partial U}{\partial Z}\right) & \frac{1}{2}\left(\frac{\partial W}{\partial Y} - \frac{\partial V}{\partial Z}\right) & 0 \end{pmatrix} = \begin{pmatrix} 0 & -\theta\hat{n}_z & \theta\hat{n}_y \\ \theta\hat{n}_z & 0 & -\theta\hat{n}_x \\ -\theta\hat{n}_y & \theta\hat{n}_z & 0 \end{pmatrix}$$

Therefore, for small deformations, the rotation angles around the $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$ and $\hat{\boldsymbol{k}}$ axes are given, respectively, by

$$\begin{aligned} \omega_x := \theta\hat{n}_x &= \frac{1}{2}\left(\frac{\partial W}{\partial Y} - \frac{\partial V}{\partial Z}\right) \\ \omega_y := \theta\hat{n}_y &= \frac{1}{2}\left(\frac{\partial U}{\partial Z} - \frac{\partial W}{\partial X}\right) \\ \omega_z := \theta\hat{n}_z &= \frac{1}{2}\left(\frac{\partial V}{\partial X} - \frac{\partial U}{\partial Y}\right); \end{aligned}$$

this relation reveals that the infinitesimal rotation field $\vec{\boldsymbol{\omega}}$ is directly related to the curl (or rotational) of the displacement field $\vec{\boldsymbol{U}}$:

$$\vec{\boldsymbol{\omega}} = \frac{1}{2}\operatorname{curl}\vec{\boldsymbol{U}} = \frac{1}{2}\vec{\nabla}\times\vec{\boldsymbol{U}}.$$

Remember that the curl operation, $\operatorname{curl}\vec{\boldsymbol{U}}$ or $\vec{\nabla}\times\vec{\boldsymbol{U}}$, is a differential operator applied to a vector field to measure its degree of local spinning.

Finally, from equations (3.27), (3.61), and (3.68), it is straightforward to show that the material gradient $\frac{\partial \vec{\boldsymbol{U}}}{\partial \vec{\boldsymbol{X}}}$ can be decomposed into its symmetric $\underline{\underline{\boldsymbol{\varepsilon}}}$ and skew-symmetric parts $\boldsymbol{\Omega}$ as follows:

$$\frac{\partial \vec{\boldsymbol{U}}}{\partial \vec{\boldsymbol{X}}} = \boldsymbol{F} - \boldsymbol{I} = \underline{\underline{\boldsymbol{\varepsilon}}} + \boldsymbol{\Omega}; \tag{3.69}$$

this equation expresses any infinitesimal motion as the sum of an infinitesimal deformation $\underline{\underline{\boldsymbol{\varepsilon}}}$ and an infinitesimal rotation $\boldsymbol{\Omega}$.

It is worth noting that the assumption $\left|\frac{\partial U_i}{\partial X_j}\right| \ll 1$, for $i,j = 1,2,3$, implies that both the strain and the rotation are small.

3.15.5 Approximations of the tensors C, U, v, ε, U^{-1}, R, and Ω in the small deformation regime

From equation (3.69), it follows that $\boldsymbol{F} = \boldsymbol{I} + \boldsymbol{\varepsilon} + \boldsymbol{\Omega}$. Then, using equation (3.42), we can express the right Cauchy-Green tensor as:

$$\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F} = (\boldsymbol{I} + \boldsymbol{\varepsilon} + \boldsymbol{\Omega})^T(\boldsymbol{I} + \boldsymbol{\varepsilon} + \boldsymbol{\Omega});$$

using the the properties $\boldsymbol{\varepsilon} = \boldsymbol{\varepsilon}^T$ (symmetric infinitesimal strain tensor) and $\boldsymbol{\Omega} = -\boldsymbol{\Omega}^T$ (skew-symmetric infinitesimal rotation tensor), we have:

$$\boldsymbol{C} = (\boldsymbol{I} + \boldsymbol{\varepsilon} - \boldsymbol{\Omega})(\boldsymbol{I} + \boldsymbol{\varepsilon} + \boldsymbol{\Omega});$$

in the case of small strains and rotations, we neglect the squares and products of $\boldsymbol{\varepsilon}$ and $\boldsymbol{\Omega}$; therefore:

$$\boldsymbol{C} = (\boldsymbol{I} + \boldsymbol{\varepsilon} - \boldsymbol{\Omega})(\boldsymbol{I} + \boldsymbol{\varepsilon} + \boldsymbol{\Omega}) = \boldsymbol{I}^2 + 2\boldsymbol{\varepsilon}\boldsymbol{I} + \underbrace{\boldsymbol{\varepsilon}^2 - \boldsymbol{\Omega}^2}_{\approx \boldsymbol{0}} \approx \boldsymbol{I} + 2\boldsymbol{\varepsilon};$$

thus, we can approximate $\boldsymbol{C} \approx \boldsymbol{I} + 2\boldsymbol{\varepsilon}$ in the small deformation regime. Additionally, $\boldsymbol{U} \approx \boldsymbol{I} + \boldsymbol{\varepsilon}$; which follows from the fact that:

$$\boldsymbol{C} = \boldsymbol{U}^2 \approx (\boldsymbol{I} + \boldsymbol{\varepsilon})(\boldsymbol{I} + \boldsymbol{\varepsilon}) = \boldsymbol{I} + 2\boldsymbol{\varepsilon} + \underbrace{\boldsymbol{\varepsilon}^2}_{\approx \boldsymbol{0}} \approx \boldsymbol{I} + 2\boldsymbol{\varepsilon}.$$

Similarly, we can show that $\boldsymbol{v} \approx \boldsymbol{I} + \boldsymbol{\varepsilon}$, and consequently $\boldsymbol{\varepsilon} \approx \boldsymbol{U} - \boldsymbol{I} \approx \boldsymbol{v} - \boldsymbol{I}$ in the small deformation regime.

To approximate the inverse of a matrix $\boldsymbol{A}$, we can use the Neumann series expansion (see Appendix A.13): $\boldsymbol{A}^{-1} \approx \sum_{i=0}^{n} (\boldsymbol{I} - \boldsymbol{A})^i$. Then, for $\boldsymbol{U}^{-1}$, we have:

$$\boldsymbol{U}^{-1} = (\boldsymbol{I} - \boldsymbol{U})^0 + (\boldsymbol{I} - \boldsymbol{U})^1 + (\boldsymbol{I} - \boldsymbol{U})^2 + (\boldsymbol{I} - \boldsymbol{U})^3 + (\boldsymbol{I} - \boldsymbol{U})^4 + (\boldsymbol{I} - \boldsymbol{U})^5 + \cdots;$$

using the fact that $\boldsymbol{U} \approx \boldsymbol{I} + \boldsymbol{\varepsilon}$ and the Python code:

```
from sympy import symbols, expand
I, E = symbols('I, E')
U = I + E
expand(I + (I-U)**1 + (I-U)**2 + (I-U)**3 + (I-U)**4 + (I-U)**5)
```

it can be shown that $\boldsymbol{U}^{-1} = \boldsymbol{I} - \boldsymbol{\varepsilon} + \boldsymbol{\varepsilon}^2 - \boldsymbol{\varepsilon}^3 + \boldsymbol{\varepsilon}^4 - \boldsymbol{\varepsilon}^5 + \cdots$. This means that we can approximate the inverse of $\boldsymbol{U}$ as $\boldsymbol{U}^{-1} \approx \boldsymbol{I} - \boldsymbol{\varepsilon}$; in consequence,

$$\boldsymbol{R} = \boldsymbol{F}\boldsymbol{U}^{-1} \approx (\boldsymbol{I} + \boldsymbol{\varepsilon} + \boldsymbol{\Omega})(\boldsymbol{I} - \boldsymbol{\varepsilon}) \approx \boldsymbol{I} + \boldsymbol{\Omega}.$$

and thus, in the case of small rotations, $\boldsymbol{\Omega} \approx \boldsymbol{R} - \boldsymbol{I}$.

In summary, in the small deformation regime, the following approximations hold:

$$\boldsymbol{C} \approx \boldsymbol{I} + 2\boldsymbol{\varepsilon} \qquad \boldsymbol{U} \approx \boldsymbol{I} + \boldsymbol{\varepsilon} \qquad \boldsymbol{v} \approx \boldsymbol{I} + \boldsymbol{\varepsilon} \qquad \boldsymbol{U}^{-1} \approx \boldsymbol{I} - \boldsymbol{\varepsilon} \qquad \boldsymbol{R} \approx \boldsymbol{I} + \boldsymbol{\Omega}$$

and

$$\boldsymbol{\varepsilon} \approx \boldsymbol{U} - \boldsymbol{I} \approx \boldsymbol{v} - \boldsymbol{I} \qquad\qquad \boldsymbol{\Omega} \approx \boldsymbol{R} - \boldsymbol{I}.$$

These approximations simplify the calculations and analyses in the small deformation regime, where the nonlinear terms involving the products and squares of the strain and rotation tensors can be neglected.

3.16 The eigendecomposition and the spectral decomposition of a matrix

The *eigendecomposition of a matrix** is a factorization of a matrix into a canonical form, whereby the matrix is represented in terms of its eigenvalues and eigenvectors.

Let us consider a square matrix $\boldsymbol{A}$ with dimensions $n \times n$; if $\boldsymbol{A}$ has n linearly independent eigenvectors $\vec{\boldsymbol{v}}_i$ with corresponding eigenvalues λ_i, for $i = 1, 2, \ldots, n$, we can write the set of equations

$$\boldsymbol{A}\vec{\boldsymbol{v}}_i = \lambda_i \vec{\boldsymbol{v}}_i$$

as

$$\boldsymbol{A}\underbrace{\begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{\boldsymbol{v}}_1 & \vec{\boldsymbol{v}}_2 & \cdots & \vec{\boldsymbol{v}}_n \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}}_{\boldsymbol{V}} = \underbrace{\begin{pmatrix} \uparrow & \uparrow & & \uparrow \\ \vec{\boldsymbol{v}}_1 & \vec{\boldsymbol{v}}_2 & \cdots & \vec{\boldsymbol{v}}_n \\ \downarrow & \downarrow & & \downarrow \end{pmatrix}}_{\boldsymbol{V}} \underbrace{\begin{pmatrix} \lambda_1 & 0 & & 0 \\ 0 & \lambda_2 & & 0 \\ & & \ddots & \\ 0 & 0 & & \lambda_n \end{pmatrix}}_{\boldsymbol{\Lambda}},$$

that is,

$$\boldsymbol{A}\boldsymbol{V} = \boldsymbol{V}\boldsymbol{\Lambda};$$

here the arrows have been employed to emphasize that matrix $\boldsymbol{V}$ is formed by the column vectors $\vec{\boldsymbol{v}}_i$. Since the eigenvectors are linearly independent, $\boldsymbol{V}$ is invertible and then matrix $\boldsymbol{A}$ can be factorized as the so-called *eigendecomposition** of matrix $\boldsymbol{A}$:

$$\boxed{\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^{-1},}$$

where $\boldsymbol{V}$ is the square $n \times n$ matrix whose i-th column is the eigenvector $\vec{\boldsymbol{v}}_i$ of $\boldsymbol{A}$, and $\boldsymbol{\Lambda}$ is the diagonal $n \times n$ matrix whose diagonal elements are the corresponding eigenvalues λ_i. The n eigenvectors $\vec{\boldsymbol{v}}_i$ are typically normalized, but this is not a requirement, inasmuch as non-normalized eigenvectors can also be used as the

columns of $\boldsymbol{V}$. This flexibility arises because the magnitudes of the eigenvectors in $\boldsymbol{V}$ gets canceled in the decomposition by the presence of $\boldsymbol{V}^{-1}$.

In the special case that $\boldsymbol{A}$ is a real symmetric matrix, its eigenvalues are real and its eigenvectors are real and orthogonal and might be normalized; in this case, $\boldsymbol{V}$ becomes an orthogonal matrix and the eigendecomposition of $\boldsymbol{A}$ simplifies to:

$$\boxed{\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T,} \tag{3.70}$$

in this last case, when $\boldsymbol{A}$ is a real symmetric matrix, the eigendecomposition is also known as the *spectral decomposition** of matrix $\boldsymbol{A}$.

The spectral decomposition can be also written alternatively as the sum of rank-one matrices:

$$\boldsymbol{A} = \sum_{i=1}^{n} \lambda_i \vec{\boldsymbol{v}}_i \vec{\boldsymbol{v}}_i^T.$$

or in terms of the dyadic product as:

$$\boldsymbol{A} = \sum_{i=1}^{n} \lambda_i \vec{\boldsymbol{v}}_i \otimes \vec{\boldsymbol{v}}_i.$$

We will see in Section 3.18, that in the context of continuum mechanics, the spectral decomposition of deformation and strain tensors provides valuable insights into the deformation state of materials. The eigenvalues and eigenvectors of these tensors represent the principal stretches and principal directions of deformation, respectively, which are essential for understanding and characterizing the material behavior under various loading conditions.

3.17 Square root of a semipositive definite matrix

*The square root of a matrix is a generalization of the notion of square root from positive real numbers to matrices. A matrix $\boldsymbol{S}$ is said to be a square root of another matrix $\boldsymbol{A}$ if $\boldsymbol{A} = \boldsymbol{S}^2 = \boldsymbol{S}\boldsymbol{S}$, that is, if the matrix product $\boldsymbol{S}\boldsymbol{S}$ is equal to $\boldsymbol{A}$. In general, a matrix can have multiple square roots. However, there is a theorem in linear algebra that states that any positive semidefinite matrix $\boldsymbol{A}$ has a unique decomposition $\boldsymbol{A} = \boldsymbol{S}\boldsymbol{S}$ such that $\boldsymbol{S}$ is also a and positive semidefinite matrix. In this case, $\boldsymbol{S}$ is the *principal square root* or *non-negative square root* of $\boldsymbol{A}$, and we denote $\boldsymbol{S}$ by $\sqrt{\boldsymbol{A}}$ or by $\boldsymbol{A}^{1/2}$. Moreover, $\sqrt{\boldsymbol{A}}$ is real, and its rank is the same as the rank of $\boldsymbol{A}$. If $\boldsymbol{A}$ is positive definite, then $\sqrt{\boldsymbol{A}}$ is called the *positive square root* of $\boldsymbol{A}$.

To determine $\sqrt{\boldsymbol{A}}$, we perform the spectral decomposition of $\boldsymbol{A}$ (see Section 3.16), i.e., $\boldsymbol{A} = \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T$, and then the principal square root of $\boldsymbol{A}$ is calculated by

$$\sqrt{\boldsymbol{A}} := \boldsymbol{V}\sqrt{\boldsymbol{\Lambda}}\boldsymbol{V}^T$$

where $\sqrt{\boldsymbol{\Lambda}} = \text{diag}\left(\sqrt{\lambda_1}, \sqrt{\lambda_2}, \ldots, \sqrt{\lambda_n}\right)$ represents the diagonal matrix formed by the square roots of the eigenvalues λ_i. This follows from the fact that:

$$\begin{aligned}
\boldsymbol{A} &= \sqrt{\boldsymbol{A}}\sqrt{\boldsymbol{A}} \\
&= \boldsymbol{V}\sqrt{\boldsymbol{\Lambda}}\underbrace{\boldsymbol{V}^T\boldsymbol{V}}_{=\boldsymbol{I}}\sqrt{\boldsymbol{\Lambda}}\boldsymbol{V}^T \\
&= \boldsymbol{V}\underbrace{\sqrt{\boldsymbol{\Lambda}}\sqrt{\boldsymbol{\Lambda}}}_{=\boldsymbol{\Lambda}}\boldsymbol{V}^T \\
&= \boldsymbol{V}\boldsymbol{\Lambda}\boldsymbol{V}^T.
\end{aligned}$$

While the spectral decomposition approach provides a straightforward way to compute the square root of a matrix, it can be computationally expensive for large matrices. In such cases, methods like the Denman-Beavers iteration or the blocked Schur algorithms can be more efficient. In fact, the latter algorithm is the one that underlies the Python function `sqrtm` from the `scipy.linalg` library:

```
from scipy.linalg import sqrtm, eigh
from numpy.random import randn
from numpy import diag, sqrt, allclose

# create a randomly generated positive semidefinite matrix A
M = randn(3,3)
A = M.T @ M

# calculate the eigenvalues L=Λ and eigenvectors V = [v1, v2, ..., vn]
L, V = eigh(A)

# calculate √Λ
sqrt_L = diag(sqrt(L))

# calculate √A using the spectral decomposition, equation (3.70)
sqrt_A1 = V @ sqrt_L @ V.T

# calculate √A using the "sqrtm" function
sqrt_A2 = sqrtm(A)

# verify that both methods provide the same result
print(allclose(sqrt_A1, sqrt_A2))    # prints True
```

Note that, in the context of the left and right Cauchy-Green tensors and according to equations (3.49) and (3.45), we have that $\boldsymbol{v} = \sqrt{\boldsymbol{b}}$ and $\boldsymbol{U} = \sqrt{\boldsymbol{C}}$.

3.18 Eigendecomposition of the deformation and strain tensors

Tensors $\boldsymbol{C}$, $\boldsymbol{C}^{-1}$, $\boldsymbol{U}$, $\boldsymbol{v}$, $\boldsymbol{b}$, $\boldsymbol{b}^{-1}$, $\boldsymbol{E}$, $\boldsymbol{e}$, and $\underline{\underline{\boldsymbol{\varepsilon}}}$ are all symmetric second-order tensors; therefore, their eigenvalues are real numbers, and their eigenvectors form orthogonal bases, representing principal components and principal directions, similar to the Cauchy stress tensor $\underline{\underline{\boldsymbol{\sigma}}}$ in the theory of elasticity. Additionally, it is possible to represent the deformation state using Mohr's circle. In the following, we will discuss the relationship between the eigenvalues and eigenvectors of these tensors.

Firstly, let us make the eigendecomposition of the stretch tensor $\boldsymbol{U}$:

$$\boldsymbol{U}\hat{\boldsymbol{N}}_i = \lambda_i \hat{\boldsymbol{N}}_i \quad \text{for } i = 1,2,3; \tag{3.71}$$

this decomposition states that the unit vector $\hat{\boldsymbol{N}}_i$ is deformed during motion by the "gradient deformation" $\boldsymbol{U}$, as in $\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\vec{\boldsymbol{X}}$ (equation [3.25]); however, $\hat{\boldsymbol{N}}_i$ does not rotate but only stretches λ_i times; here, $\hat{\boldsymbol{N}}_1$, $\hat{\boldsymbol{N}}_2$, and $\hat{\boldsymbol{N}}_3$ are the set of mutually orthogonal and normalized eigenvectors of $\boldsymbol{U}$, with corresponding eigenvalues λ_1, λ_2, λ_3, ordered by convention such that $\lambda_1 \geq \lambda_2 \geq \lambda_3 > 0$. These eigenvectors are unique as long as their corresponding eigenvalues are different.

Since $\boldsymbol{U}$ is a symmetric matrix these eigenvalues will be real numbers and since $\boldsymbol{U}$ is positive definite, all those eigenvalues will be positive, i.e. $\lambda_i > 0$. A negative eigenvalue λ_i is not physically admissible, as that would imply a reflection* in the deformation (like in a mirror).

Taking into account that $\boldsymbol{C} = \boldsymbol{U}^2$, by premultiplying (3.71) with $\boldsymbol{U}$, and again with the aid of (3.71), we arrive to

$$\boldsymbol{C}\hat{\boldsymbol{N}}_i = \boldsymbol{U}\underbrace{\boldsymbol{U}\hat{\boldsymbol{N}}_i}_{=\lambda_i\hat{\boldsymbol{N}}_i} = \lambda_i \underbrace{\boldsymbol{U}\hat{\boldsymbol{N}}_i}_{=\lambda_i\hat{\boldsymbol{N}}_i} = \lambda_i^2 \hat{\boldsymbol{N}}_i,$$

that is,

$$\boldsymbol{C}\hat{\boldsymbol{N}}_i = \lambda_i^2 \hat{\boldsymbol{N}}_i \quad \text{for } i = 1,2,3; \tag{3.72}$$

this means that $\boldsymbol{C}$ and $\boldsymbol{U}$ share the same set of orthogonal eigenvectors; however, the eigenvalues of $\boldsymbol{C}$ are the squared eigenvalues of $\boldsymbol{U}$. As explained in Section 3.10.4, we will call the eigenvectors of $\boldsymbol{U}$, namely $\hat{\boldsymbol{N}}_i$, the *principal material directions** or *principal referential directions** while their corresponding eigenvalues λ_i will be referred to as the *principal stretches*.*

It is possible to write the eigendecompositions (3.71) and (3.72) as:

$$\boldsymbol{U}\boldsymbol{N} = \boldsymbol{N}\boldsymbol{\Lambda} \qquad\qquad \boldsymbol{C}\boldsymbol{N} = \boldsymbol{N}\boldsymbol{\Lambda}^2$$

where $\boldsymbol{N} = \left[\hat{\boldsymbol{N}}_1\ \hat{\boldsymbol{N}}_2\ \hat{\boldsymbol{N}}_3\right]$ is an orthogonal matrix whose columns are the eigenvectors of $\boldsymbol{U}$ and $\boldsymbol{\Lambda} = \text{diag}\,(\lambda_1,\ \lambda_2,\ \lambda_3)$ is a diagonal matrix whose entries are the eigenvalues of $\boldsymbol{U}$. Notice that $\boldsymbol{\Lambda}^2 = \boldsymbol{\Lambda}\boldsymbol{\Lambda} = \text{diag}\left(\lambda_1^2,\ \lambda_2^2,\ \lambda_3^2\right)$. These equations lead to the spectral decompositions of $\boldsymbol{U}$ and $\boldsymbol{C}$, namely:

$$\boldsymbol{U} = \boldsymbol{N}\boldsymbol{\Lambda}\boldsymbol{N}^T \qquad\qquad \boldsymbol{C} = \boldsymbol{N}\boldsymbol{\Lambda}^2\boldsymbol{N}^T,$$

whose corresponding representations using the dyadic product are:

$$\boldsymbol{U} = \sum_{i=1}^{3} \lambda_i \hat{\boldsymbol{N}}_i \otimes \hat{\boldsymbol{N}}_i \qquad\qquad \boldsymbol{C} = \sum_{i=1}^{3} \lambda_i^2 \hat{\boldsymbol{N}}_i \otimes \hat{\boldsymbol{N}}_i.$$

The principal stretches and principal material directions of $\boldsymbol{U}$ and $\boldsymbol{C}$ can be interpreted in a different form. Let us remember that from equation (3.44), that $\lambda^2 = \hat{\boldsymbol{N}}^T\boldsymbol{C}\hat{\boldsymbol{N}}$; this formula allows us to estimate the squared stretch ratio λ^2 of any material element in the direction $\hat{\boldsymbol{N}}$; from this equation, we would like to ask the direction of $\hat{\boldsymbol{N}}$ for which λ^2 reaches its extreme values, subject to the constrain that $\left\|\hat{\boldsymbol{N}}\right\| = 1$. The solution to this question is related to the Lagrangian function:

$$f(\hat{\boldsymbol{N}}) = \hat{\boldsymbol{N}}^T\boldsymbol{C}\hat{\boldsymbol{N}} - \lambda^2\left(\hat{\boldsymbol{N}}^T\hat{\boldsymbol{N}} - 1\right)$$

where λ^2 acts as a Lagrangian multiplier; if we derive this equation with respect to $\hat{\boldsymbol{N}}$ and equalize it zero, we obtain:

$$\frac{\partial f(\hat{\boldsymbol{N}})}{\partial \hat{\boldsymbol{N}}} = 2\boldsymbol{C}\hat{\boldsymbol{N}} - 2\lambda^2\hat{\boldsymbol{N}} = 0$$

and this equation reduces to the eigenproblem $\boldsymbol{C}\hat{\boldsymbol{N}} = \lambda^2\hat{\boldsymbol{N}}$ which has already appeared as equation (3.72); consequently, the direction of $\hat{\boldsymbol{N}}$ for which λ^2 is maximum (minimum) is given by $\hat{\boldsymbol{N}}_1$ ($\hat{\boldsymbol{N}}_3$). This interpretation shows us that if we have a rubber band that is stretched, the maximum and minimum stretches will occur in those directions given by $\hat{\boldsymbol{N}}_1$ and $\hat{\boldsymbol{N}}_3$, respectively; the stretching factor in those directions, is given by λ_1 and λ_3.

In order to relate the eigenvectors and eigenvalues of $\boldsymbol{U}$ with those of $\boldsymbol{v}$, let us post-multiply $\boldsymbol{F} = \boldsymbol{v}\boldsymbol{R}$ by $\hat{\boldsymbol{N}}_i$ and apply equations (3.51) (that is $\boldsymbol{v} = \boldsymbol{R}\boldsymbol{U}\boldsymbol{R}^T$) and (3.71):

$$\boldsymbol{F}\hat{\boldsymbol{N}}_i = \boldsymbol{v}\underbrace{\boldsymbol{R}\hat{\boldsymbol{N}}_i}_{\hat{\boldsymbol{n}}_i} = \boldsymbol{R}\boldsymbol{U}\underbrace{\boldsymbol{R}^T\boldsymbol{R}}_{=\boldsymbol{I}}\hat{\boldsymbol{N}}_i = \boldsymbol{R}\underbrace{\boldsymbol{U}\hat{\boldsymbol{N}}_i}_{=\lambda_i\hat{\boldsymbol{N}}_i} = \lambda_i\underbrace{\boldsymbol{R}\hat{\boldsymbol{N}}_i}_{\hat{\boldsymbol{n}}_i}$$

that is,

$$\boldsymbol{v}\hat{\boldsymbol{n}}_i = \lambda_i\hat{\boldsymbol{n}}_i; \tag{3.73}$$

this implies that the eigenvalues of $\boldsymbol{v}$ are the same eigenvalues of $\boldsymbol{U}$ and their corresponding eigenvectors

$$\hat{\boldsymbol{n}}_i = \boldsymbol{R}\hat{\boldsymbol{N}}_i \quad \text{for } i = 1,2,3 \tag{3.74}$$

are those of $\boldsymbol{U}$ rotated by the action of tensor $\boldsymbol{R}$. Take into account that this last equation can also be written as the product

$$\boldsymbol{n} = \boldsymbol{R}\boldsymbol{N},$$

where $\boldsymbol{n} = [\hat{\boldsymbol{n}}_1\ \hat{\boldsymbol{n}}_2\ \hat{\boldsymbol{n}}_3]$ is an orthogonal matrix whose columns are the rotated eigenvectors of $\boldsymbol{U}$ or simply the eigenvectors of $\boldsymbol{v}$.

Taking into account equation (3.50), that is $\boldsymbol{b} = \boldsymbol{v}^2$, by premultiplying (3.73) with $\boldsymbol{v}$, and using again (3.73), we arrive to

$$\boldsymbol{b}\hat{\boldsymbol{n}}_i = \boldsymbol{v}\underbrace{\boldsymbol{v}\hat{\boldsymbol{n}}_i}_{=\lambda_i\hat{\boldsymbol{n}}_i} = \lambda_i\underbrace{\boldsymbol{v}\hat{\boldsymbol{n}}_i}_{=\lambda_i\hat{\boldsymbol{n}}_i} = \lambda_i^2\hat{\boldsymbol{n}}_i,$$

that is,

$$\boldsymbol{b}\hat{\boldsymbol{n}}_i = \lambda_i^2\hat{\boldsymbol{n}}_i \quad \text{for } i = 1,2,3; \tag{3.75}$$

this indicates that $\boldsymbol{b}$ and $\boldsymbol{v}$ share the same orthogonal eigenvectors; however, the eigenvalues of $\boldsymbol{b}$ are the squared eigenvalues of $\boldsymbol{v}$. Subsequently, we will call the eigenvectors of $\boldsymbol{v}$, namely $\hat{\boldsymbol{n}}_i$, the *principal spatial directions**. These equations prompt to the spectral decomposition of $\boldsymbol{v}$ and $\boldsymbol{b}$, namely:

$$\boldsymbol{v} = \boldsymbol{n}\boldsymbol{\Lambda}\boldsymbol{n}^T \qquad\qquad \boldsymbol{b} = \boldsymbol{n}\boldsymbol{\Lambda}^2\boldsymbol{n}^T,$$

whose corresponding representations using the dyadic product are:

$$\boldsymbol{v} = \sum_{i=1}^{3}\lambda_i\hat{\boldsymbol{n}}_i\otimes\hat{\boldsymbol{n}}_i \qquad\qquad \boldsymbol{b} = \sum_{i=1}^{3}\lambda_i^2\hat{\boldsymbol{n}}_i\otimes\hat{\boldsymbol{n}}_i.$$

It is the turn to estimate the eigendecomposition of $\boldsymbol{b}^{-1}$. If we premultiply equation (3.75) by $\boldsymbol{b}^{-1}$, it results that:

$$\underbrace{\boldsymbol{b}^{-1}\boldsymbol{b}}_{=\boldsymbol{I}}\hat{\boldsymbol{n}}_i = \lambda_i^2\boldsymbol{b}^{-1}\hat{\boldsymbol{n}}_i \quad \text{for } i = 1,2,3;$$

that is $\hat{\boldsymbol{n}}_i = \lambda_i^2\boldsymbol{b}^{-1}\hat{\boldsymbol{n}}_i$ and in consequence,

$$\boldsymbol{b}^{-1}\hat{\boldsymbol{n}}_i = \frac{1}{\lambda_i^2}\hat{\boldsymbol{n}}_i \quad \text{for } i = 1,2,3; \tag{3.76}$$

this mean that $\boldsymbol{b}$ and $\boldsymbol{b}^{-1}$ possess the same eigenvectors, but the eigenvalues of $\boldsymbol{b}^{-1}$ are the inverse squared principal stretches λ^{-2}.

Let us find the eigendecomposition of the Cauchy-Lagrange tensor $\boldsymbol{E}$. From equation (3.52), it follows that $\boldsymbol{C} = 2\boldsymbol{E} + \boldsymbol{I}$; if we replace this equation into (3.72), that is

$\boldsymbol{C}\hat{\boldsymbol{N}}_i = \lambda_i^2\hat{\boldsymbol{N}}_i$, it follows that $(2\boldsymbol{E}+\boldsymbol{I})\hat{\boldsymbol{N}}_i = \lambda_i^2\hat{\boldsymbol{N}}_i$ and that $2\boldsymbol{E}\hat{\boldsymbol{N}}_i = \lambda_i^2\hat{\boldsymbol{N}}_i - \hat{\boldsymbol{N}}_i$; in conclusion,

$$\boldsymbol{E}\hat{\boldsymbol{N}}_i = \frac{\lambda_i^2-1}{2}\hat{\boldsymbol{N}}_i \quad \text{for } i=1,2,3;$$

this means that $\boldsymbol{E}$ shares the same eigenvectors as $\boldsymbol{U}$ and $\boldsymbol{C}$, while its eigenvalues are $\frac{\lambda_i^2-1}{2}$ for $i=1,2,3$.

Similarly, in order to find the eigendecomposition of the Euler-Almansi tensor $\boldsymbol{e}$, take into account that from (3.54) it follows that $\boldsymbol{b}^{-1} = \boldsymbol{I}-2\boldsymbol{e}$; replacing this equation into (3.76), we obtain $(\boldsymbol{I}-2\boldsymbol{e})\hat{\boldsymbol{n}}_i = \lambda_i^{-2}\hat{\boldsymbol{n}}_i$, that is $(1-\lambda_i^{-2})\hat{\boldsymbol{n}}_i = 2\boldsymbol{e}\hat{\boldsymbol{n}}_i$ which results in:

$$\boldsymbol{e}\hat{\boldsymbol{n}}_i = \frac{1-\lambda_i^{-2}}{2}\hat{\boldsymbol{n}}_i \quad \text{for } i=1,2,3;$$

this implies that $\boldsymbol{e}$ has the same eigenvectors as $\boldsymbol{v}$, $\boldsymbol{b}$ and $\boldsymbol{b}^{-1}$, while its eigenvalues are $\frac{1-\lambda_i^{-2}}{2}$ for $i=1,2,3$.

Now, we would like to express $\boldsymbol{F}$ in terms of principal stretches and directions. Using equations $\boldsymbol{F}=\boldsymbol{RU}$, $\boldsymbol{U}=\boldsymbol{N}\boldsymbol{\Lambda}\boldsymbol{N}^T$ and $\boldsymbol{n}=\boldsymbol{RN}$ (equations [3.12], [3.18] and [3.18], respectively), we have

$$\boldsymbol{F}=\boldsymbol{RU}=\boldsymbol{RN}\boldsymbol{\Lambda}\boldsymbol{N}^T=\boldsymbol{n}\boldsymbol{\Lambda}\boldsymbol{N}^T,$$

that is,

$$\boldsymbol{F}=\boldsymbol{n}\boldsymbol{\Lambda}\boldsymbol{N}^T \tag{3.77}$$

or using the dyadic notation:

$$\boldsymbol{F}=\sum_{i=1}^{3}\lambda_i\hat{\boldsymbol{n}}_i\otimes\hat{\boldsymbol{N}}_i.$$

Finally, we would like to express $\boldsymbol{R}$ in terms of principal directions. From $\boldsymbol{F}=\boldsymbol{v}\boldsymbol{R}$ it follows that $\boldsymbol{R}=\boldsymbol{v}^{-1}\boldsymbol{F}$. Taking into account that $\boldsymbol{v}=\boldsymbol{n}\boldsymbol{\Lambda}\boldsymbol{n}^T$ and that $\boldsymbol{ABC}=\boldsymbol{C}^{-1}\boldsymbol{B}^{-1}\boldsymbol{A}^{-1}$, it results that $\boldsymbol{v}^{-1}=\boldsymbol{n}^{-T}\boldsymbol{\Lambda}^{-1}\boldsymbol{n}^{-1}$, therefore using (3.77),

$$\boldsymbol{R}=\boldsymbol{v}^{-1}\boldsymbol{F}=(\boldsymbol{n}^{-T}\boldsymbol{\Lambda}^{-1}\boldsymbol{n}^{-1})(\boldsymbol{n}\boldsymbol{\Lambda}\boldsymbol{N}^T)=\boldsymbol{n}^{-T}\boldsymbol{N}^T$$

and since $\boldsymbol{n}$ is an orthogonal matrix then $\boldsymbol{n}^{-T}=(\boldsymbol{n}^{-1})^T=(\boldsymbol{n}^T)^T=\boldsymbol{n}$ and in consequence,

$$\boldsymbol{R}=\boldsymbol{n}\boldsymbol{N}^T \tag{3.78}$$

or using the dyadic notation:

$$\boldsymbol{R}=\sum_{i=1}^{3}\hat{\boldsymbol{n}}_i\otimes\hat{\boldsymbol{N}}_i.$$

Observe how both the reference and current configurations are involved in the definitions of $\boldsymbol{F}$ and $\boldsymbol{R}$, in terms of the material and principal directions $\hat{\boldsymbol{N}}_i$ and $\hat{\boldsymbol{n}}_i$, as

revealed by equations (3.77) and (3.78), emphasizing the two-point nature of $\boldsymbol{F}$ and $\boldsymbol{R}$.

The physical interpretation of these eigendecompositions is that the deformation gradient tensor $\boldsymbol{F}$ can be viewed as a combination of stretches along the principal material directions $\hat{\boldsymbol{N}}_i$, followed by a rigid rotation given by $\boldsymbol{R}$. The principal stretches λ_i represent the stretching factors along each principal material direction $\hat{\boldsymbol{N}}_i$.

Similarly, the rotation tensor $\boldsymbol{R}$ represents a rigid rotation that aligns the principal material directions $\hat{\boldsymbol{N}}_i$ with the principal spatial directions $\hat{\boldsymbol{n}}_i$. This means that the deformation process can be conceptualized as first stretching the material along the principal directions by the corresponding stretch ratios, and then rotating the stretched material to its final configuration.

3.19 The polar decomposition revisited

The polar decomposition theorem states that any invertible second-order tensor with a positive determinant, such as the deformation gradient $\boldsymbol{F}$, can be decomposed into the product of two second-order tensors: an orthogonal tensor $\boldsymbol{R}$ and a positive definite symmetric tensor ($\boldsymbol{U}$ or $\boldsymbol{v}$), as $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{v}\boldsymbol{R}$. While the polar decomposition was introduced in Section 3.12, certain details were deferred for a later exposition. In the following subsections, we will delve into the physical interpretation of the principal stretches and directions in the context of the polar decomposition, demonstrate that $\boldsymbol{U}$ and $\boldsymbol{v}$ are positive definite tensors, prove that $\boldsymbol{R}$ is an orthogonal tensor, establish the uniqueness of the polar decomposition, and explore numerical methods for its calculation.

3.19.1 Physical interpretation of the principal stretches and directions in light of the polar decomposition

Provide an introduction ...

Equation (3.38), that is $\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}} = \boldsymbol{R}(\boldsymbol{U}\,\mathrm{d}\vec{\boldsymbol{X}})$, describes a pure stretch, without rotation, performed by the material stretch tensor $\boldsymbol{U}$, in the direction of the principal material stretches $\hat{\boldsymbol{N}}_i$ with stretches given by the eigenvalues λ_i; once the body is deformed, the rotation takes place and is performed by tensor $\boldsymbol{R}$. In other words, in the equation, $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$, the deformation process is viewed from the reference (material) configuration. The stretching along the principal material directions $\hat{\boldsymbol{N}}_i$ by the stretch ratios λ_i occurs first through $\boldsymbol{U}$, followed by the rigid rotation $\boldsymbol{R}$ to align with the current (spatial) configuration.

Equation (3.39), i.e. $\mathrm{d}\vec{\boldsymbol{x}} = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}} = \boldsymbol{v}(\boldsymbol{R}\,\mathrm{d}\vec{\boldsymbol{X}})$, describes a pure rotation carried out by $\boldsymbol{R}$ followed by three stretches given by the eigenvalues λ_i of the spatial stretch

tensor $\boldsymbol{v}$ along the directions given by its eigenvectors $\hat{\boldsymbol{n}}_i$; the eigenvectors $\hat{\boldsymbol{n}}_i$ are the rotated principal material directions, that is, $\hat{\boldsymbol{n}}_i = \boldsymbol{R}\hat{\boldsymbol{N}}_i$ according to equation (3.74). In other words, the equation, $\boldsymbol{F} = \boldsymbol{v}\boldsymbol{R}$, the deformation process is viewed from the current (spatial) configuration. First, the rigid rotation $\boldsymbol{R}$ aligns the principal material directions $\hat{\boldsymbol{N}}_i$ to the principal spatial directions $\hat{\boldsymbol{n}}_i$. Then, the stretching along these rotated directions occurs through $\boldsymbol{v}$.

When comparing both equations (3.38) and (3.39), observe that the both stretch tensors $\boldsymbol{U}$ and $\boldsymbol{v}$ provide the same amount of local stretching (or contraction), since its eigenvalues λ_i are the same; however, the directions of stretching $\hat{\boldsymbol{N}}_{\boldsymbol{i}}$ and $\hat{\boldsymbol{n}}_i$ are different. Since both sets of eigenvectors $\{\hat{\boldsymbol{N}}_1, \hat{\boldsymbol{N}}_2, \hat{\boldsymbol{N}}_3\}$ and $\{\hat{\boldsymbol{n}}_1, \hat{\boldsymbol{n}}_2, \hat{\boldsymbol{n}}_3\}$ are mutually orthogonal, they represent locally the change of shape.

Both perspectives provide different intuitive ways to visualize and interpret the deformation process, highlighting the interplay between stretching and rotational components. The choice of perspective may depend on the specific problem or the frame of reference most convenient for analysis.

3.19.2 Demonstration that U and v are positive definite tensors

Since the right Cauchy-Green deformation tensor $\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F}$ is positive definite, it has a square root, $\boldsymbol{U} = \sqrt{\boldsymbol{C}}$ which is a symmetric positive definite tensor. Therefore $\boldsymbol{C}$ can also be defined as $\boldsymbol{C} = \boldsymbol{U}\boldsymbol{U}$.

Similarly, since the left Cauchy-Green deformation tensor $\boldsymbol{b} = \boldsymbol{F}\boldsymbol{F}^T$ is positive definite, it has a square root, $\boldsymbol{v} = \sqrt{\boldsymbol{b}}$ which is a symmetric positive definite tensor. Therefore $\boldsymbol{b}$ can also be defined as $\boldsymbol{b} = \boldsymbol{v}\boldsymbol{v}$.

3.19.3 Demonstration that R is an orthogonal tensor

To demonstrate that the rotation tensor $\boldsymbol{R}$ is an orthogonal matrix, we use two additional facts:

- From $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$, and taking into account that since $\boldsymbol{U}$ is positive definite its inverse $\boldsymbol{U}^{-1}$ exits, that we can rearrange the terms to obtain $\boldsymbol{R} = \boldsymbol{F}\boldsymbol{U}^{-1}$.
- The transpose of the inverse of a symmetric matrix is also symmetric. In other words, since $\boldsymbol{U}$ is symmetric, then $\boldsymbol{U}^{-1} = (\boldsymbol{U}^{-1})^T$. To show this, consider that since $\boldsymbol{U}$ is nonsingular, $\boldsymbol{U}^{-1}$ exists. Also, we have the property that $\boldsymbol{I} = \boldsymbol{U}\boldsymbol{U}^{-1} = \boldsymbol{U}^{-1}\boldsymbol{U}$, where $\boldsymbol{I}$ is the identity matrix. Using the property that $(\boldsymbol{A}\boldsymbol{B})^T = \boldsymbol{B}^T\boldsymbol{A}^T$ we have: $\boldsymbol{U}^{-1}\boldsymbol{U} = \boldsymbol{I} = \boldsymbol{U}\boldsymbol{U}^{-1} = \boldsymbol{I}^T = (\boldsymbol{U}\boldsymbol{U}^{-1})^T = (\boldsymbol{U}^{-1})^T\boldsymbol{U}^T$. Inasmuch as $\boldsymbol{U}$ is symmetric (i.e., $\boldsymbol{U} = \boldsymbol{U}^T$), we substitute this into the right side: $\boldsymbol{U}^{-1}\boldsymbol{U} = (\boldsymbol{U}^{-1})^T\boldsymbol{U}$. Now, postmultiplying both sides by $\boldsymbol{U}^{-1}$ gives: $\boldsymbol{U}^{-1}\boldsymbol{U}\boldsymbol{U}^{-1} = (\boldsymbol{U}^{-1})^T\boldsymbol{U}\boldsymbol{U}^{-1}$ and since $\boldsymbol{U}\boldsymbol{U}^{-1} = \boldsymbol{I}$, we conclude that $\boldsymbol{U}^{-1} = (\boldsymbol{U}^{-1})^T$, which proves that the inverse of a symmetric matrix is symmetric.

We can show that $\boldsymbol{R}$ is an orthogonal matrix as follows:

$$\boldsymbol{R}^T\boldsymbol{R} = \left(\boldsymbol{F}\boldsymbol{U}^{-1}\right)^T\left(\boldsymbol{F}\boldsymbol{U}^{-1}\right) = \left(\boldsymbol{U}^{-1}\right)^T\boldsymbol{F}^T\boldsymbol{F}\boldsymbol{U}^{-1} = \left(\boldsymbol{U}^{-1}\right)^T\boldsymbol{C}\boldsymbol{U}^{-1} = \boldsymbol{U}^{-1}\boldsymbol{U}\boldsymbol{U}\boldsymbol{U}^{-1} = \boldsymbol{I}.$$

3.19.4 Uniqueness of the polar decomposition

The polar decomposition is unique, which means that for a given deformation gradient tensor $\boldsymbol{F}$, there exists only one pair of rotation tensor $\boldsymbol{R}$ and stretch tensor $\boldsymbol{U}$ (or $\boldsymbol{v}$) that satisfies the equation $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{v}\boldsymbol{R}$.

To show the uniqueness of the right polar decomposition $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$, let us assume that there is another decomposition $\boldsymbol{F} = \tilde{\boldsymbol{R}}\tilde{\boldsymbol{U}}$ with $\tilde{\boldsymbol{R}}^T\tilde{\boldsymbol{R}} = \boldsymbol{I}$ (orthogonal tensor) and $\tilde{\boldsymbol{U}}^T = \tilde{\boldsymbol{U}}$ (symmetric tensor).

Inasmuch as $\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F}$ is the right Cauchy-Green deformation tensor, we can write:

$$\boldsymbol{C} = (\boldsymbol{R}\boldsymbol{U})^T(\boldsymbol{R}\boldsymbol{U}) = \boldsymbol{U}^T\boldsymbol{R}^T\boldsymbol{R}\boldsymbol{U} = \boldsymbol{U}^2$$
$$\boldsymbol{C} = (\tilde{\boldsymbol{R}}\tilde{\boldsymbol{U}})^T(\tilde{\boldsymbol{R}}\tilde{\boldsymbol{U}}) = \tilde{\boldsymbol{U}}^T\tilde{\boldsymbol{R}}^T\tilde{\boldsymbol{R}}\tilde{\boldsymbol{U}} = \tilde{\boldsymbol{U}}^2$$

Since $\boldsymbol{C}$ is a positive definite tensor, it has only one square root; therefore, $\boldsymbol{U} = \tilde{\boldsymbol{U}} = \sqrt{\boldsymbol{C}}$, and consequently, $\tilde{\boldsymbol{R}} = \boldsymbol{R}$. This proves that the right polar decomposition is unique.

A similar proof can be constructed for the left polar decomposition $\boldsymbol{F} = \boldsymbol{v}\boldsymbol{R}$, which is left as an exercise to the reader.

3.19.5 Numerical methods for the calculation of the polar decomposition

In the following, we will present four alternative methods for calculating the polar decomposition.

Method 1

In the SciPy library of Python, the `scipy.linalg.polar` function can be used to compute the polar decomposition. This is the method that should be used in practical applications. The usage of this function will be illustrated in the code of Method 2, that is explained in the next.

Method 2

The singular value decomposition of a matrix $\boldsymbol{M} \in \mathbb{R}^{m\times n}$ is a factorization of the form $\boldsymbol{M} = \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{V}^T$, where $\boldsymbol{W} \in \mathbb{R}^{m\times m}$ and $\boldsymbol{V} \in \mathbb{R}^{n\times n}$ are orthogonal matrices and $\boldsymbol{\Sigma} \in \mathbb{R}^{m\times n}$ is a rectangular diagonal matrix with non-negative real numbers on the diagonal. It is guaranteed that such a decomposition always exists.

It can be shown that using the singular value decomposition of the deformation gradient $\boldsymbol{F}$,

$$\boldsymbol{F} = \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{V}^T,$$

the polar decomposition can be expressed as:

$$\boldsymbol{R} = \boldsymbol{W}\boldsymbol{V}^T \qquad \boldsymbol{U} = \boldsymbol{V}\boldsymbol{\Sigma}\boldsymbol{V}^T \qquad \boldsymbol{v} = \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{W}^T.$$

This follows from the fact that:

$$\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U} = \boldsymbol{W}\underbrace{\boldsymbol{V}^T\boldsymbol{V}}_{=\boldsymbol{I}}\boldsymbol{\Sigma}\boldsymbol{V}^T = \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{V}^T$$
$$\boldsymbol{F} = \boldsymbol{v}\boldsymbol{R} = \boldsymbol{W}\boldsymbol{\Sigma}\underbrace{\boldsymbol{W}^T\boldsymbol{W}}_{=\boldsymbol{I}}\boldsymbol{V}^T = \boldsymbol{W}\boldsymbol{\Sigma}\boldsymbol{V}^T.$$

This is illustrated using the following Python code:

```
from scipy.linalg import polar
from numpy.linalg import svd
from numpy.random import randn
from numpy import allclose, diag

F = randn(3,3)

W, S, VT = svd(F)

R1, v = polar(F, 'left')
R2, U = polar(F, 'right')

print(allclose(R1, R2))                    # prints True
print(allclose(R1, W @ VT))                # prints True
print(allclose(U,  VT.T @ diag(S) @ VT))   # prints True
print(allclose(v,  W @ diag(S) @ W.T))     # prints True
```

Method 3

Another form to calculate the polar decomposition of the deformation gradient tensor $\boldsymbol{F}$ takes advantage of the fact that the right stretch tensor $\boldsymbol{U}$ is the square root of the right Cauchy-Green deformation tensor $\boldsymbol{C} = \boldsymbol{F}^T\boldsymbol{F}$:

$$\boldsymbol{U} = \sqrt{\boldsymbol{C}};$$

once $\boldsymbol{U}$ is obtained, the rotation tensor $\boldsymbol{R}$ can be determined from the equation $\boldsymbol{F} = \boldsymbol{R}\boldsymbol{U}$ as follows:

$$\boldsymbol{R} = \boldsymbol{F}\boldsymbol{U}^{-1};$$

with $\boldsymbol{R}$ known, the left stretch tensor $\boldsymbol{v}$ can be obtained from the equation $\boldsymbol{F} = \boldsymbol{v}\boldsymbol{R}$ using:

$$\boldsymbol{v} = \boldsymbol{F}\boldsymbol{R}^T.$$

This procedure is implemented in the following Matlab code:

```
function [R, U, v] = polar(F)
    [m, n] = size(F);      % get matrix dimensions

    % check if F is square
    assert(m == n, 'Matrix F must be square');

    C = F' * F;            % right Cauchy-Green tensor
    U = sqrtm(C);          % material (right) stretch tensor
    R = F * inv(U);        % rotation tensor
    v = F * R';            % spatial (left) stretch tensor
end
```

Method 4

Using the spectral decomposition

3.20 Push-forward and pull-backward operations

The push-forward and pull-backward operations transform vectors and tensors between their Lagrangian and Eulerian representations.

The *push-forward** operation transforms a Lagrangian tensor to its Eulerian equivalent, and is denoted by $\phi_*[\cdot]$; in other words, it transforms a vector or tensor valued quantity defined on the reference configuration to the current configuration. For example in Section 3.8, we studied that the deformation gradient $\boldsymbol{F}$ maps a material vector element $\mathrm{d}\vec{\boldsymbol{X}}$ to its corresponding spatial vector element $\mathrm{d}\vec{\boldsymbol{x}}$ by means of the equation $\mathrm{d}\vec{\boldsymbol{x}} = \phi_*[\mathrm{d}\vec{\boldsymbol{X}}] = \boldsymbol{F}\,\mathrm{d}\vec{\boldsymbol{X}}$; this kind of mapping is an example of what is called a push forward operation.

The inverse of the push-forward operation is the *pull-backward** which transforms an Eulerian tensor to its Lagrangian representation, and is denoted by the shorthand $\phi_*^{-1}[\cdot]$; in other words, it transforms a vector or tensor valued quantity defined on the current configuration to the reference configuration. For instance, in Section 3.9, we studied that the inverse of the deformation gradient $\boldsymbol{F}^{-1}$ maps a spatial vector element $\mathrm{d}\vec{\boldsymbol{x}}$ to its corresponding material vector element $\mathrm{d}\vec{\boldsymbol{X}}$ by means of the pull-backward operation $\mathrm{d}\vec{\boldsymbol{X}} = \phi_*^{-1}[\mathrm{d}\vec{\boldsymbol{x}}] = \boldsymbol{F}^{-1}\,\mathrm{d}\vec{\boldsymbol{x}}$.

An example for a second order tensor is in order; from equation (3.56), we have that $d\vec{\boldsymbol{X}}_1^T \boldsymbol{E}\, d\vec{\boldsymbol{X}}_2 = d\vec{\boldsymbol{x}}_1^T \boldsymbol{e}\, d\vec{\boldsymbol{x}}_2$; therefore, $d\vec{\boldsymbol{X}}_1^T \boldsymbol{E}\, d\vec{\boldsymbol{X}}_2 = d\vec{\boldsymbol{X}}_1^T \boldsymbol{F}^T \boldsymbol{e}\boldsymbol{F}\, d\vec{\boldsymbol{X}}_2$ and

$$\boxed{\boldsymbol{E} = \boldsymbol{F}^T \boldsymbol{e}\boldsymbol{F}} \tag{3.79}$$

also, from $d\vec{\boldsymbol{x}}_1^T \boldsymbol{e}\, d\vec{\boldsymbol{x}}_2 = d\vec{\boldsymbol{X}}_1^T \boldsymbol{F}^{-T} \boldsymbol{E}\boldsymbol{F}^{-1}\, d\vec{\boldsymbol{X}}_2$ it follows that

$$\boxed{\boldsymbol{e} = \boldsymbol{F}^{-T} \boldsymbol{E}\boldsymbol{F}^{-1};} \tag{3.80}$$

here, equation (3.79) is a pull-back operation, while equation (3.80) is a push-forward operation. These operations might be written as $\boldsymbol{E} = \phi_*^{-1}[\boldsymbol{e}]$ and $\boldsymbol{e} = \phi_*[\boldsymbol{E}]$, respectively. Observe that the nomenclature $\phi_*[]$ denotes an operator that will function differently for distinct operands.

3.21 Time derivatives, velocity and acceleration

So far, in our exposition we have not given the appropriate attention to the influence of time in motion, despite the fact that many nonlinear phenomena are time-dependent. We will do so, by discussing time derivatives. There are two types of time derivatives of motion: the local- and the material-time derivative. The first one tells us how a quantity changes in time at a fixed point in space; the latter describes the time rate of change of some quantity (like temperature, chemical concentration or density) of a material element that is moving along a trajectory.

For example, in fluid dynamics, the velocity field is the flow velocity, and the quantity of interest might be the temperature of the fluid. In which case, the local time derivative describes the temperature change at a fixed point and the material derivative describes the temperature change of a certain fluid parcel with time, as it flows along its pathline.

3.21.1 Velocity

While in solid mechanics the motion and deformation of a body are typically expressed in terms of displacements, in fluid mechanics velocity and acceleration are the primary field values that characterize how a particle is moving. In the next, we will analyze velocity; acceleration will be dealt with in Section 3.21.4.

The *velocity** of a moving particle is defined as the instantaneous rate of change of its position with respect to time. It is also known as the *flow velocity** or *velocity field*. Mathematically, the velocity can be represented, in the material description, as:

$$\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) := \lim_{\Delta t \to 0} \frac{\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t + \Delta t) - \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)}{\Delta t} = \frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)}{\partial t}, \tag{3.81}$$

where $\vec{\boldsymbol{X}}$ is the position of the particle in the initial configuration, and $\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$ is its position as a function of time t; thus, as described in Appendix A.6, the velocity is the tangent vector of the trajectory described by the particle initially at $\vec{\boldsymbol{X}}$.

It is possible the express velocity in terms of displacements by adding and substracting $\vec{\boldsymbol{X}}$ in the numerator

$$\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) = \lim_{\Delta t \to 0} \frac{\left[\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t + \Delta t) - \vec{\boldsymbol{X}}\right] - \left[\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) - \vec{\boldsymbol{X}}\right]}{\Delta t},$$

so that, when using equation (3.3), i.e. $\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) - \vec{\boldsymbol{X}}$, we obtain:

$$\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) = \lim_{\Delta t \to 0} \frac{\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t + \Delta t) - \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)}{\Delta t} = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)}{\partial t}; \tag{3.82}$$

this means that velocity can also be defined as the instantaneous rate of change of the displacement with respect to time. Note that both equations, (3.81) and (3.82), express the velocity at time t of the particle which was at $\vec{\boldsymbol{X}}$ at time t_0.

The spatial description of velocity is found by using equation (3.2), that is $\vec{\boldsymbol{X}} = \vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t)$. So, the velocity in the spatial description can be represented as:

$$\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) := \vec{\boldsymbol{V}}\left(\vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t), t\right) = \vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t); \tag{3.83}$$

notice that $\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$ even though they represent equal quantities that are functions of different arguments; also, the SI unit for velocity is meter per second, i.e. m/s. Observe that velocity is not defined as $\frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)}{\partial t}$.

It's worth mentioning that velocity is sometimes represented, in the spatial description and with some abuse of notation, as

$$\vec{\boldsymbol{v}}(t) = \dot{\vec{\boldsymbol{x}}}(t) = \frac{\mathrm{d}\vec{\boldsymbol{x}}(t)}{\mathrm{d}t} := \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t);$$

here $\vec{\boldsymbol{x}}$ is a function of $\vec{\boldsymbol{X}}$ and t through equation (3.1), that is $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$; however, the position argument is avoided for the sake of compactness in the representation.

Bear in mind that the norm of the velocity field, either $\|\vec{\boldsymbol{V}}\|$ or $\|\vec{\boldsymbol{v}}\|$, is a scalar known as the flow *speed*[★]; it is a scalar magnitude of a velocity vector that denotes only how fast an object is moving, while velocity indicates both the object speed and direction.

In order to have a constant velocity, a particle must move at a constant speed and in a constant direction. The constant direction constraint requires the object to move along a straight trajectory. Therefore, a constant velocity implies motion in a straight line at a constant speed. For example, consider a car moving at a constant speed of 20 kilometers per hour on a circular path; although the car's speed is constant, its velocity is not constant because its direction continually changes as it follows the circular path.

Example 3.13. *Displacement and velocity of a body*

Consider a body that undergoes motion described by:

$$\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} (1+at^2)X_1 \\ X_2 \\ X_3 \end{pmatrix}$$

where a is a constant. Find the displacement and the velocity in both the material and spatial descriptions. Compare the displacement and velocity in both descriptions.

Solution:

To solve the problem, we begin by analyzing the given motion of the body and derive the displacement and velocity in both material (Lagrangian) and spatial (Eulerian) descriptions.

On the one hand, the displacement $\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)$ in the material description is the difference between the current position $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$ and the reference position $\vec{\boldsymbol{X}}$:

$$\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{x}} - \vec{\boldsymbol{X}} = \begin{pmatrix} (1+at^2)X_1 - X_1 \\ X_2 - X_2 \\ X_3 - X_3 \end{pmatrix} = \begin{pmatrix} at^2 X_1 \\ 0 \\ 0 \end{pmatrix}.$$

and the material velocity $\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)$ is the time derivative of the displacement $\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)$, holding $\vec{\boldsymbol{X}}$ constant:

$$\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t)}{\partial t} = \begin{pmatrix} \frac{\mathrm{d}}{\mathrm{d}t}\left[(1+at^2)X_1\right] \\ \frac{\mathrm{d}}{\mathrm{d}t}X_2 \\ \frac{\mathrm{d}}{\mathrm{d}t}X_3 \end{pmatrix} = \begin{pmatrix} 2atX_1 \\ 0 \\ 0 \end{pmatrix},$$

and this gives the velocity of the particle that at time t_0 occupies the position $\vec{\boldsymbol{X}}$.

On the other hand, to express quantities in terms of spatial coordinates $\vec{\boldsymbol{x}}$, we invert the motion. From $x_1 = (1+at^2)X_1$, we solve for X_1, so that, $X_1 = \frac{x_1}{1+at^2}$, $X_2 = x_2$, and $X_3 = x_3$. That is,

$$\vec{\boldsymbol{X}} = \vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t) = \begin{pmatrix} \frac{x_1}{1+at^2} \\ x_2 \\ x_3 \end{pmatrix}.$$

The displacement $\vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)$ in the spatial description is the difference between the current position $\vec{\boldsymbol{x}}$ and the reference position $\vec{\boldsymbol{X}} = \vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t)$:

$$\vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t) = \vec{\boldsymbol{x}} - \vec{\boldsymbol{X}} = \begin{pmatrix} x_1 - \frac{x_1}{1+at^2} \\ x_2 - x_2 \\ x_3 - x_3 \end{pmatrix} = \begin{pmatrix} \frac{at^2 x_1}{1+at^2} \\ 0 \\ 0 \end{pmatrix}.$$

and the spatial velocity $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$ is the found by substituting $\vec{\boldsymbol{X}}$ into the material velocity $\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)$:

$$\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \begin{pmatrix} 2at\left(\frac{x_1}{1+at^2}\right) \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{2atx_1}{1+at^2} \\ 0 \\ 0 \end{pmatrix},$$

and this gives the velocity of the particle that at time t occupies the position $\vec{\boldsymbol{x}}$. This is a good point to emphasize that velocity is not defined as $\frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}},t)}{\partial t}$; in this case,

$$\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \neq \frac{\partial \vec{\boldsymbol{u}}(\vec{\boldsymbol{x}}, t)}{\partial t} = \begin{pmatrix} \frac{2atx_1}{(1+at^2)^2} \\ 0 \\ 0 \end{pmatrix}.$$

While the expressions for displacement differ significantly between descriptions due to the time-dependent deformation, the velocity fields are physically consistent. The spatial velocity is simply the material velocity re-expressed in terms of the current configuration. This examples shows the distinction between Lagrangian (tracking particles) and Eulerian (observing fixed points) perspectives.

3.21.2 Local time derivative

In the spatial description, the *spatial time derivative*★ or *local time derivative*★ of a spatial field is quantified in the scalar case as

$$\frac{\partial h(\vec{\boldsymbol{x}}, t)}{\partial t} := \lim_{\Delta t \to 0} \frac{h(\vec{\boldsymbol{x}}, t + \Delta t) - h(\vec{\boldsymbol{x}}, t)}{\Delta t}.$$

or in the vector case as

$$\frac{\partial \vec{\boldsymbol{h}}(\vec{\boldsymbol{x}}, t)}{\partial t} := \lim_{\Delta t \to 0} \frac{\vec{\boldsymbol{h}}(\vec{\boldsymbol{x}}, t + \Delta t) - \vec{\boldsymbol{h}}(\vec{\boldsymbol{x}}, t)}{\Delta t}.$$

where the vector $\vec{\boldsymbol{x}}$ is held constant. It represents the instantaneous rate at which a field changes with time as seen by an observer currently located at a *fixed* position $\vec{\boldsymbol{x}}$ in space. For instance, the velocity as defined in equation (3.83), for a constant $\vec{\boldsymbol{x}}$, is an example of a local time derivative of a spatial vector field.

The local time derivative is the one that is used for measuring for example the velocity of a moving body at a specific point $\vec{\boldsymbol{x}}$ that does not change with time t, as it does an

anemometer with a thermometer at the top of a building, which measures the wind velocity and air temperature.

Finally, a variable that defines a behavior is said to reach a *steady-state** if its local time derivative vanishes for all $\vec{\boldsymbol{x}} \in \Omega$. For example, a motion is said to be *steady** if its velocity does not change with time, that is, the velocity $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$ is independent of time t and depends only on the position $\vec{\boldsymbol{x}}$, expressed as $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}})$. This implies that the partial derivative of velocity in the spatial description with respect to time, also known as the *local acceleration** (see Section 3.21.4), is zero, $\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\partial t} = \vec{\boldsymbol{0}}$; in other words, for a steady motion, the velocity at any given point in space remains constant over time.

3.21.3 Material time derivative

Consider a scalar material field $H(\vec{\boldsymbol{X}}, t)$ and a scalar spatial field $h(\vec{\boldsymbol{x}}, t)$ of some physical scalar, like the temperature or density. In mechanics, it is required to study how H and h vary with time for a given particle that travels along a trajectory; in order to define the time derivatives of these scalar fields, we require to assume that they are derivable functions with respect to time t.

The *material time derivative**, also known as the *total derivative** or *substantial time derivative** is a way of measuring the instantaneous rate of change of a scalar field (such as density or temperature) as the particle moves along a trajectory over time. In other words, it tells us how the temperature (or density, etc.) is affected by the particles's movement. This derivative is customarily denoted as $\frac{\mathrm{D}}{\mathrm{D}t}$, and its definition will be explained in the next.

In the material description, the *material time derivative of a material field* is quantified as

$$\frac{\mathrm{D}H(\vec{\boldsymbol{X}}, t)}{\mathrm{D}t} \equiv \dot{H}(\vec{\boldsymbol{X}}, t) \equiv \frac{\partial H(\vec{\boldsymbol{X}}, t)}{\partial t} := \lim_{\Delta t \to 0} \frac{H(\vec{\boldsymbol{X}}, t + \Delta t) - H(\vec{\boldsymbol{X}}, t)}{\Delta t}.$$

where the vector $\vec{\boldsymbol{X}}$ is held constant. This derivative estimates the instantaneous rate of change of H for that particle that at time t_0 was at position $\vec{\boldsymbol{X}}$. Thus $\dot{H}(\vec{\boldsymbol{X}}, t^*)$ represents the rate at which the material field H changes with time at an specific time t^*, as seen by an observer following the trajectory of the movement of the particle. Velocity as defined in equations (3.81) and (3.82) is an example of the material time derivative of a material field.

While in the material description it is possible to locate the particle under analysis in the reference configuration Ω_0, oftentimes in the spatial configuration this information is not available; consider for example the point of view taken in fluid mechanics at time t^*: there it is unknown the trajectory $\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$ of the particle $\vec{\boldsymbol{X}} \in \Omega_0$ under study

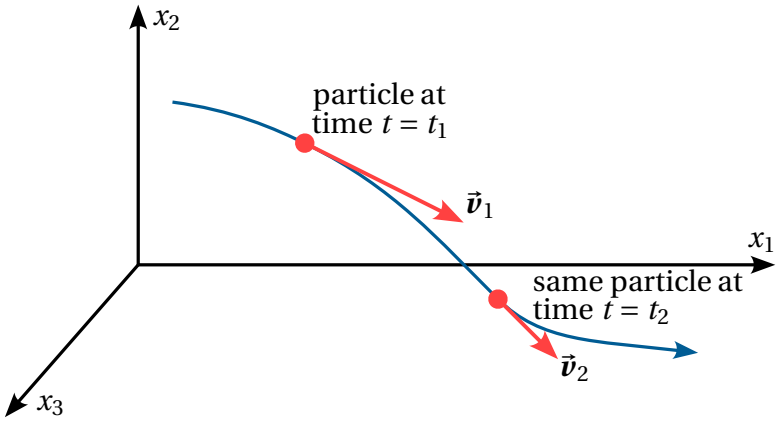


Figure 3.24: *Particle moving in along a trajectory. As the particle moves, its properties like temperature or density may change. Remember that the velocity vector $\vec{\boldsymbol{v}}$ is tangent to the trajectory at each point.*

at those times $t \in [0, t^*)$. Therefore, in the spatial description, a different formulation is required since the local time derivative $\frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial t}$ does not quantify appropriately the change of the scalar h with time given that here $\vec{\boldsymbol{x}}$ is assumed to be a fixed point in space that does not change with time t; nevertheless, for a particle that moves along a trajectory, $\vec{\boldsymbol{x}}$ changes continuously its position, that is $\vec{\boldsymbol{x}}$ changes with time t; therefore we require to calculate the total derivative $\frac{\mathrm{D}h(\vec{\boldsymbol{x}}(t),t)}{\mathrm{D}t}$, as motivated in the following.

Imagine a fluid with particle moving along a trajectory, like the one shown in the Figure 3.24. At each point $\vec{\boldsymbol{x}} := [x,\ y,\ z]^T$ in space, the velocity vector field $\vec{\boldsymbol{v}}$ and the spatial scalar field $h(\vec{\boldsymbol{x}}, t)$ are defined. Let us suppose that h represents, for instance, temperature.

At time t_1, the particle is at point $\vec{\boldsymbol{x}}_1$ and the temperature of the fluid is $h_1 = h(\vec{\boldsymbol{x}}_1, t_1)$. Later, at time t_2, the particle has moved to $\vec{\boldsymbol{x}}_2$ and the temperature of the fluid is $h_2 = h(\vec{\boldsymbol{x}}_2, t_2)$. We use a first-order Taylor series expansion (see Appendix A.14) to expand the scalar field h at point 1, so that

$$h(\vec{\boldsymbol{x}}_2, t_2) = h(\vec{\boldsymbol{x}}_1, t_1) + \frac{\partial h(\vec{\boldsymbol{x}}_1, t_1)}{\partial \vec{\boldsymbol{x}}} \cdot (\vec{\boldsymbol{x}}_2 - \vec{\boldsymbol{x}}_1) + \frac{\partial h(\vec{\boldsymbol{x}}_1, t_1)}{\partial t}(t_2 - t_1) + (\text{higher order terms}).$$

From this equation, we can estimate the average rate of change in temperature of the particle as it moves from position 1 to 2 by dividing $h_2 - h_1$ by $t_2 - t_1$ and ignoring higher order terms, so that:

$$\frac{h_2 - h_1}{t_2 - t_1} \approx \frac{\partial h(\vec{\boldsymbol{x}}_1, t_1)}{\partial t} + \frac{\partial h(\vec{\boldsymbol{x}}_1, t_1)}{\partial \vec{\boldsymbol{x}}} \cdot \frac{\vec{\boldsymbol{x}}_2 - \vec{\boldsymbol{x}}_1}{t_2 - t_1};$$

In the limit, as t_2 approaches t_1, this term becomes the material derivative: $\frac{\mathrm{D}h(\vec{\boldsymbol{x}}_1,t_1)}{\mathrm{D}t} := \lim_{t_2 \to t_1} \frac{h_2 - h_1}{t_2 - t_1}$; this represents the instantaneous rate at which the scalar field (temperature) changes as the fluid moves; here our attention is fixed on the particle's

temperature as it moves. In addition, $\frac{\vec{x}_2-\vec{x}_1}{t_2-t_1}$ becomes the flow velocity $\vec{v}(\vec{x}_1,t_1)=\frac{\mathrm{d}\vec{x}(t_1)}{\mathrm{d}t}$ at the fixed point $\vec{x}_1$. Note that $\frac{\mathrm{D}h(\vec{x}_1,t_1)}{\mathrm{D}t}$ is different from the local time derivative $\frac{\partial h(\vec{x}_1,t_1)}{\partial t}$ which measures the variation of temperature as time passes at the fixed point 1.

Thus, mathematically speaking, the *material time derivative of a spatial field* $h(\vec{x},t)$ is defined as:

$$\frac{\mathrm{D}h(\vec{x},t)}{\mathrm{D}t}=\frac{\partial h(\vec{\chi}(\vec{X},t),t)}{\partial t}:=\lim_{\Delta t\to 0}\frac{h(\vec{\chi}(\vec{X},t+\Delta t),t+\Delta t)-h(\vec{\chi}(\vec{X},t),t)}{\Delta t};$$

note that if h is defined as the spatial description of H, that is $h(\vec{x},t):=H\left(\vec{\chi}^{-1}(\vec{x},t),t\right)=H\left(\vec{X},t\right)$, then

$$\frac{\mathrm{D}h(\vec{x},t)}{\mathrm{D}t}=\frac{\mathrm{D}H(\vec{X},t)}{\mathrm{D}t}=\frac{\partial H(\vec{X},t)}{\partial t};$$

using the chain rule (see Appendix A.10), and taking into account that $\vec{x}=\vec{\chi}(\vec{X},t)$, these equations reduce to:

$$\frac{\mathrm{D}h(\vec{x},t)}{\mathrm{D}t}:=\underbrace{\frac{\partial h(\vec{x},t)}{\partial t}}_{\text{local time derivative of } h}+\underbrace{\overbrace{\frac{\partial h(\vec{x},t)}{\partial \vec{x}}}^{\operatorname{grad} h(\vec{x},t)}\cdot\overbrace{\frac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t}}^{\vec{v}(\vec{x},t)}}_{\text{convective derivative of } h}. \tag{3.84}$$

In this case $\frac{\mathrm{D}h(\vec{x},t)}{\mathrm{D}t}$ can now be determined, because the derivatives in the right-hand side can be either calculated or measured without knowing the motion.

Let us analyze each of the terms of the material time derivative $\frac{\mathrm{D}h(\vec{x},t)}{\mathrm{D}t}$ which is the time rate of change of h for a particle that is moving along with the flow with a trajectory or pathline described by $\vec{x}$; it is composed of two terms: a *local time derivative of* h which is the time rate of change of h at a fixed point and a *convective derivative** of h, that is the time rate of change due to the movement, with a velocity $\vec{v}$, of the fluid element from one location to the other, taking into account that the properties of h vary spatially. This last term appears for example when the particle moves through regions of spatially varying velocity, as in a nozzle or diffuser; in this case, the fluid speeds up as it flows along the decreasing cross-section path; the total derivative takes this phenomenon into account. The word *convection* is used physics to denote the transfer of heat from one place to another due to the movement of a fluid; a familiar example occurs in coastal regions when warm air masses over the ocean are driven inland toward mountains, carrying thermal energy with them. In short, while the local time derivative describes the intrinsic variation of the field, independent of the presence of any flow, the convective derivative describes the transport of the field in the flow.

Let us consider an example that will clarify the physical meaning of the material time derivative. Imagine that you are hiking in the mountains; if you stay under a tree for the whole afternoon, your velocity is null $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \vec{\mathbf{0}}$ and the material time derivative becomes equal to the local time derivative $\frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial t}$ which will measure the change of temperature under the tree as time advances; on the other hand, the convective derivative of h measures the change of temperature as you descend from the top of the mountain to the bottom of the valley assuming that the air temperature does not change during the whole afternoon.

In our previous discussion, we developed the concept of the material (or substantial) derivative by carefully tracking how physical quantities change as they move with fluid particles. While this approach provides valuable physical insight, we can establish the same result more directly by recognizing the connection to the total derivative from classical calculus.

Let us consider a scalar field $h(\vec{\boldsymbol{x}}, t) = h(x, y, z, t)$ that represents any continuum property (such as density, temperature, or pressure) defined at position $\vec{\boldsymbol{x}} = [x, y, z]^T$ and time t. When we follow this property along a material path $\vec{\boldsymbol{x}}(t) = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$, for $t \in T$, the total rate of change of h is given by the chain rule of differential calculus (see Appendix A.10):

$$\frac{\mathrm{d}h(x,y,z,t)}{\mathrm{d}t} = \frac{\partial h(x,y,z,t)}{\partial t} + \frac{\partial h(x,y,z,t)}{\partial x}\frac{\mathrm{d}x(t)}{\mathrm{d}t} + \frac{\partial h(x,y,z,t)}{\partial y}\frac{\mathrm{d}y(t)}{\mathrm{d}t} + \frac{\partial h(x,y,z,t)}{\partial z}\frac{\mathrm{d}z(t)}{\mathrm{d}t}.$$

Recall that as we follow a material particle, its velocity components are precisely the rates of change of its position coordinates, that is, $u(x, y, z, t) = \frac{\mathrm{d}x(t)}{\mathrm{d}t}$, $v(x, y, z, t) = \frac{\mathrm{d}y(t)}{\mathrm{d}t}$, and $w(x, y, z, t) = \frac{\mathrm{d}z(t)}{\mathrm{d}t}$. Substituting these velocity components, our expression becomes:

$$\frac{\mathrm{d}h(x,y,z,t)}{\mathrm{d}t} = \frac{\partial h(x,y,z,t)}{\partial t} + \underbrace{\begin{pmatrix} \frac{\partial h(x,y,z,t)}{\partial x} \\ \frac{\partial h(x,y,z,t)}{\partial y} \\ \frac{\partial h(x,y,z,t)}{\partial z} \end{pmatrix}}_{=\frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}} \cdot \underbrace{\begin{pmatrix} u(x,y,z,t) \\ v(x,y,z,t) \\ w(x,y,z,t) \end{pmatrix}}_{=\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}.$$

This equation can be written more compactly using vector notation:

$$\frac{\mathrm{d}h(\vec{\boldsymbol{x}},t)}{\mathrm{d}t} = \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial t} + \frac{\partial h(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}} \cdot \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) \tag{3.85}$$

Comparing equations (3.84) and (3.85), we confirm that the material derivative $\frac{\mathrm{D}h}{\mathrm{D}t}$ is mathematically equivalent to the total derivative $\frac{\mathrm{d}h}{\mathrm{d}t}$ (see Appendix A.10). While the formulation in equation (3.85) emphasizes mathematical formalism through calculus,

the approach leading to equation (3.84) offers deeper physical intuition about the material derivative's origins.

Now, let us assume that $h(\vec{x}, t)$ is a spatial vector field. In this case, the material derivative generalizes to:

$$\frac{\mathrm{D}\vec{h}(\vec{x}, t)}{\mathrm{D}t} := \frac{\partial \vec{h}(\vec{x}, t)}{\partial t} + \underbrace{\frac{\partial \vec{h}(\vec{x}, t)}{\partial \vec{x}}}_{\operatorname{grad} \vec{h}(\vec{x}, t)} \underbrace{\frac{\mathrm{d}\vec{x}(t)}{\mathrm{d}t}}_{\vec{v}(\vec{x}, t)}. \tag{3.86}$$

observe now that the term $\operatorname{grad} \vec{h}(\vec{x}, t)$ is in fact a Jacobian matrix.

In Section 3.3, we studied that in the material, also known as the Lagrangian description, the attention is on a single particle that moves along a trajectory, whereas in the spatial or Eulerian description attention is focused on a specific location in space. Note that the material time derivative allows the Eulerian framework to follow single particles.

Finally, a flow is said to be *uniform** if quantities are independent of position, and thus $\frac{\partial h(\vec{x},t)}{\partial \vec{x}}$ or $\frac{\partial \vec{h}(\vec{x},t)}{\partial \vec{x}}$ vanish for all $\vec{x} \in \Omega$. In this case, the material time derivative is equal to the local time derivative, i.e., $\frac{\mathrm{D}h(\vec{x},t)}{\mathrm{D}t} = \frac{\partial h(\vec{x},t)}{\partial t}$ and $\frac{\mathrm{D}\vec{h}(\vec{x},t)}{\mathrm{D}t} = \frac{\partial \vec{h}(\vec{x},t)}{\partial t}$. For instance, consider an ideal gas in a large, perfectly insulated container; assuming no external forces or temperature gradients, the gas molecules will distribute themselves evenly throughout the volume of the container; in this scenario, the material time derivative of any scalar or vector quantity associated with the gas flow, such as density, pressure, kinetic energy or acceleration, would be equal to the local time derivative; this is because there are no spatial gradients, and any changes in these quantities would be solely due to temporal variations, making the flow uniform.

3.21.4 Acceleration

The *acceleration field*, or simply *acceleration*,* describes the instantaneous rate of change of the velocity of a particle with respect to time. It is the material time derivative of the velocity. In the material description, acceleration at time t of a particle that was at $\vec{X}$ at time t_0 can be represented as:

$$\vec{A}(\vec{X}, t) := \frac{\mathrm{D}\vec{V}(\vec{X}, t)}{\mathrm{D}t} = \frac{\partial \vec{V}(\vec{X}, t)}{\partial t} = \frac{\partial^2 \vec{\chi}(\vec{X}, t)}{\partial t^2} = \frac{\partial^2 \vec{U}(\vec{X}, t)}{\partial t^2}.$$

In the spatial description, it is defined as

$$\vec{a}(\vec{x}, t) = \vec{A}\left(\vec{\chi}^{-1}(\vec{x}, t), t\right);$$

however, since $\vec{\boldsymbol{\chi}}^{-1}$ is not always explicitly available, we can use equation (3.86) to express the acceleration of a moving particle in terms of the derivatives of the velocity components:

$$\vec{\boldsymbol{a}}(\vec{\boldsymbol{x}},t) := \frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} = \underbrace{\frac{\partial\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial t}}_{\text{local acceleration}} + \underbrace{\frac{\partial\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial\vec{\boldsymbol{x}}}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}_{\text{convective acceleration}} \tag{3.87}$$

Here, the first term on the right-hand side, $\frac{\partial\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial t}$, represents the *local acceleration**, which is the local rate of change of the velocity field. The second term,

$$\frac{\partial\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial\vec{\boldsymbol{x}}}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) = \begin{pmatrix} \frac{\partial v_1}{\partial x_1} & \frac{\partial v_1}{\partial x_2} & \frac{\partial v_1}{\partial x_3} \\ \frac{\partial v_2}{\partial x_1} & \frac{\partial v_2}{\partial x_2} & \frac{\partial v_2}{\partial x_3} \\ \frac{\partial v_3}{\partial x_1} & \frac{\partial v_3}{\partial x_2} & \frac{\partial v_3}{\partial x_3} \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \\ v_3 \end{pmatrix}, \tag{3.88}$$

is the *convective acceleration**, which accounts for the rate of change of the velocity field due to the change in position. Here, $\frac{\partial\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial\vec{\boldsymbol{x}}}$ is a Jacobian matrix known as the *spatial velocity gradient**, which we will study in detail in Section 3.22.1. This equation is particularly useful in fluid mechanics, as it allows for calculating the acceleration $\vec{\boldsymbol{a}}(\vec{\boldsymbol{x}},t)$ without explicitly knowing the motion $\vec{\boldsymbol{\chi}}$ or the displacement $\vec{\boldsymbol{U}}$. In fact, according to equation (3.87), a small object being carried along by the fluid flow can change velocity for two reasons: first, because the velocity field at its position is changing over time, and second, because it moves to a new location where the velocity field has a different value.

Notice that $\vec{\boldsymbol{A}}(\vec{\boldsymbol{X}},t) = \vec{\boldsymbol{a}}(\vec{\boldsymbol{x}},t)$ even though they represent equal quantities that are functions of different arguments; also, the SI unit for acceleration is meter per squared second $\frac{\mathrm{m}}{\mathrm{s}^2}$.

It's worth mentioning that acceleration is sometimes represented, in the spatial description, and with some abuse of notation, as

$$\vec{\boldsymbol{a}}(t) = \dot{\vec{\boldsymbol{v}}}(t) = \ddot{\vec{\boldsymbol{x}}}(t) = \frac{\mathrm{d}\dot{\vec{\boldsymbol{x}}}(t)}{\mathrm{d}t} = \frac{\mathrm{d}^2\vec{\boldsymbol{x}}(t)}{\mathrm{d}t^2} := \vec{\boldsymbol{a}}(\vec{\boldsymbol{x}},t);$$

here $\vec{\boldsymbol{x}}$ is a function of $\vec{\boldsymbol{X}}$ and t through equation (3.1), that is $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)$; however, the position argument is avoided for the sake of compactness in the representation.

Example 3.14. *Acceleration of a body*

Find the acceleration in both the material and spatial descriptions, of the body that is analyzed in Example 3.13. Compare the acceleration in both descriptions.

Solution:

To determine the acceleration in both material and spatial descriptions, we build on the previous analysis of displacement and velocity. Let us proceed systematically.

The material acceleration $\vec{\boldsymbol{A}}(\vec{\boldsymbol{X}}, t)$ is the time derivative of the material velocity $\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)$, holding $\vec{\boldsymbol{X}}$ constant:

$$\vec{\boldsymbol{A}}(\vec{\boldsymbol{X}}, t) = \frac{\partial \vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)}{\partial t}.$$

From the material velocity:

$$\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} 2atX_1 \\ 0 \\ 0 \end{pmatrix},$$

we compute:

$$\vec{\boldsymbol{A}}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} \frac{\partial}{\partial t}(2atX_1) \\ 0 \\ 0 \end{pmatrix} = \begin{pmatrix} 2aX_1 \\ 0 \\ 0 \end{pmatrix}.$$

The spatial acceleration is obtained via the material derivative of the spatial velocity (3.87). From the spatial velocity:

$$\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \begin{pmatrix} \frac{2atx_1}{1+at^2} \\ 0 \\ 0 \end{pmatrix},$$

we compute the local acceleration $\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\partial t}$:

$$\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\partial t} = \begin{pmatrix} \frac{2ax_1(1-at^2)}{(1+at^2)^2} \\ 0 \\ 0 \end{pmatrix},$$

this results after applying the quotient rule to estimate the time derivative of v_1:

$$\frac{\partial v_1}{\partial t} = \frac{2ax_1(1+at^2) - 2atx_1(2at)}{(1+at^2)^2} = \frac{2ax_1(1-at^2)}{(1+at^2)^2}.$$

Now, the convective acceleration $\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$ is calculated as equation (3.88). Since $\vec{\boldsymbol{v}}$ only has an x_1-component and depends only on x_1:

$$\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\partial \vec{\boldsymbol{x}}} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \frac{\partial v_1}{\partial x_1} v_1 \hat{\boldsymbol{e}}_1.$$

Compute $\frac{\partial v_1}{\partial x_1}$:

$$\frac{\partial v_1}{\partial x_1} = \frac{2at}{1+at^2}.$$

Thus:

$$\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)=\frac{\partial v_1}{\partial x_1}v_1\hat{\boldsymbol{e}}_1=\left(\frac{2at}{1+at^2}\right)\left(\frac{2atx_1}{1+at^2}\right)\hat{\boldsymbol{e}}_1=\frac{4a^2t^2x_1}{(1+at^2)^2}\hat{\boldsymbol{e}}_1.$$

Summing the local and convective terms, we obtain the spatial acceleration:

$$\vec{\boldsymbol{a}}(\vec{\boldsymbol{x}},t)=\begin{pmatrix}\frac{2ax_1(1-at^2)}{(1+at^2)^2}+\frac{4a^2t^2x_1}{(1+at^2)^2}\\0\\0\end{pmatrix}=\begin{pmatrix}\frac{2ax_1}{1+at^2}\\0\\0\end{pmatrix}.$$

Note that the material acceleration $\vec{\boldsymbol{A}}(\vec{\boldsymbol{X}},t)=2aX_1\hat{\boldsymbol{e}}_1$ depends on the initial position X_1 and is constant for a given particle; now, the spatial acceleration $\vec{\boldsymbol{a}}(\vec{\boldsymbol{x}},t)=\frac{2ax_1}{1+at^2}\hat{\boldsymbol{e}}_1$ depends on the current position x_1 and time t. To reconcile the two descriptions, substitute $x_1=(1+at^2)X_1$ into the spatial acceleration:

$$\vec{\boldsymbol{a}}(\vec{\boldsymbol{x}},t)=\frac{2a(1+at^2)X_1}{1+at^2}=2aX_1\hat{\boldsymbol{e}}_1.$$

This matches the material acceleration, confirming consistency.

3.22 The rate of deformation and spin tensors

In many continuum mechanics situations, the most important kinematic attribute is not the change of shape of a body but the rate at which a body's form changes. This is especially true in fluid mechanics, where it is normally necessary to determine fluid flow in a specific region of space, and the form of the fluid body at a given moment is rarely significant.

Write a better introduction

3.22.1 Spatial and material velocity gradients

There are two types of velocity gradients: the spatial and the material.

The *spatial velocity gradient** is the second order tensor

$$\boldsymbol{l}(\vec{\boldsymbol{x}},t):=\operatorname{grad}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)=\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}=\begin{pmatrix}\frac{\partial v_1}{\partial x_1}&\frac{\partial v_1}{\partial x_2}&\frac{\partial v_1}{\partial x_3}\\\frac{\partial v_2}{\partial x_1}&\frac{\partial v_2}{\partial x_2}&\frac{\partial v_2}{\partial x_3}\\\frac{\partial v_3}{\partial x_1}&\frac{\partial v_3}{\partial x_2}&\frac{\partial v_3}{\partial x_3}\end{pmatrix};$$

it represents the rate of change of the velocity field with respect to spatial coordinates at a given point $\vec{\boldsymbol{x}}$ and time t. It quantifies how the velocity vector changes from

one spatial point to another, which is crucial in understanding phenomena like shear flows and vorticity in fluids. This Jacobian matrix plays a principal role in the definition of the convective acceleration, as seen in equation (3.88).

On the other hand, the *material velocity gradient** is the second order tensor

$$\begin{aligned}\text{Grad}\,\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t) = \frac{\partial \vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}} &= \begin{pmatrix} \frac{\partial V_1}{\partial X_1} & \frac{\partial V_1}{\partial X_2} & \frac{\partial V_1}{\partial X_3} \\ \frac{\partial V_2}{\partial X_1} & \frac{\partial V_2}{\partial X_2} & \frac{\partial V_2}{\partial X_3} \\ \frac{\partial V_3}{\partial X_1} & \frac{\partial V_3}{\partial X_2} & \frac{\partial V_3}{\partial X_3} \end{pmatrix} \\ &= \frac{\partial}{\partial \vec{\boldsymbol{X}}}\left(\frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial t}\right) = \frac{\partial}{\partial t}\left(\frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}\right) = \frac{\partial \boldsymbol{F}(\vec{\boldsymbol{X}},t)}{\partial t} = \dot{\boldsymbol{F}}(\vec{\boldsymbol{X}},t);\end{aligned}$$

that describes the rate of change of the velocity field with respect to material coordinates at a given point $\vec{\boldsymbol{X}}$ and time t. Since it is the material time derivative of the deformation gradient $\boldsymbol{F}$, it captures how the deformation gradient $\boldsymbol{F}$ evolves over time for a given material point $\vec{\boldsymbol{X}}$.

The relationship between both velocity gradients is shown in the following. Using the chain rule (see Appendix A.10), and the fact that $\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$, we can proceed in the following way

$$\begin{aligned}\boldsymbol{l}(\vec{\boldsymbol{x}},t) &= \frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}} \\ &= \frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{X}}} \underbrace{\frac{\partial \vec{\boldsymbol{X}}}{\partial \vec{\boldsymbol{x}}}}_{\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t)} && \text{using the chain rule} \\ &= \frac{\partial}{\partial \vec{\boldsymbol{X}}}\left[\frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial t}\right]\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t) && \text{since } \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) = \vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t) = \frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial t} \\ &= \frac{\partial}{\partial t}\left[\frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}\right]\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t) \\ &= \frac{\partial}{\partial t}\left[\boldsymbol{F}(\vec{\boldsymbol{X}},t)\right]\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t) && \text{using equation (3.26)} \\ &= \dot{\boldsymbol{F}}(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t)\end{aligned}$$

to show that the spatial velocity gradient is given by

$$\boxed{\boldsymbol{l}(\vec{\boldsymbol{x}},t) = \dot{\boldsymbol{F}}(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t);}$$

and the material velocity gradient by

$$\boxed{\dot{\boldsymbol{F}}(\vec{\boldsymbol{X}},t) = \boldsymbol{l}(\vec{\boldsymbol{x}},t)\boldsymbol{F}(\vec{\boldsymbol{X}},t).} \tag{3.89}$$

It is important to note that for the material and spatial velocity gradient tensors to be well-defined, the motion function $\vec{\chi}$ must be continuous and differentiable with respect to time, as demonstrated in the preceding derivations. The continuity requirement implies that the motion of material particles must be a continuous process over time, without any abrupt jumps or discontinuities. Furthermore, the differentiability condition ensures that the velocity field, which is the time derivative of the motion function, remains well-defined and smooth throughout the deformation process.

The velocity gradient is physically interpreted in Figure 3.25. Observe that in rigid body motion $\boldsymbol{F} = \boldsymbol{R}$ (see Section 3.10.2); therefore $\dot{\boldsymbol{F}} = \dot{\boldsymbol{R}}$ and $\boldsymbol{l} = \dot{\boldsymbol{F}}\boldsymbol{F}^{-1} = \dot{\boldsymbol{R}}\boldsymbol{R}^{T}$.

The displacement gradient in the material description $\operatorname{Grad}\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) = \frac{\partial \vec{\boldsymbol{U}}(\vec{\boldsymbol{X}},t)}{\partial \vec{\boldsymbol{X}}}$ is of special interest in elasticity theory, since it can be decomposed into the deformation $\underline{\underline{\boldsymbol{\varepsilon}}}$ and rotation $\boldsymbol{\Omega}$ tensors, so that $\operatorname{Grad}\vec{\boldsymbol{U}}(\vec{\boldsymbol{X}}, t) = \underline{\underline{\boldsymbol{\varepsilon}}} + \boldsymbol{\Omega}$, as stated in equation (3.69). Similarly, in fluid mechanics, spatial velocity gradient $\boldsymbol{l}$ can be additively decomposed into a symmetric $\boldsymbol{d}$ (i.e., $\boldsymbol{d} = \boldsymbol{d}^{T}$) and a skew-symmetric $\boldsymbol{w}$ (i.e., $\boldsymbol{w} = -\boldsymbol{w}^{T}$) tensor:

$$\boldsymbol{l} = \underbrace{\frac{\boldsymbol{l} + \boldsymbol{l}^{T}}{2}}_{\boldsymbol{d}} + \underbrace{\frac{\boldsymbol{l} - \boldsymbol{l}^{T}}{2}}_{\boldsymbol{w}}.$$

The symmetric part of the spatial velocity gradient $\boldsymbol{l}$ is known as the *rate of deformation tensor** or *rate of strain tensor** $\boldsymbol{d}$ and the skew-symmetric part is the *spin tensor** $\boldsymbol{w}$; they are going to be studied in the next.

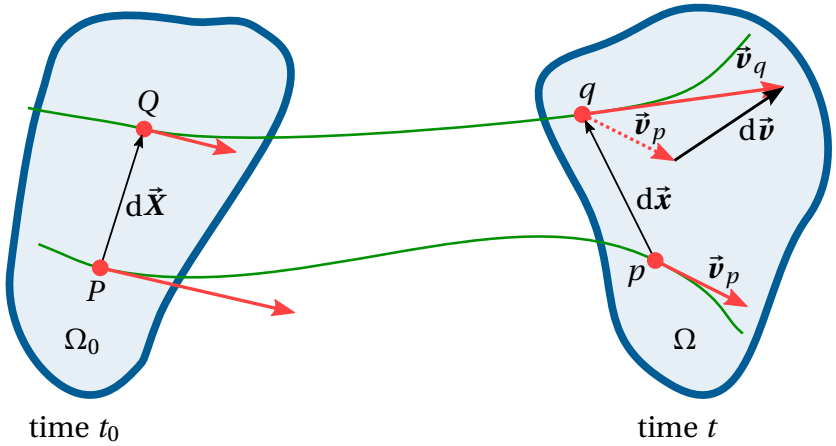


Figure 3.25: *The velocity gradient tensor $\boldsymbol{l}$ relates the velocity $\mathrm{d}\vec{\boldsymbol{v}}$ of the particle at q with respect to the one of the particle at p, through the expression $\mathrm{d}\vec{\boldsymbol{v}} = \boldsymbol{l}\left(\vec{\boldsymbol{x}}_p, t\right)\mathrm{d}\vec{\boldsymbol{x}}$; in this way, $\vec{\boldsymbol{v}}_q = \vec{\boldsymbol{v}}_p + \mathrm{d}\vec{\boldsymbol{v}}$. This tensor captures the spatial variation of the velocity field and is fundamental in understanding the deformation and flow behavior of continua. Remember that the velocity vector $\vec{\boldsymbol{v}}$ is tangent to the trajectory at each point.*

3.22.2 The rate of strain tensor

One question that we may have is which is the rate of change with time of the inner product between the spatial element vectors $d\vec{x}_1$ and $d\vec{x}_2$; this is an indirect measure of how fast the angle between $d\vec{x}_1$ and $d\vec{x}_2$ changes over time. We will see, in the following, that the rate of strain tensor plays an major role in this calculation.

Using equation (3.41)

$$d\vec{x}_1 \cdot d\vec{x}_2 = d\vec{X}_1^T \boldsymbol{C}(\vec{X}, t)\, d\vec{X}_2$$

and taking its material time derivative, it results

$$\frac{D}{Dt}(d\vec{x}_1 \cdot d\vec{x}_2) = d\vec{X}_1^T \frac{\partial \boldsymbol{C}(\vec{X}, t)}{\partial t} d\vec{X}_2$$

where

$$\dot{\boldsymbol{C}}(\vec{X}, t) = \frac{\partial \boldsymbol{C}(\vec{X}, t)}{\partial t} = \frac{\partial}{\partial t}\left[\boldsymbol{F}^T(\vec{X}, t)\boldsymbol{F}(\vec{X}, t)\right] = \boldsymbol{F}^T(\vec{X}, t)\dot{\boldsymbol{F}}(\vec{X}, t) + \dot{\boldsymbol{F}}^T(\vec{X}, t)\boldsymbol{F}(\vec{X}, t).$$

From equation (3.52) it follows that $\boldsymbol{C}(\vec{X}, t) = 2\boldsymbol{E}(\vec{X}, t) + \boldsymbol{I}$, therefore

$$\dot{\boldsymbol{C}}(\vec{X}, t) = 2\frac{\partial \boldsymbol{E}(\vec{X}, t)}{\partial t} = 2\dot{\boldsymbol{E}}(\vec{X}, t)$$

that is,

$$\dot{\boldsymbol{E}}(\vec{X}, t) = \frac{\partial \boldsymbol{E}(\vec{X}, t)}{\partial t} = \frac{\boldsymbol{F}^T(\vec{X}, t)\dot{\boldsymbol{F}}(\vec{X}, t) + \dot{\boldsymbol{F}}^T(\vec{X}, t)\boldsymbol{F}(\vec{X}, t)}{2}. \tag{3.90}$$

We will call $\dot{\boldsymbol{E}}$ the *material strain rate tensor*.*

In conclusion,

$$\frac{D}{Dt}(d\vec{x}_1 \cdot d\vec{x}_2) = d\vec{X}_1^T \dot{\boldsymbol{C}}(\vec{X}, t)\, d\vec{X}_2 = 2\, d\vec{X}_1^T \dot{\boldsymbol{E}}(\vec{X}, t)\, d\vec{X}_2.$$

This same result could have been achieve by taking the material time derivative of equation (3.56); that is, from equation:

$$\frac{d\vec{x}_1 \cdot d\vec{x}_2 - d\vec{X}_1 \cdot d\vec{X}_2}{2} = d\vec{X}_1^T \boldsymbol{E}(\vec{X}, t)\, d\vec{X}_2, = d\vec{x}_1^T \boldsymbol{e}\, d\vec{x}_2$$

it results that:

$$\frac{D}{Dt}\left(d\vec{x}_1 \cdot d\vec{x}_2 - d\vec{X}_1 \cdot d\vec{X}_2\right) = \frac{D}{Dt}\left[d\vec{x}_1 \cdot d\vec{x}_2\right] = 2\, d\vec{X}_1^T \dot{\boldsymbol{E}}(\vec{X}, t)\, d\vec{X}_2;$$

this last result emphasizes that we measuring how the inner product between any pair of line elements $d\vec{x}_1$ and $d\vec{x}_2$ changes over time, taking as a reference the inner product of the same line elements, but in the undeformed configuration.

Using the fact that $\mathrm{d}\vec{\boldsymbol{X}}_1 = \boldsymbol{F}^{-1}\,\mathrm{d}\vec{\boldsymbol{x}}_1$ and $\mathrm{d}\vec{\boldsymbol{X}}_2 = \boldsymbol{F}^{-1}\,\mathrm{d}\vec{\boldsymbol{x}}_2$, we can transform last equation to:

$$\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}\vec{\boldsymbol{x}}_1 \cdot \mathrm{d}\vec{\boldsymbol{x}}_2) = 2\,\mathrm{d}\vec{\boldsymbol{X}}_1^T \dot{\boldsymbol{E}}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\vec{\boldsymbol{X}}_2 = 2\,\mathrm{d}\vec{\boldsymbol{x}}_1^T \underbrace{\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\dot{\boldsymbol{E}}(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}}, t)}_{\boldsymbol{d}(\vec{\boldsymbol{x}}, t)}\,\mathrm{d}\vec{\boldsymbol{x}}_2 \tag{3.91}$$

where

$$\boxed{\boldsymbol{d}(\vec{\boldsymbol{x}}, t) = \boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\dot{\boldsymbol{E}}(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}}, t)}$$

is the so-called *rate of deformation tensor*,* also known as the *rate of strain* of the *strain-rate tensor*; it is the pushed-forward spatial version of $\dot{\boldsymbol{E}}$ (i.e., $\boldsymbol{d} = \phi_*[\dot{\boldsymbol{E}}]$), while $\dot{\boldsymbol{E}}$ is the pushed-backward material counterpart of $\boldsymbol{d}$ (i.e., $\dot{\boldsymbol{E}} = \phi_*^{-1}[\boldsymbol{d}]$):

$$\boxed{\dot{\boldsymbol{E}}(\vec{\boldsymbol{X}}, t) = \boldsymbol{F}(\vec{\boldsymbol{X}}, t)^T \boldsymbol{d}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}(\vec{\boldsymbol{X}}, t).} \tag{3.92}$$

Note that in rigid body motion, $\boldsymbol{E} = \boldsymbol{0}$, therefore $\dot{\boldsymbol{E}}$ is null, implying that $\dot{\boldsymbol{E}} = \boldsymbol{d} = \boldsymbol{0}$.

In conclusion,

$$\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}\vec{\boldsymbol{x}}_1 \cdot \mathrm{d}\vec{\boldsymbol{x}}_2) = 2\,\mathrm{d}\vec{\boldsymbol{X}}_1^T \dot{\boldsymbol{E}}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\vec{\boldsymbol{X}}_2 = 2\,\mathrm{d}\vec{\boldsymbol{x}}_1^T \boldsymbol{d}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\vec{\boldsymbol{x}}_2$$

shows the current rate of change of the scalar product between the spatial line elements can be estimated either using the material strain rate tensor $\dot{\boldsymbol{E}}$ in terms of the undeformed material line elements or employing the rate of deformation tensor $\boldsymbol{d}$ in terms of the deformed spatial line elements.

If we replace (3.89) into (3.90), it follows that:

$$\begin{aligned}
\dot{\boldsymbol{E}}(\vec{\boldsymbol{X}}, t) &= \frac{\boldsymbol{F}(\vec{\boldsymbol{X}}, t)^T \dot{\boldsymbol{F}}(\vec{\boldsymbol{X}}, t) + \dot{\boldsymbol{F}}(\vec{\boldsymbol{X}}, t)^T \boldsymbol{F}(\vec{\boldsymbol{X}}, t)}{2} \\
&= \frac{\boldsymbol{F}(\vec{\boldsymbol{X}}, t)^T \boldsymbol{l}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}(\vec{\boldsymbol{X}}, t) + \boldsymbol{F}(\vec{\boldsymbol{X}}, t)^T \boldsymbol{l}(\vec{\boldsymbol{x}}, t)^T \boldsymbol{F}(\vec{\boldsymbol{X}}, t)}{2} \\
&= \boldsymbol{F}(\vec{\boldsymbol{X}}, t)^T \frac{\boldsymbol{l}(\vec{\boldsymbol{x}}, t) + \boldsymbol{l}(\vec{\boldsymbol{x}}, t)^T}{2} \boldsymbol{F}(\vec{\boldsymbol{X}}, t)
\end{aligned} \tag{3.93}$$

by comparing (3.92) and (3.93), it follows that the rate of deformation tensor $\boldsymbol{d}$ is the symmetric part of the spatial velocity gradient $\boldsymbol{l}$:

$$\boxed{\boldsymbol{d}(\vec{\boldsymbol{x}}, t) = \frac{\boldsymbol{l}(\vec{\boldsymbol{x}}, t) + \boldsymbol{l}(\vec{\boldsymbol{x}}, t)^T}{2};}$$

that is, it is the symmetric tensor:

$$\boldsymbol{d} = \frac{\boldsymbol{l} + \boldsymbol{l}^T}{2} = \begin{pmatrix} \frac{\partial v_1}{\partial x_1} & \frac{1}{2}\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1}\right) & \frac{1}{2}\left(\frac{\partial v_1}{\partial x_3} + \frac{\partial v_3}{\partial x_1}\right) \\ \frac{1}{2}\left(\frac{\partial v_2}{\partial x_1} + \frac{\partial v_1}{\partial x_2}\right) & \frac{\partial v_2}{\partial x_2} & \frac{1}{2}\left(\frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_2}\right) \\ \frac{1}{2}\left(\frac{\partial v_3}{\partial x_1} + \frac{\partial v_1}{\partial x_3}\right) & \frac{1}{2}\left(\frac{\partial v_3}{\partial x_2} + \frac{\partial v_2}{\partial x_3}\right) & \frac{\partial v_3}{\partial x_3} \end{pmatrix};$$

observe that the right-hand side of this equation involves only derivatives in the current configuration; in addition, the components of $\boldsymbol{d}(\vec{\boldsymbol{x}}, t)$ are linear in the velocity components, which is a favorable situation that simplifies solving problems in fluid mechanics.

The rate-of-deformation tensor $\boldsymbol{d}(\vec{\boldsymbol{x}}, t)$ has properties analogous to the infinitesimal strain tensor $\underline{\underline{\boldsymbol{\varepsilon}}}$, such as being a second-order symmetric tensor with eigenvectors and eigenvalues representing directions and values of the largest and smallest extension rates, having a characteristic polynomial with its invariants, etc. However, $\boldsymbol{d}(\vec{\boldsymbol{x}}, t)$ differs from $\underline{\underline{\boldsymbol{\varepsilon}}}$ in that it is an exact measure of deformation rate, whereas $\underline{\underline{\boldsymbol{\varepsilon}}}$ is an approximation, as discussed in Section 3.15.1.

To understand the physical meaning of the components of the rate-of-deformation tensor, we begin with equation (3.91):

$$\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}\vec{\boldsymbol{x}}_1 \cdot \mathrm{d}\vec{\boldsymbol{x}}_2) = 2\,\mathrm{d}\vec{\boldsymbol{x}}_1^T \boldsymbol{d}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\vec{\boldsymbol{x}}_2 \tag{3.94}$$

which is the rate of change with time of the inner product between the spatial element vectors $\mathrm{d}\vec{\boldsymbol{x}}_1$ and $\mathrm{d}\vec{\boldsymbol{x}}_2$; this is an indirect measure of how fast the angle between $\mathrm{d}\vec{\boldsymbol{x}}_1$ and $\mathrm{d}\vec{\boldsymbol{x}}_2$ changes over time.

Using equation (3.94) with $\mathrm{d}\vec{\boldsymbol{x}}_1 = \mathrm{d}\vec{\boldsymbol{x}}_2 = \mathrm{d}s\hat{\boldsymbol{n}}$, it follows that:

$$\frac{1}{2}\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}\vec{\boldsymbol{x}} \cdot \mathrm{d}\vec{\boldsymbol{x}}) = \frac{1}{2}\frac{\mathrm{D}}{\mathrm{D}t}\left(\mathrm{d}s^2\right) = \mathrm{d}s\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}s) = \mathrm{d}s^2\,\hat{\boldsymbol{n}}^T \boldsymbol{d}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}};$$

and thus, using the fact that $\frac{\mathrm{d}}{\mathrm{d}x}\ln f(x) = \frac{1}{f(x)}\frac{\mathrm{d}f(x)}{\mathrm{d}x}$,

$$\hat{\boldsymbol{n}}^T \boldsymbol{d}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}} = \frac{1}{\mathrm{d}s}\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}s) = \frac{\mathrm{D}}{\mathrm{D}t}(\ln \mathrm{d}s);$$

this equation is useful to measure *rate of extension* (rate of change of length per unit length or rate of stretching/contraction) of a spatial line element that is in direction $\hat{\boldsymbol{n}}$ after deformation, as shown in Figure 3.26. So, in particular d_{11}, d_{22} and d_{33} measure the rate of extension for those line elements that after deformation are in the $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$ and $\hat{\boldsymbol{k}}$ directions, respectively.

Finally, observe that the trace of $\boldsymbol{d}$ is the divergence of the velocity field $\vec{\boldsymbol{v}}$, i.e., $\operatorname{div}\vec{\boldsymbol{v}} = d_{11} + d_{22} + d_{33}$.

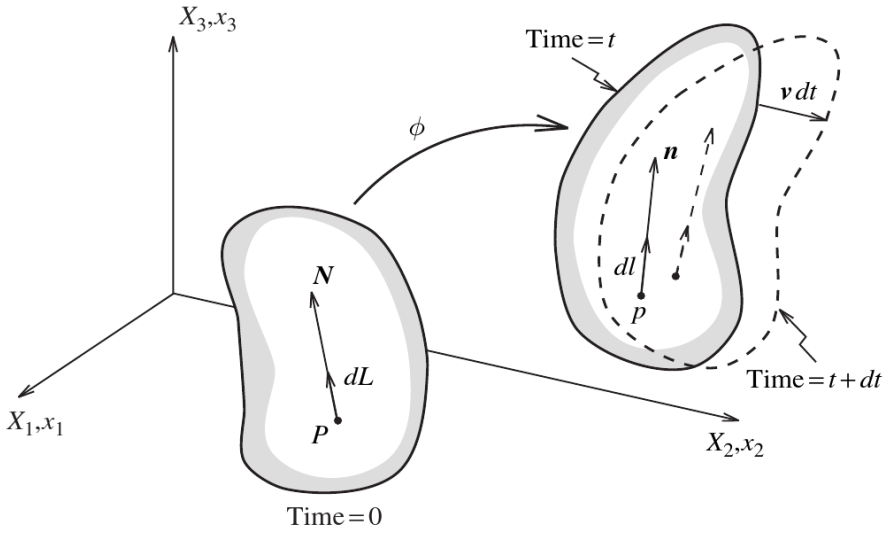
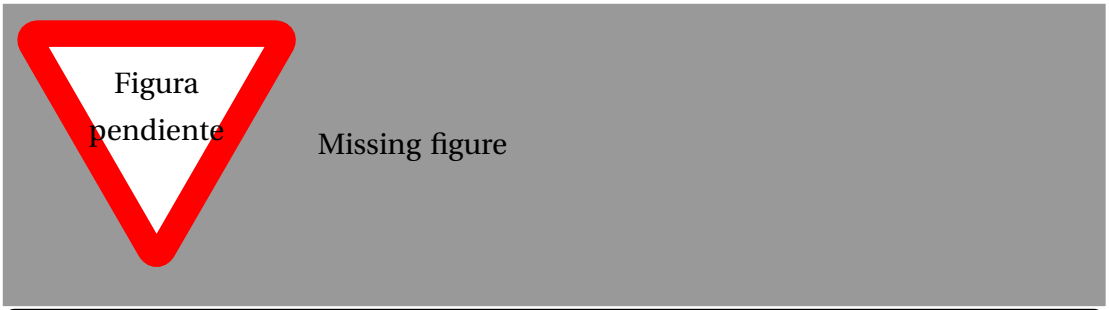


Figure 3.26: *Rate of extension.*



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Figure 3.27: *Rotating disk.*

3.22.3 The role of the curl

The *curl** operator, denoted by curl $\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}})$ or $\nabla \times \vec{\boldsymbol{f}}(\vec{\boldsymbol{x}})$, is a vector operation that describes the infinitesimal rotation or circulation of a vector field $\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}})$ around a given point. It is defined as the following vector operation:

$$\operatorname{curl} \vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}) = \nabla \times \vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}) = \begin{pmatrix} \frac{\partial f_3}{\partial x_2} - \frac{\partial f_2}{\partial x_3} \\ \frac{\partial f_1}{\partial x_3} - \frac{\partial f_3}{\partial x_1} \\ \frac{\partial f_2}{\partial x_1} - \frac{\partial f_1}{\partial x_2} \end{pmatrix}, \tag{3.95}$$

where $\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}})$ is a vector field, and x_1, x_2 and x_3 represent the spatial coordinates.

Consider a disk rotating about the vector $\hat{\boldsymbol{k}}$ at a constant angular speed ω, like the one shown in Figure 3.27. This rigid-body rotation can be described by the equation (3.96),

setting the constants C_x and C_y to zero since there is no rigid-body translation and taking into account that $\theta(t) = \omega t$ represents the angular displacement at time t:

$$\begin{pmatrix} x \\ y \\ z \end{pmatrix} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) = \begin{pmatrix} (X - X_0)\cos(\omega t) - (Y - Y_0)\sin\theta \\ (X - X_0)\sin(\omega t) + (Y - Y_0)\cos\theta \\ 0 \end{pmatrix}; \tag{3.96}$$

here the constants X_0 and Y_0 define the disk's center of rotation.

Using equation (3.81), we can calculate the velocity of each particle of the disk as:

$$\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) = \frac{\partial \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)}{\partial t} = \begin{pmatrix} -\omega\left((X - X_0)\sin(\omega t) + (Y - Y_0)\cos\theta\right) \\ \omega\left((X - X_0)\cos(\omega t) - (Y - Y_0)\sin\theta\right) \\ 0 \end{pmatrix};$$

substituting the expressions for x and y from equation (3.96), we can express the velocity in spatial coordinates as:

$$\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \begin{pmatrix} -\omega y \\ \omega x \\ 0 \end{pmatrix}.$$

Using the following Python program:

```
from sympy import symbols, diff, Matrix

# define some variables
omega   = symbols('omega')     # rotational speed
x, y, z = symbols('x, y, z')   # spatial coordinates

# spatial curl
curl = lambda f : Matrix([ diff(f[2], y) - diff(f[1], z),
                           diff(f[0], z) - diff(f[2], x),
                           diff(f[1], x) - diff(f[0], y)  ])

# spatial velocity vector
v = Matrix([ -omega*y,
              omega*x,
                    0  ])

# calculate the spatial curl of the velocity field
curl_v = curl(v)
```

it is possible to show the curl of the velocity field is:

$$\operatorname{curl}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \begin{pmatrix} 0 \\ 0 \\ 2\omega \end{pmatrix}.$$

Here, curl $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$ gives the axis of rotation, which is aligned with vector $\hat{\boldsymbol{k}}$ for the given rigid-body rotation. The magnitude of the curl, $\left\| \operatorname{curl} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \right\| = 2\omega$, is directly related to the angular speed; specifically, $\frac{1}{2}\left\| \operatorname{curl} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \right\| = \omega$ is the corresponding angular speed of the rotating disk.

In conclusion, the curl characterizes the local rotation or vorticity of the flow field. In the case of a spinning disk, the curl captures motion around the rotation axis, and its magnitude is proportional to the angular speed of rotation.

3.22.4 The spin tensor

The *spin tensor**, *rate of rotation tensor** or *vorticity tensor** is defined as:

$$\boldsymbol{w} = \frac{\boldsymbol{l} - \boldsymbol{l}^T}{2} = \begin{pmatrix} 0 & \frac{1}{2}\left(\frac{\partial v_1}{\partial x_2} - \frac{\partial v_2}{\partial x_1}\right) & \frac{1}{2}\left(\frac{\partial v_1}{\partial x_3} - \frac{\partial v_3}{\partial x_1}\right) \\ \frac{1}{2}\left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2}\right) & 0 & \frac{1}{2}\left(\frac{\partial v_2}{\partial x_3} - \frac{\partial v_3}{\partial x_2}\right) \\ \frac{1}{2}\left(\frac{\partial v_3}{\partial x_1} - \frac{\partial v_1}{\partial x_3}\right) & \frac{1}{2}\left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3}\right) & 0 \end{pmatrix}.$$

Since this is a skew-symmetric tensor, $\boldsymbol{w} = -\boldsymbol{w}^T$; also, there is an associated dual or axial vector $\vec{\boldsymbol{w}}$, the *vorticity vector** (also known as the *angular velocity vector*)

$$\vec{\boldsymbol{w}} = \begin{pmatrix} \frac{1}{2}\left(\frac{\partial v_3}{\partial x_2} - \frac{\partial v_2}{\partial x_3}\right) \\ \frac{1}{2}\left(\frac{\partial v_1}{\partial x_3} - \frac{\partial v_3}{\partial x_1}\right) \\ \frac{1}{2}\left(\frac{\partial v_2}{\partial x_1} - \frac{\partial v_1}{\partial x_2}\right) \end{pmatrix}$$

in the sense that for any vector $\vec{\boldsymbol{a}}$, $\boldsymbol{w}\vec{\boldsymbol{a}} = \vec{\boldsymbol{w}} \times \vec{\boldsymbol{a}}$; using equation (3.95), it follows that $\vec{\boldsymbol{w}} = \frac{1}{2}\operatorname{curl} \vec{\boldsymbol{v}}$; in other words, the vorticity vector $\vec{\boldsymbol{w}}$ is equal to half the curl of the velocity field $\vec{\boldsymbol{v}}$.

In consequence, the vorticity vector $\vec{\boldsymbol{w}}$ describes the rotational characteristics of a flow field. It defines an axis of rotation around which the flow rotates with an angular velocity equal to $\frac{1}{2}\|\vec{\boldsymbol{w}}\|$, measured in radians per second.

This relationship between the vorticity vector and the curl of the velocity field is particularly useful in analyzing and visualizing the rotational behavior of fluid flows. It allows for the identification of regions with high vorticity, which may indicate the presence of vortices or other rotational structures within the flow.

$\boldsymbol{w}$ is a measure of the rate of rotation of an element.

the fact that the components of $\boldsymbol{w}(\vec{\boldsymbol{x}}, t)$ are linear in the velocity components is a favorable situation that simplifies solving problems in fluid mechanics.

The spin tensor $\boldsymbol{w}$ has properties similar to the infinitesimal rotation $\boldsymbol{\Omega}$ tensor of elasticity theory, except that no approximation is involved in its derivation.

FALTA:

Using the property that for a skew-symmetric matrix $\boldsymbol{A}$, we have that $\vec{\boldsymbol{x}}^T \boldsymbol{A} \vec{\boldsymbol{x}} = 0$ (DEMUESTRE) for any $\vec{\boldsymbol{x}}$, then In conclusion,

$$\begin{aligned}\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}\vec{\boldsymbol{x}}_1 \cdot \mathrm{d}\vec{\boldsymbol{x}}_1) &= 2\,\mathrm{d}\vec{\boldsymbol{X}}_1^T \dot{\boldsymbol{E}}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\vec{\boldsymbol{X}}_1 \\ &= 2\,\mathrm{d}\vec{\boldsymbol{x}}_1^T \boldsymbol{d}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\vec{\boldsymbol{x}}_1 \\ &= 2\,\mathrm{d}\vec{\boldsymbol{x}}_1^T (\boldsymbol{d}(\vec{\boldsymbol{x}}, t) + \boldsymbol{w}(\vec{\boldsymbol{x}}, t))\,\mathrm{d}\vec{\boldsymbol{x}}_1 \\ &= 2\,\mathrm{d}\vec{\boldsymbol{x}}_1^T \boldsymbol{l}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\vec{\boldsymbol{x}}_1.\end{aligned}$$

this means that the spin tensor does not have to do with ...

and $\boldsymbol{d}$ is a suitable measure of the deformation rate.

3.22.5 Compatibility conditions for the components of the rate of deformation tensor

In this section, we will see that the components of $\boldsymbol{d}(\vec{\boldsymbol{x}}, t)$ obey compatibility relations analogous to the compatibility equations of elasticity theory (Section 2.1.1) satisfied by $\underline{\underline{\boldsymbol{\varepsilon}}}$, except that differentiation must be with respect to spatial coordinates $\vec{\boldsymbol{x}}$ instead of material coordinates $\vec{\boldsymbol{X}}$.

When three derivable velocity functions v_1, v_2 and v_3 are given, one can always determine the six rates of deformation components d_{11}, d_{22}, d_{33}, d_{12}, d_{13}, d_{23} of the rate of deformation tensor $\boldsymbol{d}$. However, if these six components are arbitrarily prescribed in a region, there may not exist three velocity functions v_1, v_2, and v_3 that satisfy the following six equations defining the rate of deformation-velocity relationships:

$$d_{11} = \frac{\partial v_1}{\partial x_1} \qquad d_{22} = \frac{\partial v_2}{\partial x_2} \qquad d_{33} = \frac{\partial v_3}{\partial x_3}$$

$$d_{12} = \frac{1}{2}\left(\frac{\partial v_1}{\partial x_2} + \frac{\partial v_2}{\partial x_1}\right) \qquad d_{13} = \frac{1}{2}\left(\frac{\partial v_1}{\partial x_3} + \frac{\partial v_3}{\partial x_1}\right) \qquad d_{23} = \frac{1}{2}\left(\frac{\partial v_2}{\partial x_3} + \frac{\partial v_3}{\partial x_2}\right)$$

The compatibility conditions for the rate of deformation components are analogous to the Saint-Venant compatibility conditions (equation [2.2]) for the infinitesimal strain components ε_x, ε_y, ..., ε_{yz}:

$$\begin{aligned}
&\frac{\partial^2 d_{11}}{\partial y^2}+\frac{\partial^2 d_{22}}{\partial x^2}=\frac{\partial^2 (2d_{12})}{\partial x \partial y} \qquad 2\frac{\partial^2 d_{11}}{\partial y \partial z}=\frac{\partial}{\partial x}\left(-\frac{\partial (2d_{23})}{\partial x}+\frac{\partial (2d_{13})}{\partial y}+\frac{\partial (2d_{12})}{\partial z}\right)\\
&\frac{\partial^2 d_{22}}{\partial z^2}+\frac{\partial^2 d_{33}}{\partial y^2}=\frac{\partial^2 (2d_{23})}{\partial y \partial z} \qquad 2\frac{\partial^2 d_{22}}{\partial x \partial z}=\frac{\partial}{\partial y}\left(+\frac{\partial (2d_{23})}{\partial x}-\frac{\partial (2d_{13})}{\partial y}+\frac{\partial (2d_{12})}{\partial z}\right)\\
&\frac{\partial^2 d_{33}}{\partial x^2}+\frac{\partial^2 d_{11}}{\partial z^2}=\frac{\partial^2 (2d_{13})}{\partial x \partial z} \qquad 2\frac{\partial^2 d_{33}}{\partial x \partial y}=\frac{\partial}{\partial z}\left(+\frac{\partial (2d_{23})}{\partial x}+\frac{\partial (2d_{13})}{\partial y}-\frac{\partial (2d_{12})}{\partial z}\right);
\end{aligned}$$

inasmuch as they impose constraints on the spatial derivatives of the rate of deformation components to ensure the existence of a continuous and differentiable velocity field that generates those deformation rates.

These compatibility conditions ensure that the prescribed rate of deformation components are consistent with the existence of a continuous velocity field. However, in practice, we often deal directly with differentiable velocity functions v_1, v_2, and v_3, and therefore it is not necessary to verify that those compatibility equations are satisfied. These conditions become relevant when the rate of deformation components are prescribed or obtained from experimental data, rather than derived from known velocity functions. In such cases, the compatibility conditions must be satisfied to guarantee the existence of a kinematically admissible velocity field.

3.22.6 The time derivative of the Jacobian determinant J

As studied in Section *** the Jacobian determinant $J = \det \boldsymbol{F}$ represents the local volume change during deformation. In the following, we will calculate its time derivative, a result that will be employed in subsequent demonstrations.

According to Jacobi's formula[10], for any invertible matrix $\boldsymbol{F}$:

$$\frac{\partial}{\partial t}\det \boldsymbol{F} = \det \boldsymbol{F}\,\mathrm{tr}\left(\boldsymbol{F}^{-1}\frac{\partial \boldsymbol{F}}{\partial t}\right)$$

Also, according to equation (3.89), $\dot{\boldsymbol{F}}(\vec{\boldsymbol{X}},t) = \boldsymbol{l}(\vec{\boldsymbol{x}},t)\boldsymbol{F}(\vec{\boldsymbol{X}},t)$, where $\boldsymbol{l}(\vec{\boldsymbol{x}},t) = \frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}$ is the spatial velocity gradient; substituting theses equations into Jacobi's formula, we obtain:

$$\frac{\partial}{\partial t}\det \boldsymbol{F} = \det \boldsymbol{F}\,\mathrm{tr}\left(\boldsymbol{F}^{-1}\boldsymbol{l}\boldsymbol{F}\right)$$

[10] See e.g. https://en.wikipedia.org/wiki/Jacobi's_formula.

Now, using the property of the trace of a matrix that states that $\mathrm{tr}(\boldsymbol{AB}) = \mathrm{tr}(\boldsymbol{BA})$, for any matrices $\boldsymbol{A}$ and $\boldsymbol{B}$, and using the identity $\boldsymbol{F}\boldsymbol{F}^{-1} = \boldsymbol{I}$ we get:

$$\frac{\partial}{\partial t} \det \boldsymbol{F} = \det \boldsymbol{F} \, \mathrm{tr}(\boldsymbol{l})$$

since the trace of the spatial velocity gradient is the divergence of velocity, that is $\mathrm{tr}(\boldsymbol{l}) = \operatorname{div} \boldsymbol{v}$, we finally obtain the requested equation:

$$\boxed{\frac{\partial}{\partial t} \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t) = \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t) \operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t);} \tag{3.97}$$

this equation can also be written as:

$$\boxed{\frac{\partial J(\vec{\boldsymbol{X}}, t)}{\partial t} = J(\vec{\boldsymbol{X}}, t) \operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t).}$$

or as

$$\boxed{\frac{\mathrm{D} J(\vec{\boldsymbol{X}}, t)}{\mathrm{D} t} = J(\vec{\boldsymbol{X}}, t) \operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t).}$$

This is an important result that is employed to understand volume changes during deformation and to deduce conservation laws in Chapter 5.

Show that $\operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \operatorname{div} \vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)$

3.23 Exercises

1. Show that for any rotation matrix $\boldsymbol{R}$, the trace must satisfy $-1 \leq \mathrm{tr}\, \boldsymbol{R} \leq 3$. What do these extreme values represent physically?



Chapter 4

Stress

This chapter deals with ...

Write a proper introduction

4.1 Cauchy's stress vector

Consider an imaginary plane embedded within a continuous material, like a thin membrane separating two adjacent regions. Within this region, the material on one side of the plane exerts a force on the material on the other side. The *Cauchy's stress vector** also known as the *Cauchy's traction vector** $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}$ at a point $\vec{\boldsymbol{x}}$ and time t is defined as the force per unit area that the material on one side of the plane exerts on the material on the other side:

$$\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t) = \lim_{\Delta S \to \vec{\boldsymbol{x}}} \frac{\Delta \vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t)}{\Delta S},$$

where $\hat{\boldsymbol{n}}$ is the unit normal vector to the plane pointing outward from the region of interest. This traction depends on the orientation $\hat{\boldsymbol{n}}$ of the surface.

If we consider the other side of the plane, the material on that side exerts an equal and opposite force on the material on the first side, so we have $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t) = -\vec{\boldsymbol{t}}^{(-\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t)$, which is a consequence of Newton's third law of motion (see Section 5.9).

In the case of a fluid at rest, the traction vector is normal to the surface and its magnitude is given by the hydrostatic pressure p, so $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})} = -p\hat{\boldsymbol{n}}$. In the case of a solid,

the traction vector can have both normal and tangential components, depending on the deformation of the material.

To fully characterize the internal forces within the material, we need to know the traction vector for all possible orientations $\hat{\boldsymbol{n}}$. In the next, we will see that the traction vector can be expressed in terms of a second-order tensor field, the Cauchy stress tensor $\boldsymbol{\sigma}$, which fully characterizes the internal forces within the material.

4.2 Cauchy's fundamental theorem of stress

Cauchy's fundamental theorem of stress* states that the traction vector at a point $\vec{\boldsymbol{x}}$ and time t is given by

$$\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t) = \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t), \tag{4.1}$$

where $\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)$ is the so-called *Cauchy stress tensor**, a second-order tensor field that fully characterizes the internal forces throughout the material. In component form, for a Cartesian basis,

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix} \equiv \begin{pmatrix} \sigma_{11} & \sigma_{12} & \sigma_{13} \\ \sigma_{21} & \sigma_{22} & \sigma_{23} \\ \sigma_{31} & \sigma_{32} & \sigma_{33} \end{pmatrix}.$$

Here, σ_{ij} represents the force per unit area in the i-direction acting across a surface whose outward normal points in the j-direction. For instance, $\sigma_{xz} = \sigma_{13}$ is the x_1-component of traction on a surface normal to the x_3-axis. Tensor $\boldsymbol{\sigma}$ encodes tractions $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}$ for *all* possible surface orientations $\hat{\boldsymbol{n}}$ via equation (4.1).

The proof of equation (4.1) results from the application of Newton's second law of motion (the principle of conservation of linear momentum) to an infinitesimally small tetrahedron carved out of the material, with one face oriented in the direction of $\hat{\boldsymbol{n}}$ and the other three faces aligned with the coordinate planes. By analyzing the balance of forces on this element and taking the limit as its size approaches zero, we derive this relationship.

The Cauchy stress tensor is frame-indifferent (transforms as $\boldsymbol{\sigma}' = \boldsymbol{R}\boldsymbol{\sigma}\boldsymbol{R}^T$ under rotation $\boldsymbol{R}$) and symmetric ($\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$) by angular momentum balance, as we'll derive later, in Sections 5.11.3 and 5.11.4.

4.3 Decomposition of the Cauchy's stress tensor

It is important to note that the stress tensor is decomposed as $\boldsymbol{\sigma} = -p\boldsymbol{I} + \boldsymbol{\tau}$ where p is the hydrostatic pressure and $\boldsymbol{\tau}$ the deviatoric (viscous) stress tensor. In the case of a fluid at rest, only hydrostatic pressure acts, so $\boldsymbol{\tau} = 0$ and $\boldsymbol{\sigma} = -p\boldsymbol{I}$. Flow introduces

velocity gradients, generating viscous drag, that is $\boldsymbol{\tau}$. For a Newtonian fluid (linear viscosity, like air or oil),

$$\boldsymbol{\tau} = \lambda(\operatorname{tr}\boldsymbol{D})\boldsymbol{I} + 2\mu\boldsymbol{D},$$

where μ is shear viscosity, λ bulk viscosity, and $\boldsymbol{D} = \frac{1}{2}(\nabla\vec{\boldsymbol{v}} + (\nabla\vec{\boldsymbol{v}})^T)$ the symmetric rate-of-deformation tensor. Observe that in the case of incompressible fluids $\operatorname{tr}\boldsymbol{D} = 0$; this implies $\boldsymbol{\tau} = 2\mu\boldsymbol{D}$.

In the case of solids, this decomposition is necessary to study plasticity and material failure; here the deviatoric stress $\boldsymbol{\tau}$ is responsible for plastic deformation.

This is a good point to mention that in the case of solids, the stress tensor is related to the deformation of the material through a constitutive relation, which depends on the material properties. For example, in linear elasticity, the stress tensor is linearly related to the strain tensor through Lame's equation:

$$\boldsymbol{\sigma} = \lambda(\operatorname{tr}\boldsymbol{\varepsilon})\boldsymbol{I} + 2\mu\boldsymbol{\varepsilon},$$

where λ and μ are the Lame parameters, and $\boldsymbol{\varepsilon}$ is the strain tensor.

4.4 Piola-Kirchhoff and the Kirchhoff stress tensors

The Cauchy stress tensor $\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)$ is defined using the differential area $\mathrm{d}S$ in the current (deformed) configuration. However, it is often convenient to formulate problems in terms of the undeformed configuration, leading to the definition of alternative stress tensors that employ the undeformed area $\mathrm{d}S_0$. These are the *Piola-Kirchhoff stress tensors**, and we will define them in the following.

The stress vector can be defined as the intensity of a force acting on a given area. With reference to Figure 4.1, the stress vector at a point $\vec{\boldsymbol{x}}$ in the deformed configuration can be written using an area $\Delta S(\vec{\boldsymbol{x}}, t)$ on the deformed configuration, the force $\Delta\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t)$ acting on it and the unit normal vector $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)$, using the Cauchy stress tensor $\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)$ (see Section 4.2) as:

$$\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t) = \lim_{\Delta S \to \vec{\boldsymbol{x}}} \frac{\Delta\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t)}{\Delta S(\vec{\boldsymbol{x}}, t)} = \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t); \tag{4.2}$$

where $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t)$ is the *Cauchy* (or *true*) *traction vector** or *Cauchy stress vector**; it acts on the current configuration. Note that this equation refers to the current deformed geometry as a reference for both the force and the area. It represents the true stress state in the deformed configuration.

Another perspective defines the stress vector at a point $\vec{\boldsymbol{X}} \in \Omega_0$ using the same force $\Delta\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t)$ but acting on an area $\Delta S_0(\vec{\boldsymbol{X}})$ on the undeformed configuration and the unit normal vector $\vec{\boldsymbol{N}}(\vec{\boldsymbol{X}})$ as:

$$\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{N}})}(\vec{\boldsymbol{X}}, t) = \lim_{\Delta S \to \vec{\boldsymbol{X}}} \frac{\Delta\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t)}{\Delta S_0(\vec{\boldsymbol{X}})} = \boldsymbol{P}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}}); \tag{4.3}$$

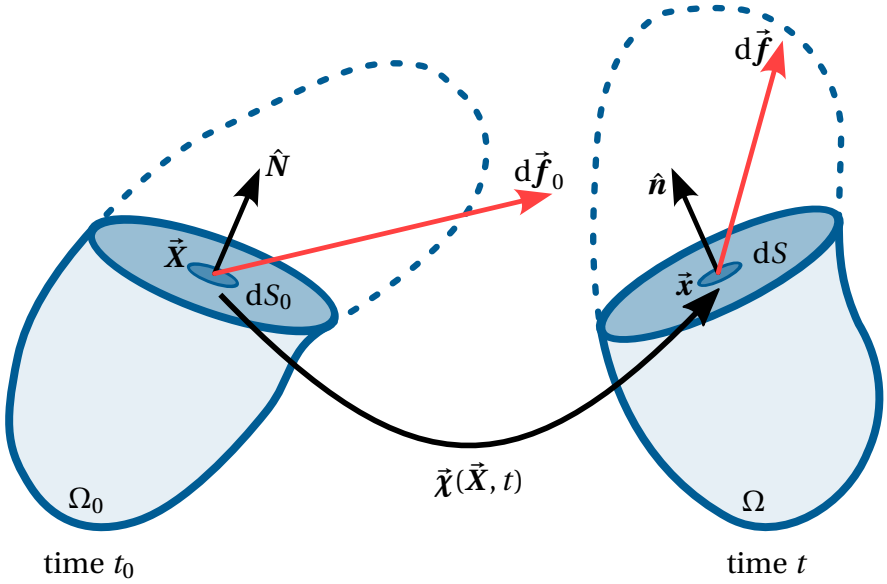


Figure 4.1: *Differential force vectors* $d\vec{f}_0$ *and* $d\vec{f}$ *acting on the undeformed and deformed configurations, respectively; they are related through equation* $d\vec{f} = \boldsymbol{F}\, d\vec{f}_0$ *. The normals* $\vec{\boldsymbol{N}}$ *and* $\vec{\boldsymbol{n}}$ *are associated with the undeformed and deformed configurations, respectively. The areas* dS_0 *and* dS *are the corresponding surface elements; they are related through Nanson's formula* $dS\vec{\boldsymbol{n}} = J(\boldsymbol{F}^{-1})^T \vec{\boldsymbol{N}}\, dS_0$*. The first Piola-Kirchhoff stress tensor* $\boldsymbol{P}$ *relates the force true differential force* $d\vec{f}$ *in the current geometry to the area in the initial geometry* dS_0*, while the second Piola-Kirchhoff stress tensor* $\boldsymbol{S}$ *relates the force* $d\vec{f}_0$ *in the undeformed geometry to the area in the initial geometry* dS_0*. The Cauchy stress tensor* $\boldsymbol{\sigma}$ *relates the true differential force* $d\vec{f}$ *in the current geometry to the area in the current geometry* dS*.*

here $\boldsymbol{P}(\vec{\boldsymbol{x}}, t)$ is the so-called *first Piola-Kirchhoff stress tensor*, also known as the *Lagrangian stress tensor* or the *nominal stress tensor* and $\vec{\boldsymbol{t}}^{(\hat{N})}(\vec{\boldsymbol{X}}, t)$ is the *first Piola-Kirchhoff* (or *nominal*) *traction vector*, which acts on the undeformed configuration. This tensor relates the force $\Delta\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t)$ in the current geometry to the area in the initial geometry $\Delta S_0(\vec{\boldsymbol{X}})$; since it is defined on the undeformed area, $\vec{\boldsymbol{t}}^{(\hat{N})}(\vec{\boldsymbol{X}}, t)$ does not describe the actual intensity of the force acting on the deformed area.

In both equations (4.2) and (4.3), the applied force $\Delta\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t)$ is the same, therefore, we can relate its differential counterparts as:

$$d\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}, t) = \underbrace{\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)}_{\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t)}\, dS = \underbrace{\boldsymbol{P}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})}_{\vec{\boldsymbol{t}}^{(\hat{N})}(\vec{\boldsymbol{X}}, t)}\, dS_0$$

where dS_0, dS, $\vec{\boldsymbol{N}}(\vec{\boldsymbol{X}})$, and $\vec{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)$ are the differential areas and normal vectors associated to ΔS_0 and ΔS, respectively; note that both traction vectors $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}}, t)$ and $\vec{\boldsymbol{t}}^{(\hat{N})}(\vec{\boldsymbol{X}}, t)$ share the same direction, although they have different magnitudes, and

act over surfaces with different normal vectors. This last equation shows that $\boldsymbol{P}$ is a two-point tensor relating two vectors in different configurations.

Nanson's equation (3.37) relates the unit vectors $\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})$ and $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)$ in the undeformed and the deformed configuration as $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S = J(\vec{\boldsymbol{X}},t)\,\mathrm{d}S_0\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})$, where $J(\vec{\boldsymbol{X}},t) := \det\big(\boldsymbol{F}(\vec{\boldsymbol{X}},t)\big)$ is the volume ratio; it follows that:

$$\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S = \underbrace{\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)J(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}},t)}_{\boldsymbol{P}(\vec{\boldsymbol{X}},t)}\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S_0,$$

and we obtain a formula that relates the first Piola-Kirchhoff stress tensor with the Cauchy stress tensor:

$$\boxed{\boldsymbol{P}(\vec{\boldsymbol{X}},t) = J(\vec{\boldsymbol{X}},t)\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}},t)} \tag{4.4}$$

observe that the first Piola-Kirchhoff stress tensor is not symmetric, and is described by 9 independent components.

The force $\mathrm{d}\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}},t)$ acts on the current configuration; one could use the inverse of the deformation gradient to retrieve its associated vector in the undeformed configuration as $\mathrm{d}\vec{\boldsymbol{f}}_0(\vec{\boldsymbol{X}},t) = \boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}},t)$; this is the pull-backward operation $\mathrm{d}\vec{\boldsymbol{f}}_0(\vec{\boldsymbol{X}},t) = \phi_*^{-1}[\mathrm{d}\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}},t)]$ (see Section 3.20):

$$\begin{aligned}\vec{\boldsymbol{t}}_0^{(\hat{\boldsymbol{N}})}(\vec{\boldsymbol{X}},t) &= \lim_{\Delta S_0\to\vec{\boldsymbol{X}}}\frac{\Delta\vec{\boldsymbol{f}}_0(\vec{\boldsymbol{X}},t)}{\Delta S_0(\vec{\boldsymbol{X}})}\\ &= \lim_{\Delta S_0\to\vec{\boldsymbol{X}}}\frac{\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}},t)}{\Delta S_0(\vec{\boldsymbol{X}})}\\ &= \underbrace{\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t)\boldsymbol{P}(\vec{\boldsymbol{X}},t)}_{\boldsymbol{S}(\vec{\boldsymbol{x}},t)}\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}});\end{aligned}$$

and so, we obtain a symmetric tensor, the so-called *second Piola-Kirchhoff stress tensor* $\overset{\star}{\boldsymbol{S}}(\vec{\boldsymbol{x}},t)$:

$$\boxed{\boldsymbol{S}(\vec{\boldsymbol{x}},t) = \boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t)\boldsymbol{P}(\vec{\boldsymbol{X}},t) = J(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{-1}(\vec{\boldsymbol{x}},t)\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}},t)}$$

unfortunately, the stress components of $\boldsymbol{S}$ have no physical interpretation in terms of surface tractions.

If Piola-Kirchhoff tensors are known, we can retrieve the Cauchy stress tensor as:

$$\boxed{\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t) = \frac{1}{J(\vec{\boldsymbol{X}},t)}\boldsymbol{P}(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{T}(\vec{\boldsymbol{X}},t) = \frac{1}{J(\vec{\boldsymbol{X}},t)}\boldsymbol{F}(\vec{\boldsymbol{X}},t)\boldsymbol{S}(\vec{\boldsymbol{x}},t)\boldsymbol{F}^{T}(\vec{\boldsymbol{X}},t).} \tag{4.5}$$

The calculation of the determinant associated with the volume ratio $J(\vec{\boldsymbol{X}}, t)$ can be cumbersome, as it depends on the deformation. To circumvent this inconvenience, an unnormalized new stress measure can be defined as:

$$\boldsymbol{\tau}(\vec{\boldsymbol{x}}, t) = J(\vec{\boldsymbol{X}}, t)\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) = \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\boldsymbol{S}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}^T(\vec{\boldsymbol{X}}, t)$$

this new stress tensor, $\boldsymbol{\tau}(\vec{\boldsymbol{x}}, t)$, is known as the *Kirchhoff stress tensor**. It is a symmetric tensor widely used in numerical algorithms for metal plasticity, where there is no change in volume during plastic deformation. Alternatively, it can be referred to as the *weighted Cauchy stress tensor*.

It must bear in mind that the Piola-Kirchhoff tensors and the Kirchhoff tensor reduce to $\boldsymbol{\sigma}$ in the case of infinitesimal deformations, i.e., $\boldsymbol{P} \approx \boldsymbol{S} \approx \boldsymbol{\tau} \approx \boldsymbol{\sigma}$; this is because there is no distinction between the initial and the deformed geometries.

The Piola-Kirchhoff and the Kirchhoff stress tensors appear in many derivations in continuum mechanics, driven by the convenience of writing the derivations in terms of the initial (undeformed) configuration instead of the current (deformed) configuration.

4.5 Some mathematical results

In the following, we provide some mathematical results that will be useful in the derivations of the balance equations in the next chapters.

Example 4.1. *Relationship between the divergence of the first Piola-Kirchhoff stress tensor and divergence of the Cauchy stress tensor*

Show that:

$$\nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t) = J(\vec{\boldsymbol{X}}, t)\nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t). \tag{4.6}$$

Solution:

As a starting point, let's integrate $\nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t)$ over an arbitrary reference subvolume $\mathscr{V}_0 \subseteq \Omega_0$ and apply the divergence theorem:

$$\iiint_{\mathscr{V}_0} \nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V}_0 = \oiint_{\partial\mathscr{V}_0} \boldsymbol{P}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\, \mathrm{d}S.$$

Using the definition (4.4) of the first Piola-Kirchhoff stress tensor, i.e., $\boldsymbol{P}(\vec{\boldsymbol{X}}, t) = J(\vec{\boldsymbol{X}}, t)\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)$, and the Nanson formula (3.37), which relates the normal vectors and surface elements in the reference and current configurations as $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}s = J(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\, \mathrm{d}S$, we can rewrite the surface integral as:

$$\oiint_{\partial\mathscr{V}_0} \boldsymbol{P}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\, \mathrm{d}S = \oiint_{\partial\mathscr{V}_0} J(\vec{\boldsymbol{X}}, t)\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\, \mathrm{d}S$$

$$= \oiint_{\partial\mathscr{V}_0} \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\, \underbrace{J(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S}_{=\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}s}$$
$$= \oiint_{\partial\mathscr{V}_t} \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}s.$$

Here, $\partial\mathscr{V}_t = \vec{\boldsymbol{\chi}}(\partial\mathscr{V}_0, t)$ is the current configuration of the boundary of $\mathscr{V}_0$, and $J(\vec{\boldsymbol{X}}, t)$ is the determinant of the deformation gradient $\boldsymbol{F}(\vec{\boldsymbol{X}}, t)$. Since $\partial\mathscr{V}_t$ is an arbitrary surface in the current configuration, we can apply the divergence theorem to $\boldsymbol{\sigma}$ over the volume $\mathscr{V}_t = \vec{\boldsymbol{\chi}}(\mathscr{V}_0, t)$:

$$\oiint_{\partial\mathscr{V}_t} \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}s = \iiint_{\mathscr{V}_t} \nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}v_t.$$

Under the change of variables $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$, the volume elements are related by $\mathrm{d}v_t = J(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\mathscr{V}_0$, according to equation (3.35), so:

$$\iiint_{\mathscr{V}_t} \nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}v_t = \iiint_{\mathscr{V}_0} \nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) J(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\mathscr{V}_0,$$

where $\nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}$ is understood to be evaluated at $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$. Assembling all steps into a single chain:

$$\iiint_{\mathscr{V}_0} \nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\mathscr{V}_0 = \iiint_{\mathscr{V}_0} J(\vec{\boldsymbol{X}}, t)\nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\mathscr{V}_0,$$

which can be rearranged as:

$$\iiint_{\mathscr{V}_0} \left[\nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t) - J(\vec{\boldsymbol{X}}, t)\nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\right] \mathrm{d}\mathscr{V}_0 = \vec{\boldsymbol{0}}.$$

Since $\mathscr{V}_0 \subseteq \Omega_0$ is arbitrary and the integrand is continuous, the localization theorem (see Appendix A.18) demands that the integrand vanishes pointwise everywhere in Ω_0:

$$\nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t) = J(\vec{\boldsymbol{X}}, t)\,\nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t).$$



Chapter 5

Conservation laws and balance equations

The fundamental principles of classical physics often take the form of *conservation laws**, that is, statements that certain physical quantities like mass, energy, or momentum remain constant over time. These conservation laws are universal, applying to all materials regardless of their specific properties, magnitude of displacements or deformations. Consequently, their mathematical formulations, also known as *balance equations**, result in equations that must always be satisfied. The three most fundamental physical principles governing continuum mechanics are:

1. **Conservation of mass:** Mass can neither be created nor destroyed, only transformed or moved.

2. **Newton's laws of motion:** They relate a body's motion and the forces acting upon it.

3. **Conservation of energy:** Energy can be transformed but never created or destroyed.

While we'll develop these concepts using fluid mechanics as our primary example, remember that these principles apply equally well to solids undergoing small or large deformations. This universality is what makes continuum mechanics such a powerful framework.

5.1 Control volumes, control masses and material elements

Before formulating conservation laws rigorously, it is essential to define the framework or region in which they operate. This brings us to the concept of a system. A *system** is defined as a specific, identifiable quantity of matter, or a specific region in space, that define the region or set of particles chosen for study. The complement of a system, that is the region outside the system, is called the *surroundings**, which includes everything outside the system. The *boundary** is the interface that separates the system from its surroundings.

A system can be classified into three types based on the exchange of mass and energy (in the form of work or heat) with its surroundings:

- *Control volume* or *open system*: A system that can exchange both mass and energy with its surroundings. For example, a boiling pot of water is an open system because it allows steam (mass) to escape and heat (energy) to be transferred.
- *Control mass* or *closed system*: A system that can exchange energy but not mass with its surroundings. An example of a closed system is a sealed container of gas, which can transfer heat but does not allow the gas to escape.
- *Isolated system*: A system that cannot exchange either mass or energy with its surroundings. An example of an isolated system is an insulated thermos flask, which ideally prevents any heat transfer and does not allow mass to enter or leave. Note that no physical system is truly isolated.

Understanding the differences, purposes, and significance of these three types of systems is necessary for formulating and solving problems in fluid dynamics, solid mechanics, thermodynamics, and beyond. Let's delve into the details of open and closed systems. Isolated systems will be analyzed in Section ***.

On the one hand, a *control mass** $\mathscr{V}_t^{\mathrm{cm}} \subseteq \Omega_t$, also known as a *closed system** or *material volume**, refers to a fixed collection of material (i.e., a *fixed amount of mass*) that we follow as it moves with the flow deforming to ensure that it always contains the same set of particles; every point on its boundary $\partial\mathscr{V}_t^{\mathrm{cm}}$ travels at the local material velocity $\vec{\boldsymbol{v}}$, so no new particles will come inside $\mathscr{V}_t^{\mathrm{cm}}$; here, we analyze the motion of the particles that were initially within $\mathscr{V}_0^{\mathrm{cm}} \subseteq \Omega_0$ at time t_0. The boundary $\partial\mathscr{V}_t^{\mathrm{cm}}$ of the control mass encloses a specific amount of matter, and there is no mass transfer across these boundaries during the process under consideration; however, energy can be transferred into or out of the system in the form of work or heat. While the shape and volume of $\mathscr{V}_t^{\mathrm{cm}}$ may change with time t, it always contains the same matter, that is, its mass remains constant. To emphasize this time-dependent nature, we denote our control volume as $\mathscr{V}_t^{\mathrm{cm}}$, where the subscript t explicitly indicates that the

volume's shape and position may change with time. By focusing on a fixed group of particles, this framework allows us to apply conservation laws directly to the material itself without worrying about inflows or outflows. For example, if you consider a sealed piston-cylinder assembly containing gas, the gas itself constitutes the control mass; in this case, the amount of gas does not change in quantity but may change in shape, temperature and pressure. This approach follows a *Lagrangian* perspective, focusing on the motion of specific particles within the control volume. When using this framework, the resulting equations are said to be in *non-conservation form**.

On the other hand, a *control volume** $\mathscr{V}^{\text{cv}} \subset \Omega_t$, also called an *open system**, is a fixed region in space, with an unchanging shape, which encloses a finite portion of continuum, through which mass and energy may flow; this region has a *fixed amount of volume*; it can be understood as a window for observation of the flow. Unlike a control mass, the boundaries of a control volume, known as the *control surface** $\partial\mathscr{V}^{\text{cv}}$, allow for the transfer of mass across their boundaries; this means that both mass and energy (in the form of work or heat) can enter and exit the control volume during operation. This makes it easier to account for the flux rate at which mass, momentum, or energy cross the control surface and relate them to changes occurring inside the control volume. A classic example of a control volume is a fixed section of a pipe or channel through which water flows; here, the focus is on what happens within the defined spatial region $\mathscr{V}^{\text{cv}}$ rather than tracking individual molecules of water. Control volumes provide us with an *Eulerian* perspective, where our focus is on how material crosses the stationary boundary $\partial\mathscr{V}^{\text{cv}}$, and the equations derived using this approach are said to be in *conservation form**.

Moreover, depending on the size and behavior of the region under consideration, we may refer to an arbitrarily large amount of continuum or an infinitesimal amount; thus, we classify control masses and control volumes as either *finite* or *infinitesimal*. For finite control volumes and masses, the governing conservation laws are integral equations, and they describe the *global* cumulative/average behavior over an arbitrarily large region. These integral equations are the starting point for formulating numerical methods like the finite element method or the finite volume method.

Also, when studying phenomena at a smaller scale, we use infinitesimal control volumes and masses, represented as tiny parallelepipeds $\mathrm{d}\mathscr{V}^{\text{cv}}$ or $\mathrm{d}\mathscr{V}_t^{\text{cm}}$. Although infinitesimal, these volumes are sufficiently large to encompass many particles, consistent with the continuum assumption introduced in Chapter 1. In this case, the resulting equations are expressed as partial differential equations, capturing the *local* behavior of the material at a point in space. These differential equations can be solved numerically using the computer and methods like the finite difference method.

We will see that the integral and differential forms of the conservation laws are mathematically equivalent under appropriate smoothness conditions. The integral form directly relates the accumulation of a quantity within a region to the fluxes across its

boundaries, which is a natural way to express conservation principles. The differential form, on the other hand, provides a pointwise description of how quantities change at specific locations. However, in cases where solutions may be non-differentiable (e.g., shock waves in fluid dynamics), the equivalence between the integral and differential forms can break down, as integral equations can admit non-smooth solutions that do not satisfy the corresponding differential equations in the classical sense, while the derivatives in the differential form may not exist everywhere due to discontinuities or singularities in the solution; thus integral formulations lead to the concept of weak solutions, because it requires less smoothness of the solution, making them more appropriate for analysis.

Regions (control mass vs. control volume) and their sizes (finite vs. infinitesimal) are different aspects that we have to take into consideration in our analysis. This creates a combination resulting in the four distinct cases shown in Figure 5.1. These cases provide the basis for analyzing conservation laws in different scales and perspectives. In the next, we will describe those four models.

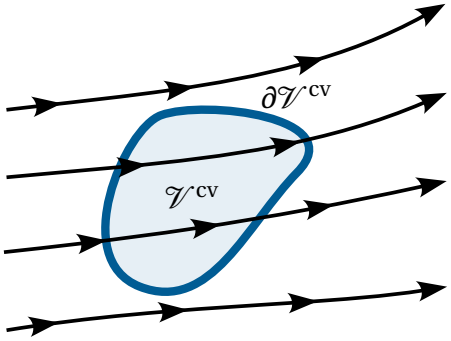
5.1.1 Case 1: finite control volume

A finite control volume $\mathscr{V}^{\text{cv}}$ is particularly suited for analyzing systems where mass crosses boundaries, such devices with inlets and outlets like pipes, pumps, nozzles, and turbines. They allow us to write conservation laws in their integral form, which relates the accumulation of a quantity inside the volume to the fluxes across its boundaries. This is especially useful when dealing with complex geometries where detailed pointwise solutions are unnecessary.

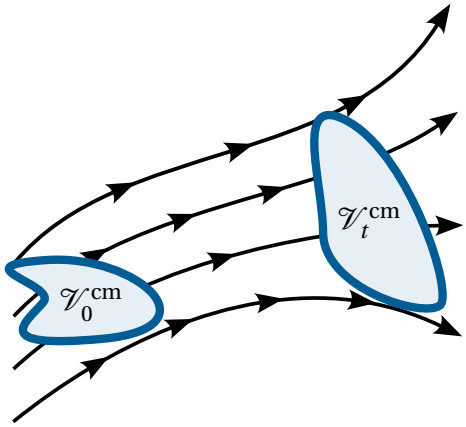
5.1.2 Case 2: finite control mass

A finite control mass $\mathscr{V}_t^{\text{cm}}$ is typically used when we are tracking a specific group of particles, like in the case of a weather balloon as it rises through the atmosphere, a rigid body under external forces, a parcel of gas, or a droplet of liquid flowing down a river. In thermodynamics, it is used when analyzing the expansion or compression of a gas inside a closed piston-cylinder system. It is often used to derive global conservation laws for systems where the entire material content is considered as a single entity. For instance, it is used when we want to apply Newton's laws of motion on a falling raindrop.

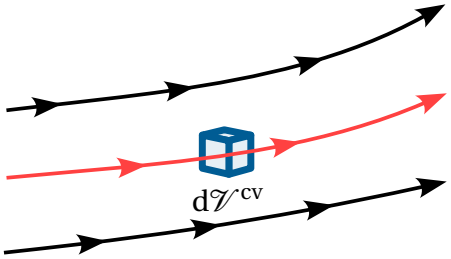
Since the volume and shape of the control mass may change with time, we write $\mathscr{V}_t^{\text{cm}}$ to specify the form of the control mass at time t; consequently, $\mathscr{V}_0^{\text{cm}}$ stands for the control mass at time t_0.



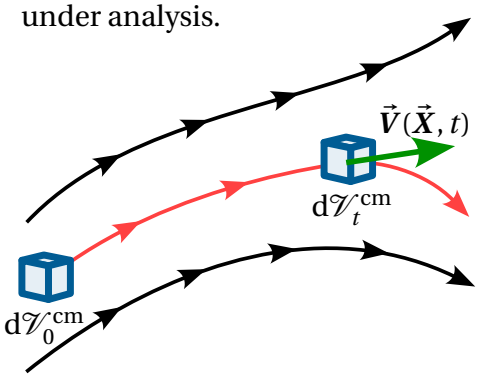
Case 1: finite control volume fixed in space with the material moving through it



Case 2: finite control mass moving with the flow such that the same material particles are always under analysis.



Case 3: infinitesimal control volume fixed in space with the material moving through it.



Case 4: infinitesimal control mass moving along a streamline with velocity $\vec{V}$ equal to the local flow velocity at each point.

Figure 5.1: *This illustration consider a flow of material that is represented by the* streamlines**, that is curves that are everywhere tangent to the velocity vector of the flow. On this flow, we will characterize four cases that depict finite and infinitesimal regions in physical space, that may remain stationary or may move with the flow. In cases 1 and 3, we employ a fixed amount of volume, known as a* control volume*, that allows mass and energy to pass over the boundary (also called a* control surface*). In contrast, in cases 2 and 4, we consider a fixed amount of mass that travels with the flow, referred to as a* control mass *that allows only energy to cross the boundary.*

5.1.3 Case 3: infinitesimal control volume

An infinitesimal control volume $d\mathscr{V}^{cv}$, also known as *spatial element**, remains fixed in space, allowing the fluid to flow through it. Like case 1, this is an Eulerian perspective, and the derived equations are said to be in conservation form. Here, the focus shifts

to local behavior, such as how the energy varies at a specific point. This approach is crucial for deriving the governing equations of continuum mechanics in their differential form. By shrinking the control volume to a point, we can relate spatial gradients of quantities (e.g., pressure, velocity, temperature) to local changes.

5.1.4 Case 4: infinitesimal control mass

Infinitesimal control masses $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ are also known as *material elements** or *fluid parcels**. They move along streamlines with a velocity $\vec{\boldsymbol{V}}(\vec{\boldsymbol{x}}, t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$ that matches the local flow velocity at every point. This Lagrangian perspective examines the motion of individual particles over time. The derived equations are said to be in non-conservation form and reflect the evolution of properties, such as density, velocities or stresses, within the infinitesimal volume as it moves.

Define density of a material element as the mass per unit volume of the material element. Holzapfel, p.133

5.2 Flux, flow, and the transport of physical quantities

Every balance law in continuum mechanics states that whatever quantity we are tracking, like mass, momentum, or energy, can only change inside a region either because it is produced there or because it crosses the boundary. In this last case, we use two words to describe boundary crossing: *flux* and *flow*. Let us define these terms precisely:

Flow* refers to the bulk movement of the continuum through a region of space. It is a qualitative description of the fact that material particles are in motion: a river flows, air flows over a wing, blood flows through an artery. Flow is characterized by the velocity field $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$, which assigns a velocity vector to every point $\vec{\boldsymbol{x}}$ in the continuum at time t.

Flux* is a quantitative measure of how much of a given physical quantity crosses a surface per unit time per unit area; in other words, if $\boldsymbol{Q}$ is the quantity of interest (mass, momentum, energy, etc.), then the flux of $\boldsymbol{Q}$ through a surface element $\mathrm{d}S$ with unit normal $\hat{\boldsymbol{n}}$ tells us the rate at which $\boldsymbol{Q}$ is transported across that surface, per unit area. Flux therefore has dimensions of

$$[\text{flux of } \boldsymbol{Q}] = \frac{[\text{amount of } \boldsymbol{Q}]}{[\text{time}] \cdot [\text{area}]}.$$

With this distinction established, we will build several concepts, related to flow and flux.

Volumetric flow rate

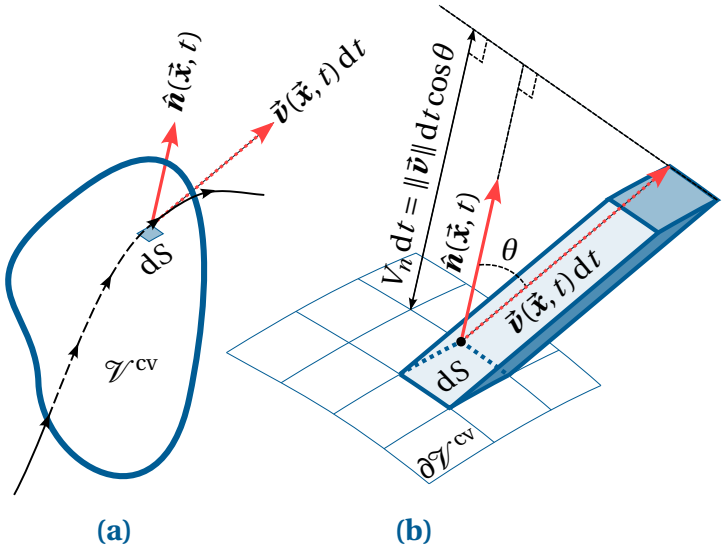


Figure 5.2: *Volume rate of flow through an arbitrary surface. Figure (a) shows a differential* $\mathrm{d}S$*, which is located at* $\vec{\boldsymbol{x}}$*, on the surface* $\partial\mathscr{V}^{cv}$*, together with the streamline that crosses this differential; the normal of* $\partial\mathscr{V}^{cv}$ *at* $\vec{\boldsymbol{x}}$ *is* $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)$*; the particle at the differential* $\mathrm{d}S$ *travels at a velocity* $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$*. Figure (b) shows that the volume that flows through* $\mathrm{d}S$ *during a time interval* $\mathrm{d}t$ *is* $V_n\,\mathrm{d}S\,\mathrm{d}t = \|\vec{\boldsymbol{v}}\|\cos\theta\,\mathrm{d}S\,\mathrm{d}t = \vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}S\,\mathrm{d}t$.

With reference to Figure 5.2a, consider a control volume $\mathscr{V}^{\mathrm{cv}}$ that is fixed in space and surrounded by its control surface $\partial\mathscr{V}^{\mathrm{cv}}$. The continuum moves through it with velocity field $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$. Our first question is: *how much volume of material passes through the surface per unit time?*

Let us fix our attention on a single infinitesimal surface element $\mathrm{d}S$ located at a point $\vec{\boldsymbol{x}}$ on $\partial\mathscr{V}^{\mathrm{cv}}$, and let $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)$ be the outward-pointing unit normal to $\partial\mathscr{V}^{\mathrm{cv}}$ at that point. We write the vectorial surface element as $\mathrm{d}\vec{\boldsymbol{S}} = \hat{\boldsymbol{n}}\,\mathrm{d}S$. In general, the velocity $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$ is not perpendicular to $\mathrm{d}S$; suppose it makes an angle θ with $\hat{\boldsymbol{n}}$.

In a short time interval $\mathrm{d}t$, the material that will have swept through $\mathrm{d}S$ occupies a slanted parallelepiped with base $\mathrm{d}S$ and oblique side $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}t$ (see Figure 5.2b). The volume of this parallelepiped is

$$\mathrm{d}\mathscr{V}_{\mathrm{swept}}(\vec{\boldsymbol{x}},t) = V_n(\vec{\boldsymbol{x}},t)\,\mathrm{d}S\,\mathrm{d}t = \big(\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\big)\,\mathrm{d}S\,\mathrm{d}t = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}}\,\mathrm{d}t. \tag{5.1}$$

The factor $V_n \equiv \vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}} = \|\vec{\boldsymbol{v}}\|\cos\theta$ is the normal component of the velocity at $\vec{\boldsymbol{x}}$: the component perpendicular to the surface, which is the only component responsible for transport through the surface. The tangential components of $\vec{\boldsymbol{v}}$ are parallel to the surface and contribute nothing to the volume crossing it.

Dividing (5.1) by $\mathrm{d}t$, we obtain the *volumetric flow rate** (or *volume rate of flow**) through the surface element $\mathrm{d}S$:

$$\mathrm{d}Q = \frac{\mathrm{d}\mathscr{V}_{\text{swept}}}{\mathrm{d}t} = (\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\, \mathrm{d}S = \vec{\boldsymbol{v}} \cdot \mathrm{d}\vec{\boldsymbol{S}}. \tag{5.2}$$

This quantity is measured in m^3/s and represents the volume of material crossing the element $\mathrm{d}S$ per unit time.

The sign of $\mathrm{d}Q$ carries important directional information. By convention, $\hat{\boldsymbol{n}}$ is the *outward* unit normal, so:

- $\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}} > 0$: the flow has a component directed outward; material is *leaving* $\mathscr{V}^{\text{cv}}$ through $\mathrm{d}S$; this is called an *outflow**.
- $\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}} < 0$: the flow has a component directed inward; material is *entering* $\mathscr{V}^{\text{cv}}$ through $\mathrm{d}S$; this is called an *inflow**.
- $\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}} = 0$: the flow is tangential to the surface $\partial\mathscr{V}^{\text{cv}}$; no material crosses $\mathrm{d}S$.

Thus, we will use the dot product $\vec{\boldsymbol{v}} \cdot \mathrm{d}\vec{\boldsymbol{S}}$ (or $\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}}$) rather than $\|\vec{\boldsymbol{v}}\|\, \mathrm{d}S$ because it automatically encodes the direction of transport without any additional sign conventions.

To find the total volume of material crossing the entire closed surface $\partial\mathscr{V}^{\text{cv}}$ per unit time, we integrate (5.2) over all surface elements:

$$Q(t) = \oiint_{\partial\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}}.$$

The quantity $Q(t)$ is the *net volumetric flow rate** or *net volumetric outflow** through $\partial\mathscr{V}^{\text{cv}}$ at time t. Because the outward normal convention is built into the sign of each $\vec{\boldsymbol{v}} \cdot \mathrm{d}\vec{\boldsymbol{S}}$, outflowing contributions add positively and inflowing contributions add negatively, and $Q(t)$ gives the algebraic balance; this justifies the use of the word "net". If $Q > 0$, more volume leaves $\mathscr{V}^{\text{cv}}$ than enters; if $Q < 0$, more volume enters than leaves.

Mass rate of flow and mass flux

Volume alone is not a conserved quantity in a compressible continuum, because the density $\rho(\vec{\boldsymbol{x}}, t)$ varies from place to place and time to time. The physically meaningful transported quantity is the "mass". Multiplying the swept volume by the local density gives the mass that crosses $\mathrm{d}S$ in time $\mathrm{d}t$:

$$\mathrm{d}m = \rho(\vec{\boldsymbol{x}}, t)\, (\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\, \mathrm{d}S\, \mathrm{d}t = \rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}}\, \mathrm{d}t.$$

Dividing by $\mathrm{d}t$, the *mass rate of flow** (or *mass flow rate**) through $\mathrm{d}S$ is

$$\begin{array}{l}\text{mass flow rate across}\\ \text{the differential}\\ \text{surface } \mathrm{d}S \text{ at time } t\end{array} \quad = \quad \rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}} \quad = \quad \rho(\vec{\boldsymbol{x}}, t)\, (\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\, \mathrm{d}S, \tag{5.3}$$

a quantity that is measured in kg/s. Integrating (5.3) over the entire closed surface, we obtain the *net mass outflow*★ (or *net mass flow rate*★) from $\mathscr{V}^{\text{cv}}$ at time t:

$$\frac{\mathrm{d}m(t)}{\mathrm{d}t} = \oiint_{\partial\mathscr{V}^{\text{cv}}} \rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}} = \oiint_{\partial\mathscr{V}^{\text{cv}}} \rho\, (\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\, \mathrm{d}S. \tag{5.4}$$

This is the rate at which mass leaves $\mathscr{V}^{\text{cv}}$ in excess of the rate at which it enters; it is measured in kg/s. If $\mathrm{d}m/\mathrm{d}t > 0$, more mass leaves than enters; if $\mathrm{d}m/\mathrm{d}t < 0$, more mass enters than leaves.

Momentum flux and its difference from mass flux

Momentum★ is a concept representing the "quantity of motion" an object possesses, calculated as mass multiplied by velocity:

$$\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}.$$

It is a vector quantity (having both magnitude and direction) measured in kilogram meters per second (kg·m/s). It measures how difficult an object is to stop, with higher mass or velocity increasing momentum. We will study momentum with more detail in Section 5.7.

In continuum mechanics, we often work with *momentum density*★ $\rho\vec{\boldsymbol{v}}$, which is the momentum per unit volume. It is a momentum vector field representing the momentum per unit volume at each point in the continuum. It has units of $(\mathrm{kg}\cdot\mathrm{m/s})/\mathrm{m}^3$ (do not perform simplifications).

Do not confuse momentum density with mass flux. *Mass flux*★ $\rho\vec{\boldsymbol{v}}$ is a vector field representing the mass transported per unit area per unit time with units of $\mathrm{kg}/(\mathrm{m}^2\cdot\mathrm{s})$. Observe that both momentum density and mass flux share the same mathematical form $\rho\vec{\boldsymbol{v}}$, although they represent different physical quantities and have different units.

In conclusion, the same mathematical expression, $\rho\vec{\boldsymbol{v}}$, can represent different physical concepts depending on the context and interpretation.

Flow not only carries mass, but also carries momentum. By the same geometric argument as before, the momentum transported through $\mathrm{d}S$ in time $\mathrm{d}t$ is

$$\mathrm{d}\vec{\boldsymbol{p}} = (\rho\vec{\boldsymbol{v}})\,(\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\,\mathrm{d}S\,\mathrm{d}t,$$

and the *momentum flow rate* (momentum crossing $\mathrm{d}S$ per unit time) is the vector

$$\begin{array}{c}\text{momentum flow rate}\\ \text{across the differential}\\ \text{surface } \mathrm{d}S \text{ at time } t\end{array} \quad = \quad (\rho\vec{\boldsymbol{v}})\,(\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\,\mathrm{d}S. \tag{5.5}$$

We will call the term $(\rho\vec{\boldsymbol{v}})\,(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})$ the *convective momentum flux density*: it is a vector representing the momentum per unit area per unit time being transported across the surface by the flow of the material itself. Remember that the word *convection** refers to the transfer of heat through the physical movement of fluids. This is contrast to *conduction**, which is the transfer of heat through a material without any bulk movement, and to *radiation**, which is the transfer of energy through electromagnetic waves. Now, in continuum mechanics, the term *convective flux** refers to the transport of any physical quantity by the bulk motion of the material (like when a river carries logs downstream), with no reference to diffusion (i.e. no molecular motion), conduction (i.e. no thermal motion), or contact (surface) forces.

Let us study the role of velocity $\vec{\boldsymbol{v}}$ in the expressions for mass and momentum flow rates:

- In the mass flow rate $\rho(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S$, the velocity $\vec{\boldsymbol{v}}$ determines only the *transport rate*, i.e., how fast material sweeps through the surface. Mass is a scalar and has no directional content; once we know how many material crosses, we know how many mass crosses.
- In the momentum flow rate $(\rho\vec{\boldsymbol{v}})\,(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S$, the velocity $\vec{\boldsymbol{v}}$ plays two simultaneous roles: one copy of $\vec{\boldsymbol{v}}$ appears inside $\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}$, determining the transport rate (exactly as for mass), while a second copy of $\vec{\boldsymbol{v}}$ appears inside $\rho\vec{\boldsymbol{v}}$, represents the momentum density being transported. It is the same physical velocity doing two different jobs, and the result is a quadratic expression in $\vec{\boldsymbol{v}}$.

Continuing with the convective contribution, integrating (5.5) over the entire control surface $\partial\mathscr{V}^{\mathrm{cv}}$ gives the net rate at which momentum is being carried out of $\mathscr{V}^{\mathrm{cv}}$ by the flow:

$$\begin{array}{c}\text{net momentum flow}\\ \text{rate across the closed}\\ \text{surface } \partial\mathscr{V}^{\mathrm{cv}} \text{ at time } t\end{array} \quad = \quad \frac{\mathrm{d}\vec{\boldsymbol{p}}(t)}{\mathrm{d}t} \quad = \quad \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} (\rho\vec{\boldsymbol{v}})\,(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S. \tag{5.6}$$

This net outflow represents the rate at which momentum is being carried away from $\mathscr{V}^{\mathrm{cv}}$ by the flow of material itself. The faster and denser the flow, the larger this convective outflow, and the harder it is for forces to change the state of motion inside $\mathscr{V}^{\mathrm{cv}}$.

In order to express the surface integral (5.6) into a volume integral over $\mathscr{V}^{\mathrm{cv}}$, we apply the divergence theorem (see Appendix A.16), which states that for any smooth vector field $\vec{\boldsymbol{f}}$,

$$\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{f}}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}S = \iiint_{\mathscr{V}^{\mathrm{cv}}} \nabla\cdot\vec{\boldsymbol{f}}\,\mathrm{d}v.$$

The challenge is that the integrand in (5.6) is not a scalar dot product of a vector with $\hat{\boldsymbol{n}}$, but rather the *vector* $\rho\vec{\boldsymbol{v}}$ multiplied by the scalar $\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}$. To apply the divergence theorem, we must recast the integrand in the form $(\times)\cdot\hat{\boldsymbol{n}}$ where $(\times)$ is a tensor. To

do this, we introduce the *outer product** (also known as the dyadic product* or *tensor product** – See Appendix A.17) of two vectors. For vectors $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$, the dyadic product $\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}$ is the second-order tensor with Cartesian components $(\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}})_{ij} = a_i b_j$. Its fundamental action on an arbitrary vector $\hat{\boldsymbol{n}}$ is

$$(\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}) \cdot \hat{\boldsymbol{n}} = \vec{\boldsymbol{a}} (\vec{\boldsymbol{b}} \cdot \hat{\boldsymbol{n}}), \tag{5.7}$$

which follows immediately from the component calculation:

$$\left[(\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}) \cdot \hat{\boldsymbol{n}}\right]_i = \sum_{j=1}^{3} (\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}})_{ij}\, n_j = \sum_{j=1}^{3} a_i b_j n_j = a_i\, (\vec{\boldsymbol{b}} \cdot \hat{\boldsymbol{n}}) = \left[\vec{\boldsymbol{a}} (\vec{\boldsymbol{b}} \cdot \hat{\boldsymbol{n}})\right]_i.$$

Applying (5.7) with $\vec{\boldsymbol{a}} = \vec{\boldsymbol{v}}$ and $\vec{\boldsymbol{b}} = \vec{\boldsymbol{v}}$, and absorbing the density:

$$\rho \vec{\boldsymbol{v}} (\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}}) = (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) \cdot \hat{\boldsymbol{n}}. \tag{5.8}$$

The surface integral (5.6) now has exactly the structure required by the tensorial generalization of the divergence theorem: for any smooth second-order tensor $\boldsymbol{T}$,

$$\oiint_{\partial \mathscr{V}^{\mathrm{cv}}} \boldsymbol{T} \cdot \hat{\boldsymbol{n}}\, \mathrm{d}S = \iiint_{\mathscr{V}^{\mathrm{cv}}} \nabla \cdot \boldsymbol{T}\, \mathrm{d}\mathscr{v}, \tag{5.9}$$

where $(\nabla \cdot \boldsymbol{T})_i = \sum_{j=1}^{3} \partial_j T_{ij}$. Substituting (5.8) into (5.6) and applying (5.9) with $\boldsymbol{T} = \rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}$ gives:

$$\oiint_{\partial \mathscr{V}^{\mathrm{cv}}} \rho \vec{\boldsymbol{v}} (\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\, \mathrm{d}S = \oiint_{\partial \mathscr{V}^{\mathrm{cv}}} (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) \cdot \hat{\boldsymbol{n}}\, \mathrm{d}S = \iiint_{\mathscr{V}^{\mathrm{cv}}} \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}})\, \mathrm{d}\mathscr{v}. \tag{5.10}$$

The vector field $\nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}})$ is the *convective term* in its local, pointwise form; its i-th Cartesian component is

$$\left[\nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}})\right]_i = \sum_{j=1}^{3} \frac{\partial}{\partial x_j} (\rho v_i v_j),$$

This vector field, defined at every interior point of $\mathscr{V}^{\mathrm{cv}}$, encodes the local rate at which momentum is being carried away from that point by the flow.

5.3 Local and material time derivatives of a volume integral

In continuum mechanics, we often study physical quantities such as mass, energy, or momentum that are distributed throughout a volume. These quantities can be scalar, vectorial, or tensorial in nature. Let us denote such a quantity by $\boldsymbol{Q}$, and let $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)$ represent its corresponding volumetric density (the amount of $\boldsymbol{Q}$ per unit

volume) at a point $\vec{x}$ in space and time t. In the following, we will assume that $\boldsymbol{\mu}$ is a continuously differentiable function of position $\vec{x}$ and time t. For example, $\boldsymbol{\mu}$ could represent mass density, energy density, momentum density or chemical concentration. Mathematically, this is expressed as:

$$\boldsymbol{\mu}(\vec{x}, t) = \frac{\text{amount of } \boldsymbol{Q}}{\text{unit volume}} = \lim_{\Delta v \to 0} \frac{\Delta \boldsymbol{Q}}{\Delta v}.$$

Consider an arbitrary volume $\mathscr{V} \subseteq \Omega_t$ in space. At a specific time t, the total amount of the property $\boldsymbol{Q}$ contained within $\mathscr{V}$ is given by the volume integral:

$$\boldsymbol{Q}_{\mathscr{V}}(t) = \iiint_{\mathscr{V}} \boldsymbol{\mu}(\vec{x}, t)\, \mathrm{d}v, \tag{5.11}$$

where $\mathrm{d}v$ denotes the differential volume element at position $\vec{x}$ and time t.

To understand how $\boldsymbol{Q}_{\mathscr{V}}(t)$ evolves over time, we need to compute its time derivative. However, the method of computing this derivative depends on whether the volume $\mathscr{V}$ is fixed in space (a control volume, $\mathscr{V} = \mathscr{V}^{\mathrm{cv}}$) or moves with the material (a control mass, $\mathscr{V} = \mathscr{V}_t^{\mathrm{cm}}$). This distinction leads us to two fundamental concepts: the local time derivative and the material time derivative of a volume integral.

5.3.1 Local time derivative of a volume integral over a control volume

When the volume $\mathscr{V}$ in integral (5.11) is a control volume $\mathscr{V}^{\mathrm{cv}}$, it remains fixed in space and does not move or deform with the material. In this case, integral (5.11) that represents the total amount of $\boldsymbol{Q}$ within $\mathscr{V} = \mathscr{V}^{\mathrm{cv}}$ at time t, becomes:

$$\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cv}}}(t) = \iiint_{\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\mu}(\vec{x}, t)\, \mathrm{d}v.$$

To determine how $\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cv}}}(t)$ changes over time within this fixed volume, we compute its local time derivative:

$$\begin{aligned}
\frac{\mathrm{d}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cv}}}(t)}{\mathrm{d}t} &= \frac{\mathrm{d}}{\mathrm{d}t} \iiint_{\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\mu}(\vec{x}, t)\, \mathrm{d}v \\
&= \lim_{\Delta t \to 0} \frac{\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cv}}}(t + \Delta t) - \boldsymbol{Q}_{\mathscr{V}^{\mathrm{cv}}}(t)}{\Delta t} \\
&= \lim_{\Delta t \to 0} \frac{1}{\Delta t} \left[\iiint_{\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\mu}(\vec{x}, t + \Delta t)\, \mathrm{d}v - \iiint_{\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\mu}(\vec{x}, t)\, \mathrm{d}v \right] \\
&= \iiint_{\mathscr{V}^{\mathrm{cv}}} \lim_{\Delta t \to 0} \frac{\boldsymbol{\mu}(\vec{x}, t + \Delta t) - \boldsymbol{\mu}(\vec{x}, t)}{\Delta t}\, \mathrm{d}v \\
&= \iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial \boldsymbol{\mu}(\vec{x}, t)}{\partial t}\, \mathrm{d}v.
\end{aligned}$$

This shows that, since the control volume $\mathscr{V}^{\text{cv}}$ is fixed in space, the domain of integration does not change with time. This allows us to interchange the time derivative and the integral, yielding:

$$\boxed{\frac{\mathrm{d}\boldsymbol{Q}_{\mathscr{V}^{\text{cv}}}(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\iiint_{\mathscr{V}^{\text{cv}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\upsilon = \iiint_{\mathscr{V}^{\text{cv}}} \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)}{\partial t}\,\mathrm{d}\upsilon.} \tag{5.12}$$

This result tells us that the rate of change of $\boldsymbol{Q}_{\mathscr{V}^{\text{cv}}}(t)$ within a fixed control volume is simply the integral of the local rate of change of $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)$ over the volume.

5.3.2 Material time derivative of a volume integral over a control mass: the Reynolds' transport formula

When the volume $\mathscr{V}$ in integral (5.11) is a control mass $\mathscr{V}_t^{\text{cm}}$, it moves and deforms with the material. In this case, integral (5.11) that represents the total amount of $\boldsymbol{Q}$ within $\mathscr{V}$ at time t, becomes:

$$\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t) = \iiint_{\mathscr{V}_t^{\text{cm}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\upsilon_t. \tag{5.13}$$

Here, both the integrand $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)$ and the domain of integration $\mathscr{V}_t^{\text{cm}}$ depend on time. To compute the rate of change of $\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t)$, we employ the material time derivative $\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t)}{\mathrm{D}t}$, since it accounts for the motion of the control mass. This derivative measures how $\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t)$ changes as the control mass $\mathscr{V}_t^{\text{cm}}$ moves through space. Mathematically, it is defined as:

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t)}{\mathrm{D}t} = \lim_{\Delta t \to 0} \frac{\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t + \Delta t) - \boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t)}{\Delta t}.$$

To handle the moving domain $\mathscr{V}_t^{\text{cm}}$ of the integral, we pull it back to a fixed reference configuration $\mathscr{V}_0^{\text{cm}} \subset \Omega_0$ using the motion $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$. The volume elements change, according to equation (3.35), as:

$$\mathrm{d}\upsilon_t = \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\mathscr{V}_0, \tag{5.14}$$

where $\det \boldsymbol{F}$ is the Jacobian determinant, representing the local volume dilation or contraction. Substituting this into $\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t)$, like we did with equation (3.36), we transform integral (5.13) into:

$$\boldsymbol{Q}_{\mathscr{V}^{\text{cm}}}(t) = \iiint_{\mathscr{V}_0^{\text{cm}}} \boldsymbol{\mu}_0(\vec{\boldsymbol{X}}, t) \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)\,\mathrm{d}\mathscr{V}_0,$$

where $\boldsymbol{\mu}_0(\vec{\boldsymbol{X}}, t) = \boldsymbol{\mu}(\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t), t)$ expresses the density in material coordinates.

Since the domain $\mathscr{V}_0^{\mathrm{cm}}$ is fixed, we can apply the local time derivative result from equation (5.12) to compute the material time derivative:

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \frac{\partial}{\partial t}\Big[\boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)\det \boldsymbol{F}(\vec{\boldsymbol{X}},t)\Big]\,\mathrm{d}\mathscr{V}_0.$$

Expanding the time derivative using the product rule, we obtain:

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \Big[\frac{\partial \boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)}{\partial t}\det \boldsymbol{F}(\vec{\boldsymbol{X}},t) + \boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)\frac{\partial}{\partial t}\det \boldsymbol{F}(\vec{\boldsymbol{X}},t)\Big]\,\mathrm{d}\mathscr{V}_0. \tag{5.15}$$

The rate of change of the Jacobian determinant is related to the velocity field $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$ through equation (3.97):

$$\frac{\partial}{\partial t}\det \boldsymbol{F}(\vec{\boldsymbol{X}},t) = \operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\det \boldsymbol{F}(\vec{\boldsymbol{X}},t).$$

Here, the divergence $\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$ quantifies the volumetric strain rate (expansion or compression) of the material. Substituting this result and equation (3.83), that is $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) = \vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)$, into (5.15), we obtain:

$$\begin{aligned}\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} &= \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \Big[\frac{\partial \boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)}{\partial t}\det \boldsymbol{F}(\vec{\boldsymbol{X}},t) + \boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)\operatorname{div}\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)\det \boldsymbol{F}(\vec{\boldsymbol{X}},t)\Big]\,\mathrm{d}\mathscr{V}_0 \\ &= \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \Big[\frac{\partial \boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)}{\partial t} + \boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)\operatorname{div}\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)\Big]\det \boldsymbol{F}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0.\end{aligned}$$

Returning to the current configuration $\mathscr{V}_t^{\mathrm{cm}}$, we replace $\det \boldsymbol{F}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0$ with $\mathrm{d}v_t$ using equation (5.14). Noting that $\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$, $\boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t) = \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)$, and $\frac{\partial \boldsymbol{\mu}_0(\vec{\boldsymbol{X}},t)}{\partial t} = \frac{\mathrm{D}\boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t}$, it results:

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\frac{\mathrm{D}\boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} + \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right]\mathrm{d}v_t, \tag{5.16}$$

where the material time derivative of $\boldsymbol{\mu}(\vec{\boldsymbol{x}},t)$ is given by equation (3.84):

$$\frac{\mathrm{D}\boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} = \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\partial t} + \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}\cdot\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t).$$

Now, substituting $\frac{\mathrm{D}\boldsymbol{\mu}}{\mathrm{D}t}$ we obtain:

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\partial t} + \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\partial \vec{\boldsymbol{x}}}\cdot\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) + \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right]\mathrm{d}v_t$$

and using the product rule for the divergence (A.4), that is $\operatorname{div}(\boldsymbol{\mu}\vec{\boldsymbol{v}}) = \nabla\boldsymbol{\mu}\cdot\vec{\boldsymbol{v}} + \boldsymbol{\mu}\operatorname{div}\vec{\boldsymbol{v}}$, we rewrite the integrand:

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\partial t} + \operatorname{div}\big(\boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big)\right]\mathrm{d}v_t$$

$$= \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)}{\partial t} \, \mathrm{d}v_t + \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \mathrm{div}\left(\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\right) \mathrm{d}v_t .$$

Here, the term $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)$ denotes the flux of $\boldsymbol{Q}$ passing through point $\vec{\boldsymbol{x}}$ at time t; it denotes the amount of $\boldsymbol{Q}$ flowing per unit time, through a unit area. It is measured in [units of $\boldsymbol{Q}$] $\cdot$ [time]$^{-1}$ $\cdot$ [area]$^{-1}$.

Applying the divergence theorem (see Appendix A.16) to the second term, we convert the volume integral into a surface integral over the boundary $\partial \mathscr{V}_t^{\mathrm{cm}}$:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \mathrm{div}\left(\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\right) \mathrm{d}v_t = \oiint_{\partial \mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}}$$

and in consequence,

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)}{\partial t} \, \mathrm{d}v_t + \oiint_{\partial \mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}}.$$

Finally, using (5.13), we arrive to the so called *Reynolds' transport formula** or *Reynolds' transport theorem**:

$$\begin{aligned} \frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} &= \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t) \, \mathrm{d}v_t\right) \\ &= \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)}{\partial t} \, \mathrm{d}v_t + \oiint_{\partial \mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}} \end{aligned} \tag{5.17}$$

Remember that this equation requires the function $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)$ to be continuously differentiable; moreover, $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)$ can be scalar-, vector- or tensor-valued.

This equation is named after the British engineer and physicist Osborne Reynolds (1842 – 1912), who proposed this equation in 1903. It is a fundamental result in continuum mechanics, fluid dynamics, and transport phenomena, as it provides a mathematical framework for analyzing how physical quantities change within a moving and deforming material volume. The Reynolds' transport formula tells us that the rate of change of the total quantity $\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)$ within a moving material volume $\mathscr{V}_t^{\mathrm{cm}}$, that travels with velocity $\vec{\boldsymbol{v}}$, consists of two contributions: a local rate of change of $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t)$ integrated over the volume $\mathscr{V}_t^{\mathrm{cm}}$ and the flux of $\boldsymbol{Q}$ across the moving boundary $\partial \mathscr{V}_t^{\mathrm{cm}}$ of the control mass (compare this contour integral with equation [5.4], making $\boldsymbol{\mu} = \rho$). This theorem will be used to derive the balance laws of continuum mechanics, such as the conservation of mass, momentum, and energy.

5.3.3 Reynolds' transport formula as a generalization of the Leibniz integral rule

Consider a function $f(x,t)$ such that $f(x,t)$ and its partial derivative $\frac{\partial f(x,t)}{\partial t}$ are continuous in x and t in the region bounded by $a(t) \le x \le b(t)$ and $t_0 \le t \le t_1$. Also suppose that the functions $-\infty < a(t)$ and $b(t) < \infty$ are both continuous and both have continuous derivatives for $t_0 \le t \le t_1$. The *Leibniz integral rule*[★] states that:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\int_{a(t)}^{b(t)} f(x,t)\,\mathrm{d}x\right) = \int_{a(t)}^{b(t)} \frac{\partial f(x,t)}{\partial t}\,\mathrm{d}x + f\big(b(t),t\big)\frac{\mathrm{d}b(t)}{\mathrm{d}t} - f\big(a(t),t\big)\frac{\mathrm{d}a(t)}{\mathrm{d}t}.$$

for $t_0 \le t \le t_1$. That is, the Leibniz integral rule allows differentiation under the integral sign. This rule is named after the German polymath and co-inventor of calculus Gottfried Wilhelm Leibniz (1646 – 1716).

Leibniz's rule can be seen as a special case of the Reynolds' transport formula. In fact, if we consider a one-dimensional control mass $\mathscr{V}_t^{\mathrm{cm}} = [a(t), b(t)]$ and let $\boldsymbol{\mu}(x,t) = f(x,t)$, then the Reynolds' transport formula reduces to the Leibniz integral rule. In this case, the boundary $\partial\mathscr{V}_t^{\mathrm{cm}}$ consists of two points $x = a(t)$ and $x = b(t)$, and the flux term in the Reynolds' transport formula corresponds to the contributions from these two moving boundaries.

Reynolds' transport formula can be generalized to an arbitrary moving and deforming domain $\mathscr{V}(t)$ whose boundary $\partial\mathscr{V}(t)$ moves with a relative velocity $\vec{\boldsymbol{v}}_r(\vec{\boldsymbol{x}},t)$ with respect to the fluid velocity $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$. In this case, the convective velocity in the flux term becomes the boundary velocity $\vec{\boldsymbol{v}}_r$, and the formula reads:

$$\frac{\mathrm{D}}{\mathrm{D}t}\iiint_{\mathscr{V}(t)} \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon_t = \iiint_{\mathscr{V}(t)} \frac{\partial \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\partial t}\,\mathrm{d}\upsilon_t + \oiint_{\partial\mathscr{V}(t)} \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\big(\vec{\boldsymbol{v}}_r(\vec{\boldsymbol{x}},t)\cdot\hat{\boldsymbol{n}}\big)\,\mathrm{d}S.$$

The reader interested in the derivation of this more general version of the Reynolds' transport formula can consult, for example, White (2015).

5.4 Conservation of mass

A fundamental attribute of every continuum body is its mass, denoted by the positive scalar m. Historically, *mass*[★] has been associated with the quantity of matter in a body and with its inertia, i.e., its resistance to changes in motion. In the International System of Units (SI), mass is measured in kilograms (kg). In this chapter, we adopt the continuum hypothesis, according to which mass is continuously distributed over any region Ω at each time t; therefore, concentrated masses of the type used in classical Newtonian mechanics are excluded.

The *principle of conservation of mass** is one of the most fundamental laws in physics. In its simplest form, it states that in any control mass, that is in any volume isolated from all exchanges of matter, the total mass must remain constant over time; in other words, it is a *conserved quantity**. For example, with reference to case 2:

$$\boxed{\begin{array}{c}\text{total mass inside the} \\ \text{control mass } \mathscr{V}_0^{\text{cm}}\end{array} \quad = \quad \begin{array}{c}\text{total mass inside the} \\ \text{control mass } \mathscr{V}_t^{\text{cm}}\end{array} \quad > \quad 0} \tag{5.18}$$

for all times $t \in T$; here T stands for the interval of time under study.

Moreover, mass can neither be created nor destroyed; it can only be redistributed or transformed. This principle is central to continuum mechanics and provides the foundation for analyzing fluid motion, deformation, and transport phenomena.

This principle also dictates that any mass leaving a control volume must correspond to a decrease in the mass contained within it. Mathematically, this is written for case 1 as (observe the minus sign):

$$\boxed{\begin{array}{c}\text{net mass outflow through} \\ \text{the control surface } \partial\mathscr{V}^{\text{cv}}\end{array} \quad = \quad - \begin{array}{c}\text{rate of mass} \\ \text{accumulation inside the} \\ \text{control volume } \mathscr{V}^{\text{cv}}.\end{array}} \tag{5.19}$$

and this is also equal to the rate of mass depletion inside the control volume $\mathscr{V}^{\text{cv}}$. Equations (5.18) and (5.19) are called the *mass balance equations**. The first one applies to control masses, whereas the latter is related to control volumes.

Example: toothpaste, tub with a water filtration

While modern physics (specifically, special relativity) has shown through Einstein's famous equation $E = mc^2$ that mass m and energy E can be interchanged, in the realm of continuum mechanics where velocities are much smaller than the speed of light c, classical mass conservation holds to extremely high precision. Also, we are neglecting nuclear reactions where mass is converted into energy.

In our analysis of mass conservation, we will also assume that no internal mass sources or sinks are present, so mass is neither created nor destroyed within the system and can only be transported; typical examples of sources and sinks are fluid injection into a reservoir and fluid extraction through production wells.

In this section, we apply the conservation-of-mass principle to each of the four flow models introduced in Section 5.1. Although this yields four apparently different expressions, we will later show in Section 5.4.6 that they are mathematically equivalent formulations of the same physical law. For brevity, we will write $m_0(\vec{\boldsymbol{X}})$ instead of $m_0(\vec{\boldsymbol{X}}, t_0)$ and $\rho_0(\vec{\boldsymbol{X}})$ instead of $\rho_0(\vec{\boldsymbol{X}}, t_0)$.

5.4.1 Case 1: conservation of mass for a finite control volume fixed in space

Consider the finite control volume $\mathscr{V}^{\text{cv}}$ that is shown in Figures 5.1 (case 1) and 5.3. This is a region of arbitrary shape and finite size that remains fixed in space while fluid moves through it. The boundary of this volume, the control surface $\partial\mathscr{V}^{\text{cv}}$, is where we will analyze the flow of mass into and out of our region of interest.

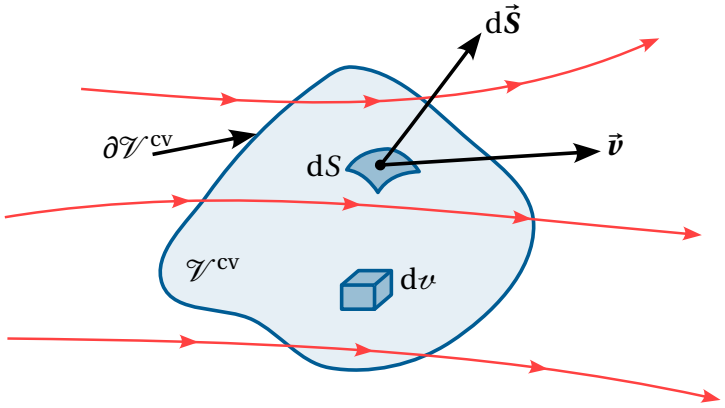


Figure 5.3: *Schematic representation of a fixed control volume $\mathscr{V}^{cv}$ bounded by control surface $\partial\mathscr{V}^{cv}$; this figure shows a differential volume element $\mathrm{d}\mathscr{v}$, a differential surface element $\mathrm{d}S \subset \partial\mathscr{V}^{cv}$ with its corresponding normal vector $\mathrm{d}\vec{\boldsymbol{S}} = \mathrm{d}S\hat{\boldsymbol{n}}$, a velocity vector $\vec{\boldsymbol{v}}$, and streamlines crossing the control surface.*

According to equation (5.4), the net mass flow rate through the control surface $\partial\mathscr{V}^{\text{cv}}$ is the surface integral:

$$\begin{pmatrix}\text{net mass outflow through}\\ \text{the control surface } \partial\mathscr{V}^{\text{cv}}\end{pmatrix} = \oiint_{\partial\mathscr{V}^{\text{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}}; \tag{5.20}$$

here $\rho(\vec{\boldsymbol{x}},t)$ represents the mass density in the spatial description.

At the same time, we must consider the mass m contained within the control volume itself. At any instant, the mass of fluid in a small volume element $\mathrm{d}\mathscr{v}$ is $\rho\,\mathrm{d}\mathscr{v}$. Thus, the total mass within the control volume $m(t)$ is the volume integral:

$$m(t) = \iiint_{\mathscr{V}^{\text{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v}. \tag{5.21}$$

The rate at which this mass changes, that is the rate of mass accumulation, is given by the time derivative:

$$\begin{pmatrix}\text{rate of mass accumulation}\\ \text{inside the control volume } \mathscr{V}^{\text{cv}}\end{pmatrix} = \frac{\mathrm{d}m(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\text{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v}\right). \tag{5.22}$$

By the principle of conservation of mass, any mass leaving the control volume must be accounted for by a decrease in the mass contained within it. Mathematically, and according to equation (5.19), the net mass outflow (5.20) must equal the negative rate of mass accumulation (5.22):

$$\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}} = -\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right)$$

or simply,

$$\boxed{\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}} = 0;} \tag{5.23}$$

this is the *conservative integral form of the continuity equation for mass conservation* deduced on the basis of a finite control volume $\mathscr{V}^{\mathrm{cv}}$ fixed in space.

Let us interpret each of the terms of equation (5.23):

- The first term represents the rate at which mass accumulates within the control volume $\mathscr{V}^{\mathrm{cv}}$; physically, it quantifies whether the mass inside the control volume is increasing or decreasing due to changes in density. If this term is positive, mass is accumulating inside the control volume; if negative, mass is leaving the control volume.
- The second term characterizes the net rate at which mass is flowing out of the control volume through its control surface $\partial\mathscr{V}^{\mathrm{cv}}$.

The sum of these two terms equals zero because mass is conserved. Any change in the total mass inside the control volume $\mathscr{V}^{\mathrm{cv}}$ must be balanced by the net mass flow across its boundary $\partial\mathscr{V}^{\mathrm{cv}}$. In other words, mass cannot be created or destroyed within the control volume; it can only enter or leave through the control surface.

This equation is particularly useful for analyzing systems where mass crosses boundaries, such as pipes, nozzles, turbines, and heat exchangers.

Example

5.4.2 Case 2: conservation of mass for a finite control mass moving with the flow

Let us now examine case 2, where our control mass $\mathscr{V}_t^{\mathrm{cm}}$ moves and deforms with the flow. This approach offers a different perspective from the fixed control volume we studied earlier, as we now follow the same group of particles through space and time.

Consider an infinitesimal volume element in the spatial description $\mathrm{d}\upsilon$ within $\mathscr{V}_t^{\mathrm{cm}}$ (Figure 5.1, case 2) that moves with the flow and that is located on position $\vec{\boldsymbol{x}}$ at time t. At the initial time t_0, this element was located at position $\vec{\boldsymbol{X}}$. As time progresses,

this element moves and possibly deforms, but it always contains the same particles. The mass of this infinitesimal volume element is given by $\rho(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\mathscr{v}$, where $\rho(\vec{\boldsymbol{x}}, t)$ represents the mass density at time t. Integrating over the entire control mass $\mathscr{V}_t^{\mathrm{cm}}$ gives us the total mass $m(t)$:

$$\begin{array}{c}\text{total mass inside the}\\ \text{control mass } \mathscr{V}_t^{\mathrm{cm}}\end{array} \quad = \quad m(t) \quad = \quad \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\mathscr{v}$$

This expression looks similar to what we saw in case 1, but there is an important difference. In our previous analysis with the fixed control volume, as denoted by equation (5.21), the region of integration $\mathscr{V}^{\mathrm{cv}}$ remained constant. Here, $\mathscr{V}_t^{\mathrm{cm}}$ changes with time as it follows the flow, meaning both its shape and size may vary. However, since we are following the same set of particles, one physical fact remains constant: the total mass of $\mathscr{V}_t^{\mathrm{cm}}$ cannot change with time.

Now, let us remember that the physical meaning of the material time derivative, that we studied in Section 3.21.3. This operator, denoted as $\frac{\mathrm{D}}{\mathrm{D}t}$, expresses how properties change in time for a specific group of particles as they move with the flow. Since our control mass contains the same particles throughout time, and mass must be conserved, the material time derivative of the total mass $\frac{\mathrm{D}m(t)}{\mathrm{D}t}$ must be zero:

$$\boxed{\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\mathscr{v}\right) = 0.} \tag{5.24}$$

for all $t \in T$. This equation is known as the *non-conservative integral form of the continuity equation for mass conservation* for a moving control mass.

The simplicity of equation (5.24) masks its power: it tells us that even as our control mass distorts and moves through space, the mass within it remains absolutely constant. This formulation emphasizes the Lagrangian perspective, where we follow the motion of specific material particles rather than observing changes in a fixed spatial region.

This equation is particularly useful when analyzing systems where the focus is on tracking a specific group of particles, such as: a parcel of fluid moving through a pipe, a droplet of liquid evaporating or condensing, or a solid object undergoing deformation.

Example

5.4.3 Case 3: conservation of mass for an infinitesimal control volume fixed in space

Let us now examine the flow model shown in Figure 5.1 (case 3), where we consider an infinitesimal control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ fixed in space with fluid moving through it.

The flow configuration is depicted in Figure 5.4, where the control volume is a small rectangular parallelepiped of dimensions $\mathrm{d}x$, $\mathrm{d}y$, and $\mathrm{d}z$, oriented along the Cartesian axis $\hat{\boldsymbol{i}}$, $\hat{\boldsymbol{j}}$, and $\hat{\boldsymbol{k}}$.

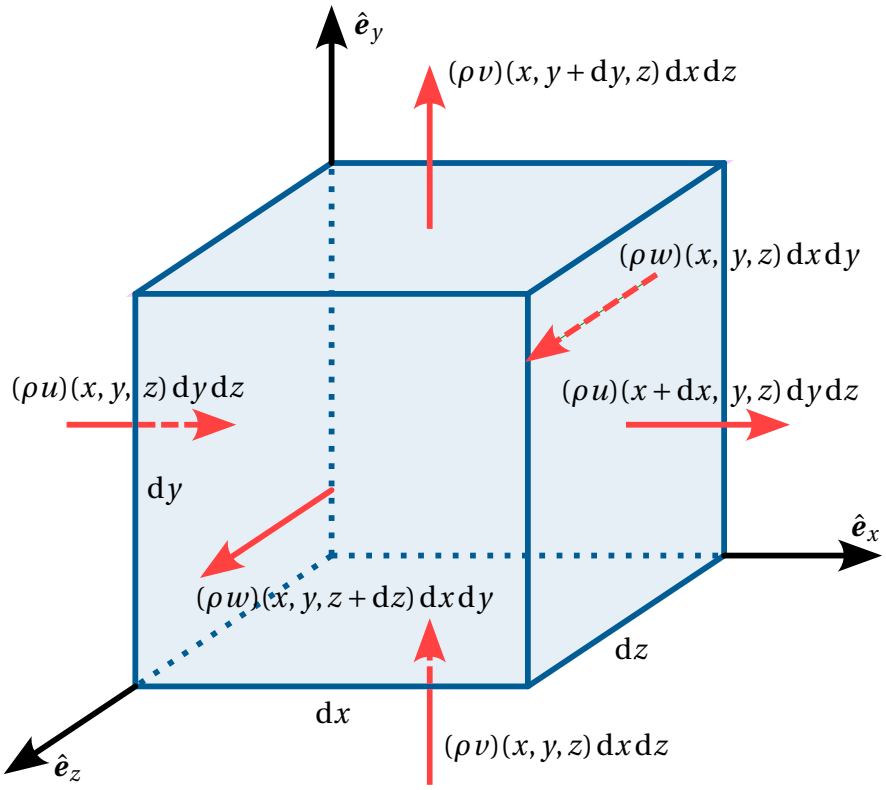


Figure 5.4: *An infinitesimal control volume fixed in space. The arrows indicate the inflow and outflow of mass through the faces of the control volume. The velocity field* $\vec{\boldsymbol{v}} = [u, v, w]^T$ *and density* ρ *vary with position* (x, y, z) *and time* t.

In the following, we will assume that the velocity field $\vec{\boldsymbol{v}} = [u, v, w]^T$ and density ρ are functions of both position $\vec{\boldsymbol{x}} = [x, y, z]^T$ and time t. In particular, we will suppose that both quantities are continuously differentiable functions, which is not always the case; this idealization, while powerful, does not universally apply to all physical phenomena. Some notable exceptions include:

Shock waves: they are a type of propagating disturbance that moves faster than the local speed of sound in the medium; they occur for example in explosions from bombs, supersonic flights, or meteor entries; they manifest as extremely thin regions across which physical properties change abruptly. Across a shock front, quantities such as density ρ, pressure p, and velocity $\vec{\boldsymbol{v}}$ experience discontinuous jumps, violating our continuity assumption.

Multiphase flows: when multiple phases coexist (e.g., liquid-gas systems), the interfaces between phases introduce natural discontinuities. Consider, for instance,

the interface between crude oil and natural gas in a petroleum reservoir, or the boundary between air and seawater in oceanographic studies. At these interfaces, properties such as density can change by orders of magnitude over distances approaching molecular scales.

For a differential surface element, the mass flow rate is given by $\rho V_n \, \mathrm{d}S$, where V_n is the velocity component normal to the surface. Let us analyze the flow through each pair of opposite faces.

Through the faces perpendicular to the x-axis, the mass flow entering through the left face is $\rho(t,x,y,z)\,u(t,x,y,z)\,\mathrm{d}y\,\mathrm{d}z$, while the flow leaving through the right face is $\rho(t,x+\mathrm{d}x,y,z)\,u(t,x+\mathrm{d}x,y,z)\,\mathrm{d}y\,\mathrm{d}z$. Now, the difference of mass flow between the left and the right face can be expressed as:

$$\Big[\rho(x+\mathrm{d}x,y,z,t)\,u(x+\mathrm{d}x,y,z,t)-\rho(x,y,z,t)\,u(x,y,z,t)\Big]\,\mathrm{d}y\,\mathrm{d}z$$

or equivalently, in terms of a partial derivative,

$$\begin{array}{c}\text{net mass outflow}\\ \text{in } x \text{ direction}\end{array} \quad = \quad \frac{\partial}{\partial x}\big(\rho(x,y,z,t)\,u(x,y,z,t)\big)\,\mathrm{d}\upsilon,$$

where $\mathrm{d}\upsilon = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ is the infinitesimal volume of $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ in the spatial description.

Similar expressions arise for the y and z directions. For example, the difference of mass flow between the bottom and the top faces can be expressed as:

$$\begin{array}{c}\text{net mass outflow}\\ \text{in } y \text{ direction}\end{array} \quad = \quad \frac{\partial}{\partial y}\big(\rho(x,y,z,t)\,v(x,y,z,t)\big)\,\mathrm{d}\upsilon,$$

and the difference of mass flow between the front and the back faces can be expressed as:

$$\begin{array}{c}\text{net mass outflow}\\ \text{in } z \text{ direction}\end{array} \quad = \quad \frac{\partial}{\partial z}\big(\rho(x,y,z,t)\,w(x,y,z,t)\big)\,\mathrm{d}\upsilon.$$

Thus, net mass outflow through the control volume is the sum of the contributions from all three directions is given by,

$$\begin{array}{c}\text{net mass outflow}\\ \text{through the control}\\ \text{surface } \partial\,\mathrm{d}\mathscr{V}^{\mathrm{cv}}\end{array} \quad = \quad \left[\frac{\partial(\rho u)}{\partial x}+\frac{\partial(\rho v)}{\partial y}+\frac{\partial(\rho w)}{\partial z}\right]\mathrm{d}\upsilon \quad = \quad \operatorname{div}\big(\rho\vec{\boldsymbol{v}}\big)\,\mathrm{d}\upsilon; \quad (5.25)$$

remember that the units of this quantity are kilograms per second, i.e., kg/s.

The mass of the control volume is $\mathrm{d}m(x,y,z,t) = \rho(x,y,z,t)\,\mathrm{d}\upsilon$. The rate at which this mass changes, that is the rate of mass accumulation, is given by the time derivative:

$$\begin{array}{c}\text{rate of mass}\\ \text{accumulation inside}\\ \text{the control volume}\\ \mathrm{d}\mathscr{V}^{\mathrm{cv}}\end{array} \quad = \quad \frac{\partial}{\partial t}\big(\mathrm{d}m(x,y,z,t)\big) \quad = \quad \frac{\partial\rho(x,y,z,t)}{\partial t}\,\mathrm{d}\upsilon. \qquad (5.26)$$

The principle of mass conservation dictates that any mass leaving the control volume must correspond to a decrease in the mass contained within it. Mathematically, and according to equation (5.19), the net mass outflow of the control volume (5.25) must equal the negative rate of mass accumulation within it (5.26):

$$\operatorname{div}\left(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right)\mathrm{d}\upsilon = -\frac{\partial\rho(\vec{\boldsymbol{x}},t)}{\partial t}\,\mathrm{d}\upsilon.$$

Dividing through by $\mathrm{d}\upsilon$ and rearranging terms, we obtain:

$$\boxed{\frac{\partial\rho(\vec{\boldsymbol{x}},t)}{\partial t} + \operatorname{div}\left(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right) = 0;} \tag{5.27}$$

this is the *conservative partial differential equation of the continuity equation for mass conservation* deduced on the basis of an infinitesimal control volume fixed in space.

This equation provides a local description of mass conservation within the fluid flow, valid at every point in the flow field, that is, it is valid for all $\vec{\boldsymbol{x}} \in \Omega_t$ at all times t. This is a form of the continuity equation that provides a spatial or Eulerian description of mass conservation, complementing the material or Lagrangian description that we will deduce in case 4.

Let us interpret each of the terms of equation (5.27):

- The term $\frac{\partial\rho}{\partial t}$ accounts for the local rate of change of density at a fixed point $\vec{\boldsymbol{x}}$ in space Ω_t over time. Physically, it quantifies whether the material at a specific location $\vec{\boldsymbol{x}}$ is becoming more dense (mass is accumulating) or less dense (mass is diminishing). If this term is positive, the density at that point is increasing; if negative, the density is decreasing.
- The term $\operatorname{div}\left(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right)$ represents the net rate at which mass is flowing out of a small region around the point $\vec{\boldsymbol{x}}$. The quantity $\rho\vec{\boldsymbol{v}}$ is the mass flux, which describes the amount of mass transported per unit area per unit time by the velocity field. The divergence operator $\operatorname{div}\left(\rho\vec{\boldsymbol{v}}\right)$ measures the net outward flow of mass from the point $\vec{\boldsymbol{x}}$. If $\operatorname{div}\left(\rho\vec{\boldsymbol{v}}\right) > 0$, mass is leaving the region; if $\operatorname{div}\left(\rho\vec{\boldsymbol{v}}\right) < 0$, mass is entering the region.

The sum of these two terms equals zero because mass is conserved. Any local change in density $\frac{\partial\rho}{\partial t}$ must be balanced by the net mass flux divergence $\operatorname{div}\left(\rho\vec{\boldsymbol{v}}\right)$. In other words, if mass is accumulating at a point (density increasing), it must be due to a net inflow of mass into the region; conversely, if mass is leaving the region (density decreasing), there must be a net outflow.

Example

5.4.4 Case 4: conservation of mass for an infinitesimal control mass moving with the flow

In this case, we consider the infinitesimal volume element $\mathrm{d}v_t := \mathrm{d}\mathscr{V}_t^{\mathrm{cm}} := \mathrm{d}v(\vec{\boldsymbol{x}}, t) = \mathrm{d}\mathscr{V}(\vec{\boldsymbol{x}}, t)$ that represents a control mass which moves with the fluid flow and that was located on position $\vec{\boldsymbol{X}}$ at time t_0. (Figure 5.1, case 4). We assume that the mass of this element, denoted by $\mathrm{d}m$, can be calculated as the product of its density ρ and its volume $\mathrm{d}v$, that is, $\mathrm{d}m(\vec{\boldsymbol{x}}, t) = \rho(\vec{\boldsymbol{x}}, t)\,\mathrm{d}v(\vec{\boldsymbol{x}}, t)$; using this definition, we can state the differential form of equation (5.18) as:

$$\mathrm{d}m_0(\vec{\boldsymbol{X}}, t_0) = \mathrm{d}m(\vec{\boldsymbol{x}}, t) > 0$$

for all times $t \in T$; according to the principle of conservation of mass, if we follow an infinitesimal volume of material through its motion, its volume $\mathrm{d}\mathscr{V}(\vec{\boldsymbol{x}}, t)$ and density ρ may vary, but its total mass $\rho\,\mathrm{d}\mathscr{V}(\vec{\boldsymbol{x}}, t)$ will stay constant as the differential element moves along with the flow; so, expressing the mass of a differential element in terms of the initial $\mathrm{d}\mathscr{V}_0 := \mathrm{d}\mathscr{V}(\vec{\boldsymbol{X}}, t_0)$ and current $\mathrm{d}v_t := \mathrm{d}v(\vec{\boldsymbol{x}}, t)$ volumes, and in terms of the initial $\rho_0(\vec{\boldsymbol{X}}, t_0)$ and current $\rho(\vec{\boldsymbol{x}}, t)$ mass densities, we obtain:

$$\rho_0(\vec{\boldsymbol{X}}, t_0) \underbrace{\mathrm{d}\mathscr{V}(\vec{\boldsymbol{X}}, t_0)}_{=\mathrm{d}\mathscr{V}_0} = \rho_0(\vec{\boldsymbol{x}}, t) \underbrace{\mathrm{d}v(\vec{\boldsymbol{x}}, t)}_{=\mathrm{d}v_t}.$$

This equation says that for a constant mass, when density increases, the volume decreases. Thus, using the definition of the volume ratio $J(\vec{\boldsymbol{x}}, t) = \mathrm{d}v_t / \mathrm{d}\mathscr{V}_0 = \det \boldsymbol{F}(\vec{\boldsymbol{x}}, t) > 0$ (see Section 3.11.3), the principle of conservation of mass requires that

$$\boxed{\rho_0(\vec{\boldsymbol{X}}) = \rho_0(\vec{\boldsymbol{x}}, t) \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t),} \tag{5.28}$$

which holds for all $\vec{\boldsymbol{X}} \in \Omega_0$. Since $\rho_0(\vec{\boldsymbol{x}}, t) = \rho_0(\vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}}, t), t) =: \rho(\vec{\boldsymbol{x}}, t)$, we can write the last equation in terms of the density in the spatial description as

$$\boxed{\rho_0(\vec{\boldsymbol{X}}) = \rho(\vec{\boldsymbol{x}}, t) \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t),} \tag{5.29}$$

Let us take the material time derivative of the material field $\rho_0(\vec{\boldsymbol{X}}, t_0)$. Since $\rho_0(\vec{\boldsymbol{X}}, t_0)$ does not change with time,

$$\frac{\mathrm{D}\rho_0(\vec{\boldsymbol{X}}, t_0)}{\mathrm{D}t} = \frac{\mathrm{D}\big(\rho_0(\vec{\boldsymbol{x}}, t) \det \boldsymbol{F}(\vec{\boldsymbol{x}}, t)\big)}{\mathrm{D}t} = \frac{\mathrm{D}\big(\rho(\vec{\boldsymbol{x}}, t) \det \boldsymbol{F}(\vec{\boldsymbol{x}}, t)\big)}{\mathrm{D}t} = 0.$$

Now, from equation (5.28), using the product rule, we obtain that:

$$\frac{\mathrm{D}\rho_0(\vec{\boldsymbol{X}}, t_0)}{\mathrm{D}t} = \frac{\partial \rho_0(\vec{\boldsymbol{X}}, t_0)}{\partial t}$$

$$
\begin{aligned}
&= \frac{\partial\left(\rho_0(\vec{\boldsymbol{x}},t)\det \boldsymbol{F}(\vec{\boldsymbol{x}},t)\right)}{\partial t} \\
&= \frac{\partial \rho_0(\vec{\boldsymbol{x}},t)}{\partial t}\det \boldsymbol{F}(\vec{\boldsymbol{x}},t) + \frac{\partial \det \boldsymbol{F}(\vec{\boldsymbol{x}},t)}{\partial t}\rho_0(\vec{\boldsymbol{x}},t) \\
&= 0
\end{aligned}
$$

replacing equation (3.97), that is $\frac{\partial}{\partial t}\det \boldsymbol{F}(\vec{\boldsymbol{X}},t) = \det \boldsymbol{F}(\vec{\boldsymbol{X}},t)\,\mathrm{div}\,\vec{\boldsymbol{V}}(\vec{\boldsymbol{x}},t)$, we obtain:

$$
\begin{aligned}
\frac{\partial \rho_0(\vec{\boldsymbol{X}},t_0)}{\partial t} &= \frac{\partial \rho_0(\vec{\boldsymbol{x}},t)}{\partial t}\det \boldsymbol{F}(\vec{\boldsymbol{x}},t) + \det \boldsymbol{F}(\vec{\boldsymbol{X}},t)\,\mathrm{div}\,\vec{\boldsymbol{V}}(\vec{\boldsymbol{x}},t)\rho_0(\vec{\boldsymbol{x}},t) \\
&= \left(\frac{\partial \rho_0(\vec{\boldsymbol{x}},t)}{\partial t} + \rho_0(\vec{\boldsymbol{x}},t)\,\mathrm{div}\,\vec{\boldsymbol{V}}(\vec{\boldsymbol{x}},t)\right)\det \boldsymbol{F}(\vec{\boldsymbol{x}},t) \\
&= 0
\end{aligned}
$$

and since in general $\det \boldsymbol{F}(\vec{\boldsymbol{x}},t) > 0$,

$$
\boxed{\frac{\partial \rho_0(\vec{\boldsymbol{x}},t)}{\partial t} + \rho_0(\vec{\boldsymbol{x}},t)\,\mathrm{div}\,\vec{\boldsymbol{V}}(\vec{\boldsymbol{x}},t) = 0} \tag{5.30}
$$

this is the *non-conservative partial differential equation of the continuity equation* deduced on the basis of an infinitesimal control volume moving with flow, which provides a material local description of mass conservation within the fluid flow, valid at every point in the flow field.

Equation (5.30) can be written in the spatial description by making the equivalences $\frac{\partial \rho_0(\vec{\boldsymbol{x}},t)}{\partial t} = \frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t}$, $\rho_0(\vec{\boldsymbol{x}},t) = \rho(\vec{\boldsymbol{\chi}}^{-1}(\vec{\boldsymbol{x}},t),t) = \rho(\vec{\boldsymbol{x}},t)$, and $\vec{\boldsymbol{V}}(\vec{\boldsymbol{x}},t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$, so that:

$$
\boxed{\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} + \rho(\vec{\boldsymbol{x}},t)\,\mathrm{div}\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) = 0.} \tag{5.31}
$$

Equations (5.30) and (5.31) are forms of the continuity equation that provide a material or Lagrangian description of mass conservation, complementing the spatial or Eulerian description of case 3.

Let us interpret each of the terms of equations (5.30) and (5.31):

- The terms $\frac{\partial \rho_0(\vec{\boldsymbol{x}},t)}{\partial t}$ and $\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t}$ represent the rate of change of density as observed by an observer moving with the material particle. Physically, it quantifies how the density of a specific parcel of material changes over time as it moves through space. If these terms are nonzero, they indicates that the material is either compressing (density increasing) or expanding (density decreasing).

- The term $\operatorname{div}\vec{\boldsymbol{V}}(\vec{\boldsymbol{x}},t)$ is the divergence of the velocity field, which measures the net rate of expansion ($\operatorname{div}\vec{\boldsymbol{V}}>0$) or contraction ($\operatorname{div}\vec{\boldsymbol{V}}<0$) of the material point labelled as $\vec{\boldsymbol{X}}$. If $\operatorname{div}\vec{\boldsymbol{V}}>0$, the material is diverging (expanding), leading to a decrease in density. If $\operatorname{div}\vec{\boldsymbol{V}}<0$, the material is converging (compressing), leading to an increase in density. Multiplying this divergence by the density ρ_0 scales the effect of expansion or compression by the amount of mass present in the region. A similar interpretation can be done for $\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$. We will discuss in detail the physical interpretation of the divergence of the velocity field in Section 5.5.

The sum of these two terms equals zero because mass is conserved. Any change in density $\frac{\mathrm{D}\rho}{\mathrm{D}t}$ must be balanced by the effects of velocity divergence $\operatorname{div}\vec{\boldsymbol{V}}$. In other words, if the material expands (positive divergence), its density decreases, and vice versa. These changes are coupled in such a way that the total mass remains constant.

Example

5.4.5 Particularization of the conservation of mass equation for special cases

Fix this subsection. It has errors.

In the preceding sections, we derived several forms of the continuity equation that express the principle of mass conservation. Now, we will examine how these equations simplify under specific physical conditions that frequently arise in practical applications. We will examine the implications of homogeneity, incompressibility, and steady-state flow on the conservation of mass equation for each of the four cases discussed earlier.

Let us first clarify these important concepts:

- A medium is *homogeneous** when its physical properties (such as density, elasticity modulus, thermal conductivity, etc.) are identical at all locations within the material. In the context of mass conservation, this means the reference density does not vary with position, i.e., $\frac{\partial\rho_0(\vec{\boldsymbol{X}},t_0)}{\partial\vec{\boldsymbol{X}}}=\vec{\boldsymbol{0}}$, so that $\rho_0(\vec{\boldsymbol{X}},t_0)=\rho_0=\text{const}$; this means that all material points share the same density in the undeformed state; take into account that the reference density is typically defined at t_0, and that homogeneity is defined based on the *Lagrangian description*. In a homogeneous medium, even though $\rho_0(\vec{\boldsymbol{X}},t_0)$ is constant with respect to $\vec{\boldsymbol{X}}$, the *Eulerian density field* $\rho(\vec{\boldsymbol{x}},t)$ might still vary spatially if the deformation itself is non-homogeneous. This follows directly from equation (5.29), which relates the two densities through the deformation gradient $\boldsymbol{F}$, so that

$\rho(\vec{\boldsymbol{x}}, t) = \rho_0(\vec{\boldsymbol{X}}, t_0)/\det \boldsymbol{F}(\vec{\boldsymbol{X}}, t) = \rho_0/\det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)$. If the deformation is non-homogeneous, meaning different parts of the body are compressed or stretched by different amounts, then $\det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)$ varies from point to point, and consequently $\rho(\vec{\boldsymbol{x}}, t)$ is spatially non-uniform even though ρ_0 is constant. In conclusion, homogeneity of the material must be defined in the Lagrangian description.

- A medium is *incompressible** when the density of each material particle remains constant along its trajectory; this constraint implies that the *material time derivative* of the density must vanish, i.e., $\frac{\mathrm{D}\rho_0(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} = \frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} = 0$; thus, if we follow a specific fluid particle, its density never changes regardless of the deformations it experiences. Incompressibility does not imply rigidity; the material can deform, but its volume remains constant. Such deformations are characteristic of *isochoric* (volume-preserving) motions; in this case, the determinant of the deformation gradient satisfies $\det \boldsymbol{F}(\vec{\boldsymbol{X}}, t) = 1$, for all $\vec{\boldsymbol{X}} \in \Omega_0$ and all $t \in T$. Many liquids, such as water under normal conditions, can be treated as incompressible with excellent accuracy. Even some solids may be modeled as incompressible when volume changes are negligible compared to shape changes.

- A flow is said to be in a *steady-state** if its *local time derivatives* vanish for all $\vec{\boldsymbol{x}} \in \Omega_t$; in other words, the properties of the flow, such as density and velocity, at a specific point $\vec{\boldsymbol{x}}$ do not change with time. Mathematically, this means that $\frac{\partial \rho(\vec{\boldsymbol{x}},t)}{\partial t} = 0$ and $\frac{\partial \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\partial t} = \vec{\boldsymbol{0}}$, and this implies that $\rho(\vec{\boldsymbol{x}}, t) = \rho(\vec{\boldsymbol{x}})$ and $\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}})$. Under steady-state conditions, the continuity equation simplifies significantly, as the time-dependent terms vanish.

Let us now see how the conservation of mass equations derived in the previous sections simplify under these conditions.

Case 1: fixed finite control volume

Recall the conservative integral form of the continuity equation for a fixed control volume $\mathscr{V}^{\mathrm{cv}}$ (equation [5.23]):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\mathscr{v}\right) + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}}, t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\cdot \mathrm{d}\vec{\boldsymbol{S}} = 0.$$

Particularization for incompressible flow: When the fluid is incompressible, $\rho(\vec{\boldsymbol{x}}, t) = \rho_0$ is constant for all $\vec{\boldsymbol{x}} \in \Omega_t$ and all $t \in T$. Therefore, ρ_0 can be factored out of both integrals:

$$\rho_0 \frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \mathrm{d}\mathscr{v}\right) + \rho_0 \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\cdot \mathrm{d}\vec{\boldsymbol{S}} = 0.$$

Since $\mathscr{V}^{\text{cv}}$ is fixed in space, its volume does not change with time, so $\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\text{cv}}} \mathrm{d}\upsilon\right) = 0$. Dividing through by $\rho_0 \neq 0$, we obtain:

$$\boxed{\oiint_{\partial\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \cdot \mathrm{d}\vec{\boldsymbol{S}} = 0.} \tag{5.32}$$

Equation (5.32) implies that for incompressible flow through a fixed control volume, the net volume flow rate across the control surface must be zero. In practical terms, this means that the volume of fluid entering the control volume must exactly equal the volume of fluid exiting it, a principle familiar to anyone who has observed water flowing through a pipe system.

Particularization for steady-state conditions: In steady-state conditions, the flow properties do not change with time. Mathematically, this means local time derivatives are null: $\frac{\partial \rho}{\partial t} = 0$ and $\frac{\partial \vec{\boldsymbol{v}}}{\partial t} = \vec{\boldsymbol{0}}$. Under these conditions, the first term in equation (5.23) vanishes, yielding:

$$\boxed{\oiint_{\partial\mathscr{V}^{\text{cv}}} \rho(\vec{\boldsymbol{x}})\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}) \cdot \mathrm{d}\vec{\boldsymbol{S}} = 0.} \tag{5.33}$$

Note that we have dropped the explicit time dependence since the flow is steady. Equation (5.33) tells us that in steady-state flow, the net mass flux across the control surface is zero, the rate at which mass enters the control volume exactly equals the rate at which it exits.

Particularization for steady-state, incompressible flow conditions: When both conditions apply, we combine equations (5.32) and (5.33) to get:

$$\oiint_{\partial\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}) \cdot \mathrm{d}\vec{\boldsymbol{S}} = 0.$$

This equation emphasizes that for steady, incompressible flow, the principle of volume conservation applies at all times.

Case 2: moving finite control mass

For a finite control mass $\mathscr{V}_t^{\text{cm}}$ moving with the flow, we derived the non-conservative integral form (equation [5.24]):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\text{cm}}} \rho(\vec{\boldsymbol{x}}, t)\, \mathrm{d}\upsilon\right) = 0.$$

Particularization for incompressible flow: When the fluid is incompressible, $\rho(\vec{\boldsymbol{x}}, t) = \rho_0$ is constant, so:

$$\rho_0 \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \mathrm{d}\upsilon\right) = 0.$$

Since $\rho_0 \neq 0$, this implies:

$$\boxed{\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \mathrm{d}\upsilon\right) = 0.} \tag{5.34}$$

Equation (5.34) expresses a fundamental property of incompressible flow: the volume of any material region remains constant as it moves and deforms with the flow. This is the mathematical statement of volume conservation for a control mass, which is a direct consequence of mass conservation when density is constant. This reflects the physical reality of fluids like water or liquid metals, which resist compression even under pressure. Mathematically, this is equivalent to $\det(\boldsymbol{F}) = 1$, ensuring local volume preservation.

Case 3: fixed infinitesimal control volume

For an infinitesimal control volume fixed in space, we derived the conservative partial differential equation (equation [5.27]):

$$\frac{\partial \rho(\vec{\boldsymbol{x}}, t)}{\partial t} + \operatorname{div}\left(\rho(\vec{\boldsymbol{x}}, t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\right) = 0.$$

Particularization for incompressible flow: When the fluid is incompressible, $\rho(\vec{\boldsymbol{x}}, t) = \rho_0$ is constant. Therefore:

$$\frac{\partial \rho_0}{\partial t} + \operatorname{div}\left(\rho_0\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\right) = 0.$$

Since ρ_0 is constant, $\frac{\partial \rho_0}{\partial t} = 0$ and $\operatorname{div}\left(\rho_0\,\vec{\boldsymbol{v}}\right) = \rho_0 \operatorname{div}\vec{\boldsymbol{v}}$. Dividing through by $\rho_0 \neq 0$, we obtain:

$$\boxed{\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = 0.} \tag{5.35}$$

Equation (5.35) is the differential form of the continuity equation for incompressible flow. It states that the velocity field must be divergence-free, which is a local expression of volume conservation. Physically, it means that at every point in the flow, there can be no net accumulation or depletion of fluid.

Particularization for steady-state conditions: In steady-state conditions, $\frac{\partial \rho}{\partial t} = 0$. The continuity equation then reduces to:

$$\boxed{\operatorname{div}\left(\rho(\vec{\boldsymbol{x}})\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}})\right) = 0.} \tag{5.36}$$

Equation (5.36) states that for steady flow, the divergence of the mass flux must be zero at every point. This ensures that the net mass flux into any region equals the net mass flux out.

Particularization for steady-state, incompressible flow conditions: Combining the conditions of steady-state and incompressibility, we get:

$$\operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}) = 0.$$

This equation, identical in form to equation (5.35) but without time dependence, represents the most simplified form of the continuity equation for the common case of steady, incompressible flow.

Case 4: moving infinitesimal control mass

For an infinitesimal control mass $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ moving with the flow, we derived the non-conservative partial differential equation (equation [5.31]):

$$\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} + \rho(\vec{\boldsymbol{x}}, t) \operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = 0.$$

Particularization for incompressible flow: For incompressible flow, $\frac{\mathrm{D}\rho}{\mathrm{D}t} = 0$ by definition. The continuity equation then simplifies to:

$$\boxed{\rho(\vec{\boldsymbol{x}}) \operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = 0.} \tag{5.37}$$

Since $\rho(\vec{\boldsymbol{x}}) > 0$ for any physical fluid, equation (5.37) implies:

$$\boxed{\operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = 0.} \tag{5.38}$$

Interestingly, we arrive at the same mathematical condition as in the fixed infinitesimal control volume case. Equation (5.38) represents a kinematic constraint on the velocity field for incompressible flow, regardless of whether we adopt a Lagrangian or Eulerian perspective. This convergence of perspectives highlights the fundamental nature of the divergence-free condition for incompressible flows.

In the material description, equation (5.38) can be written as:

$$\operatorname{div} \vec{\boldsymbol{V}}(\vec{\boldsymbol{x}}, t) = 0,$$

which characterizes a volume-preserving (isochoric) motion in the material description.

Physical interpretation and practical significance

The simplifications we have derived for special cases are not merely mathematical conveniences; they reflect profound physical principles that govern fluid behavior. Let us reflect on their broader significance:

The divergence-free condition: The condition $\operatorname{div} \vec{\boldsymbol{v}} = 0$ for incompressible flow is arguably one of the most important equations in fluid mechanics. It serves as a constraint on permissible velocity fields and is central to numerous analytical and computational methods for solving flow problems.

Volume conservation vs. mass conservation: For incompressible flows, mass conservation manifests as volume conservation. This principle finds application in diverse fields, from hydraulic engineering to biofluid mechanics.

Mass flux continuity: In steady flows, the condition $\operatorname{div}(\rho \vec{\boldsymbol{v}}) = 0$ ensures that mass flux is conserved across any closed surface. This principle is foundational for analyzing steady flow systems, such as pipe networks, blood circulation, and ventilation systems.

The simplified equations for incompressible and steady flows significantly reduce computational complexity in numerical simulations. For instance, consider water flowing through a pipe with varying cross-sectional area $A(x)$. If we model water as incompressible and the flow as steady, then equation (5.32) implies:

$$\sum_{\text{inputs}} Q_{\text{in}} = \sum_{\text{outputs}} Q_{\text{out}},$$

where $Q = Av$ is the volumetric flow rate. For a pipe with one inlet and one outlet, this simplifies to $A_1 v_1 = A_2 v_2$, a result known as the continuity equation in elementary fluid mechanics. This explains why water speeds up when flowing through a narrowed section of pipe, a phenomenon we experience when partially blocking a garden hose with our thumb.

It is worth noting that while many liquids can be accurately modeled as incompressible under normal conditions, all real fluids exhibit some degree of compressibility. For water, the bulk modulus K is approximately 2.2 GPa, meaning that a pressure p increase of 1 atmosphere (that is, 101325 Pa) causes only about a $p/K \approx 0.005\%$

decrease in volume. This justifies the incompressible approximation for most water flow problems. However, in situations involving high pressures, shock waves, or multiphase flow, compressibility effects become significant and must be accounted for in the analysis.

5.4.6 Mathematical equivalence between all four cases

The four forms of the continuity equation we have derived, and that are summarized in Figure 5.5, represent different perspectives on mass conservation, yet they are mathematically equivalent expressions for the continuity equation. Let us explore their relationships and demonstrate their equivalence systematically.

Our four equations can be classified along two important dimensions:

- By mathematical form: Integral equations (5.23), (5.24) vs. Differential equations (5.27), (5.31)
- By formulation type: Conservation form (5.23), (5.27) vs. Non-conservation form (5.24), (5.31)

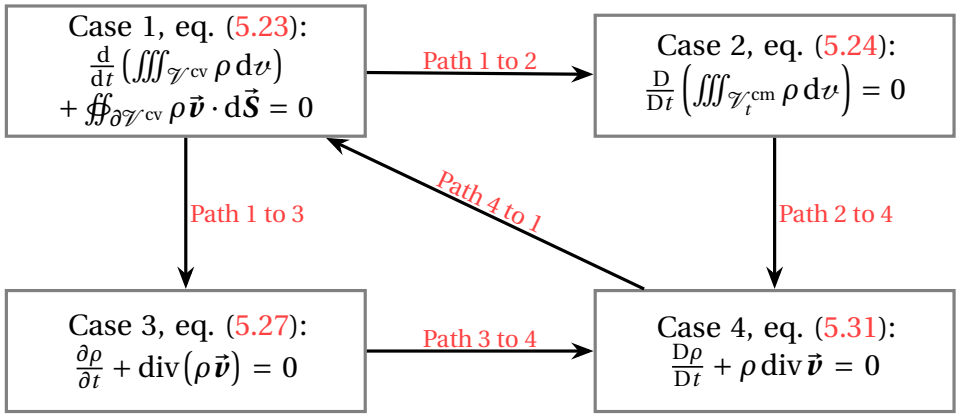


Figure 5.5: *The four different variants of the continuity equation.*

Any of these equations can be deduced by mathematical manipulation from any of the others, as we will illustrate in the next, for the paths shown in Figure 5.5.

Path 1 to 3: from equation (5.23) to equation (5.27)

In this demonstration we will illustrate that conservative (Eulerian) forms of the continuity equation are equivalent when we pass from the integral form (5.23) to the partial differential equation form (5.27).

We begin with the integral conservation form (5.23):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}} = 0.$$

The proof proceeds in four key steps: firstly, since the control volume is fixed in space, and does not change with time, we can move the time derivative inside the volume integral, using (5.12):

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial \rho(\vec{\boldsymbol{x}}, t)}{\partial t} \,\mathrm{d}\upsilon + \oiint_{\partial \mathscr{V}^{\mathrm{cv}}} \big(\rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\big) \cdot \mathrm{d}\vec{\boldsymbol{S}} = 0;$$

secondly, we apply the divergence theorem (Appendix A.16) to transform the surface integral into a volume integral:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial \rho(\vec{\boldsymbol{x}}, t)}{\partial t} \,\mathrm{d}\upsilon + \iiint_{\mathscr{V}^{\mathrm{cv}}} \mathrm{div}\big(\rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\big) \,\mathrm{d}\upsilon = 0;$$

thirdly, since both integrals are over the same volume $\mathscr{V}^{\mathrm{cv}}$, we can combine them:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \left[\frac{\partial \rho(\vec{\boldsymbol{x}}, t)}{\partial t} + \mathrm{div}\big(\rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\big) \right] \mathrm{d}\upsilon = 0;$$

finally, since the integration region $\mathscr{V}^{\mathrm{cv}}$ is arbitrarily drawn from Ω, then according to the localization theorem (see Appendix A.18), the only way for the integral to be zero for all $\mathscr{V}^{\mathrm{cv}} \subseteq \Omega$, is that the integrand is null at every point $\vec{\boldsymbol{x}} \in \Omega$; so, it follows the conservative partial differential equation of the continuity equation (5.27):

$$\frac{\partial \rho(\vec{\boldsymbol{x}}, t)}{\partial t} + \mathrm{div}\big(\rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\big) = 0.$$

Path 3 to 4: from equation (5.27) to equation (5.31)

Now, let us illustrate that the partial differential equation forms of the continuity equations are equivalent when we pass from the conservation form to its non-conservation counterpart.

To show that the conservation form (5.27):

$$\frac{\partial \rho(\vec{\boldsymbol{x}}, t)}{\partial t} + \mathrm{div}\big(\rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\big) = 0$$

is mathematically equivalent to the non-conservation form (5.31), we use the product rule for divergence (A.4), that is,

$$\mathrm{div}(\rho\, \vec{\boldsymbol{v}}) = \nabla \rho \cdot \vec{\boldsymbol{v}} + \rho\, \mathrm{div}\, \vec{\boldsymbol{v}}.$$

Substituting this expression into (5.27), we arrive to:

$$\frac{\partial \rho(\vec{\boldsymbol{x}}, t)}{\partial t} + \nabla \rho(\vec{\boldsymbol{x}}, t) \cdot \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) + \rho(\vec{\boldsymbol{x}}, t)\, \mathrm{div}\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = 0;$$

recognizing that the first two terms in our equation constitute the material time derivative of the density ρ, that is $\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t}$ according to equation (3.84), we arrive to the non-conservation form (5.31):

$$\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} + \rho(\vec{\boldsymbol{x}}, t)\, \mathrm{div}\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = 0.$$

Path 1 to 2: from equation (5.23) to equation (5.24)

We begin with the integral form of the conservation of mass equation expressed over a fixed control volume $\mathscr{V}^{\mathrm{cv}}$, that is equation (5.23), which is rewritten here for convenience:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}} = 0.$$

Since the control volume $\mathscr{V}^{\mathrm{cv}}$ is fixed in space, its boundaries do not move or deform with time. This allows us to interchange the time derivative and the volume integral, yielding, according to equation (5.12):

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial\rho(\vec{\boldsymbol{x}},t)}{\partial t}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}} = 0.$$

To recover the material description, we need to rewrite this equation in terms of a control mass $\mathscr{V}_t^{\mathrm{cm}}$, which follows the material particles as they move and deform. Since the previous equation holds for every fixed spatial region $\mathscr{V}^{\mathrm{cv}} \subset \Omega$, it holds in particular for the region instantaneously occupied by $\mathscr{V}_t^{\mathrm{cm}}$ at each time t, allowing us to replace $\mathscr{V}^{\mathrm{cv}}$ by $\mathscr{V}_t^{\mathrm{cm}}$:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial\rho(\vec{\boldsymbol{x}},t)}{\partial t}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}} = 0.$$

Here, the domain of integration $\mathscr{V}_t^{\mathrm{cm}}$ is a control mass that changes with time because it moves with the material. However, the physical meaning of the equation remains the same: the total rate of change of mass inside the control mass balances the net mass flux across its boundary.

To simplify the analysis, we use the Reynolds' transport formula (5.17), which relates the material derivative of an integral over a moving control mass to integrals over the same domain. Specifically, the theorem states:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial\rho(\vec{\boldsymbol{x}},t)}{\partial t}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}}.$$

By comparing this result with the earlier equation, we see that both expressions are identical. Thus, the conservation of mass equation can be written compactly as equation (5.24):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) = 0.$$

Path 2 to 4: from equation (5.24) to equation (5.31)

We now demonstrate how the material integral form (5.24):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) = 0.$$

transforms into the material differential form (5.31). This transformation is particularly interesting as it connects integral and differential perspectives while maintaining the material description.

In this demonstration, the control mass $\mathscr{V}_t^{\mathrm{cm}}$ is not fixed in space but moves with the material particles. Therefore, we employ equations (5.13) and (5.16) substituting $\boldsymbol{\mu}$ by ρ:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} + \rho(\vec{\boldsymbol{x}},t)\,\mathrm{div}\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right]\mathrm{d}\upsilon = 0.$$

Since the integration region $\mathscr{V}_t^{\mathrm{cm}}$ is arbitrarily drawn from Ω_t, the only way for the second integral to be zero for all $\mathscr{V}_t^{\mathrm{cm}} \subseteq \Omega_t$, is that the integrand is null at every point $\vec{\boldsymbol{x}} \in \Omega_t$, according to the localization theorem. Thus we arrive to the non-conservation form (5.31):

$$\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} + \rho(\vec{\boldsymbol{x}},t)\,\mathrm{div}\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) = 0.$$

Path 4 to 1: from equation (5.31) to equation (5.23)

Starting with equation (5.31):

$$\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} + \rho(\vec{\boldsymbol{x}},t)\,\mathrm{div}\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) = 0,$$

we integrate over a fixed control volume $\mathscr{V}^{\mathrm{cv}}$:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \left[\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} + \rho(\vec{\boldsymbol{x}},t)\,\mathrm{div}\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right]\mathrm{d}\upsilon = 0. \tag{5.39}$$

The transformation proceeds through the following steps: firstly, we use the definition of the material time derivative (3.84), so that

$$\frac{\mathrm{D}\rho}{\mathrm{D}t} = \frac{\partial\rho}{\partial t} + \vec{\nabla}\rho\cdot\vec{\boldsymbol{v}};$$

then, we substitute this result into (5.39):

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \left[\frac{\partial\rho}{\partial t} + \vec{\nabla}\rho\cdot\vec{\boldsymbol{v}} + \rho\,\mathrm{div}\,\vec{\boldsymbol{v}}\right]\mathrm{d}\upsilon = 0.$$

and using the product rule for the divergence (A.4), that is $\mathrm{div}(\rho\vec{\boldsymbol{v}}) = \nabla\rho\cdot\vec{\boldsymbol{v}} + \rho\,\mathrm{div}\,\vec{\boldsymbol{v}}$, we get

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial\rho}{\partial t}\,\mathrm{d}\upsilon + \iiint_{\mathscr{V}^{\mathrm{cv}}} \mathrm{div}(\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon = 0.$$

now, using the divergence theorem, we rewrite the second volume integral as a surface integral:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial\rho}{\partial t}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}\cdot\mathrm{d}\vec{\boldsymbol{S}} = 0.$$

and since the integration region is constant, we can extract the differentiation operation from the integral according to equation (5.12), to obtain equation (5.23):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right)+\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}}=0.$$

Other conversions

Here we have performed the 5 out of 12 possible conversions; however, we will not go on making demonstrations of the equivalences and the other 7 cases are left as exercises to the reader.

5.4.7 Conservation of mass balance equations in the material description

Write this subsection

In the previous sections ***

Case 2: linear momentum equation for a finite control mass moving with the flow

In Section ***,

Case 4: linear momentum equation for an infinitesimal control mass moving with the flow

In Section ***,

$$\rho_0(\vec{\boldsymbol{X}})=\rho(\vec{\boldsymbol{chi}}(\vec{\boldsymbol{X}},t),t)\det\boldsymbol{F}(\vec{\boldsymbol{X}},t)$$

5.4.8 Final notes

In this section, we have demonstrated that, under the assumption of smoothness of the mass density ρ, the integral and differential forms of the continuity equation are entirely equivalent. However, the integral form is more general than its differential counterpart because it accommodates discontinuities in ρ within finite control masses or volumes. In contrast, the differential form assumes that flow properties are differentiable, and therefore continuous, limiting its applicability to smooth flows. This highlights that the integral form of the continuity equation is more fundamental, as it remains valid even in the presence of discontinuities. This distinction is particularly important in scenarios involving flows with abrupt changes, such as shock waves, where the integral form proves indispensable.

5.5 Physical interpretation of the divergence of the velocity field

Having established the mathematical framework for mass transport, we now turn to the physical interpretation of the divergence of the velocity field. This interpretation connects our mathematical formalism with intuitive physical understanding, providing deeper insight into flow deformation. In the following, we will provide two different interpretations of $\operatorname{div}\vec{\boldsymbol{v}}$:

Interpretation 1: the divergence as a change of density Rearranging equation (5.31) to isolate $\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$, we obtain:

$$\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t) = -\frac{1}{\rho(\vec{\boldsymbol{x}},t)}\frac{\mathrm{D}\rho(\vec{\boldsymbol{x}},t)}{\mathrm{D}t}.$$

According to this formula, the divergence the velocity field, i.e. $\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)$, determines the density change along the particle path, normalized by the density itself. It describes the local rate of volumetric expansion or contraction of an infinitesimal material fluid element as it moves through a flow field. Let us explain this physical interpretation.

In the context of local volume expansion or compression, if $\operatorname{div}\vec{\boldsymbol{v}} > 0$, then $\frac{\mathrm{D}\rho}{\mathrm{D}t} < 0$, which means the mass density is decreasing following a material particle. This corresponds to a *local expansion* over time of the material, as the same amount of mass is spreading over a larger volume. On the contrary, if $\operatorname{div}\vec{\boldsymbol{v}} < 0$, then $\frac{\mathrm{D}\rho}{\mathrm{D}t} > 0$, meaning the density is increasing over time. This corresponds to a *local compression* of the material, as the same mass is being concentrated into a smaller volume. Now, in an *incompressible flow* (e.g., water flowing through a pipe), the mass density ρ is constant along a material path, meaning $\frac{\mathrm{D}\rho}{\mathrm{D}t} = 0$, which directly implies that $\operatorname{div}\vec{\boldsymbol{v}} = 0$. This means that locally, the volume of material elements does not change; this indicates that the flow is locally an isochoric (volume-preserving) flow.

In other words, a positive divergence at a point means that fluid particles are moving apart, like in a region of flow expansion (e.g., behind a diverging nozzle), and a negative divergence implies fluid elements are coming together, indicating a compressing region (e.g., a converging nozzle or a shock wave in supersonic flow).

Interpretation 2: the divergence as a change of volume An alternative interpretation of the divergence of the velocity field is in order. Let us rewrite equation (5.16):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\,\mathrm{d}v_t\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\frac{\mathrm{D}\boldsymbol{\mu}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} + \boldsymbol{\mu}(\vec{\boldsymbol{x}},t)\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\right]\mathrm{d}v_t.$$

In case $\boldsymbol{\mu}(\vec{\boldsymbol{x}}, t) = 1$, this equation simplifies to:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \mathrm{d}\upsilon_t\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\upsilon_t;$$

this formula states that the rate of change of volume of a control mass $\mathscr{V}_t^{\mathrm{cm}}$ as it moves with the flow is determined by the integral of the divergence of the velocity field over that control mass.

To extract a local interpretation, we consider the limiting case where the control mass $\mathscr{V}_t^{\mathrm{cm}}$ shrinks down to the infinitesimal material element of volume $\mathrm{d}\upsilon_t$, which is located at $\vec{\boldsymbol{x}}$. In this limit, the previous equation becomes:

$$\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}\upsilon_t) = \operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\upsilon_t,$$

and rearranging to isolate the divergence term:

$$\boxed{\operatorname{div}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) = \frac{1}{\mathrm{d}\upsilon_t}\frac{\mathrm{D}}{\mathrm{D}t}(\mathrm{d}\upsilon_t)}$$

this equation provides an alternative physical interpretation of the velocity divergence: the divergence of the velocity field at any point $\vec{\boldsymbol{x}} \in \Omega$ represents the instantaneous rate of relative volume change (volumetric dilatation rate) of an infinitesimal fluid element passing through that point. More precisely, it is the time rate of change of volume of a moving material element per unit volume.

These two interpretations connect the kinematics of the flow (velocity field) with its density and relative volume change, providing insights into how fluid elements expand, contract, or remain incompressible during their motion.

5.6 Reynolds' lemma

There is a useful mathematical result that can be derived from the Reynolds' transport theorem, which is known as *Reynolds' lemma**. This lemma provides a powerful tool for analyzing the time evolution of integrals over moving control masses, particularly when the integrand is a product of density and another quantity. To derive Reynolds' lemma, we start by substituting $\boldsymbol{\mu} := \rho\boldsymbol{\psi}$ in equation (5.16), where $\boldsymbol{\psi}$ is a scalar, vector or tensor quantity:

$$\frac{\mathrm{D}\boldsymbol{Q}_{\mathscr{V}^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}}, t)\boldsymbol{\psi}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\upsilon\right)$$

$$= \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\frac{\mathrm{D}}{\mathrm{D}t} \big(\rho(\vec{\boldsymbol{x}}, t) \boldsymbol{\psi}(\vec{\boldsymbol{x}}, t) \big) + \rho(\vec{\boldsymbol{x}}, t) \boldsymbol{\psi}(\vec{\boldsymbol{x}}, t) \operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \right] \mathrm{d}\upsilon. \quad (5.40)$$

Using the product rule for differentiation and the continuity equation (conservation of mass) (5.31), we find:

$$\begin{aligned} \frac{\mathrm{D}}{\mathrm{D}t} \big(\rho(\vec{\boldsymbol{x}}, t) \boldsymbol{\psi}(\vec{\boldsymbol{x}}, t) \big) &= \rho(\vec{\boldsymbol{x}}, t) \frac{\mathrm{D}\boldsymbol{\psi}(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} + \frac{\mathrm{D}\rho(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} \boldsymbol{\psi}(\vec{\boldsymbol{x}}, t) \\ &= \rho(\vec{\boldsymbol{x}}, t) \frac{\mathrm{D}\boldsymbol{\psi}(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} - \rho(\vec{\boldsymbol{x}}, t) \boldsymbol{\psi}(\vec{\boldsymbol{x}}, t) \operatorname{div} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t); \end{aligned}$$

replacing this result back into (5.40), we arrive to the following relation, which is known as *Reynolds' lemma*:

$$\boxed{\frac{\mathrm{D}}{\mathrm{D}t} \left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}}, t) \boldsymbol{\psi}(\vec{\boldsymbol{x}}, t) \, \mathrm{d}\upsilon \right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}}, t) \frac{\mathrm{D}\boldsymbol{\psi}(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} \, \mathrm{d}\upsilon.} \quad (5.41)$$

This equation will be used in the next sections to derive the conservation of linear and angular momentum equations, as well as the conservation of energy equation.

5.7 Momentum

Classical mechanics defines *momentum* as the "amount of motion" of a body. In other words, it is a measure of the effort required to bring a moving body to rest. In continuous media, we deal with two distinct, though dynamically related, forms of momentum: linear momentum and angular momentum.

5.7.1 Linear momentum

Linear momentum, also known as *translational momentum*, reflects a body's tendency to continue moving in a straight line at a constant speed.

For a discrete particle of mass m and velocity $\vec{\boldsymbol{v}}$, its linear momentum $\vec{\boldsymbol{p}}$ (from Latin *pellere* that means "push" or "drive") is defined as the vector $\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}$. This is a vector quantity, possessing a magnitude and a direction. In the International System of Units (SI), the unit of measurement of momentum is the kilogram meter per second (kg m/s), which is dimensionally equivalent to the newton-second (N·s).

In the context of a continuous medium, the total linear momentum of a control mass occupying a spatial volume $\mathscr{V}_t^{\mathrm{cm}}$ at time t is the volume integral of the momentum density $\rho\vec{\boldsymbol{v}}$:

$$\vec{\boldsymbol{p}}_{\mathscr{V}_t^{\mathrm{cm}}}(t) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}}, t) \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t) \, \mathrm{d}\upsilon; \quad (5.42)$$

in the material description, the total linear momentum of a control mass $\mathscr{V}_t^{\mathrm{cm}}$ is given by:

$$\vec{\boldsymbol{p}}_{\mathscr{V}_t^{\mathrm{cm}}}(t) = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \rho_0(\vec{\boldsymbol{X}})\,\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}.$$

5.7.2 Angular momentum

Angular momentum*, sometimes called *rotational momentum** or *moment of momentum**, represents a body's tendency to continue rotating about a specific reference point or axis.

For a discrete particle, the angular momentum $\vec{\boldsymbol{h}}$ is defined as the cross product of the position vector $\vec{\boldsymbol{r}}$ and the linear momentum $\vec{\boldsymbol{p}}$, that is $\vec{\boldsymbol{h}} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{p}}$. This is a vector quantity, possessing a magnitude and a direction. In the International System of Units (SI), the unit of measurement of angular momentum is the kilogram meter squared per second (kg m^2/s), which is dimensionally equivalent to the newton-meter-second (N·m·s). Just as linear momentum can be written as $\vec{\boldsymbol{p}} = m\vec{\boldsymbol{v}}$, where m is the scalar mass, an analogous relation exists for angular momentum. For a shape that is restricted to rotate about a single axis, the angular momentum can be expressed as $\vec{\boldsymbol{h}} = I\vec{\boldsymbol{\omega}}$, where I is the so-called *scalar moment of inertia** and $\vec{\boldsymbol{\omega}}$ is the angular velocity vector. This scalar moment of inertia I is a measure of how the mass of the body is distributed with respect to the axis of rotation, and it quantifies the body's resistance to changes in its rotational motion. Mathematically, it is defined as the integral of the mass density ρ multiplied by the square of the distance from the axis of rotation, integrated over the entire volume of the body:

$$I = \iiint_{\mathscr{V}} \rho(\vec{\boldsymbol{x}})\, r(\vec{\boldsymbol{x}})^2\,\mathrm{d}\upsilon,$$

where $r(\vec{\boldsymbol{x}})$ is the distance from the axis of rotation to the point $\vec{\boldsymbol{x}}$ in the body. Now, if a body is allowed to rotate in all three dimensions, its moment of inertia becomes a symmetric second-order tensor, known as the *moment of inertia tensor** (also called the *inertia tensor**). The inertia tensor $\boldsymbol{I}$ is a 3×3 tensor that encapsulates the distribution of mass in three-dimensional space and how it affects the body's resistance to rotational motion about any axis. The angular momentum vector $\vec{\boldsymbol{h}}$ can then be expressed as the product of the inertia tensor and the angular velocity vector,[11] that is, $\vec{\boldsymbol{h}} = \boldsymbol{I}\vec{\boldsymbol{\omega}}$. The Dutch polymath Christiaan Huygens (1629–1695), introduced the concept of moment of inertia in his work *Horologium Oscillatorium* in 1673, where he analyzed the dynamics of pendulums. However, the term moment of inertia (*momentum inertiae* in Latin) was coined by Leonhard Euler in his book *Theoria motus*

[11] The interested reader is referred to https://en.wikipedia.org/wiki/Moment_of_inertia for more information.

corporum solidorum seu rigidorum in 1765, where he developed the mathematical framework for rigid body dynamics and introduced the concept of the inertia tensor.

For a continuous medium, the total angular momentum of a control mass about a fixed point $\tilde{\vec{x}}_0$ is obtained by integrating the moment caused by the linear momentum density, i.e. $\vec{r} \times (\rho \vec{v})$:

$$\vec{h}_{\mathscr{V}_t^{\mathrm{cm}}}(t) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{r}(\vec{x}, t) \times \big(\rho(\vec{x}, t)\vec{v}(\vec{x}, t)\big)\, \mathrm{d}v. \tag{5.43}$$

Here, $\vec{r}(\vec{x}, t) = \vec{x}(t) - \tilde{\vec{x}}_0$ is the position vector of a material point $\vec{x}$ measured from a fixed point $\tilde{\vec{x}}_0$. The cross product $\vec{r}(\vec{x}, t) \times \big(\rho \vec{v}(\vec{x}, t)\big)$ is therefore the angular momentum density per unit volume, which encodes, at each point $\vec{x}$, the tendency of that material element to rotate about $\tilde{\vec{x}}_0$.

In the material description, the total angular momentum of a control mass $\mathscr{V}_t^{\mathrm{cm}}$ about a fixed point $\tilde{\vec{x}}_0$ is given by:

$$\vec{h}_{\mathscr{V}_t^{\mathrm{cm}}}(t) = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{r}(\vec{X}, t) \times \big(\rho_0(\vec{X})\vec{V}(\vec{X}, t)\big)\, \mathrm{d}\mathscr{V}.$$

where $\vec{r}(\vec{X}, t) = \vec{\chi}(\vec{X}, t) - \tilde{\vec{x}}_0$ is the position vector of a material point $\vec{X}$ measured from the fixed point $\tilde{\vec{x}}_0$.

Whereas linear momentum governs translation, angular momentum dictates rotational motion. Momentum is a conserved quantity, meaning that if a control mass is not affected by external forces, its total momentum does not change; in other words, the total momentum of a control mass remains constant unless acted upon by an external force or torque. This fact is known as the *law of conservation of momentum*.★

We will analyze the conservation of linear momentum in Section 5.10 and the conservation of angular momentum in Section 5.11.

5.8 Body and surface forces and torques

In continuum mechanics, we classify the forces that act on a continuum $\mathscr{V} \subseteq \Omega_t$ into two main types: body forces and surface forces. These forces are fundamental in formulating the equations of motion and understanding the behavior of materials. Additionally, we can also define the associated torques that arise from these forces, which are crucial for analyzing rotational dynamics.

5.8.1 Body and surface forces

Body forces,★ also known as *volume forces*,★ are forces that act throughout the entire volume of a material body $\mathscr{V}$. They arise from fields that permeate space, such as

gravitational, electric or magnetic fields. Inertial forces such as the centrifugal force are also considered body forces. Mathematically, a body force can be represented as a force density $\vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t)$, which is a vector field defined at every point $\vec{\boldsymbol{x}}$ in the material volume and at every time t. Do not confuse vector $\vec{\boldsymbol{b}}$ with the left Cauchy-Green deformation tensor $\boldsymbol{b}$ which was defined in Section 3.13.2. In the International System of Units (SI), the units of body force density are newtons per cubic meter ($\mathrm{N/m^3}$). The total body force acting on a region $\mathscr{V}$ can be calculated by integrating the body force density over the volume of $\mathscr{V}$:

$$\vec{\boldsymbol{f}}_{\text{body}}(t) = \iiint_{\mathscr{V}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}v.$$

After a change of variables, we can also express the total body force in the material description as an integral over the reference configuration $\mathscr{V}_0$ associated to $\mathscr{V}$:

$$\vec{\boldsymbol{f}}_{\text{body}}(t) = \iiint_{\mathscr{V}_0} \vec{\boldsymbol{b}}(\vec{\boldsymbol{X}}, t) J(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V} = \iiint_{\mathscr{V}_0} \vec{\boldsymbol{B}}(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V}.$$

where $\vec{\boldsymbol{B}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{b}}(\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t), t) J(\vec{\boldsymbol{X}}, t)$ is the body force density in the material description (mass per unit volume in the reference configuration), $J(\vec{\boldsymbol{X}}, t) = \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)$ is the Jacobian determinant of the deformation gradient $\boldsymbol{F}$, which accounts for the change in volume during deformation, and $\mathscr{V}_0 = \vec{\boldsymbol{\chi}}^{(-1)}(\mathscr{V}, t)$ is the reference configuration associated to $\mathscr{V}$.

Surface forces[★], also known as *contact forces*[★], are forces that act on the surface of a the region $\mathscr{V}$ due to contact with other bodies or fluids. They include normal forces (perpendicular to the surface) and shear forces (tangential to the surface). Mathematically, surface forces can be represented as a traction vector $\vec{\boldsymbol{t}}(\vec{\boldsymbol{x}}, t)$ defined on the boundary $\partial\mathscr{V}$ of the control mass. In the International System of Units (SI), the units of surface force density are newtons per square meter ($\mathrm{N/m^2}$). The total surface force acting on $\mathscr{V}$ can be calculated by integrating the traction vector over the surface of the control mass:

$$\vec{\boldsymbol{f}}_{\text{surface}}(t) = \oiint_{\partial\mathscr{V}} \vec{\boldsymbol{t}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}S.$$

According to Cauchy's stress principle (see Section 4.2), the traction vector $\vec{\boldsymbol{t}}(\vec{\boldsymbol{x}}, t)$ can be expressed as the product of the Cauchy stress tensor $\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)$ and the outward unit normal vector $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)$ to the surface at point $\vec{\boldsymbol{x}}$, that is $\vec{\boldsymbol{t}}(\vec{\boldsymbol{x}}, t) = \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)$. Thus, the total surface force can also be written as:

$$\vec{\boldsymbol{f}}_{\text{surface}}(t) = \oiint_{\partial\mathscr{V}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}S.$$

The total force acting on the control mass $\mathscr{V}$ is the sum of the body and surface forces:

$$\vec{\boldsymbol{f}}_{\text{total}}(t) = \vec{\boldsymbol{f}}_{\text{body}}(t) + \vec{\boldsymbol{f}}_{\text{surface}}(t).$$

After a change of variables, using Nanson's formula (3.37) $\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S = J(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S_0$, we can also express the total surface force in the material description as an integral over the reference configuration $\mathscr{V}_0$ associated to $\mathscr{V}$:

$$\begin{aligned}\vec{\boldsymbol{f}}_{\text{surface}}(t) &= \oiint_{\partial\mathscr{V}_0} \vec{\boldsymbol{\sigma}}(\vec{\boldsymbol{x}},t)J(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S_0 \\ &= \oiint_{\partial\mathscr{V}_0} \boldsymbol{P}(\vec{\boldsymbol{X}},t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}},t)\,\mathrm{d}S_0\end{aligned}$$

where

$$\boldsymbol{P}(\vec{\boldsymbol{X}},t) = J(\vec{\boldsymbol{X}},t)\boldsymbol{\sigma}(\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}},t),t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{X}},t)$$

is the first Piola-Kirchhoff stress tensor, $\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}},t)$ is the outward unit normal vector to the surface $\partial\mathscr{V}_0$ in the reference configuration, and $\partial\mathscr{V}_0 = \vec{\boldsymbol{\chi}}^{(-1)}(\mathscr{V},t)$ is the reference configuration associated to $\mathscr{V}$.

5.8.2 Body and surface torques

The torques associated with body and surface forces can be calculated by taking the cross product of the position vector $\vec{\boldsymbol{r}}(\vec{\boldsymbol{x}},t)$ with the respective force densities:

$$\vec{\boldsymbol{m}}_{\text{body}}(t) = \iiint_{\mathscr{V}} \vec{\boldsymbol{r}}(\vec{\boldsymbol{x}},t)\times\vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}v, \tag{5.44}$$

$$\vec{\boldsymbol{m}}_{\text{surface}}(t) = \oiint_{\partial\mathscr{V}} \vec{\boldsymbol{r}}(\vec{\boldsymbol{x}},t)\times\vec{\boldsymbol{t}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S. \tag{5.45}$$

Here, $\vec{\boldsymbol{r}}(\vec{\boldsymbol{x}},t) = \vec{\boldsymbol{x}}(t) - \tilde{\vec{\boldsymbol{x}}}_0$ is the position vector of a material point $\vec{\boldsymbol{x}}$ measured from a fixed point $\tilde{\vec{\boldsymbol{x}}}_0$. The total torque acting on the control mass $\mathscr{V}$ is the sum of the body and surface torques:

$$\vec{\boldsymbol{m}}_{\text{total}}(t) = \vec{\boldsymbol{m}}_{\text{body}}(t) + \vec{\boldsymbol{m}}_{\text{surface}}(t).$$

Remark: In the formulation above, we have considered only the classical body forces and surface tractions as sources of torque. However, in certain physical situations, additional contributions to the angular momentum balance must be accounted for:

Intrinsic body moments, *also called *body couples**, are distributed torques that act throughout the volume of a body, independently of any force. They arise when the material itself carries an intrinsic angular momentum density (spin), as happens, for instance, when an external magnetic field acts on a polarized or magnetized material (e.g., ferrofluids, liquid crystals, or polar dielectrics). In such cases, the magnetic torque per unit volume $\vec{\boldsymbol{c}}(\vec{\boldsymbol{x}},t)$ contributes an additional term to the angular momentum balance:

$$\vec{\boldsymbol{m}}_{\text{body, intrinsic}}(t) = \iiint_{\mathscr{V}} \vec{\boldsymbol{c}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}v.$$

Surface couples, ★also called *couple stresses*★, are distributed torques acting on the boundary of a region. They appear in generalized continuum theories, such as the *Cosserat (micropolar) theory*★, where material points are endowed with an independent rotational degree of freedom. In such theories, a couple stress tensor $\boldsymbol{M}$ supplements the Cauchy stress tensor, and gives rise to an additional surface torque:

$$\vec{\boldsymbol{m}}_{\text{surface, couple}}(t) = \oiint_{\partial\mathscr{V}} \boldsymbol{M}(\vec{\boldsymbol{x}}, t)\,\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}S.$$

Therefore, body and surface torques generalize to:

$$\vec{\boldsymbol{m}}_{\text{body}}(t) = \iiint_{\mathscr{V}} \Big[\vec{\boldsymbol{r}}(\vec{\boldsymbol{x}}, t) \times \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t) + \vec{\boldsymbol{c}}(\vec{\boldsymbol{x}}, t)\Big]\,\mathrm{d}v,$$
$$\vec{\boldsymbol{m}}_{\text{surface}}(t) = \oiint_{\partial\mathscr{V}} \Big[\vec{\boldsymbol{r}}(\vec{\boldsymbol{x}}, t) \times \vec{\boldsymbol{t}}(\vec{\boldsymbol{x}}, t) + \boldsymbol{M}(\vec{\boldsymbol{x}}, t)\,\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\Big]\,\mathrm{d}S.$$

In classical continuum mechanics, both intrinsic body moments and surface couples are assumed to be absent; a continuum without these moments and couples is called a *classical continuum*★ or *non-polar continuum*★. Under this assumption, the angular momentum balance reduces to the requirement that the Cauchy stress tensor $\boldsymbol{\sigma}$ be *symmetric*, i.e., $\sigma_{ij} = \sigma_{ji}$, as we shall demonstrate in Section 5.11. When body couples or couple stresses are present, however, this symmetry no longer holds in general, and more general continuum theories like the Cosserat (micropolar) theory must be employed.

5.9 Newton's and Euler's laws of motion

Newton's laws of motion★ form the basis of classical mechanics. They describe the relationship between the motion of an object and the forces acting on it. Since these rules are designed for point or particle masses, they apply to idealized things with mass but no volume, which prevents them from deforming or experiencing flow. For many physical systems, when the motion of internal components can be disregarded and the distance between bodies is significantly greater than their respective sizes, this assumption is an acceptable approximation. For example, both the Earth and the Sun can be roughly described as pointlike when taking into account the former's orbit around the latter, while the Earth is not pointlike when taking into account activity on its surface.

Newton's laws of motion can be paraphrased as follows:

1. **First law (law of inertia):** A body remains at rest, or in motion at a constant speed in a straight line, unless it is acted upon by a force.

2. **Second law (law of momentum and force):** At any instant of time, the net force $\vec{\boldsymbol{f}}$ acting on a body is equal to the rate at which the body's momentum is changing with time:

$$\vec{\boldsymbol{f}} = \frac{\mathrm{D}\vec{\boldsymbol{p}}}{\mathrm{D}t}.$$

   In case mass is constant, this equation can be rewritten as $\vec{\boldsymbol{f}} = m\vec{\boldsymbol{a}}$, where $\vec{\boldsymbol{a}} = \frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t}$ is the acceleration of the body.

3. **Third law (law of action and reaction):** If two bodies exert forces on each other, these forces have the same magnitude but opposite directions.

The three laws of motion were first stated by Isaac Newton in his *Philosophiæ Naturalis Principia Mathematica* (Mathematical Principles of Natural Philosophy), originally published in 1687. Newton used them to investigate and explain the motion of many physical objects and systems. In the time since Newton, new insights, especially around the concept of energy, built the field of classical mechanics on his foundations. In modern times, limitations to Newton's laws have been discovered; new theories were consequently developed, such as quantum mechanics and relativity to address the physics of objects which are either moving at speeds approaching the speed of light, or whose behavior is governed at the subatomic scale.

However, because continuous bodies are deformable and extended in space, they cannot be perfectly modeled as mere point masses. The robust generalization of Newton's second law to continuous, extended bodies (both solid and fluid) is attributed to Leonhard Euler, resulting in *Euler's laws of motion**, which are the fundamental principles governing the dynamics of continuous media. These laws can be stated as follows:

1. **Euler's first law (balance of linear momentum):** The rate of change of the total linear momentum of any control mass $\mathscr{V}_t^{\mathrm{cm}}$ is equal to the sum of all external forces acting upon that material volume. In mathematical terms, this can be expressed as:

$$\frac{\mathrm{D}\vec{\boldsymbol{p}}_{\mathscr{V}_t^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \vec{\boldsymbol{f}}_{\mathrm{body}}(t) + \vec{\boldsymbol{f}}_{\mathrm{surface}}(t), \tag{5.46}$$

   where $\vec{\boldsymbol{p}}_{\mathscr{V}_t^{\mathrm{cm}}}(t)$ is the total linear momentum of the control mass $\mathscr{V}_t^{\mathrm{cm}}$ at time t as defined by (5.42), $\vec{\boldsymbol{f}}_{\mathrm{body}}(t)$ is the total body force acting on $\mathscr{V}_t^{\mathrm{cm}}$ at time t (equation [5.8.1]), and $\vec{\boldsymbol{f}}_{\mathrm{surface}}(t)$ is the total surface traction acting on $\mathscr{V}_t^{\mathrm{cm}}$ at time t (equation [5.8.1]). Note that internal forces between the particles that make up a body do not contribute to changing the momentum of the body as there is an equal and opposite force resulting in no net effect.

2. **Euler's second law (balance of angular momentum):** The rate of change of the total angular momentum of any material volume about a fixed point $\vec{\tilde{\boldsymbol{x}}}_0$ is equal

to the resultant torque (or moment) of all external forces acting on it about that same point. In mathematical terms, this can be expressed as:

$$\frac{\mathrm{D}\vec{\boldsymbol{h}}_{\mathscr{V}_t^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \vec{\boldsymbol{m}}_{\mathrm{body}}(t) + \vec{\boldsymbol{m}}_{\mathrm{surface}}(t), \tag{5.47}$$

where $\vec{\boldsymbol{h}}_{\mathscr{V}_t^{\mathrm{cm}}}(t)$ is the total angular momentum of the control mass $\mathscr{V}_t^{\mathrm{cm}}$ at time t about $\tilde{\vec{\boldsymbol{x}}}_0$ as defined by (5.43), $\vec{\boldsymbol{m}}_{\mathrm{body}}(t)$ is the total body torque acting on $\mathscr{V}_t^{\mathrm{cm}}$ (equation [5.44]), and $\vec{\boldsymbol{m}}_{\mathrm{surface}}(t)$ is the total surface torque acting on $\mathscr{V}_t^{\mathrm{cm}}$ (equation [5.45]); both torques are evaluated at time t about the fixed point $\tilde{\vec{\boldsymbol{x}}}_0$.

These generalizations were formulated by Leonhard Euler in the 1740s/1750s/1760s, about 60 years after Isaac Newton formulated his laws.

5.10 Conservation of linear momentum

In this section, we formulate the conservation of linear momentum from multiple perspectives. We will explore its integral forms for both a fixed control volume in space (Eulerian specification) and a moving control mass (Lagrangian specification). By applying the Reynolds transport formula and the divergence theorem discussed in previous sections, we will localize these integral balances. This localization will lead to the Cauchy momentum equation (5.74), the fundamental local differential equation of motion that must be satisfied at every point within the continuum, serving as a basis for both fluid dynamics (e.g., the Navier-Stokes equations) and solid mechanics (e.g., elastodynamics).

5.10.1 Case 1: linear momentum equation for a finite control volume fixed in space

Case 1 reflects the Eulerian perspective for a finite control volume $\mathscr{V}^{\mathrm{cv}}$, where we fix our attention on a region of space and ask what happens to momentum within it. This is the natural framework for engineering analysis, where instruments are stationary and where fluxes of matter through control surfaces are central to the physics.

In this approach, we apply Euler's first law to the associated control mass $\mathscr{V}_t^{\mathrm{cm}}$ that happens to coincide with a fixed control volume $\mathscr{V}^{\mathrm{cv}}$ at a particular time t. We must account for the fact that mass enters and leaves $\mathscr{V}^{\mathrm{cv}}$, carrying momentum with it.

Let $\mathscr{V}^{\mathrm{cv}}$ be a finite region fixed in space, with outward unit normal $\hat{\boldsymbol{n}}$ on its closed boundary $\partial\mathscr{V}^{\mathrm{cv}}$. Denote by $\mathrm{d}v$ a volume element and by $\mathrm{d}S$ a surface element.

At time t the volume defined by the control mass $\mathscr{V}_t^{\mathrm{cm}}$ coincides with the one occupied by $\mathscr{V}^{\mathrm{cv}}$ at that instant t. So, its total linear momentum (5.42) becomes

$$\vec{\boldsymbol{p}}_{\mathscr{V}_t^{\mathrm{cm}}}(t) = \iiint_{\mathscr{V}^{\mathrm{cv}}} \rho(\vec{\boldsymbol{x}}, t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\,\mathrm{d}v = \iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\,\vec{\boldsymbol{v}}\,\mathrm{d}v\Big|_t.$$

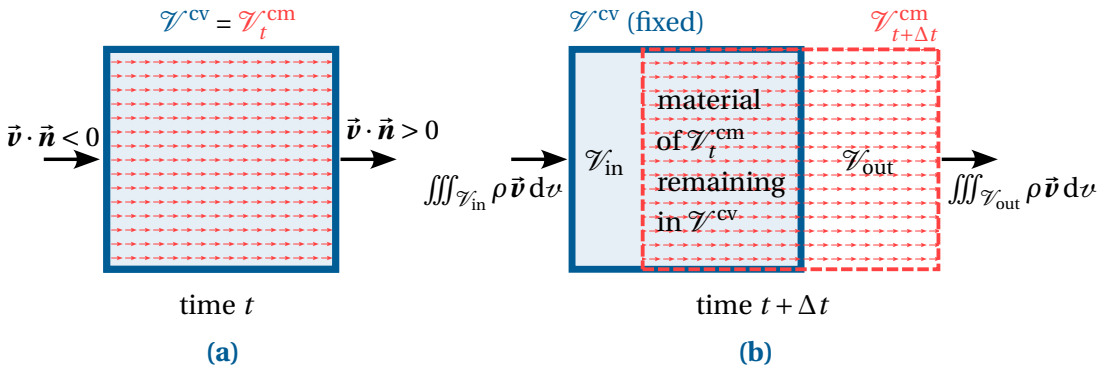


Figure 5.6: *Figure (a) depicts the instant at time t, where the control mass $\mathscr{V}_t^{cm}$ coincides with the control volume $\mathscr{V}^{cv}$; here the inflow carries momentum in and the outflow carries it out. Figure (b) shows the instant at time $t+\Delta t$; during a short time interval Δt, mass enters and leaves $\mathscr{V}^{cv}$, sweeping the volumes $\mathscr{V}_{in}$ and $\mathscr{V}_{out}$ in and out, respectively. At time $t+\Delta t$, the control mass $\mathscr{V}_{t+\Delta t}^{cm}$ occupies the region $\left(\mathscr{V}^{cv}\setminus\mathscr{V}_{in}\right)\cup\mathscr{V}_{out}$.*

Let's follow $\mathscr{V}_t^{\mathrm{cm}}$ for a short interval Δt. During Δt the mass crosses $\partial\mathscr{V}^{\mathrm{cv}}$. At a point $\vec{\boldsymbol{x}} \in \partial\mathscr{V}^{\mathrm{cv}}$, where $\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}} > 0$, a thin sheet of volume $\mathrm{d}\mathscr{v} = (\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}\Delta t)\,\mathrm{d}S$ leaves; where $\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}} < 0$, a sheet enters. Call the swept-out volume $\mathscr{V}_{\mathrm{out}}$ and the swept-in volume $\mathscr{V}_{\mathrm{in}}$.

At $t+\Delta t$ the control mass no longer coincides with $\mathscr{V}^{\mathrm{cv}}$. It occupies (see Figure 5.6):

$$\mathscr{V}_t^{\mathrm{cm}} = \left(\mathscr{V}^{\mathrm{cv}}\setminus\mathscr{V}_{\mathrm{in}}\right)\cup\mathscr{V}_{\mathrm{out}}.$$

Its momentum is therefore

$$\vec{\boldsymbol{p}}_{\mathscr{V}_{t+\Delta t}^{\mathrm{cm}}} = \iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t+\Delta t} - \iiint_{\mathscr{V}_{\mathrm{in}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t+\Delta t} + \iiint_{\mathscr{V}_{\mathrm{out}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t+\Delta t}.$$

The change of momentum in a time interval Δt is given by

$$\begin{aligned}\vec{\boldsymbol{p}}_{\mathscr{V}_{t+\Delta t}^{\mathrm{cm}}} - \vec{\boldsymbol{p}}_{\mathscr{V}_t^{\mathrm{cm}}} = &\left[\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t+\Delta t} - \iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t}\right] \\ &- \iiint_{\mathscr{V}_{\mathrm{in}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t+\Delta t} + \iiint_{\mathscr{V}_{\mathrm{out}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t+\Delta t}. \qquad (5.48)\end{aligned}$$

Now, the last two volume integrals can be expressed as surface fluxes. For an outward face, where $\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}$ is positive:

$$\iiint_{\mathscr{V}_{\mathrm{out}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v} = \Delta t \oiint_{\partial\mathscr{V}^{\mathrm{cv}}:\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}>0} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

and for an inward face, where $\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}$ is negative,

$$\iiint_{\mathscr{V}_{\mathrm{in}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v} = -\Delta t \oiint_{\partial\mathscr{V}^{\mathrm{cv}}:\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}<0} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

Subtracting both gives the single closed integral

$$\iiint_{\mathscr{V}_{\text{out}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v} - \iiint_{\mathscr{V}_{\text{in}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v} = \Delta t \oiint_{\partial\mathscr{V}^{\text{cv}}} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

Replacing this expression in the change of momentum (5.48), we have

$$\vec{\boldsymbol{p}}_{\mathscr{V}^{\text{cm}}_{t+\Delta t}} - \vec{\boldsymbol{p}}_{\mathscr{V}^{\text{cm}}_{t}} = \left[\iiint_{\mathscr{V}^{\text{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t+\Delta t} - \iiint_{\mathscr{V}^{\text{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\Big|_{t}\right] + \Delta t \oiint_{\partial\mathscr{V}^{\text{cv}}} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S\Big|_{t+\Delta t}.$$

Dividing this expression by Δt and letting $\Delta t \to 0$:

$$\frac{\mathrm{d}\vec{\boldsymbol{p}}_{\mathscr{V}^{\text{cm}}_{t}}(t)}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\text{cv}}} \rho(\vec{\boldsymbol{x}},t)\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v}\right) + \oiint_{\partial\mathscr{V}^{\text{cv}}} \rho(\vec{\boldsymbol{x}},t)\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S. \tag{5.49}$$

The left side is the rate of change of momentum of the control mass $\mathscr{V}^{\text{cm}}_t$, so Euler's first law (5.46) applies. At time t the control mass $\mathscr{V}^{\text{cm}}_t$ coincides with the fixed control volume $\mathscr{V}^{\text{cv}}$, hence the forces on it are exactly the forces acting on the fluid instantaneously inside the control volume $\mathscr{V}^{\text{cv}}$. The right side is the time derivative of the momentum contained in $\mathscr{V}^{\text{cv}}$ plus the net flux of momentum across its control surface.

By Euler's first law (5.46),

$$\frac{\mathrm{d}\vec{\boldsymbol{p}}_{\mathscr{V}^{\text{cm}}_{t}}(t)}{\mathrm{d}t} = \vec{\boldsymbol{f}}_{\text{body}}(t) + \vec{\boldsymbol{f}}_{\text{surface}}(t),$$

the left side must equal the total force acting on the material that constitutes $\mathscr{V}^{\text{cm}}_t$ at time t. This force has two contributions: body forces and contact forces.

The body force, with $\vec{\boldsymbol{b}}$ force per unit mass, contributes

$$\vec{\boldsymbol{f}}_{\text{body}}(t) = \iiint_{\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v}.$$

The surface force is governed by Cauchy's fundamental theorem of stress (4.1), that is, $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}},t) = \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)$, where $\boldsymbol{\sigma}$ is the Cauchy stress tensor. Integrating this traction over the entire bounding surface gives the total surface force:

$$\vec{\boldsymbol{f}}_{\text{surface}}(t) = \oiint_{\partial\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S = \oiint_{\partial\mathscr{V}^{\text{cv}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S.$$

Thus,

$$\frac{\mathrm{d}\vec{\boldsymbol{p}}_{\mathscr{V}^{\text{cm}}_{t}}(t)}{\mathrm{d}t} = \iiint_{\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}^{\text{cv}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S. \tag{5.50}$$

Equating both expressions for $\mathrm{d}\vec{\boldsymbol{p}}_{\mathscr{V}_t^{\mathrm{cm}}}/\mathrm{d}t$, that is equations (5.49) and (5.50), and isolating the time derivative of the momentum contained in the fixed volume, we arrive at the *integral form of the linear momentum balance for a fixed control volume*:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\,\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\right) = -\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S + \iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\sigma}\hat{\boldsymbol{n}}\,\mathrm{d}S. \tag{5.51}$$

This is the equation upon which the analysis of jets, pipe flows, turbomachinery, and aerodynamic bodies is ultimately founded.

Let us unpack the physical meaning of each term in this equation:

- The left-hand side, $\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\,\vec{\boldsymbol{v}}\,\mathrm{d}\mathscr{v}\right)$, is the accumulation term that measures how rapidly the total linear momentum inside $\mathscr{V}^{\mathrm{cv}}$ is changing.

- The term $-\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S$ is the convective flux term (5.6) that represents momentum transport by the flow itself. Wherever fluid crosses the control surface $\partial\mathscr{V}^{\mathrm{cv}}$, it carries momentum with it. The negative sign ensures the correct bookkeeping: outflow ($\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}} > 0$) removes momentum from $\mathscr{V}^{\mathrm{cv}}$, while inflow ($\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}} < 0$) deposits momentum into $\mathscr{V}^{\mathrm{cv}}$. This term has no analog in the Lagrangian formulation, where no fluid ever crosses the boundary. It is the feature of the Eulerian description. For example, in a rocket engine, the convective flux term accounts for the momentum carried out of the engine by the exhaust gases, which is what propels the rocket forward.

- The integral $\iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v}$ is the total body force acting on all material instantaneously residing inside the control volume; it acts regardless of whether the material is in motion.

- The term $\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\sigma}\hat{\boldsymbol{n}}\,\mathrm{d}S$ is the resultant of all contact (surface) forces that the surroundings exert on the material through the control surface $\partial\mathscr{V}^{\mathrm{cv}}$. This term includes pressure forces, viscous forces, and solid-wall reaction forces.

5.10.2 Case 2: linear momentum equation for a finite control mass moving with the flow

Case 2 reflects a Lagrangian perspective for a finite control mass, where we follow a specific collection of material as it moves through space and ask how its momentum changes. This is the natural framework for theoretical analysis, where we seek to understand the fundamental physics of the material itself, without reference to any particular spatial location.

Here, according to Euler's first law of motion (5.46), for a material volume, the rate of change of its total linear momentum equals the sum of all forces, both body and surface, acting upon it:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v}\right) = \vec{\boldsymbol{f}}_{\mathrm{body}}(t) + \vec{\boldsymbol{f}}_{\mathrm{surface}}(t). \tag{5.52}$$

The forces acting on the control mass are of two types: body forces, which act throughout the volume, and surface forces, which act on the boundary.

The body force per unit mass is denoted $\vec{\boldsymbol{b}}$, so the total body force is

$$\vec{\boldsymbol{f}}_{\mathrm{body}}(t) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v}. \tag{5.53}$$

Examples include gravitational forces and electromagnetic forces.

The surface force is governed by Cauchy's fundamental theorem of stress (4.1), that is, $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}},t) = \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)$. Integrating this traction over the entire bounding surface gives the total surface force:

$$\vec{\boldsymbol{f}}_{\mathrm{surface}}(t) = \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S = \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S. \tag{5.54}$$

This surface forces arise from the mechanical interaction of the material within $\mathscr{V}_t^{\mathrm{cm}}$ with the material outside it, that is, the material on its surroundings. This includes pressure forces and viscous forces as well as solid-wall reaction forces wherever $\mathscr{V}_t^{\mathrm{cm}}$ passes through a boundary.

Substituting (5.53) and (5.54) back into the momentum postulate (5.52) yields the *integral form of the momentum equation for a material volume*:

$$\boxed{\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v}\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S.} \tag{5.55}$$

In words, this equation states that the rate of change of linear momentum of any material volume $\mathscr{V}_t^{\mathrm{cm}}$ is equal to the sum of the body forces acting on it and the surface forces exerted by its surroundings. This is the equation upon which the analysis of material deformation, wave propagation, and fluid flow is ultimately founded.

5.10.3 Case 3: linear momentum equation for an infinitesimal control volume fixed in space

We will work again in the Eulerian framework, now at the level of an infinitesimal volume element fixed in space, tracking every flux and every force face by face.

Let us first define a Cartesian coordinate system with unit basis vectors $\hat{\boldsymbol{e}}_x$, $\hat{\boldsymbol{e}}_y$, $\hat{\boldsymbol{e}}_z$ along the x-, y-, and z-axes respectively. At an arbitrary point $\vec{\boldsymbol{x}} = (x, y, z)$ in the continuum, we carve out an infinitesimal control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$: a rectangular parallelepiped with side lengths $\mathrm{d}x$, $\mathrm{d}y$, $\mathrm{d}z$ aligned with the coordinate axes; the infinitesimal volume of $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ is:

$$\mathrm{d}\upsilon = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z. \tag{5.56}$$

The six faces of this element are labeled by the axis to which their outward normal is parallel and by the sign of that normal. We denote by $\mathscr{F}_x^+$ the face at $x + \mathrm{d}x$ (outward normal $+\hat{\boldsymbol{e}}_x$), and by $\mathscr{F}_x^-$ the face at x (outward normal $-\hat{\boldsymbol{e}}_x$), and analogously for the y- and z-faces. The density ρ, velocity $\vec{\boldsymbol{v}}$, and Cauchy stress $\boldsymbol{\sigma}$ are evaluated at $\vec{\boldsymbol{x}}$ and extended to the faces using a first-order Taylor expansion. Terms of second order in the infinitesimals $\mathrm{d}x$, $\mathrm{d}y$, $\mathrm{d}z$ will be neglected throughout, a procedure that is made exact in the limit as the element shrinks to a point.

Let's start by stating Euler's first law (5.46) for an infinitesimal control mass $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ that coincides exactly with $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ at time t:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}\right) = \mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{body}}(t) + \mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{surface}}(t), \tag{5.57}$$

where $\mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{body}}$ and $\mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{surface}}$ are the body and surface forces acting on the material element at instant t. Expanding the left-hand side, we have

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}\right) = \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}} \rho\,\vec{\boldsymbol{v}}\,\mathrm{d}\upsilon_t\right). \tag{5.58}$$

The problem is that the material derivative $\mathrm{D}/\mathrm{D}t$ follows the moving parcel, but control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ is fixed. We need to re-express that material derivative in terms of what happens inside and through the fixed box. Let's apply Reynolds transport formula (5.17) to the momentum of the control mass (5.58), by making $\boldsymbol{\mu} = \rho\vec{\boldsymbol{v}}$:

$$\begin{aligned}\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}\right) &= \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon_t\right)\\ &= \iiint_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial\big(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big)}{\partial t}\,\mathrm{d}\upsilon_t + \oiint_{\partial(\mathrm{d}\mathscr{V}_t^{\mathrm{cm}})} \big(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big)\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\cdot\mathrm{d}\vec{\boldsymbol{S}}.\end{aligned}$$

Taking into account that the control mass $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ coincides with the fixed control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ at time t, we can replace $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ by $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ in the integrals, and express the surface integral in terms of the outward unit normal $\hat{\boldsymbol{n}}$:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}\right) = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \frac{\partial(\rho\vec{\boldsymbol{v}})}{\partial t}\,\mathrm{d}\upsilon + \oiint_{\partial(\mathrm{d}\mathscr{V}^{\mathrm{cv}})} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

Using equation (5.10), the surface integral can be converted into a volume integral:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}\right) = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \frac{\partial(\rho\vec{\boldsymbol{v}})}{\partial t}\,\mathrm{d}\upsilon + \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon.$$

This equation applied at the infinitesimal level becomes:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}\right) = \frac{\partial(\rho\vec{\boldsymbol{v}})}{\partial t}\,\mathrm{d}\upsilon + \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon.$$

In conclusion, equation (5.57) for the infinitesimal control mass $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ that coincides with the fixed control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ at time t can be rewritten as:

$$\underbrace{\frac{\partial(\rho\vec{\boldsymbol{v}})}{\partial t}\,\mathrm{d}\upsilon}_{\text{rate of change of momentum}} + \underbrace{\nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon}_{\text{convective momentum outflow}} = \underbrace{\mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{body}}(t)}_{\text{body force}} + \underbrace{\mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{surface}}(t)}_{\text{surface force}} \tag{5.59}$$

This is Euler's first law of motion for an infinitesimal control mass $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ that coincides with the fixed control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ at time t. It states that the sum of the local rate of change of momentum and the net convective outflow of momentum through the faces of $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ must equal the total force acting on the material inside $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ at time t. In other words, the rate of change of momentum stored in $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ plus the net rate of momentum brought out by convection through the six faces, equals the body force acting on the mass inside $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ plus the net surface force exerted on those faces by the surrounding continuum.

In the next, we will analyze the convective outflow term, which is the most subtle and least intuitive of the terms in equation (5.59). We will then analyze the body and surface forces on the right-hand side, which are more straightforward.

Convective momentum outflow through the six faces

Let's try to understand the nature of the convective momentum outflow term $\nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon$ in more detail. Momentum is a vector, and it is transported across any surface element by the flow itself. The momentum flow rate (see equation [5.5]) *leaving* a face with area $\mathrm{d}S$ and outward unit normal $\hat{\boldsymbol{n}}$ is $(\rho\vec{\boldsymbol{v}})(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S$. The net convective momentum outflow is therefore the sum of outflow rates over all six faces.

On the face $\mathscr{F}_x^+$, located at $x+\mathrm{d}x$, the outward normal is $+\hat{\boldsymbol{e}}_x$, and the fields are approximated by their Taylor expansions about $\vec{\boldsymbol{x}}$. The momentum outflow leaving through $\mathscr{F}_x^+$ is:

$$\left[(\rho\vec{\boldsymbol{v}})\,v_x + \frac{\partial(\rho\vec{\boldsymbol{v}}\;v_x)}{\partial x}\,\mathrm{d}x\right]\mathrm{d}y\,\mathrm{d}z, \tag{5.60}$$

where all fields are evaluated at $\vec{\boldsymbol{x}}$ and $v_x = \vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{e}}_x$. On $\mathscr{F}_x^-$, the outward normal is $-\hat{\boldsymbol{e}}_x$, and the leaving flow is:

$$-(\rho\vec{\boldsymbol{v}})\;v_x\,\mathrm{d}y\,\mathrm{d}z. \tag{5.61}$$

The net outward convective momentum flow rate through the x-face pair is the sum of (5.60) and (5.61):

$$\frac{\partial(\rho\vec{\boldsymbol{v}}\;v_x)}{\partial x}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \frac{\partial(\rho\vec{\boldsymbol{v}}\;v_x)}{\partial x}\,\mathrm{d}\upsilon.$$

This pattern repeats identically for the y- and z-face pairs. The net outward convective momentum flow rates through those pairs are, respectively:

$$\frac{\partial(\rho\vec{\boldsymbol{v}}\, v_y)}{\partial y}\,\mathrm{d}\mathscr{v}, \qquad \text{and} \qquad \frac{\partial(\rho\vec{\boldsymbol{v}}\, v_z)}{\partial z}\,\mathrm{d}\mathscr{v}.$$

Summing all three contributions, the net *convective momentum outflow** is (see equation [5.10]):

$$\left[\frac{\partial(\rho\vec{\boldsymbol{v}}\, v_x)}{\partial x} + \frac{\partial(\rho\vec{\boldsymbol{v}}\, v_y)}{\partial y} + \frac{\partial(\rho\vec{\boldsymbol{v}}\, v_z)}{\partial z}\right]\mathrm{d}\mathscr{v} = \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\,\mathrm{d}\mathscr{v}.$$

This is the net rate at which momentum is being carried out of the control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ by the flow itself. If this term is positive, more momentum is leaving than entering, and the momentum stored in $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ decreases; if it is negative, more momentum is entering than leaving, and the momentum stored in $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ increases.

Let us remember the meaning of the symbol $\otimes$: it denotes the *outer product* of two vectors (see Appendix A.17), resulting in a second-order tensor. In this case, $\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}}$ is a tensor whose ij-th component is $\rho v_i v_j$. The divergence of this tensor, $\nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})$, is a vector whose i-th component is the sum of the partial derivatives of the ij-th components of the tensor with respect to the spatial coordinates, that is, $\sum_{j=1}^{3}\partial(\rho v_i v_j)/\partial x_j$ (see equation [5.2]); in other words,

$$\left[\nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\right]_i = \frac{\partial(\rho v_i v_x)}{\partial x} + \frac{\partial(\rho v_i v_y)}{\partial y} + \frac{\partial(\rho v_i v_z)}{\partial z}.$$

In conclusion, the convective momentum outflow $\nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})$ represents the net rate at which momentum is being carried out of the control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ by the flow itself. This term has no analog in the Lagrangian formulation, where no fluid ever crosses the boundary. It is the feature of the Eulerian description. For example, in a rocket engine, the convective flux term accounts for the momentum carried out of the engine by the exhaust gases, which is what propels the rocket forward.

Body forces on the infinitesimal element

The body force per unit mass is denoted $\vec{\boldsymbol{b}}$, so the total body force on the infinitesimal element $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ is:

$$\mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{body}} = \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v}. \tag{5.62}$$

This is the force that acts at a distance on the material inside $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$, regardless of whether it is in motion or not. Examples include gravitational forces and electromagnetic forces.

Surface forces through the six faces

The mechanical traction exerted by the surrounding continuum on each face of the element is governed by Cauchy's fundamental theorem of stress (see Section 4.2). On a face whose outward unit normal points in a given coordinate direction, the stress tensor $\boldsymbol{\sigma}$ supplies three traction components, one normal and two shearing, each of which acts as a force on that face. The nine components of $\boldsymbol{\sigma}$ are denoted σ_{ij}, where the first index i refers to the face (identified by the direction of its outward normal) and the second index j refers to the direction in which that traction component acts; in other words σ_{ij} acts in the direction of $\hat{\boldsymbol{e}}_j$. All components are evaluated at the reference corner $\vec{\boldsymbol{x}} = (x, y, z)$, and the Taylor expansion to first order in $\mathrm{d}x$, $\mathrm{d}y$, $\mathrm{d}z$ is used to express the stress on the displaced faces.

Figure 5.7 illustrates the convention for the x-components of the traction on all three face pairs; the same bookkeeping applies to the y- and z-components by complete analogy.

On the face $\mathscr{F}_x^+$, located at $x+\mathrm{d}x$ with outward normal in the positive x-direction, the three traction components have been corrected by one Taylor step from their values at $\vec{\boldsymbol{x}}$. The force vector exerted on that face is therefore:

$$\left[\left(\sigma_{xx}+\frac{\partial\sigma_{xx}}{\partial x}\,\mathrm{d}x\right)\hat{\boldsymbol{e}}_x+\left(\sigma_{xy}+\frac{\partial\sigma_{xy}}{\partial x}\,\mathrm{d}x\right)\hat{\boldsymbol{e}}_y+\left(\sigma_{xz}+\frac{\partial\sigma_{xz}}{\partial x}\,\mathrm{d}x\right)\hat{\boldsymbol{e}}_z\right]\mathrm{d}y\,\mathrm{d}z. \tag{5.63}$$

On the opposite face $\mathscr{F}_x^-$, located at x with outward normal in the negative x-direction, the traction reverses sign and no Taylor correction is needed (this face passes through $\vec{\boldsymbol{x}}$). The force on that face is:

$$\left[-\sigma_{xx}\hat{\boldsymbol{e}}_x-\sigma_{xy}\hat{\boldsymbol{e}}_y-\sigma_{xz}\hat{\boldsymbol{e}}_z\right]\mathrm{d}y\,\mathrm{d}z. \tag{5.64}$$

Adding (5.63) and (5.64), the zeroth-order terms cancel and the net force from the x-face pair is:

$$\mathrm{d}\vec{\boldsymbol{f}}_{x\text{-faces}}=\left(\frac{\partial\sigma_{xx}}{\partial x}\hat{\boldsymbol{e}}_x+\frac{\partial\sigma_{xy}}{\partial x}\hat{\boldsymbol{e}}_y+\frac{\partial\sigma_{xz}}{\partial x}\hat{\boldsymbol{e}}_z\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z. \tag{5.65}$$

This procedure repeats identically for the y- and z-face pairs. The net force from the y-face pair is:

$$\mathrm{d}\vec{\boldsymbol{f}}_{y\text{-faces}}=\left(\frac{\partial\sigma_{yx}}{\partial y}\hat{\boldsymbol{e}}_x+\frac{\partial\sigma_{yy}}{\partial y}\hat{\boldsymbol{e}}_y+\frac{\partial\sigma_{yz}}{\partial y}\hat{\boldsymbol{e}}_z\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z. \tag{5.66}$$

and the net force from the z-face pair is:

$$\mathrm{d}\vec{\boldsymbol{f}}_{z\text{-faces}}=\left(\frac{\partial\sigma_{zx}}{\partial z}\hat{\boldsymbol{e}}_x+\frac{\partial\sigma_{zy}}{\partial z}\hat{\boldsymbol{e}}_y+\frac{\partial\sigma_{zz}}{\partial z}\hat{\boldsymbol{e}}_z\right)\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z. \tag{5.67}$$

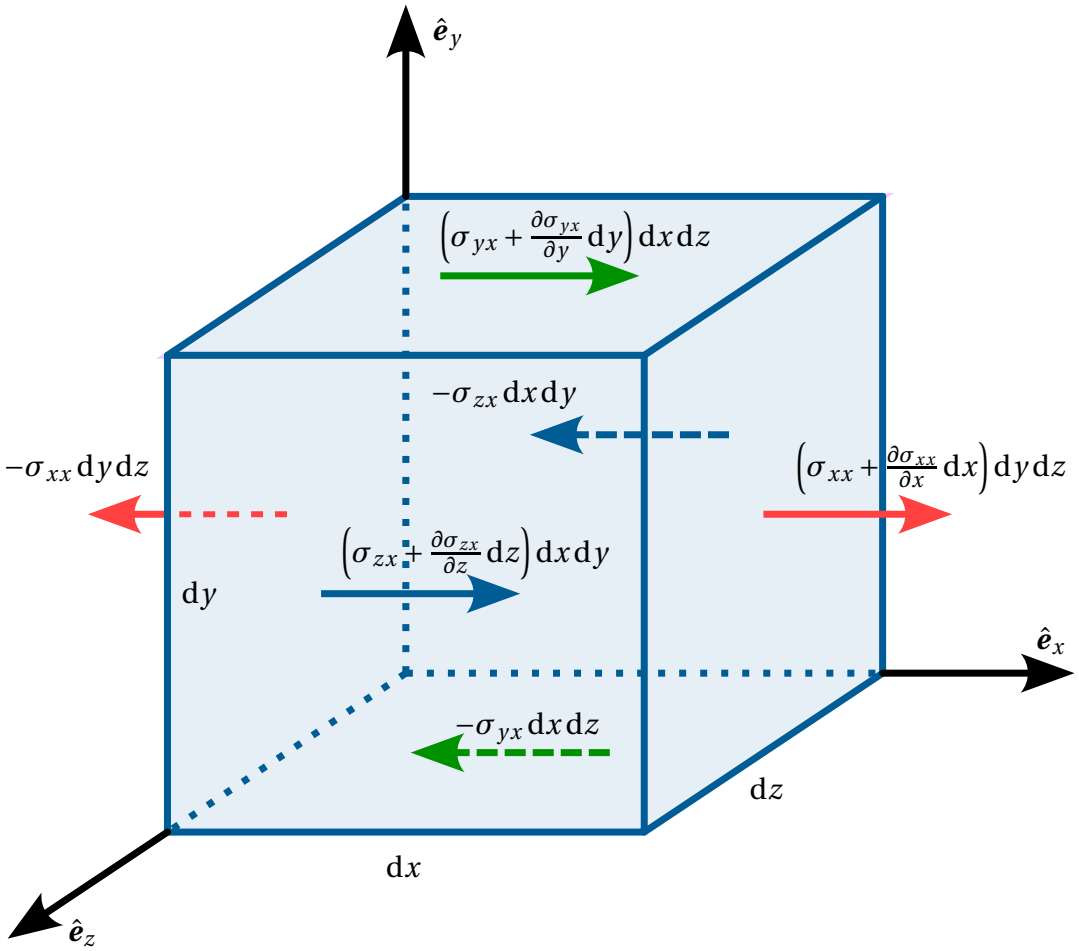


Figure 5.7: *Surface forces that act on the x-direction on the infinitesimal element* $\mathrm{d}\mathscr{V}^{cv}$. *Red arrows: normal tractions from the x-face pair. Green arrows: shear tractions from the y-face pair. Blue arrows: shear tractions from the z-face pair. The face at* $x + \mathrm{d}x$, $y + \mathrm{d}y$, *or* $z + \mathrm{d}z$ *carries the Taylor-corrected value; the opposing face carries the value at* $\vec{\boldsymbol{x}}$ *with reversed sign.*

Adding the three contributions (5.65), (5.66), and (5.67), collecting the $\hat{\boldsymbol{e}}_x$-, $\hat{\boldsymbol{e}}_y$-, and $\hat{\boldsymbol{e}}_z$-components separately, and taking into account that $\mathrm{d}\mathscr{v} = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, as indicated by equation (5.56), the total net surface force on the element is:

$$\mathrm{d}\vec{\boldsymbol{f}}_{\text{surf}} = \underbrace{\left(\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + \frac{\partial \sigma_{zx}}{\partial z}\right)}_{\text{net force per unit volume in } x}\hat{\boldsymbol{e}}_x\,\mathrm{d}\mathscr{v} + \underbrace{\left(\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zy}}{\partial z}\right)}_{\text{net force per unit volume in } y}\hat{\boldsymbol{e}}_y\,\mathrm{d}\mathscr{v}$$

$$+\underbrace{\left(\frac{\partial \sigma_{xz}}{\partial x}+\frac{\partial \sigma_{yz}}{\partial y}+\frac{\partial \sigma_{zz}}{\partial z}\right)}_{\text{net force per unit volume in } z}\hat{\boldsymbol{e}}_z \,\mathrm{d}\mathcal{v}. \quad (5.68)$$

Each scalar coefficient in (5.68) is precisely the corresponding component of the divergence of the stress tensor $\boldsymbol{\sigma}$. We may therefore write the result compactly as:

$$\mathrm{d}\vec{\boldsymbol{f}}_{\text{surf}} = \nabla \cdot \boldsymbol{\sigma} \,\mathrm{d}\mathcal{v}, \quad (5.69)$$

where, the divergence of the stress tensor is the vector field defined by:

$$\nabla \cdot \boldsymbol{\sigma} = \begin{bmatrix} \frac{\partial \sigma_{xx}}{\partial x}+\frac{\partial \sigma_{yx}}{\partial y}+\frac{\partial \sigma_{zx}}{\partial z} \\ \frac{\partial \sigma_{xy}}{\partial x}+\frac{\partial \sigma_{yy}}{\partial y}+\frac{\partial \sigma_{zy}}{\partial z} \\ \frac{\partial \sigma_{xz}}{\partial x}+\frac{\partial \sigma_{yz}}{\partial y}+\frac{\partial \sigma_{zz}}{\partial z} \end{bmatrix}.$$

This equation states that the net surface force on the infinitesimal element $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ is the divergence of the stress tensor at $\vec{\boldsymbol{x}}$ times the volume of the element. The divergence of the stress tensor is a vector field that represents the net force per unit volume exerted by the surrounding continuum on the material at each point in space. It is important to note that the surface forces are transmitted through the material even when it is at rest, and that they can be nonzero even in the absence of any convective transport of momentum.

Assembling the momentum balance

We are now in a position to substitute every term into equation (5.59). Inserting (5.62) and (5.69), the balance becomes:

$$\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}})\,\mathrm{d}\mathcal{v} + \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}})\,\mathrm{d}\mathcal{v} = \nabla \cdot \boldsymbol{\sigma}\,\mathrm{d}\mathcal{v} + \vec{\boldsymbol{b}}\,\mathrm{d}\mathcal{v}.$$

The volume element $\mathrm{d}\mathcal{v} = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$ is common to every term and strictly positive. Dividing through by $\mathrm{d}\mathcal{v}$ immediately yields the *linear momentum equation for an infinitesimal control volume fixed in space*:

$$\boxed{\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}}) + \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) = \nabla \cdot \boldsymbol{\sigma} + \vec{\boldsymbol{b}},} \quad (5.70)$$

or explicitly, we obtain a family of three equations, one for each component of the velocity vector:

$$\frac{\partial}{\partial t}(\rho v_x) + \frac{\partial(\rho v_x v_x)}{\partial x} + \frac{\partial(\rho v_x v_y)}{\partial y} + \frac{\partial(\rho v_x v_z)}{\partial z} = \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + \frac{\partial \sigma_{zx}}{\partial z} + b_x$$

$$\frac{\partial}{\partial t}(\rho v_y) + \frac{\partial(\rho v_y v_x)}{\partial x} + \frac{\partial(\rho v_y v_y)}{\partial y} + \frac{\partial(\rho v_y v_z)}{\partial z} = \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zy}}{\partial z} + b_y$$
$$\frac{\partial}{\partial t}(\rho v_z) + \frac{\partial(\rho v_z v_x)}{\partial x} + \frac{\partial(\rho v_z v_y)}{\partial y} + \frac{\partial(\rho v_z v_z)}{\partial z} = \frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + b_z.$$

These equations are upon which the analysis of fluid flow, wave propagation, and material deformation is ultimately founded. Observe that convective transport is nonlinear (it involves $\vec{\boldsymbol{v}}$ quadratically), while stress transport is linear in $\boldsymbol{\sigma}$ and communicates forces even when the fluid is at rest.

5.10.4 Case 4: linear momentum equation for an infinitesimal control mass moving with the flow

In this Lagrangian viewpoint, we analyze an specific infinitesimal control mass, i.e. a material element, and ride with it, as it moves, deforms, and interacts with its surroundings. This control mass is defined once, at some reference instant, and it keeps the same molecules forever. No mass crosses its boundary, so Euler's first law applies directly to it without any flux bookkeeping.

This material element is small enough that velocity, density and stress are essentially uniform over it, but large enough to be a continuum. At any instant t, this element occupies a small region of current space centered at $\vec{\boldsymbol{x}}$, with volume $\mathrm{d}\mathcal{v}_t$ and mass $\mathrm{d}m = \rho(\vec{\boldsymbol{x}}, t)\,\mathrm{d}\mathcal{v}_t$.

This infinitesimal control mass volume moves and deforms with the material. By construction $\mathrm{d}m$ is constant in time, thus its material derivative is null, that is $\frac{\mathrm{D}(\mathrm{d}m)}{\mathrm{D}t} = 0$, because we are always following the same matter.

The linear momentum carried by this material element is the product of its invariant mass and its instantaneous velocity, $\vec{\boldsymbol{p}}_{\mathrm{d}\mathcal{V}_t^{\mathrm{cm}}} = \mathrm{d}m\vec{\boldsymbol{v}} = \rho\vec{\boldsymbol{v}}\,\mathrm{d}\mathcal{v}_t$.

Euler's first law (5.46) says that the rate of change of linear momentum of this material element equals the sum all external body forces and surface forces acting on it at that instant; in other words, the momentum balance for this material element is:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathcal{V}_t^{\mathrm{cm}}}\right) = \mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{body}}(t) + \mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{surface}}(t), \tag{5.71}$$

where $\mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{body}}$ and $\mathrm{d}\vec{\boldsymbol{f}}_{\mathrm{surface}}$ are the body and surface forces acting on the material element at instant t. Because $\mathrm{d}m$ does not change, the left side collapses to mass times acceleration following the particle:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\vec{\boldsymbol{p}}_{\mathrm{d}\mathcal{V}_t^{\mathrm{cm}}}\right) = \mathrm{d}m\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} = \rho\,\mathrm{d}\mathcal{v}_t\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t};$$

here $\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t}$ stands for the material acceleration of the particle (see Section 3.21.4).

Now, what about the body force $\mathrm{d}\vec{\boldsymbol{f}}_{\text{body}}(t)$ and contact force $\mathrm{d}\vec{\boldsymbol{f}}_{\text{surface}}(t)$? The former act at a distance, distributed throughout the volume of the element; the latter are exerted by the surrounding continuum through the element's boundary. We analyze each in turn:

Body force: with $\vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t)$ the body force per unit mass, the contribution is distributed through the volume,

$$\mathrm{d}\vec{\boldsymbol{f}}_{\text{body}} = \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v}_t. \tag{5.72}$$

Surface force: The surrounding continuum exerts contact forces on the element through its boundary $\partial\,\mathrm{d}\mathscr{V}_t^{\text{cm}}$, and these forces depend on both the local state of stress and the orientation of each surface element. By Cauchy's theorem, the traction vector $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}$ on any surface element with outward unit normal $\hat{\boldsymbol{n}}$ is given by $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})} = \boldsymbol{\sigma}\hat{\boldsymbol{n}}$, where $\boldsymbol{\sigma}$ is the Cauchy stress tensor and $\hat{\boldsymbol{n}}$ the outward normal. Adding the tractions over the closed surface and using the divergence theorem for an infinitesimal region gives the net contact force as the divergence of stress times the current volume,

$$\mathrm{d}\vec{\boldsymbol{f}}_{\text{surface}} = \oiint_{\partial\,\mathrm{d}\mathscr{V}_t^{\text{cm}}} \boldsymbol{\sigma}\hat{\boldsymbol{n}}\,\mathrm{d}S = \iiint_{\mathrm{d}\mathscr{V}_t^{\text{cm}}} \nabla\cdot\boldsymbol{\sigma}\,\mathrm{d}\mathscr{v}_t \approx (\nabla\cdot\boldsymbol{\sigma})\,\mathrm{d}\mathscr{v}_t, \tag{5.73}$$

where $\nabla\cdot\boldsymbol{\sigma}$ is the vector sum of all the tiny pulls on opposite faces; it is the net force per unit volume that the surrounding continuum exerts on the material element. The approximation in the last step becomes exact in the limit as the element shrinks to a point, where $\nabla\cdot\boldsymbol{\sigma}$ is uniform over the element. Alternatively, we can estimate $\mathrm{d}\vec{\boldsymbol{f}}_{\text{surface}}$ using a procedure analogous to the one used for the demonstration of equation (5.69).

Substituting (5.72) and (5.73) into the right-hand side of Euler's first law (5.71), we obtain:

$$\rho\,\mathrm{d}\mathscr{v}_t\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} = (\nabla\cdot\boldsymbol{\sigma})\,\mathrm{d}\mathscr{v}_t + \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v}_t,$$

and dividing through by $\mathrm{d}\mathscr{v}_t$, which is everywhere strictly positive for a material element in a regular continuum, we arrive at the *local, pointwise form of the linear momentum balance for a material element*:

$$\boxed{\rho(\vec{\boldsymbol{x}}, t)\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} = \nabla\cdot\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) + \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t),} \tag{5.74}$$

or explicitly, we obtain a family of three equations, one for each component of the velocity vector:

$$\rho\frac{\mathrm{D}v_x}{\mathrm{D}t} = \frac{\partial\sigma_{xx}}{\partial x} + \frac{\partial\sigma_{yx}}{\partial y} + \frac{\partial\sigma_{zx}}{\partial z} + b_x$$

$$\rho \frac{\mathrm{D}v_y}{\mathrm{D}t} = \frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zy}}{\partial z} + b_y$$
$$\rho \frac{\mathrm{D}v_z}{\mathrm{D}t} = \frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + b_z.$$

Equation (5.74) is the *Cauchy's equation of motion** (also known as the *Cauchy's momentum equation**), the local, Lagrangian statement of linear momentum balance for a material point in a continuum. It applies at every point in the continuum, and it is the starting point for deriving the Navier-Stokes equations for fluid flow, as well as the equations of elastodynamics for solid mechanics. Observe that in this equation no flux terms appear because the system is closed.

When the material is at rest, the left-hand side of equation (5.74) vanishes, and the balance reduces to $\nabla \cdot \boldsymbol{\sigma} + \vec{\boldsymbol{b}} = \vec{\mathbf{0}}$:

$$\frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{yx}}{\partial y} + \frac{\partial \sigma_{zx}}{\partial z} + b_x = 0$$
$$\frac{\partial \sigma_{xy}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{zy}}{\partial z} + b_y = 0$$
$$\frac{\partial \sigma_{xz}}{\partial x} + \frac{\partial \sigma_{yz}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} + b_z = 0,$$

which are the *Cauchy's static equilibrium equations** for a continuum. This means that the divergence of the stress tensor must balance the body forces at every point in space for the material to be in static equilibrium.

5.10.5 Mathematical equivalence between all four cases

The four forms of the linear momentum balance equation we have derived, and that are summarized in Figure 5.8, represent the same physical principle of conservation of linear momentum, but they are expressed in different mathematical forms and in different reference frames. Let us explore their relationships and demonstrate their equivalence systematically.

Our four equations can be classified along two important dimensions:

- By mathematical form: Integral equations (5.51), (5.55) vs. Differential equations (5.70), (5.74)
- By formulation type: Conservation form (5.51), (5.70) vs. Non-conservation form (5.55), (5.74)

Any of these equations can be deduced by mathematical manipulation from any of the others, as we will illustrate in the next, for the paths shown in Figure 5.8.

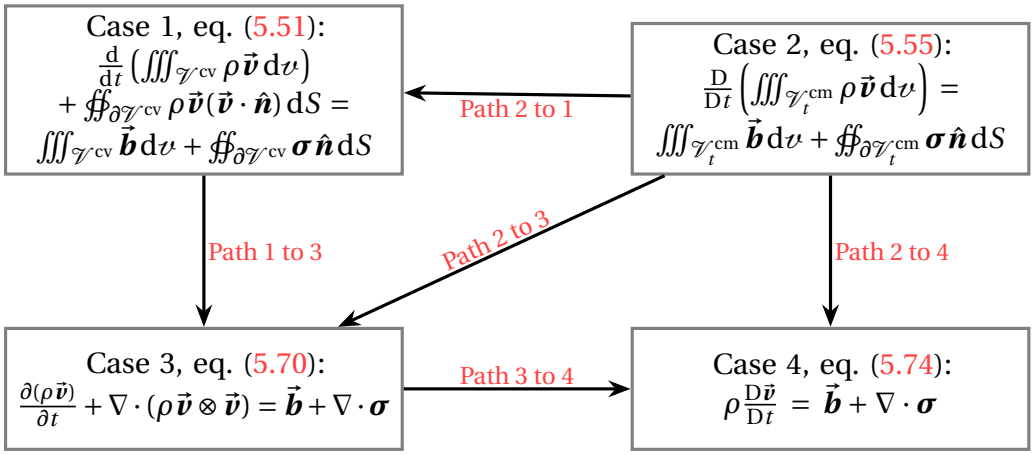


Figure 5.8: *The four different variants of the linear momentum balance equation.*

Path 2 to 1: from equation (5.55) to equation (5.51)

We will make $\boldsymbol{\mu} = \rho\vec{\boldsymbol{v}}$ in the Reynolds transport formula (5.17), which gives:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\upsilon\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

This equation states that the material time derivative of the linear momentum of the control mass $\mathscr{V}_t^{\mathrm{cm}}$ at time t is equal to the sum of two contributions: the local rate of change of linear momentum within the control mass $\mathscr{V}_t^{\mathrm{cm}}$, and the net flux of momentum across the boundary $\partial\mathscr{V}_t^{\mathrm{cm}}$ of the control mass due to the motion of the material through it; in other words, the second integral on the right-hand side accounts for the net rate at which linear momentum is being *carried out* of the control volume by the flowing material.

Now, we will apply this result to the left-hand side of the integral from of the linear momentum balance equation for a control mass (5.55):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\upsilon\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\sigma}\hat{\boldsymbol{n}}\,\mathrm{d}S.$$

obtaining:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\sigma}\hat{\boldsymbol{n}}\,\mathrm{d}S. \tag{5.75}$$

If we consider a control volume $\mathscr{V}^{\mathrm{cv}}$ that is fixed in space and that exactly coincides with $\mathscr{V}_t^{\mathrm{cm}}$ at a particular time t, we have:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\sigma}\hat{\boldsymbol{n}}\,\mathrm{d}S.$$

Using equation (5.12) for a control volume fixed in space, we can rewrite the first integral on the left-hand side as:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon = \frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\upsilon\right).$$

Substituting this back into the previous equation and rearranging terms gives:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\upsilon\right) = -\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S + \iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\sigma}\hat{\boldsymbol{n}}\,\mathrm{d}S.$$

This is precisely the integral form of the linear momentum balance equation for a fixed control volume, which is case 1, equation (5.51). Thus, we have shown that the integral form of the momentum equation for a material volume (case 2) reduces to the integral form of the momentum equation for a fixed control volume (case 1) when the control volume $\mathscr{V}^{\mathrm{cv}}$ is chosen to coincide with the material volume $\mathscr{V}_t^{\mathrm{cm}}$ at a particular time t.

Path 2 to 3: from equation (5.55) to equation (5.70)

Let us continue our demonstration from (5.75). Applying (5.8), that is $\rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}) = (\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\cdot\hat{\boldsymbol{n}}$, to the second integral of the left-hand side and the divergence theorem, for a second order tensor (see Appendix A.16), to both surface integrals gives an expression in terms of volume integrals:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon + \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \nabla\cdot\boldsymbol{\sigma}\,\mathrm{d}\upsilon.$$

Here, we are implicitly assuming that the fields ρ, $\vec{\boldsymbol{v}}$, and $\boldsymbol{\sigma}$ are sufficiently smooth in $\mathscr{V}_t^{\mathrm{cm}}$ so that all the integrals and derivatives are well-defined and the theorems of vector calculus apply.

Rearranging terms into a single volume integral gives:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}}) + \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}}) - \vec{\boldsymbol{b}} - \nabla\cdot\boldsymbol{\sigma}\right]\mathrm{d}\upsilon = 0.$$

Now, since this holds for any arbitrary control mass at any time $t \in T$, we can apply the localization theorem (see Appendix A.18) to conclude that the integrand must vanish at every point in space and time, yielding the *differential form of the linear momentum balance equation* in the spatial description (5.70):

$$\frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}}) + \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}}) = \vec{\boldsymbol{b}} + \nabla\cdot\boldsymbol{\sigma}.$$

We have thus shown that the integral form of the linear momentum balance equation for a material volume (case 2) reduces to the differential form of the linear momentum balance equation for a fixed control volume (case 3) when we assume sufficient smoothness of ρ, $\vec{\boldsymbol{v}}$, and $\boldsymbol{\sigma}$.

Path 2 to 4: from equation (5.55) to equation (5.74)

In the following, we will assume that the fields ρ, $\vec{\boldsymbol{v}}$, and $\boldsymbol{\sigma}$ are sufficiently smooth in $\mathscr{V}_t^{\mathrm{cm}}$ so that all the integrals and derivatives are well-defined and the theorems of vector calculus apply.

Using equation (5.41) with $\boldsymbol{\psi} = \vec{\boldsymbol{v}}$, we obtain:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t}\,\mathrm{d}\upsilon;$$

Replacing this result into (5.55) gives:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t}\,\mathrm{d}\upsilon = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S.$$

Applying the divergence theorem to the surface integral, we obtain:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}},t)\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t}\,\mathrm{d}\upsilon = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon + \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \nabla\cdot\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon.$$

Rearranging terms into a single volume integral gives:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \left[\rho(\vec{\boldsymbol{x}},t)\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)}{\mathrm{D}t} - \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t) - \nabla\cdot\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\right]\mathrm{d}\upsilon = 0.$$

Since this holds for any arbitrary control mass at any time $t \in T$, we can apply the localization theorem (see Appendix A.18) to conclude that the integrand must vanish at every point in space and time, yielding the *differential form of the linear momentum balance equation for a material volume*, also known as the *Cauchy's momentum equation* (5.74):

$$\rho\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} = \nabla\cdot\boldsymbol{\sigma} + \vec{\boldsymbol{b}}.$$

Thus, we have shown that the integral form of the linear momentum balance equation for a material volume (case 2) reduces to the differential form of the linear momentum balance equation for a material volume (case 4) when we assume sufficient smoothness of ρ, $\vec{\boldsymbol{v}}$, and $\boldsymbol{\sigma}$.

Path 1 to 3: from equation (5.51) to equation (5.70)

Applying (5.8), that is $\rho\vec{\boldsymbol{v}}(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}}) = (\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\cdot\hat{\boldsymbol{n}}$, and the divergence theorem to the surface integrals in (5.51) yields:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \rho\vec{\boldsymbol{v}}\,\mathrm{d}\upsilon\right) = -\iiint_{\mathscr{V}^{\mathrm{cv}}} \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon + \iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \iiint_{\mathscr{V}^{\mathrm{cv}}} \nabla\cdot\boldsymbol{\sigma}\,\mathrm{d}\upsilon.$$

Rearranging terms into a single volume integral gives:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \left[\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}}) + \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) - \vec{\boldsymbol{b}} - \nabla \cdot \boldsymbol{\sigma} \right] \mathrm{d}v = 0.$$

Since this holds for any arbitrary control volume at any time $t \in T$, we can apply the localization theorem (see Appendix A.18) to conclude that the integrand must vanish at every point in space and time, yielding the *differential form of the linear momentum equation* in the spatial description (5.70):

$$\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}}) + \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) = \vec{\boldsymbol{b}} + \nabla \cdot \boldsymbol{\sigma}.$$

We have thus shown that the integral form of the linear momentum balance equation for a fixed control volume (case 1) reduces to the differential form of the linear momentum balance equation for a fixed control volume (case 3) when we assume sufficient smoothness of ρ, $\vec{\boldsymbol{v}}$, and $\boldsymbol{\sigma}$.

Path 3 to 4: from equation (5.70) to equation (5.74)

Our starting point is the differential form of the linear momentum equation for a fixed control volume, which is case 3, equation (5.70):

$$\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}}) + \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) = \vec{\boldsymbol{b}} + \nabla \cdot \boldsymbol{\sigma}.$$

We expand the two terms on the left-hand side using the product rule, equation (A.2) for the divergence of an outer product (i.e., $\nabla \cdot (\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}) = (\nabla \cdot \vec{\boldsymbol{b}})\vec{\boldsymbol{a}} + \frac{\partial \vec{\boldsymbol{a}}}{\partial \vec{\boldsymbol{x}}}\vec{\boldsymbol{b}}$), and equation (A.3) for the gradient of a scalar-vector product (i.e., $\nabla(\phi \vec{\boldsymbol{a}}) = \phi \nabla \vec{\boldsymbol{a}} + \vec{\boldsymbol{a}} \otimes \nabla \phi$):

$$\begin{aligned}
\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}}) &= \rho \frac{\partial \vec{\boldsymbol{v}}}{\partial t} + \vec{\boldsymbol{v}} \frac{\partial \rho}{\partial t}, \\
\nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) &= (\nabla \cdot \vec{\boldsymbol{v}})\rho \vec{\boldsymbol{v}} + \frac{\partial (\rho \vec{\boldsymbol{v}})}{\partial \vec{\boldsymbol{x}}} \vec{\boldsymbol{v}} \qquad\qquad = (\nabla \cdot \vec{\boldsymbol{v}})\rho \vec{\boldsymbol{v}} + \left(\rho \frac{\partial \vec{\boldsymbol{v}}}{\partial \vec{\boldsymbol{x}}} + \vec{\boldsymbol{v}} \otimes \frac{\partial \rho}{\partial \vec{\boldsymbol{x}}} \right) \vec{\boldsymbol{v}} \\
&= (\nabla \cdot \vec{\boldsymbol{v}})\rho \vec{\boldsymbol{v}} + \rho \frac{\partial \vec{\boldsymbol{v}}}{\partial \vec{\boldsymbol{x}}} \vec{\boldsymbol{v}} + \left(\vec{\boldsymbol{v}} \otimes \frac{\partial \rho}{\partial \vec{\boldsymbol{x}}} \right) \vec{\boldsymbol{v}} = (\nabla \cdot \vec{\boldsymbol{v}})\rho \vec{\boldsymbol{v}} + \rho \frac{\partial \vec{\boldsymbol{v}}}{\partial \vec{\boldsymbol{x}}} \vec{\boldsymbol{v}} + \vec{\boldsymbol{v}} \left(\frac{\partial \rho}{\partial \vec{\boldsymbol{x}}} \cdot \vec{\boldsymbol{v}} \right)
\end{aligned}$$

Adding these two equations gives:

$$\rho \underbrace{\left(\frac{\partial \vec{\boldsymbol{v}}}{\partial t} + \frac{\partial \vec{\boldsymbol{v}}}{\partial \vec{\boldsymbol{x}}} \vec{\boldsymbol{v}} \right)}_{= \frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} \text{ by (3.87)}} + \vec{\boldsymbol{v}} \frac{\partial \rho}{\partial t} + \vec{\boldsymbol{v}} \underbrace{\left((\nabla \cdot \vec{\boldsymbol{v}})\rho + \nabla \rho \cdot \vec{\boldsymbol{v}} \right)}_{= \nabla \cdot (\rho \vec{\boldsymbol{v}}) \text{ by (A.4)}} = \rho \frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} + \vec{\boldsymbol{v}} \underbrace{\left(\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \vec{\boldsymbol{v}}) \right)}_{= 0 \text{ by (5.27)}}.$$

The first term in the left-hand side is the product of the density and the material acceleration $\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t}$, while the third term is the velocity multiplied by the expression

in parentheses. This expression in parentheses is the divergence of the mass flux $\rho\vec{\boldsymbol{v}}$, according to equation (A.4), that is, $\nabla\cdot(\rho\vec{\boldsymbol{v}}) = \nabla\rho\cdot\vec{\boldsymbol{v}} + \rho\nabla\cdot\vec{\boldsymbol{v}}$. Now, the second term in the right-hand side is the continuity equation (5.27), which states that mass is conserved. Since mass conservation holds, this expression must be zero at every point in space and time, and therefore the second term in the right-hand side vanishes.

This result reduces (5.70) to the Cauchy's equation of motion (5.74):

$$\rho\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} = \nabla\cdot\boldsymbol{\sigma} + \vec{\boldsymbol{b}}.$$

Thus, we have shown that the differential form of the linear momentum balance equation for a fixed control volume (case 3) reduces to the differential form of the linear momentum balance equation for a material volume (case 4).

Other conversions

Here we have performed the 5 out of 12 possible conversions; however, we will not go on making demonstrations of the equivalences and the other 7 cases are left as exercises to the reader.

5.10.6 Linear momentum balance equations in the material description

In the previous sections we have derived the linear momentum balance equations in the spatial description , which are expressed in terms of the spatial coordinates $\vec{\boldsymbol{x}}$ and time t. However, it is also possible to express these equations in the material description, which uses the material coordinates $\vec{\boldsymbol{X}}$ and time t. The material description is not used in fluid mechanics since in fluids there is no natural reference configuration Ω_0, but it is the natural framework for finite elasticity and nonlinear solid mechanics, where the reference configuration is known and fixed.

We will see that the material formulations will be defined in the reference configuration, where the normals, areas, and volume integrals are taken in the undeformed body. Also, it will employ the first Piola-Kirchhoff stress tensor $\boldsymbol{P}$, to express the traction on reference area elements, instead of the Cauchy stress tensor $\boldsymbol{\sigma}$, which is more convenient to express the traction on current area elements.

In the following, we will express in the material description the linear momentum balance equation for the Lagrangian formulations, i.e cases 2 and 4.

To rewrite the formulations in the material description Ω_0, let's recall equation (3.1), $\vec{\boldsymbol{x}} = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t)$, which relates the spatial coordinates $\vec{\boldsymbol{x}}$ to the material coordinates $\vec{\boldsymbol{X}}$ through the equation of motion $\vec{\boldsymbol{\chi}}$. Now, let's establish the correspondence between quantities in each description:

Conservation of mass: The local form of mass conservation relates the reference density $\rho_0(\vec{\boldsymbol{X}})$ to the current density $\rho(\vec{\boldsymbol{x}}, t)$ through equation (5.29), that is:

$$\rho_0(\vec{\boldsymbol{X}}) = J(\vec{\boldsymbol{X}}, t)\, \rho(\vec{\boldsymbol{x}}, t).$$

Velocity: The material velocity $\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{v}}(\vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t), t)$ is simply the spatial velocity $\vec{\boldsymbol{v}}$ expressed in material coordinates. Their material time derivatives are thus equivalent:

$$\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} = \frac{\mathrm{D}\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)}{\mathrm{D}t}.$$

Determinant of the deformation gradient: The determinant J of the deformation gradient $\boldsymbol{F}$, i.e. $J(\vec{\boldsymbol{X}}, t) = \det \boldsymbol{F}(\vec{\boldsymbol{X}}, t)$, relates the volume elements in the current and reference configurations (equation [3.35]):

$$\mathrm{d}\mathscr{v}_t = J(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V}_0.$$

Body force per unit reference volume: The body force $\vec{\boldsymbol{b}}$ per unit current volume and $\vec{\boldsymbol{B}}$ per unit reference volume are related by:

$$\vec{\boldsymbol{B}}(\vec{\boldsymbol{X}}, t) = J(\vec{\boldsymbol{X}}, t)\vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t).$$

Relationship between divergences: We will invoke the transformation (4.6), which relates the first Piola-Kirchhoff stress tensor $\boldsymbol{P}$ to the Cauchy stress tensor $\boldsymbol{\sigma}$:

$$\nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t) = J(\vec{\boldsymbol{X}}, t)\nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t).$$

Nanson's formula: The normal vector $\hat{\boldsymbol{N}}$ and area element $\mathrm{d}S_0$ in the reference configuration are related to the normal vector $\hat{\boldsymbol{n}}$ and area element $\mathrm{d}S$ in the current configuration by Nanson's formula (3.37):

$$\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}S = J(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\, \mathrm{d}S_0.$$

Relationship between the stress tensors: The first Piola-Kirchhoff stress tensor $\boldsymbol{P}$ is related to the Cauchy stress tensor $\boldsymbol{\sigma}$ by equation (4.4):

$$\boldsymbol{P}(\vec{\boldsymbol{X}}, t) = J(\vec{\boldsymbol{X}}, t)\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{x}}, t).$$

Case 2: linear momentum equation for a finite control mass moving with the flow

In Section 5.10.2, we showed that the integral form of the linear momentum balance equation for a material volume is given by equation (5.55):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho(\vec{\boldsymbol{x}}, t)\, \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}\mathscr{v}\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\, \hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}S.$$

This equation is written in the spatial description Ω_t.

Let's express this equation in the material description Ω_0. Using the relations established above, we can rewrite the left-hand side as:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\rho(\vec{\boldsymbol{x}},t)J(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0\right) = \frac{\partial}{\partial t}\left(\iiint_{\mathscr{V}_0^{\mathrm{cm}}} \rho_0(\vec{\boldsymbol{X}})\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0\right).$$

Here, we have used the fact that the material time derivative of a quantity expressed in material coordinates is simply the partial derivative with respect to time, since the material coordinates are fixed, and the shape of the reference control mass $\mathscr{V}_0^{\mathrm{cm}}$ does not change in time.

Now, we can express the volume integral right-hand side in the material description, by using the relation between the body force per unit current volume $\vec{\boldsymbol{b}}$ and the body force per unit reference volume $\vec{\boldsymbol{B}}$:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}v = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)J(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0 = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{B}}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0.$$

Similarly, we can express the surface integral right-hand side in the material description, by means of Nanson's formula and the transformation relating the Cauchy and first Piola-Kirchhoff stress tensors:

$$\begin{aligned}\oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}S &= \oiint_{\partial\mathscr{V}_0^{\mathrm{cm}}} \boldsymbol{\sigma}(\vec{\boldsymbol{x}},\vec{\boldsymbol{t}})J(\vec{\boldsymbol{X}},t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{X}},t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S \\ &= \oiint_{\partial\mathscr{V}_0^{\mathrm{cm}}} \boldsymbol{P}(\vec{\boldsymbol{X}},t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S_0.\end{aligned}$$

Putting everything together, we can express the integral form of the linear momentum balance equation for a material volume in the material description Ω_0:

$$\boxed{\frac{\partial}{\partial t}\left(\iiint_{\mathscr{V}_0^{\mathrm{cm}}} \rho_0(\vec{\boldsymbol{X}})\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0\right) = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{B}}(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0 + \oiint_{\partial\mathscr{V}_0^{\mathrm{cm}}} \boldsymbol{P}(\vec{\boldsymbol{X}},t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\,\mathrm{d}S_0.} \tag{5.76}$$

This equation is now expressed in the material description, as it is written in terms of the material coordinates $\vec{\boldsymbol{X}}$ and time t. The form of the equation is similar to the one in the spatial description, but the variables and derivatives are now expressed in terms of the material coordinates and the deformation gradient.

Observe that there is no convective flux term in this equation, since the control mass is moving with the flow, and therefore there is no flux of linear momentum across its

boundary. This is a key difference between the integral form of the linear momentum balance for a material volume and the integral form of the linear momentum balance for a fixed control volume. Also the traction vector $\vec{\boldsymbol{t}}^{(\hat{\boldsymbol{N}})}(\vec{\boldsymbol{X}}, t) = \boldsymbol{P}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})$ gives the actual (current) force vector acting on a reference area element $\mathrm{d}S_0$ with outward normal $\hat{\boldsymbol{N}}$.

Case 4: linear momentum equation for an infinitesimal control mass moving with the flow

In Section 5.10.4, we showed that the differential form of the linear momentum balance equation for an infinitesimal material volume, also known as Cauchy's momentum equation, is written in the spatial description Ω_t, and is given by equation (5.74):

$$\rho(\vec{\boldsymbol{x}}, t)\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} = \nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) + \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t).$$

Let's multiply this equation by $J(\vec{\boldsymbol{X}}, t)$:

$$J(\vec{\boldsymbol{X}}, t)\rho(\vec{\boldsymbol{x}}, t)\frac{\mathrm{D}\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}}, t)}{\mathrm{D}t} = J(\vec{\boldsymbol{X}}, t)\nabla_{\vec{\boldsymbol{x}}} \cdot \boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) + J(\vec{\boldsymbol{X}}, t)\vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t);$$

now, we can substitute the relations established above to express this equation in terms of the material coordinates $\vec{\boldsymbol{X}}$ and time t:

$$\boxed{\rho_0(\vec{\boldsymbol{X}})\frac{\mathrm{D}\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)}{\mathrm{D}t} = \nabla_{\vec{\boldsymbol{X}}} \cdot \boldsymbol{P}(\vec{\boldsymbol{X}}, t) + \vec{\boldsymbol{B}}(\vec{\boldsymbol{X}}, t).} \tag{5.77}$$

This equation is the material form of Cauchy's momentum equation. Observe that its form is similar to the one in the spatial description, but the variables and derivatives are now expressed in terms of the material coordinates $\vec{\boldsymbol{X}}$.

Finally, it is worth mentioning that equation (5.77) could have been deduced directly from the integral form of the linear momentum balance for a material volume in the material description, equation (5.76), by applying the localization theorem to the volume integral, as we did in Section 5.10.5 (Path 2 to 4) to deduce Cauchy's momentum equation from the integral form of the linear momentum balance for a material volume in the spatial description.

5.11 Conservation of angular momentum

Angular momentum is a vector quantity that characterizes the rotational motion of a particle or a collection of particles. The principle of conservation of angular momentum states that the total angular momentum of a control mass remains unchanged in

the absence of external torques. This principle accounts for the angular acceleration observed in an ice skater when they pull their arms and legs closer to their vertical axis of rotation. By reducing the distance of some of their body mass from the axis, they lower their moment of inertia. Since angular momentum is the product of moment of inertia and angular velocity, if angular momentum is conserved, the skater's angular velocity (rotational speed) must increase.

In continuum mechanics, this principle can be expressed in different forms, depending on the choice of the region of reference: control mass or control volume. In the local case, the key new content that emerges is not a new field equation but rather a constraint on Cauchy's stress tensor: $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$. We will explore these different forms and their mathematical equivalence in this section.

5.11.1 Case 1: angular momentum equation for a finite control volume fixed in space

In Section 5.11.6, we will show that the integral form of the angular momentum balance for a fixed control volume $\mathscr{V}^{\text{cv}}$ can be derived from the integral form of the linear momentum balance for a fixed control volume, by taking the cross product of the position vector $\vec{\boldsymbol{r}} = \vec{\boldsymbol{x}} - \vec{\tilde{\boldsymbol{x}}}_0$ with the linear momentum density $\rho\vec{\boldsymbol{v}}$, and applying the Reynolds transport theorem to the resulting expression. The resulting equation is:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}v\right) + \oiint_{\partial\mathscr{V}^{\text{cv}}} \left(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\right)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}v + \oiint_{\partial\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S. \tag{5.78}$$

This is the *angular momentum balance for a finite control volume fixed in space*. It must be satisfied at every instant for any choice of the fixed control volume $\mathscr{V}^{\text{cv}}$.

The left hand side of this equation represents the rate of change of angular momentum of the material system $\mathscr{V}_t^{\text{cm}}$ as it evolves through the fixed control volume $\mathscr{V}^{\text{cv}}$. It consists of two terms:

- The *storage term* $\frac{\mathrm{d}}{\mathrm{d}t}\iiint_{\mathscr{V}^{\text{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}v$ measures the rate at which angular momentum is accumulating within the fixed region. In a steady flow this term vanishes, but in any transient process, like the acceleration of a turbine, it captures the temporal evolution of the angular momentum inventory within $\mathscr{V}^{\text{cv}}$. For example, if a turbine impeller is accelerating, the angular momentum within the control volume will be increasing, and this term will be positive; if the impeller is decelerating, the angular momentum within the control volume will be decreasing, and this term will be negative.

- The *convective flux term* $\oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)(\vec{\boldsymbol{v}} \cdot \hat{\boldsymbol{n}})\,\mathrm{d}S$ represents the net rate at which angular momentum is transported out of the control volume by the moving material. Where material exits, angular momentum leaves the control volume; where material enters, angular momentum is enters the control volume. This term represents the swirling flow of angular momentum that a turbine impeller induces in the surrounding fluid: as the impeller spins, it imparts angular momentum to the fluid, which then convects outward through the control volume boundary, carrying angular momentum with it. This term has no counterpart in the Lagrangian description, where by definition no mass crosses the boundary.

The right-hand side consists of two terms as well: the first term is the total body torque acting on the instantaneous contents of $\mathscr{V}^{\mathrm{cm}}$, and the second term is the total surface torque acting on $\mathscr{V}^{\mathrm{cm}}$ due to surface tractions.

5.11.2 Case 2: angular momentum equation for a finite control mass moving with the flow

According to Euler's second law of motion (5.47), the rate of change of the total angular momentum of any control mass $\mathscr{V}_t^{\mathrm{cm}}$ about a fixed point $\tilde{\vec{\boldsymbol{x}}}_0$ is equal to the sum of the moments caused by both body and surface forces, acting on $\mathscr{V}_t^{\mathrm{cm}}$ about the same point. This can be expressed mathematically as:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}v\right) = \vec{\boldsymbol{m}}_{\mathrm{body}}(t) + \vec{\boldsymbol{m}}_{\mathrm{surface}}(t).$$

Here, $\vec{\boldsymbol{r}} = \vec{\boldsymbol{x}} - \tilde{\vec{\boldsymbol{x}}}_0$ is the position vector from the fixed point $\tilde{\vec{\boldsymbol{x}}}_0$ to the material point $\vec{\boldsymbol{x}}$, $\vec{\boldsymbol{m}}_{\mathrm{body}}(t)$ and $\vec{\boldsymbol{m}}_{\mathrm{surface}}(t)$ are the total body and surface torques acting on $\mathscr{V}_t^{\mathrm{cm}}$ at time t. In a classical, or non-polar continuum (one in which there are no distributed body couples and no surface couple stresses) the total body and surface torques can be expressed in terms of the body force per unit volume $\vec{\boldsymbol{b}}$ and the Cauchy stress tensor $\boldsymbol{\sigma}$, according to equations (5.44) and (5.45), respectively; that is:

$$\vec{\boldsymbol{m}}_{\mathrm{body}}(t) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}v,$$

$$\vec{\boldsymbol{m}}_{\mathrm{surface}}(t) = \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

Substituting the expressions for $\vec{\boldsymbol{m}}_{\mathrm{body}}(t)$ and $\vec{\boldsymbol{m}}_{\mathrm{surface}}(t)$ back into the equation for Euler's second law of motion, yields the *integral form of the angular momentum balance for a control mass*:

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}v\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}v + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S. \tag{5.79}$$

This equation states that the rate of change of the total angular momentum of the material system $\mathscr{V}_t^{\mathrm{cm}}$ is equal to the sum of the total body torque and the total surface torque acting on $\mathscr{V}_t^{\mathrm{cm}}$ at time t. It is important to note that this equation is valid for any choice of the fixed point $\tilde{\vec{x}}_0$. Observe that in this equation there is no convective flux term, since by definition no mass crosses the boundary of a control mass. The only term on the left-hand side is the storage term, which captures the temporal evolution of the angular momentum within $\mathscr{V}_t^{\mathrm{cm}}$.

5.11.3 Case 3: angular momentum equation for an infinitesimal control volume fixed in space

The two integral statements presented above (Cases 1 and 2, for a fixed control volume and for a material control mass) are global angular momentum balances that constrain the total torque and the total angular momentum over finite regions.

In Section 5.11.6, we will show that the local, or infinitesimal, implication of those balances is that for a classical non-polar continuum (no distributed body couples and no surface couple tractions), both integral forms lead to the same pointwise condition on the stress tensor, which is the symmetry of the Cauchy stress tensor:

$$\boxed{\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) = \boldsymbol{\sigma}^T(\vec{\boldsymbol{x}}, t).} \tag{5.80}$$

This condition reduces the nine components of Cauchy's stress tensor to six independent components (three normal and three shear). Because the argument is local, the condition must hold at every point $\vec{\boldsymbol{x}} \in \Omega_t$ and every instant t, independently of the particular choice of $\mathscr{V}^{\mathrm{cv}}$ or $\mathscr{V}^{\mathrm{cm}}$. Physically, this expresses that the continuum possesses no intrinsic source of rotational momentum at any point, and that the stress field, whatever its magnitude or spatial variation, both in the static and dynamic cases, cannot by itself set matter spinning at the infinitesimal level. It also implies that the shear stress σ_{ij} acting in direction $\hat{\boldsymbol{e}}_j$ on the face with normal $\hat{\boldsymbol{e}}_i$ must be balanced by an equal shear stress σ_{ji} in direction $\hat{\boldsymbol{e}}_i$ on the face with normal $\hat{\boldsymbol{e}}_j$. This important result will shape theory of elasticity, the Navier-Stokes equations of fluid mechanics, and every other branch of continuum mechanics.

Note: In generalized continua (e.g., Cosserat or micropolar media), body couples and couple stresses are allowed. In those theories, angular momentum balance does *not* generally reduce to $\boldsymbol{\sigma} = \boldsymbol{\sigma}^T$; instead, antisymmetric parts of stress are balanced by additional couple-stress terms. The symmetry in (5.80) is therefore a signature of the classical continuum hypothesis adopted throughout this book.

5.11.4 Case 4: angular momentum equation for an infinitesimal control mass moving with the flow

Consider a differential control mass $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ centered at $P = \left(x + \frac{\mathrm{d}x}{2}, y + \frac{\mathrm{d}y}{2}, z + \frac{\mathrm{d}z}{2}\right)$, with edge lengths $\mathrm{d}x, \mathrm{d}y, \mathrm{d}z$ and volume $\mathrm{d}v = \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z$, aligned with $\{\hat{\boldsymbol{e}}_x, \hat{\boldsymbol{e}}_y, \hat{\boldsymbol{e}}_z\}$, and shown in Figure 5.9. In this case, we will assume a classical, or non-polar continuum (one in which there are no distributed body couples and no surface couple stresses).

Euler's second law of motion (5.47) states that the rate of change of the total angular momentum $\vec{\boldsymbol{h}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}$ of a control mass $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ about a fixed point $\vec{\tilde{\boldsymbol{x}}}_0$ equals the net

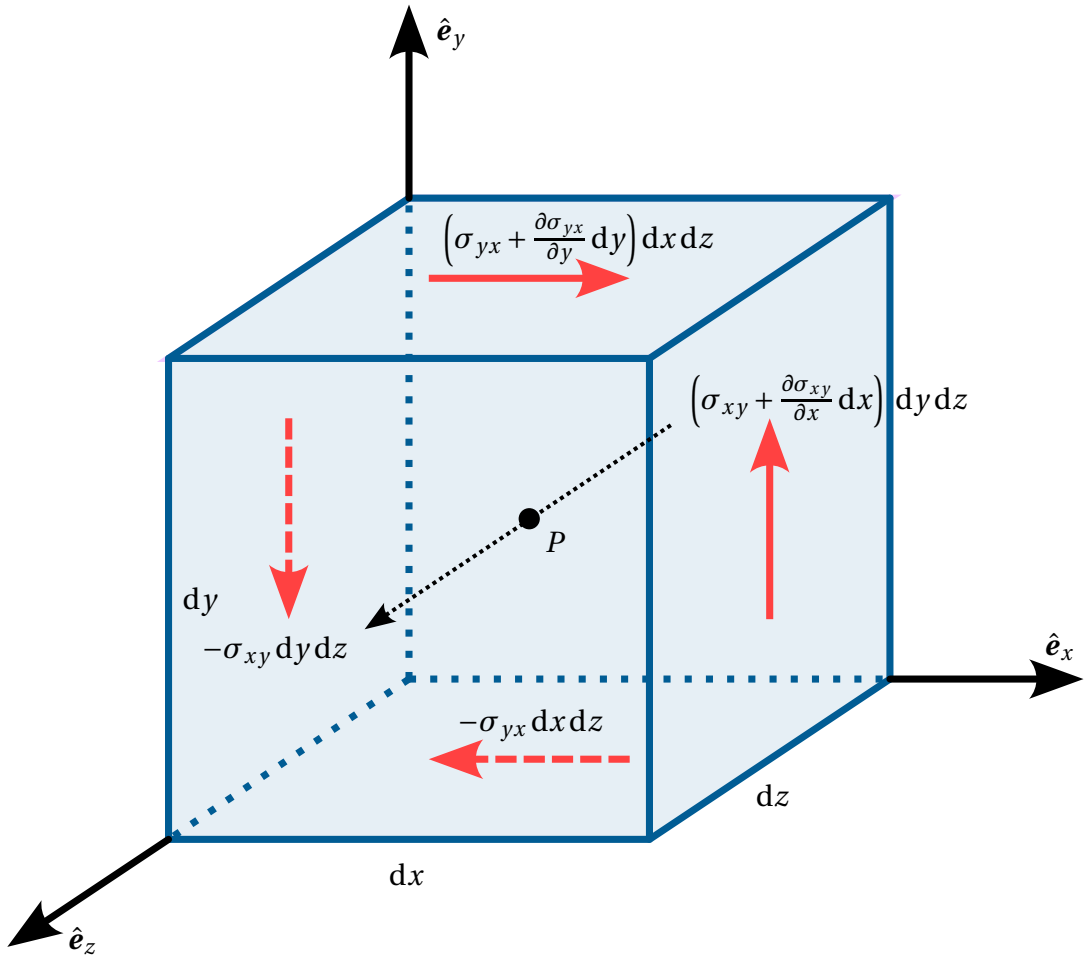


Figure 5.9: *Shear surface forces that act on the infinitesimal element* $\mathrm{d}\mathscr{V}_t^{cm}$. *Faces at* $x + \mathrm{d}x$, $y + \mathrm{d}y$, *or* $z + \mathrm{d}z$ *carry the Taylor-corrected value; the opposing face carries the value at* $\vec{\boldsymbol{x}}$ *with reversed sign. In this figure, only those forces that produce a torque around the axis parallel to* $\hat{\boldsymbol{e}}_z$ *and that passes through the centroid* P *are shown. The other forces that produce torques around the axes parallel to* $\hat{\boldsymbol{e}}_x$ *and* $\hat{\boldsymbol{e}}_y$ *are not shown for simplicity.*

torque acting on $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ about that same point, which consists of contributions from body forces, $\vec{\boldsymbol{m}}_{\mathrm{body}}$, and surface forces, $\vec{\boldsymbol{m}}_{\mathrm{surface}}$. This balance reads:

$$\frac{\mathrm{D}\vec{\boldsymbol{h}}_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \vec{\boldsymbol{m}}_{\mathrm{body}}(t) + \vec{\boldsymbol{m}}_{\mathrm{surface}}(t). \tag{5.81}$$

The angular momentum of the element $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ around $\tilde{\vec{\boldsymbol{x}}}_0$ can be calculated using equation (5.43):

$$\vec{\boldsymbol{h}}_{\mathscr{V}_t^{\mathrm{cm}}}(t) = \iiint_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}}(\vec{\boldsymbol{x}},t) \times \big(\rho(\vec{\boldsymbol{x}},t)\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big)\,\mathrm{d}v \approx \vec{\boldsymbol{r}}(\vec{\boldsymbol{x}},t) \times \big(\rho(\vec{\boldsymbol{x}},t)\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.$$

Here, $\vec{\boldsymbol{r}} = \vec{\boldsymbol{x}} - \tilde{\vec{\boldsymbol{x}}}_0$ is the position vector from the fixed point $\tilde{\vec{\boldsymbol{x}}}_0$ to the material point $\vec{\boldsymbol{x}}$. We will make $\tilde{\vec{\boldsymbol{x}}}_0 = P$, so that $\vec{\boldsymbol{r}}$ is the position vector from the centroid of the element to the material point $\vec{\boldsymbol{x}}$.

Since $\vec{\boldsymbol{r}}$ scales as $\mathscr{O}(\ell)$ and the volume of the element scales as $\mathscr{O}(\ell^3)$, the angular momentum of the element $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ scales as $\mathscr{O}(\rho v \ell^4)$, where ℓ is a characteristic size of the element and v is the magnitude of the velocity.

The body-force torques can be calculated using equation (5.44):

$$\vec{\boldsymbol{m}}_{\mathrm{body}} = \iiint_{\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}v \approx \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z,$$

where $\vec{\boldsymbol{b}}$ is the body force per unit mass. Since $\vec{\boldsymbol{r}}$ scales as $\mathscr{O}(\ell)$ and the volume of the element scales as $\mathscr{O}(\ell^3)$, the body-force torques scale as $\mathscr{O}(b\ell^4)$, where b is the magnitude of the body force.

Now, the surface-force torques can be calculated using equation (5.45):

$$\vec{\boldsymbol{m}}_{\mathrm{surface}} = \oiint_{\partial\,\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{t}}^{(\hat{\boldsymbol{n}})}\,\mathrm{d}S.$$

Let's take moments about the centroid P of the element $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}}$ to evaluate the surface-force torques. About the axis $\hat{\boldsymbol{e}}_z$, we have:

$$\begin{aligned}[\vec{\boldsymbol{m}}_{\mathrm{surface}}]_z = &\left(\frac{\mathrm{d}x}{2}\hat{\boldsymbol{e}}_x\right) \times \left(\left(\sigma_{xy} + \frac{\partial \sigma_{xy}}{\partial x}\,\mathrm{d}x\right)\mathrm{d}y\,\mathrm{d}z\hat{\boldsymbol{e}}_y\right) + \left(-\frac{\mathrm{d}x}{2}\hat{\boldsymbol{e}}_x\right) \times \left(-\sigma_{xy}\,\mathrm{d}y\,\mathrm{d}z\hat{\boldsymbol{e}}_y\right) \\ &+ \left(\frac{\mathrm{d}y}{2}\hat{\boldsymbol{e}}_y\right) \times \left(\left(\sigma_{yx} + \frac{\partial \sigma_{yx}}{\partial y}\,\mathrm{d}y\right)\mathrm{d}x\,\mathrm{d}z\hat{\boldsymbol{e}}_x\right) + \left(-\frac{\mathrm{d}y}{2}\hat{\boldsymbol{e}}_y\right) \times \left(-\sigma_{yx}\,\mathrm{d}x\,\mathrm{d}z\hat{\boldsymbol{e}}_x\right).\end{aligned}$$

In this procedure, we have discarded the contributions of the normal tractions σ_{xx} and σ_{yy}, since they do not generate a torque about the axis parallel to $\hat{\boldsymbol{e}}_z$ because they act along the line of action that passes through the centroid P.

If we discard the contributions of the Taylor-corrected values of the shear tractions, as they generate torques that scale as $\mathscr{O}(\sigma\,\ell^4)$, where σ is the magnitude of the shear

stress, and that are negligible compared to the leading terms that scale as $\mathcal{O}(\sigma\,\ell^3)$, we obtain:

$$[\vec{\boldsymbol{m}}_{\text{surface}}]_z = \left(\sigma_{xy} - \sigma_{yx}\right) \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\hat{\boldsymbol{e}}_z.$$

Making a similar calculation for the torques about the axes that pass through the centroid P gives:

- About axis $\hat{\boldsymbol{e}}_x$:

$$[\vec{\boldsymbol{m}}_{\text{surface}}]_x = \left(\sigma_{zy} - \sigma_{yz}\right) \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\hat{\boldsymbol{e}}_x = 0.$$

- About axis $\hat{\boldsymbol{e}}_y$:

$$[\vec{\boldsymbol{m}}_{\text{surface}}]_y = \left(\sigma_{xz} - \sigma_{zx}\right) \mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z\hat{\boldsymbol{e}}_y = 0.$$

In conclusion, the surface-force torques can be expressed as

$$\begin{aligned}\vec{\boldsymbol{m}}_{\text{surface}} &= \left[\left(\sigma_{yz} - \sigma_{zy}\right)\hat{\boldsymbol{e}}_x + \left(\sigma_{zx} - \sigma_{xz}\right)\hat{\boldsymbol{e}}_y + \left(\sigma_{xy} - \sigma_{yx}\right)\hat{\boldsymbol{e}}_z\right]\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \\ &= \begin{pmatrix}\sigma_{yz} - \sigma_{zy} \\ \sigma_{zx} - \sigma_{xz} \\ \sigma_{xy} - \sigma_{yx}\end{pmatrix}\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z.\end{aligned}$$

Replacing these expressions for the rate of change of angular momentum, the body-force torques, and the surface-force torques back into the angular momentum balance equation (5.81) gives:

$$\underbrace{\frac{\mathrm{D}\vec{\boldsymbol{h}}_{\mathrm{d}\mathcal{V}_t^{\text{cm}}}(t)}{\mathrm{D}t}}_{\mathcal{O}(\rho v \ell^4)} = \underbrace{\vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z}_{\mathcal{O}(b\ell^4)} + \underbrace{\begin{pmatrix}\sigma_{yz} - \sigma_{zy} \\ \sigma_{zx} - \sigma_{xz} \\ \sigma_{xy} - \sigma_{yx}\end{pmatrix}\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z}_{\mathcal{O}(\sigma\ell^3)}.$$

Note that inertia torques scale with $\mathcal{O}(\rho v \ell^4)$, body-force torques scale with $\mathcal{O}(b\ell^4)$, and surface-force torques scale with $\mathcal{O}(\sigma\ell^3)$, for a characteristic size ℓ. Since ℓ is a characteristic size of the element, in the limit as $\ell \to 0$, the leading terms in the angular momentum balance are those generated by the shear tractions on opposite faces, while the inertia torques and body-force torques become negligible. In conclusion,

$$\begin{pmatrix}\sigma_{yz} - \sigma_{zy} \\ \sigma_{zx} - \sigma_{xz} \\ \sigma_{xy} - \sigma_{yx}\end{pmatrix}\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = \vec{\boldsymbol{0}}.$$

Since $\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z \neq 0$, the surface-torque balance implies

$$\sigma_{yz} = \sigma_{zy}, \qquad \sigma_{zx} = \sigma_{xz}, \qquad \sigma_{xy} = \sigma_{yx},$$

or, equivalently,

$$\boxed{\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)=\boldsymbol{\sigma}^T(\vec{\boldsymbol{x}},t).} \tag{5.82}$$

Therefore, even in dynamics, the local angular momentum balance of a classical (non-polar) continuum implies symmetry of the Cauchy stress tensor. This is a fundamental result in continuum mechanics, as it constrains the form of the stress tensor and has important implications for the behavior of materials under deformation.

5.11.5 Some results before continuing

We in the next section we will show that the four variants of the angular momentum balance equation are mathematically equivalent, and that they can be derived from one another by mathematical manipulation. Before we do that, let us first derive some results that will be useful for the demonstration of the equivalence. In all of these demonstrations we will resort to indicial notation, since it makes the algebraic manipulations more straightforward than using vector notation.

Example 5.1. *Integral representation of the surface integral of* $\big(\vec{\boldsymbol{r}}\times(\rho\vec{\boldsymbol{v}})\big)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})$

Show that

$$\oiint_{\partial\,\mathrm{d}\mathscr{V}^{\mathrm{cv}}}\big(\vec{\boldsymbol{r}}\times(\rho\vec{\boldsymbol{v}})\big)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S=\iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}}\vec{\boldsymbol{r}}\times\big(\nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\big)\,\mathrm{d}\upsilon. \tag{5.83}$$

Solution:

The easiest way to verify this identity is to use indicial notation. The first equality is a direct application of the divergence theorem, while the second equality follows from a pointwise identity that can be verified by direct expansion and simplification. The details are as follows.

Let's rewrite the first integral in indicial notation:

$$\oiint_{\partial\,\mathrm{d}\mathscr{V}^{\mathrm{cv}}}\big(\vec{\boldsymbol{r}}\times(\rho\vec{\boldsymbol{v}})\big)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S=\oiint_{\partial\,\mathrm{d}\mathscr{V}^{\mathrm{cv}}}\epsilon_{ijk}r_j(\rho v_k)v_l n_l\,\mathrm{d}S$$

where ϵ_{ijk} is the third-order Levi-Civita alternating tensor. Applying the divergence theorem, we get:

$$\oiint_{\partial\,\mathrm{d}\mathscr{V}^{\mathrm{cv}}}\epsilon_{ijk}r_j(\rho v_k)v_l n_l\,\mathrm{d}S=\iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}}\frac{\partial}{\partial x_l}\big(\epsilon_{ijk}r_j(\rho v_k)v_l\big)\,\mathrm{d}\upsilon$$

Next, we can expand the divergence inside the integral:

$$\frac{\partial}{\partial x_l}\big(\epsilon_{ijk}r_j\rho v_k v_l\big)=\epsilon_{ijk}\left(\frac{\partial r_j}{\partial x_l}(\rho v_k v_l)+r_j\frac{\partial(\rho v_k v_l)}{\partial x_l}\right)$$

Since $r_j = x_j - x_j^0$ (where x_j^0 is the j-th component of the reference position), then $\frac{\partial r_j}{\partial x_l} = \delta_{jl}$ (the Kronecker delta), and we can simplify this to:

$$\epsilon_{ijk}\left(\delta_{jl}(\rho v_k v_l) + r_j \frac{\partial(\rho v_k v_l)}{\partial x_l}\right) = \epsilon_{ijk}\left(\rho v_k v_j + r_j \frac{\partial(\rho v_k v_l)}{\partial x_l}\right)$$

The first term $\rho\epsilon_{ijk} v_j v_k = \rho\vec{\boldsymbol{v}} \times \vec{\boldsymbol{v}}$ vanishes because $\vec{\boldsymbol{v}} \times \vec{\boldsymbol{v}} = \vec{\boldsymbol{0}}$. Therefore, we are left with:

$$\epsilon_{ijk} r_j \frac{\partial(\rho v_k v_l)}{\partial x_l} = \vec{\boldsymbol{r}} \times \left(\nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}})\right)$$

Thus, we have shown that:

$$\iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \nabla \cdot \left[\left(\vec{\boldsymbol{r}} \times (\rho \vec{\boldsymbol{v}})\right) \otimes \vec{\boldsymbol{v}}\right] \mathrm{d}\upsilon = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times \left(\nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}})\right) \mathrm{d}\upsilon$$

This completes the demonstration.

Example 5.2. *Integral representation of the surface integral of* $\vec{\boldsymbol{r}} \times \boldsymbol{\sigma}\hat{\boldsymbol{n}}$

Show that

$$\oiint_{\partial\mathscr{V}} \vec{\boldsymbol{r}} \times \boldsymbol{\sigma}\hat{\boldsymbol{n}}\, \mathrm{d}S = \iiint_{\mathscr{V}} \vec{\boldsymbol{r}} \times (\nabla \cdot \boldsymbol{\sigma})\, \mathrm{d}\upsilon + \iiint_{\mathscr{V}} \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\, \mathrm{d}\upsilon, \tag{5.84}$$

where $\boldsymbol{\epsilon}$ is the Levi-Civita symbol and the colon denotes the double contraction of tensors.

Solution:

To show equation (5.84), we can use the divergence theorem and properties of the Levi-Civita symbol.

First, we can express the left-hand side of the equation in indicial notation as:

$$\oiint_{\partial\mathscr{V}} \vec{\boldsymbol{r}} \times \boldsymbol{\sigma}\hat{\boldsymbol{n}}\, \mathrm{d}S = \oiint_{\partial\mathscr{V}} \epsilon_{ijk} r_j \sigma_{kl} n_l\, \mathrm{d}S$$

where ϵ_{ijk} is the Levi-Civita symbol (see Appendix ***), r_j is the j-th component of the position vector $\vec{\boldsymbol{r}}$, σ_{kl} is the kl-th component of the stress tensor $\boldsymbol{\sigma}$, and n_l is the l-th component of the normal vector $\hat{\boldsymbol{n}}$. Next, we can apply the divergence theorem to convert the surface integral into a volume integral:

$$\oiint_{\partial\mathscr{V}} \epsilon_{ijk} r_j \sigma_{kl} n_l\, \mathrm{d}S = \iiint_{\mathscr{V}} \nabla \cdot (\epsilon_{ijk} r_j \sigma_{kl})\, \mathrm{d}\upsilon = \iiint_{\mathscr{V}} \frac{\partial}{\partial x_l}(\epsilon_{ijk} r_j \sigma_{kl})\, \mathrm{d}\upsilon$$

Let's expand the divergence using the product rule:

$$\frac{\partial}{\partial x_l}(\epsilon_{ijk} r_j \sigma_{kl}) = \epsilon_{ijk} \frac{\partial}{\partial x_l}(r_j \sigma_{kl}) = \epsilon_{ijk}\left[r_j \frac{\partial \sigma_{kl}}{\partial x_l} + \sigma_{kl} \frac{\partial r_j}{\partial x_l}\right]$$

Since $r_j = x_j - x_j^0$ (where x_j^0 is the j-th component of the reference position), then $\frac{\partial r_j}{\partial x_l} = \delta_{jl}$ (the Kronecker delta), and we can simplify this to:

$$\frac{\partial}{\partial x_l}(\epsilon_{ijk} r_j \sigma_{kl}) = \epsilon_{ijk}\left[r_j \frac{\partial \sigma_{kl}}{\partial x_l} + \sigma_{kl}\delta_{jl}\right] = \epsilon_{ijk} r_j \frac{\partial \sigma_{kl}}{\partial x_l} + \epsilon_{ijk}\sigma_{kl}\delta_{jl}$$

Let's remember that the divergence of a second-order tensor is defined as $(\nabla \cdot \boldsymbol{\sigma})_k = \frac{\partial \sigma_{kl}}{\partial x_l}$. Thus, the first term can be rewritten as:

$$\epsilon_{ijk} r_j \frac{\partial \sigma_{kl}}{\partial x_l} = \vec{\boldsymbol{r}} \times (\nabla \cdot \boldsymbol{\sigma})$$

The second term can be simplified using the properties of Kronecker delta ($\sigma_{kl}\delta_{jl} = \sigma_{kj}$) and the Levi-Civita symbol:

$$\epsilon_{ijk}\sigma_{kl}\delta_{jl} = \epsilon_{ijk}\sigma_{kj} = \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T$$

where $\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T$ denotes the double contraction of the Levi-Civita symbol with the transpose of the stress tensor. Remember that the double contraction of two tensors $\boldsymbol{A}$ and $\boldsymbol{B}$ is defined as $\boldsymbol{A} : \boldsymbol{B} = A_{ij}B_{ij}$ (summation over repeated indices).

Putting it all together, we have:

$$\oiint_{\partial \mathscr{V}} \vec{\boldsymbol{r}} \times \boldsymbol{\sigma}\hat{\boldsymbol{n}}\, \mathrm{d}S = \iiint_{\mathscr{V}} \vec{\boldsymbol{r}} \times (\nabla \cdot \boldsymbol{\sigma})\, \mathrm{d}v + \iiint_{\mathscr{V}} \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\, \mathrm{d}v$$

This completes the proof.



Example 5.3. *Contraction of the Levi-Civita symbol with the transpose of the stress tensor*

Show that

$$\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = \begin{bmatrix} \sigma_{32} - \sigma_{23} \\ \sigma_{13} - \sigma_{31} \\ \sigma_{21} - \sigma_{12} \end{bmatrix} \tag{5.85}$$

where $\boldsymbol{\epsilon}$ is the Levi-Civita symbol and $\boldsymbol{\sigma}^T$ is the transpose of the Cauchy stress tensor $\boldsymbol{\sigma}$.

Solution:

Let's express $\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = \epsilon_{ijk}\sigma_{kj}$ as a vector in terms of the components of the stress tensor. We can define a vector $\vec{\boldsymbol{v}}$ such that its components are given by:

$$v_i = \left[\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\right]_i = \epsilon_{ijk}\sigma_{kj}$$

Each component is a sum over $j,k \in \{1,2,3\}$. Terms with repeated indices in ϵ_{ijk} are zero (e.g., $\epsilon_{i11} = \epsilon_{2j2} = \epsilon_{33k} = 0$, and similarly whenever any two indices are equal), so only permutations of $(1,2,3)$ survive:

$$\begin{aligned}
v_1 &= \epsilon_{111}\sigma_{11} + \epsilon_{112}\sigma_{21} + \epsilon_{113}\sigma_{31} + \epsilon_{121}\sigma_{12} + \epsilon_{122}\sigma_{22} + \epsilon_{123}\sigma_{32} \\
&\qquad + \epsilon_{131}\sigma_{13} + \epsilon_{132}\sigma_{23} + \epsilon_{133}\sigma_{33} \\
&= 0 + 0 + 0 + 0 + 0 + \sigma_{32} + 0 - \sigma_{23} + 0 \\
&= \sigma_{32} - \sigma_{23} \\
v_2 &= \epsilon_{211}\sigma_{11} + \epsilon_{212}\sigma_{21} + \epsilon_{213}\sigma_{31} + \epsilon_{221}\sigma_{12} + \epsilon_{222}\sigma_{22} + \epsilon_{223}\sigma_{32} \\
&\qquad + \epsilon_{231}\sigma_{13} + \epsilon_{232}\sigma_{23} + \epsilon_{233}\sigma_{33} \\
&= 0 + 0 - \sigma_{31} + 0 + 0 + 0 + \sigma_{13} + 0 + 0 \\
&= \sigma_{13} - \sigma_{31} \\
v_3 &= \epsilon_{311}\sigma_{11} + \epsilon_{312}\sigma_{21} + \epsilon_{313}\sigma_{31} + \epsilon_{321}\sigma_{12} + \epsilon_{322}\sigma_{22} + \epsilon_{323}\sigma_{32} \\
&\qquad + \epsilon_{331}\sigma_{13} + \epsilon_{332}\sigma_{23} + \epsilon_{333}\sigma_{33} \\
&= 0 + \sigma_{21} + 0 - \sigma_{12} + 0 + 0 + 0 + 0 + 0 \\
&= \sigma_{21} - \sigma_{12}
\end{aligned}$$

Thus, the vector $\vec{\boldsymbol{v}}$ can be expressed in terms of the components of the stress tensor as

$$\vec{\boldsymbol{v}} = \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = \begin{bmatrix} \sigma_{32} - \sigma_{23} \\ \sigma_{13} - \sigma_{31} \\ \sigma_{21} - \sigma_{12} \end{bmatrix}$$

This vector $\vec{\boldsymbol{v}}$ is expressed in terms of the antisymmetric part of the stress tensor, which is related to the rotational effects in the material.

5.11.6 Mathematical equivalence between all four cases

The four forms of the angular momentum balance equation we have derived, and that are summarized in Figure 5.10, represent the same physical principle of conservation of angular momentum, but they are expressed in different mathematical forms and in different reference frames. Let us explore their relationships and demonstrate their equivalence systematically.

Our four equations can be classified along two important dimensions:

- By mathematical form: Integral equations (5.78), (5.79) vs. Differential equations (5.80), (5.82)

- By formulation type: Conservation form (5.78), (5.80) vs. Non-conservation form (5.79), (5.82)

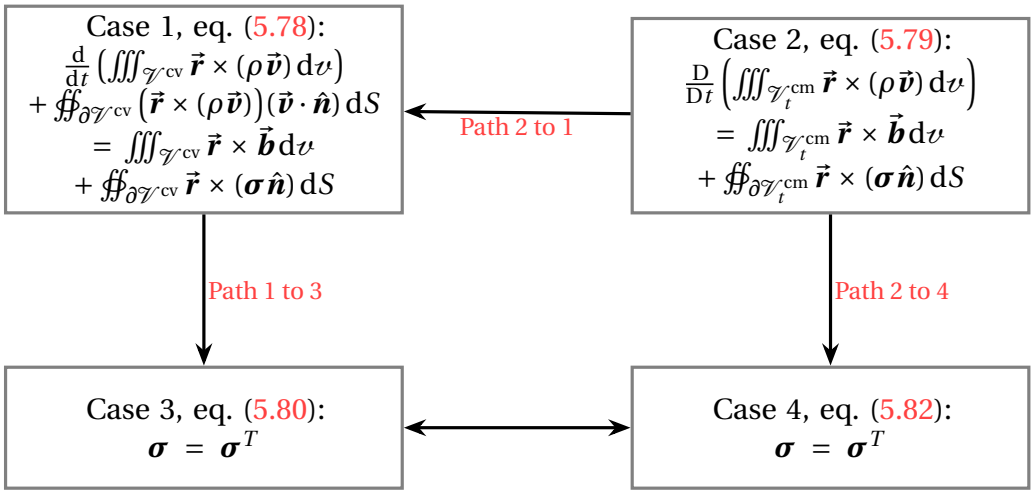


Figure 5.10: *The four different variants of the angular momentum balance equation.*

Any of these equations can be deduced by mathematical manipulation from any of the others, as we will illustrate in the next, for the paths shown in Figure 5.10.

Path 2 to 1: from equation (5.79) to equation (5.78)

We begin by considering the angular momentum balance for a control mass $\mathscr{V}_t^{\mathrm{cm}}$ as expressed in equation (5.79)

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

At time $t + \Delta t$, the material particles have moved: some have left $\mathscr{V}^{\mathrm{cv}}$ through $\partial\mathscr{V}^{\mathrm{cv}}$, others from outside have entered, and the material volume $\mathscr{V}_{t+\Delta t}^{\mathrm{cm}}$ no longer coincides with $\mathscr{V}^{\mathrm{cv}}$. It is precisely this difference between the evolving material volume and the fixed control volume that the Reynolds transport formula quantifies.

Consider the rate of change of angular momentum of the material system over a short time interval Δt:

$$\frac{\mathrm{D}\vec{\boldsymbol{h}}_{\mathscr{V}_t^{\mathrm{cm}}}(t)}{\mathrm{D}t} = \lim_{\Delta t\to 0} \frac{1}{\Delta t}\left[\iiint_{\mathscr{V}_{t+\Delta t}^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon - \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon\right];$$

according to the Reynolds transport formula (5.17), by making $\boldsymbol{\mu} = \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})$, this can be expressed as:

$$\frac{\mathrm{D}\vec{\boldsymbol{h}}_{\mathscr{V}_t^{\mathrm{cv}}}(t)}{\mathrm{D}t} = \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon\right)$$

$$= \iiint_{\mathscr{V}_t^{\mathrm{cv}}} \frac{\partial}{\partial t}\big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}_t^{\mathrm{cv}}} \big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

This means that the Reynolds transport formula decomposes the material rate of change of angular momentum into the rate at which angular momentum accumulates within the control mass $\mathscr{V}_t^{\mathrm{cv}}$, and the net rate at which the flowing material sweeps angular momentum outward through the boundary $\partial\mathscr{V}_t^{\mathrm{cv}}$.

Substituting this expression back into the angular momentum balance equation for the control volume (5.79), we obtain:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \frac{\partial}{\partial t}\big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S. \quad (5.86)$$

Now let's fix a control volume $\mathscr{V}^{\mathrm{cv}}$ in space, with outward unit normal $\hat{\boldsymbol{n}}$. At the instant t we choose, a material system $\mathscr{V}_t^{\mathrm{cm}}$ that coincides exactly with $\mathscr{V}^{\mathrm{cv}}$, so $\mathscr{V}_t^{\mathrm{cm}} = \mathscr{V}^{\mathrm{cv}}$. At this instant, the angular momentum balance for the control mass $\mathscr{V}_t^{\mathrm{cm}}$ can be expressed as:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial}{\partial t}\big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

Using equation (5.12) for a control volume fixed in space, we can rewrite the first term on the left-hand side as:

$$\iiint_{\mathscr{V}^{\mathrm{cv}}} \frac{\partial}{\partial t}\big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)\,\mathrm{d}\mathscr{v} = \frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\mathscr{v}\right).$$

Substituting this result back into equation (5.86), we obtain equation (5.78):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\mathscr{v}\right) + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \big(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\big)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}\mathscr{v} + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

This is the *angular momentum balance for a finite control volume fixed in space*. It must be satisfied at every instant for any choice of the fixed control volume $\mathscr{V}^{\mathrm{cv}}$.

Path 1 to 3: from equation (5.78) to equation (5.80)

Let's start from the angular momentum balance for a finite control volume $\mathscr{V}^{\mathrm{cv}}$ fixed in space with outward unit normal $\hat{\boldsymbol{n}}$, equation (5.78):

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon\right) + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \left(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\right)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

Let's pick as a control volume the smallest Eulerian box we can still treat as a continuum, a fixed infinitesimal control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ centered at the position $\vec{\boldsymbol{x}}$, with faces locked to the Cartesian coordinate directions $\hat{\boldsymbol{e}}_x$, $\hat{\boldsymbol{e}}_y$, $\hat{\boldsymbol{e}}_z$ and outward normal $\vec{\boldsymbol{n}}$. The angular momentum balance for this infinitesimal control volume is

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon\right) + \oiint_{\partial\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \left(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\right)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}\,\mathrm{d}\upsilon + \oiint_{\partial\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S.$$

The goal is to drive this integral statement toward a pointwise identity. Let's start by simplifying the time derivative of the angular momentum contained in the infinitesimal control volume. Since $\mathrm{d}\mathscr{V}^{\mathrm{cv}}$ is fixed in space, using equation (5.12), we can express the first integral as:

$$\frac{\mathrm{d}}{\mathrm{d}t}\left(\iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon\right) = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \frac{\partial}{\partial t}\left(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\right)\mathrm{d}\upsilon = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times \frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}})\,\mathrm{d}\upsilon.$$

This step employed the vector identity (A.5):

$$\frac{\partial}{\partial t}(\vec{\boldsymbol{r}} \times \vec{\boldsymbol{s}}) = \frac{\partial\vec{\boldsymbol{r}}}{\partial t} \times \vec{\boldsymbol{s}} + \vec{\boldsymbol{r}} \times \frac{\partial\vec{\boldsymbol{s}}}{\partial t},$$

taking into account that $\vec{\boldsymbol{r}}$ is the fixed position vector of the control volume, so $\partial\vec{\boldsymbol{r}}/\partial t = \vec{\boldsymbol{0}}$.

Now, we will use the divergence theorem to convert the surface integrals into volume integrals. Using equation (5.83), the first surface integral becomes:

$$\oiint_{\partial\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \left(\vec{\boldsymbol{r}} \times (\rho\vec{\boldsymbol{v}})\right)(\vec{\boldsymbol{v}}\cdot\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times \left(\nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}})\right)\mathrm{d}\upsilon.$$

and employing (5.84) the second surface integral transforms into:

$$\oiint_{\partial\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \vec{\boldsymbol{r}} \times (\nabla\cdot\boldsymbol{\sigma})\,\mathrm{d}\upsilon + \iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\,\mathrm{d}\upsilon$$

where $\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T$ is the double contraction of the Levi-Civita tensor $\boldsymbol{\epsilon}$ with the transpose of the Cauchy stress tensor $\boldsymbol{\sigma}$.

Let's collect all terms under a common integration, and then invoke the localization theorem. Rearranging the angular momentum balance for the infinitesimal control volume, we have

$$\iiint_{\mathrm{d}\mathscr{V}^{\mathrm{cv}}} \left(\vec{\boldsymbol{r}} \times \left[\frac{\partial}{\partial t}(\rho\vec{\boldsymbol{v}}) + \nabla\cdot(\rho\vec{\boldsymbol{v}}\otimes\vec{\boldsymbol{v}}) - \vec{\boldsymbol{b}} - \nabla\cdot\boldsymbol{\sigma}\right] - \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\right)\mathrm{d}\upsilon = 0.$$

According to the localization theorem, since this must hold for every infinitesimal control volume $\mathrm{d}\mathscr{V}^{\mathrm{cv}} \subset \Omega_t$, the integrand must vanish pointwise:

$$\vec{\boldsymbol{r}} \times \left[\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}}) + \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) - \vec{\boldsymbol{b}} - \nabla \cdot \boldsymbol{\sigma} \right] - \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = 0.$$

We identify the bracket as the linear momentum balance in Eulerian form, equation (5.70):

$$\frac{\partial}{\partial t}(\rho \vec{\boldsymbol{v}}) + \nabla \cdot (\rho \vec{\boldsymbol{v}} \otimes \vec{\boldsymbol{v}}) = \vec{\boldsymbol{b}} + \nabla \cdot \boldsymbol{\sigma}.$$

For a non-polar continuum, the linear momentum balance must hold, so the bracket vanishes identically. This leaves us with the condition $\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = \vec{\boldsymbol{0}}$ for every $\mathrm{d}\mathscr{V}^{\mathrm{cv}} \subset \Omega_t$. According to equation (5.85), this expression can be written in terms of the components of the Cauchy stress tensor as

$$\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = \begin{bmatrix} \sigma_{32} - \sigma_{23} \\ \sigma_{13} - \sigma_{31} \\ \sigma_{21} - \sigma_{12} \end{bmatrix} = \vec{\boldsymbol{0}}.$$

And this implies the symmetry of the Cauchy stress tensor, that is $\sigma_{ij} = \sigma_{ji}$ for all i, j; hence we obtain equation (5.80):

$$\boldsymbol{\sigma} = \boldsymbol{\sigma}^T.$$

for every point $\vec{\boldsymbol{x}} \in \Omega_t$.

Path 2 to 4: from equation (5.79) to equation (5.82)

Let's start from the angular momentum balance for a finite control mass $\mathscr{V}_t^{\mathrm{cm}}$ moving with the flow, with outward normal $\hat{\boldsymbol{n}}$ on its boundary $\partial\mathscr{V}_t^{\mathrm{cm}}$, equation (5.79):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho \vec{\boldsymbol{v}}) \, \mathrm{d}v \right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}} \, \mathrm{d}v + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma} \hat{\boldsymbol{n}}) \, \mathrm{d}S.$$

Since ρ is a scalar, we can use equation (5.41) with $\boldsymbol{\psi} = \vec{\boldsymbol{r}} \times \vec{\boldsymbol{v}}$ to move the material derivative inside the integral on the left-hand side, giving

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho \vec{\boldsymbol{v}}) \, \mathrm{d}v \right) = \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho \vec{\boldsymbol{r}} \times \vec{\boldsymbol{v}} \, \mathrm{d}v \right) = \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho \frac{\mathrm{D}(\vec{\boldsymbol{r}} \times \vec{\boldsymbol{v}})}{\mathrm{D}t} \, \mathrm{d}v \right)$$

The material derivative of the cross product is evaluated by the product rule (A.5):

$$\frac{\mathrm{D}}{\mathrm{D}t}(\vec{\boldsymbol{r}} \times \vec{\boldsymbol{v}}) = \frac{\mathrm{D}\vec{\boldsymbol{r}}}{\mathrm{D}t} \times \vec{\boldsymbol{v}} + \vec{\boldsymbol{r}} \times \frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t}.$$

The first term on the right vanishes instantly: since $\mathrm{D}\vec{\boldsymbol{r}}/\mathrm{D}t = \mathrm{D}(\vec{\boldsymbol{x}} - \hat{\vec{\boldsymbol{x}}}_0))/\mathrm{D}t = \vec{\boldsymbol{v}}$, this term is $\vec{\boldsymbol{v}} \times \vec{\boldsymbol{v}} = \vec{\boldsymbol{0}}$, since the cross product of any vector with itself is identically zero. We are therefore left with the expression

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times (\rho \vec{\boldsymbol{v}}) \, \mathrm{d}v \right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \rho \left(\vec{\boldsymbol{r}} \times \frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} \right) \mathrm{d}v.$$

Let's convert now the surface integral on the right-hand side of the original equation into a volume integral. According to equation (5.84), we have:

$$\oiint_{\partial\mathscr{V}} \vec{\boldsymbol{r}} \times (\boldsymbol{\sigma}\hat{\boldsymbol{n}})\,\mathrm{d}S = \iiint_{\mathscr{V}} \vec{\boldsymbol{r}} \times (\nabla\cdot\boldsymbol{\sigma})\,\mathrm{d}v + \iiint_{\mathscr{V}} \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\,\mathrm{d}v,$$

where $\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T$ is the double contraction of the Levi-Civita tensor $\boldsymbol{\epsilon}$ with the transpose of the Cauchy stress tensor $\boldsymbol{\sigma}$.

Putting everything together, into equation (5.79), the angular momentum balance for the control mass $\mathscr{V}_t^{\mathrm{cm}}$ can be written as

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \left[\rho\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} - \vec{\boldsymbol{b}} - \nabla\cdot\boldsymbol{\sigma}\right]\mathrm{d}v = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\,\mathrm{d}v.$$

We identify the bracketed term on the left as the combination of terms that appears in Cauchy's equation of motion (5.74):

$$\rho\frac{\mathrm{D}\vec{\boldsymbol{v}}}{\mathrm{D}t} = \vec{\boldsymbol{b}} + \nabla\cdot\boldsymbol{\sigma}.$$

The linear momentum balance has already been established as an independent postulate, and it holds pointwise throughout the continuum. Therefore the integrand in square brackets vanishes identically, and the entire left-hand side is zero. We are left with the remarkably simple statement,

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \boldsymbol{\epsilon} : \boldsymbol{\sigma}^T\,\mathrm{d}v = \vec{\boldsymbol{0}}.$$

According to the localization theorem (see Appendix A.18), if the integral of a continuous field vanishes over every sub-region of the domain, the field itself must vanish pointwise. We therefore conclude that for every material point $\mathrm{d}\mathscr{V}_t^{\mathrm{cm}} \subset \Omega_t$ and at every instant t, the following must hold:

$$\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = \vec{\boldsymbol{0}}.$$

According to equation (5.85), this expression can be written in terms of the components of the Cauchy stress tensor as

$$\boldsymbol{\epsilon} : \boldsymbol{\sigma}^T = \begin{bmatrix} \sigma_{32} - \sigma_{23} \\ \sigma_{13} - \sigma_{31} \\ \sigma_{21} - \sigma_{12} \end{bmatrix} = \vec{\boldsymbol{0}}.$$

This implies the symmetry of the Cauchy stress tensor, that is $\sigma_{ij} = \sigma_{ji}$ for all i, j, and for every point $\vec{\boldsymbol{x}} \in \Omega_t$, as specified by equation (5.82):

$$\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) = \boldsymbol{\sigma}^T(\vec{\boldsymbol{x}}, t).$$

This is the *local form of the angular momentum balance* for a classical, non-polar continuum. It states that the Cauchy stress tensor must be symmetric at every point in the body.

5.11.7 Angular momentum balance equation in the material description

In the previous sections, we derived the angular momentum balance equations in the spatial description, expressed in terms of the current coordinates $\vec{\boldsymbol{x}}$ and time t. These equations can also be written in the material description, which uses material coordinates $\vec{\boldsymbol{X}}$ and time t. Although this description is less common in fluid mechanics (because there is no preferred fixed reference configuration), it is the natural framework for finite elasticity and nonlinear solid mechanics, where a reference configuration Ω_0 is available.

In the material formulation, normals, areas, and volume integrals are taken over the undeformed body. The traction on reference area elements is represented by the first Piola–Kirchhoff stress tensor $\boldsymbol{P}$, instead of the Cauchy stress tensor $\boldsymbol{\sigma}$, which is associated with current area elements.

In what follows, we rewrite the angular momentum balance in material form for the Lagrangian cases 2 and 4, using the relations already established in Section 5.10.6.

Case 2: angular momentum equation for a finite control mass moving with the flow

In Section 5.11.2, we showed that the integral form of the angular momentum balance equation, about a fixed point $\tilde{\vec{\boldsymbol{x}}}_0 \in \Omega_t$, for a material volume is given by equation (5.79):

$$\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big)\,\mathrm{d}\upsilon\right) = \iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}},t)\,\mathrm{d}\upsilon + \oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\boldsymbol{\sigma}(\vec{\boldsymbol{x}},t)\,\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}},t)\big)\,\mathrm{d}S.$$

Here, $\vec{\boldsymbol{r}} = \vec{\boldsymbol{x}} - \tilde{\vec{\boldsymbol{x}}}_0$ is the position vector from a fixed point $\tilde{\vec{\boldsymbol{x}}}_0$ (fixed in space) to the material point $\vec{\boldsymbol{x}}$; this is the spatial form.

We now express this equation in the material description. First, we rewrite the volume integral on the left-hand side over the reference configuration:

$$\begin{aligned}\frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big)\,\mathrm{d}\upsilon\right) &= \frac{\mathrm{D}}{\mathrm{D}t}\left(\iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\rho(\vec{\boldsymbol{x}},t)\,\vec{\boldsymbol{v}}(\vec{\boldsymbol{x}},t)\big) J(\vec{\boldsymbol{X}},t)\,\mathrm{d}\mathscr{V}_0\right)\\ &= \frac{\partial}{\partial t}\left(\iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\rho_0(\vec{\boldsymbol{X}})\,\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}},t)\big)\,\mathrm{d}\mathscr{V}_0\right).\end{aligned}$$

Here we used that, for a material control mass written in material coordinates, the material time derivative reduces to a partial derivative with respect to t, since $\mathscr{V}_0^{\mathrm{cm}}$ is fixed in the reference configuration Ω_0.

Next, using $\vec{\boldsymbol{B}} = J\vec{\boldsymbol{b}}$ (body force per unit reference volume), the body-force moment term becomes:

$$\iiint_{\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t)\, \mathrm{d}v = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{b}}(\vec{\boldsymbol{x}}, t) J(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V}_0 = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{B}}(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V}_0.$$

Similarly, using Nanson's formula $\hat{\boldsymbol{n}}\, \mathrm{d}S = J\boldsymbol{F}^{-T}\hat{\boldsymbol{N}}\, \mathrm{d}S_0$ and the transformation relating the Cauchy and first Piola-Kirchhoff stress tensors $\boldsymbol{P} = J\boldsymbol{\sigma}\boldsymbol{F}^{-T}$, the surface moment term becomes:

$$\begin{aligned}\oiint_{\partial\mathscr{V}_t^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t)\hat{\boldsymbol{n}}(\vec{\boldsymbol{x}}, t)\big)\, \mathrm{d}S &= \oiint_{\partial\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) J(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^{-T}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\big)\, \mathrm{d}S_0 \\ &= \oiint_{\partial\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\boldsymbol{P}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\big)\, \mathrm{d}S_0.\end{aligned}$$

Putting everything together, the angular momentum balance for a finite material control mass is:

$$\begin{aligned}&\frac{\partial}{\partial t}\left(\iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\rho_0(\vec{\boldsymbol{X}})\vec{\boldsymbol{V}}(\vec{\boldsymbol{X}}, t)\big)\, \mathrm{d}\mathscr{V}_0\right) \\ &\qquad = \iiint_{\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \vec{\boldsymbol{B}}(\vec{\boldsymbol{X}}, t)\, \mathrm{d}\mathscr{V}_0 + \oiint_{\partial\mathscr{V}_0^{\mathrm{cm}}} \vec{\boldsymbol{r}} \times \big(\boldsymbol{P}(\vec{\boldsymbol{X}}, t)\hat{\boldsymbol{N}}(\vec{\boldsymbol{X}})\big)\, \mathrm{d}S_0.\end{aligned}$$

Although this balance is integrated over $\mathscr{V}_0^{\mathrm{cm}}$, the moment arm is the current vector $\vec{\boldsymbol{r}}(\vec{\boldsymbol{X}}, t) = \vec{\boldsymbol{\chi}}(\vec{\boldsymbol{X}}, t) - \tilde{\vec{\boldsymbol{x}}}_0$, since torques are evaluated in the current configuration.

This is the material form of the same balance law. Its structure is analogous to the spatial form, but all fields are expressed in terms of $\vec{\boldsymbol{X}}$ and t. No convective term appears because there is no flux of angular momentum across the boundary $\partial\mathscr{V}_t^{\mathrm{cm}}$ in the current configuration.

Case 4: angular momentum equation for an infinitesimal control mass moving with the flow

In Section 5.11.4, we obtained the local angular momentum balance for a classical (non-polar) continuum: the Cauchy stress tensor $\boldsymbol{\sigma}$ is symmetric at every point $\vec{\boldsymbol{x}} \in \Omega_t$ and every time t, according to equation (5.82):

$$\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) = \boldsymbol{\sigma}^T(\vec{\boldsymbol{x}}, t).$$

To express this in material form, using the equation (4.5):

$$\boldsymbol{\sigma}(\vec{\boldsymbol{x}}, t) = \frac{1}{J(\vec{\boldsymbol{X}}, t)}\boldsymbol{P}(\vec{\boldsymbol{X}}, t)\boldsymbol{F}^T(\vec{\boldsymbol{X}}, t),$$

we can write the symmetry condition in terms of the first Piola-Kirchhoff stress tensor $\boldsymbol{P}$ as:

$$\frac{1}{J(\vec{\boldsymbol{X}},t)}\boldsymbol{P}(\vec{\boldsymbol{X}},t)\boldsymbol{F}^T(\vec{\boldsymbol{X}},t) = \left(\frac{1}{J(\vec{\boldsymbol{X}},t)}\boldsymbol{P}(\vec{\boldsymbol{X}},t)\boldsymbol{F}^T(\vec{\boldsymbol{X}},t)\right)^T.$$

Since $(\boldsymbol{AB})^T = \boldsymbol{B}^T\boldsymbol{A}^T$, and $J > 0$, this is equivalent to:

$$\boldsymbol{P}(\vec{\boldsymbol{X}},t)\boldsymbol{F}^T(\vec{\boldsymbol{X}},t) = \boldsymbol{F}(\vec{\boldsymbol{X}},t)\boldsymbol{P}^T(\vec{\boldsymbol{X}},t).$$

This is the local angular momentum balance in material form. It must hold for every $\vec{\boldsymbol{X}} \in \Omega_0$ and every t. In general, $\boldsymbol{P}$ is not symmetric (it is a two-point tensor mapping reference normals to current forces, so symmetry is not even geometrically meaningful), but $\boldsymbol{PF}^T$ must be symmetric.

5.12 Conservation of energy

5.13 When do we use each of the four variants of the balance equations?

This section is a first draft, and it is still quite rough. It needs to be expanded and refined, and the guidelines need to be made more precise and specific.

Check that the following guidelines are consistent with the examples and applications in the rest of the book.

The four variants of the balance equations that we have derived are all mathematically equivalent, but they are expressed in different forms and reference frames, and they are useful in different contexts. The choice of which form to use depends on the specific problem at hand, the available data, and the computational or analytical tools at our disposal. Here are some general guidelines for when to use each form:

Case 1 (integral conservation form for a fixed control volume): This form is particularly useful for analyzing problems where the control volume is fixed in space and we are interested in the net fluxes of momentum across its boundaries. It is commonly used in fluid mechanics for analyzing flow through pipes, around objects, or in open channels, where the control volume can be chosen to enclose the region of interest and the fluxes can be measured or estimated.

Case 2 (integral non-conservation form for a moving control mass): This form is useful when we want to analyze the motion of a specific parcel of fluid or material as it moves through space. It is often used in Lagrangian descriptions of fluid flow, where we track the motion of individual fluid particles, and in solid

mechanics for analyzing the deformation of materials. It can also be useful in numerical methods that use a Lagrangian framework, such as smoothed particle hydrodynamics (SPH).

Case 3 (differential conservation form for a fixed control volume): This form is the most commonly used in computational fluid dynamics (CFD) and in the derivation of the Navier-Stokes equations. It is particularly useful for problems where the flow is complex and we need to solve for the velocity and pressure fields throughout the domain. It is also useful for analyzing the local behavior of the flow, such as turbulence, boundary layers, and shock waves.

Case 4 (differential non-conservation form for a moving control mass): This form is the most fundamental and is often used in theoretical analyses of fluid and solid mechanics. It is the starting point for deriving the equations of motion for both fluids and solids, and it is particularly useful for analyzing the behavior of materials under deformation, such as in elastodynamics and plasticity. It is also useful for problems where the material properties are changing with time, such as in phase transitions or chemical reactions.

In practice, the choice of which form to use is often guided by the specific problem and the available data. For example, if we have measurements of velocity and stress at specific points in space, we might prefer to use the differential forms (Cases 3 or 4). If we have measurements of fluxes across boundaries, we might prefer the integral forms (Cases 1 or 2). Ultimately, all four forms are valid and can be used to analyze the same physical phenomena, but they offer different perspectives and tools for understanding and solving problems in continuum mechanics.

5.14 The principle of virtual work

Appendix A

Review of mathematical concepts

This chapter presents a compendium of formulas and concepts that were most likely studied in mathematics courses prior to studying this text. Therefore, its purpose is not to present these concepts to the reader for the first time, but rather to compile the equations and definitions so that the reader can quickly recall the topics that have been referred to throughout the book.

A.1 Injective, surjective and bijective function

Let us consider a function $f : X \to Y$, where X is the domain and Y is the codomain. We can classify this function based on its mapping properties:

- *Injective* (or *one-to-one*)*: A function f is injective if it maps distinct elements of its domain X to distinct elements of its codomain Y, that is, every element of Y is the image of at most one element $x \in X$. Mathematically, this means that for all elements $x_1 \neq x_2$ in the domain X, $f(x_1) \neq f(x_2)$, or equivalently, $f(x_1) = f(x_2)$ implies $x_1 = x_2$. This property ensures that distinct inputs always yield distinct outputs.

- *Surjective* (or *onto*)*: A function f is surjective if, for every element y in the codomain Y, there exists at least one element x in the domain X such that $f(x) = y$. This property guarantees that every element in the codomain is "reached" or "covered" by the function.

- *Bijective**: A bijection, bijective function, or one-to-one correspondence between two mathematical sets is a function that is both injective and surjective. This

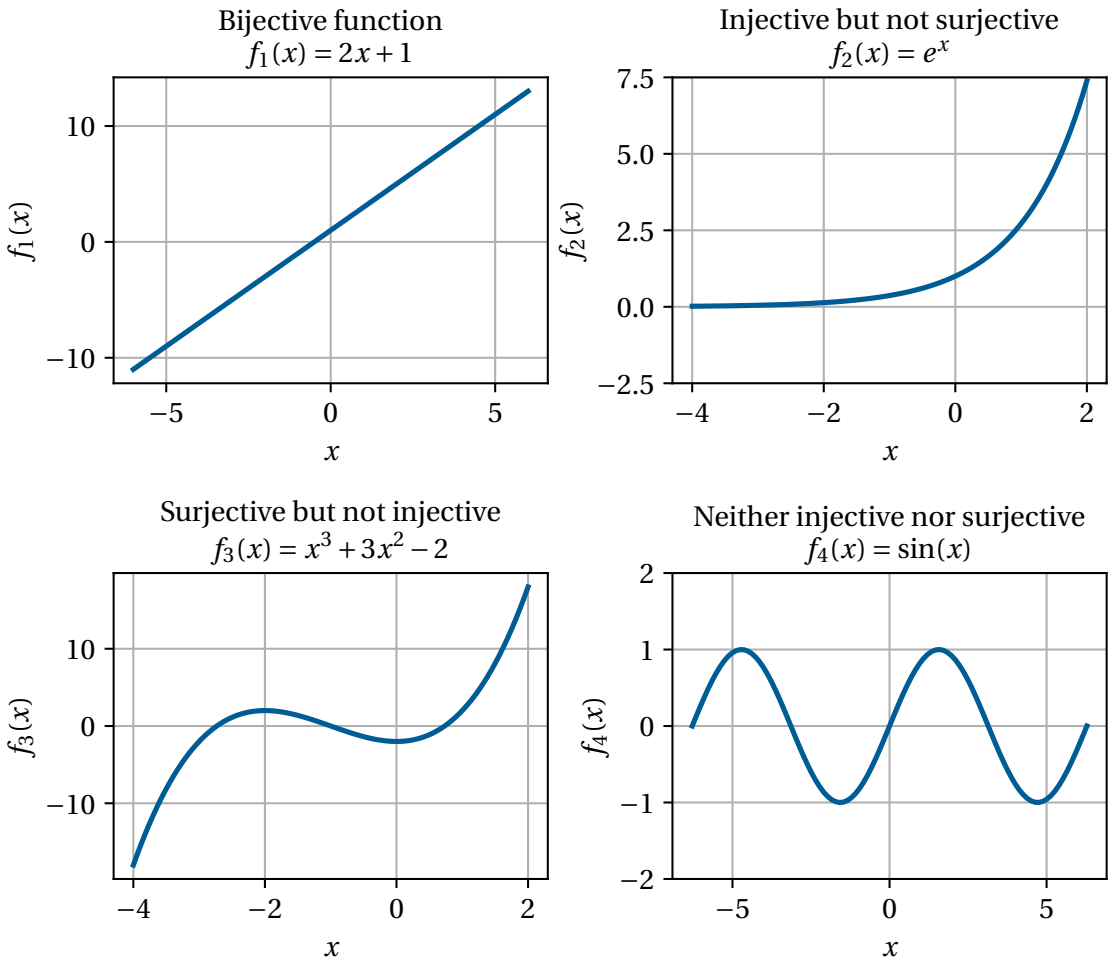


Figure A.1: *Showcase of the concepts of bijectivity, injectivity, and surjectivity in functions.*

means that every element in the codomain is mapped to exactly one element in the domain.

Bijective functions possess several important properties:

- They establish a perfect one-to-one correspondence between elements of the domain and codomain.
- They are invertible, meaning there exists an inverse function $f^{-1} : Y \to X$. In fact, a function is invertible if and only if it is bijective.
- They imply that the domain and codomain have the same cardinality (number of elements).

To illustrate these concepts, let's examine some examples. Consider the following functions, shown in figure A.1, all mapping from $\mathbb{R}$ to $\mathbb{R}$:

- Bijective function: $f_1(x) = 2x + 1$. This function is both injective (different inputs always yield different outputs) and surjective (every real number is "hit" by some input). In addition, since this function is bijective, its inverse exists: $f_1^{-1}(y) = \frac{y-1}{2}$.
- Injective but not surjective: $f_2(x) = e^x$. This function is injective (each output corresponds to a unique input) but not surjective (negative numbers in the codomain are never reached). Take into account that f_2 becomes bijective if we change its codomain to $Y = (0, \infty)$.
- Surjective but not injective: $f_3(x) = x^3 + 3x^2 - 2$. This function is surjective (each output is always reached) but not injective (since the function has three roots: -1, $-\sqrt{3} - 1$ and $+\sqrt{3} - 1$, and therefore $f_3(-1) = f_3(\pm\sqrt{3} - 1) = 0$).
- Neither injective nor surjective: $f_4(x) = \sin(x)$. This function is neither injective (e.g., $\sin(0) = \sin(k\pi) = 0$ for all $k \in \mathbb{Z}$) nor surjective (its range is limited to $[-1, 1]$, never reaching values outside this interval).

A.2 Image through a function

The *image** of a set $A \subseteq X$ through a function $f : X \to Y$, denoted as $f(A)$, is the set of all elements in Y that are obtained by applying f to elements in A; mathematically, this is expressed as

$$f(A) = \{f(x) \in Y \mid x \in A\}.$$

Important properties:

- The image is always a subset of the codomain Y: $f(A) \subseteq Y$.
- A function f is surjective if and only if $f(X) = Y$.
- $A \subseteq B \subseteq X$ implies that $f(A) \subseteq f(B) \subseteq Y$.

A.3 Inverse function

A function $f : X \to Y$ is called *invertible** if there exists a function $g : Y \to X$ such that $g(f(x)) = x$ for all $x \in X$ and $f(g(y)) = y$ for all $y \in Y$. When such a function g exists, it is uniquely determined and is termed the *inverse** of f, denoted by f^{-1}. Alternatively, the inverse function $f^{-1} : Y \to X$ is defined by:

$$f^{-1}(y) \coloneqq \{x \in X \mid f(x) = y\},$$

where the set X contains exactly one element x for every $y \in Y$. Not all functions have inverses. A function is invertible if and only if it is bijective.

A.4 Projection of a vector over another vector

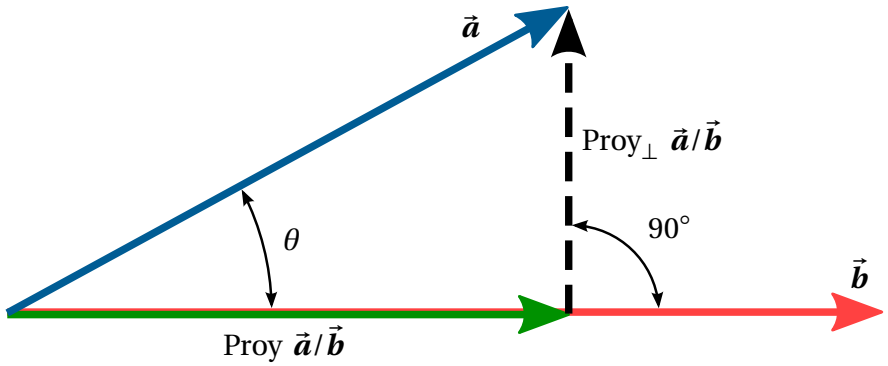


Figure A.2: *Projection of vector $\vec{a}$ onto vector $\vec{b}$.*

With reference to Figure A.2, the *vector projection** of vector $\vec{a}$ onto vector $\vec{b}$, denoted by $\text{Proj}\ \vec{a}/\vec{b}$, is a vector with the same direction as $\vec{b}$ and with magnitude $\|\vec{a}\|\cos\theta$, where θ is the angle between vectors $\vec{a}$ and $\vec{b}$. When θ is unknown, it can be calculated using:

$$\cos\theta = \frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\,\|\vec{b}\|}.$$

Therefore, the magnitude of $\text{Proj}\ \vec{a}/\vec{b}$ is given by:

$$\|\vec{a}\|\cos\theta = \|\vec{a}\|\frac{\vec{a}\cdot\vec{b}}{\|\vec{a}\|\|\vec{b}\|} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{b}\|}.$$

Since $\text{Proj}\ \vec{a}/\vec{b}$ has the same direction as $\vec{b}$, we have $\text{Proj}\ \vec{a}/\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{b}\|}\hat{b}$, where $\hat{b} := \frac{\vec{b}}{\|\vec{b}\|}$ is the unit vector in the direction of $\vec{b}$. In conclusion:

$$\text{Proj}\ \vec{a}/\vec{b} = \frac{\vec{a}\cdot\vec{b}}{\|\vec{b}\|^2}\vec{b}.$$

Finally, the *orthogonal projection** (also called *vector rejection**) of vector $\vec{a}$ onto vector $\vec{b}$, denoted by $\text{Proj}_{\perp}\ \vec{a}/\vec{b}$, is computed as:

$$\text{Proj}_{\perp}\ \vec{a}/\vec{b} = \vec{a} - \text{Proj}\ \vec{a}/\vec{b} = \vec{a} - \frac{\vec{a}\cdot\vec{b}}{\|\vec{b}\|^2}\vec{b}.$$

A.5 Parametric curve

A *parametric curve*[★] is a way of representing a curve in space by expressing the coordinates of points on the curve as functions of a single parameter. For example, a curve in three-dimensional space can be represented by three parametric equations:

$$x = f(t) \qquad y = g(t) \qquad z = h(t),$$

where x, y, and z are the Cartesian coordinates of points on the curve, and t is the parameter.

A.6 Derivative of a parametric curve

With reference to Figure A.3, let us consider a parametric curve $\vec{\boldsymbol{r}} : \Xi \to \mathbb{R}^3$, which maps a parameter ξ from the domain $\Xi \subseteq \mathbb{R}$ to a position vector $\vec{\boldsymbol{r}}(\xi)$ in three-dimensional space. The derivative of this parametric curve with respect to the parameter ξ is given by:

$$\vec{\boldsymbol{r}}'(\xi) = \frac{\mathrm{d}\vec{\boldsymbol{r}}(\xi)}{\mathrm{d}\xi} = \lim_{\Delta\xi \to 0} \frac{\vec{\boldsymbol{r}}(\xi + \Delta\xi) - \vec{\boldsymbol{r}}(\xi)}{\Delta\xi}.$$

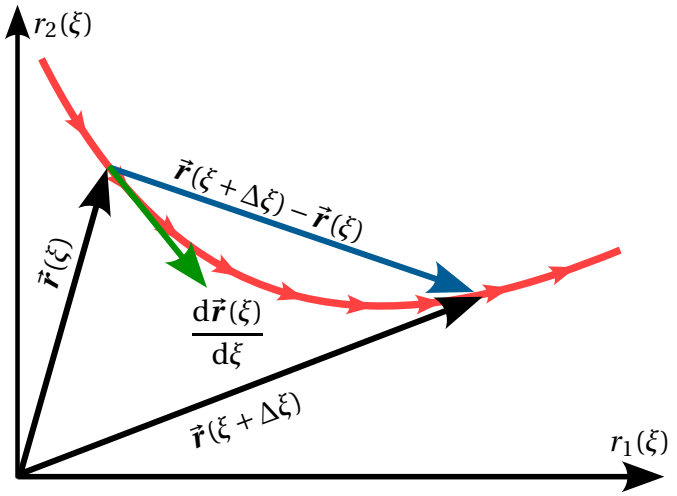


Figure A.3: *The derivative of a parametric curve $\vec{\boldsymbol{r}}(\xi)$ at a point ξ is the tangent vector $\vec{\boldsymbol{r}}'(\xi)$, which represents the direction of the curve at that point. The tangent vector points in the direction of increasing ξ along the curve.*

The vector $\vec{\boldsymbol{r}}'(\xi)$ represents the tangent vector to the curve $\vec{\boldsymbol{r}}(\xi)$ at the point corresponding to the parameter value ξ. This tangent vector points in the direction of increasing ξ along the curve.

A.7 Jacobian matrix

Consider a vector-valued function

$$\vec{\boldsymbol{f}}(x_1, x_2, ..., x_n) = \left[f_1(x_1, ..., x_n),\ f_2(x_1, ..., x_n),\ ...,\ f_m(x_1, ..., x_n)\right]^T.$$

The *Jacobian matrix* $\boldsymbol{J}_{\vec{f}}$ of $\vec{\boldsymbol{f}}$ is an $m \times n$ matrix composed of all first-order partial derivatives of the components of $\vec{\boldsymbol{f}}$:

$$\boldsymbol{J}_{\vec{f}} = \frac{\partial \vec{\boldsymbol{f}}}{\partial \vec{\boldsymbol{x}}} = \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial f_m}{\partial x_1} & \cdots & \frac{\partial f_m}{\partial x_n} \end{bmatrix}.$$

Here, the ij-th entry of this matrix, $\frac{\partial f_i}{\partial x_j}$, describes how a small change in each input variable x_j affects each output f_i. In this context, each row refers to the gradient of the output function f_i, while each column refers to the rate that function $\vec{\boldsymbol{f}}$ changes with respect to a small change in the single input x_j. In the particular case when $m = n = 1$, the Jacobian matrix simplifies to the usual derivative.

Example A.1. *Jacobian matrix*

Calculate the Jacobian matrix of the vector-valued function

$$\vec{\boldsymbol{f}}(x, y) = (x^2 + y,\ x - y^2).$$

Solution:

The Jacobian matrix of this function is:

$$\boldsymbol{J}_{\vec{f}} = \begin{bmatrix} \frac{\partial}{\partial x}(x^2 + y) & \frac{\partial}{\partial y}(x^2 + y) \\ \frac{\partial}{\partial x}(x - y^2) & \frac{\partial}{\partial y}(x - y^2) \end{bmatrix} = \begin{bmatrix} 2x & 1 \\ 1 & -2y \end{bmatrix}.$$

A.8 Inverse function theorem

The inverse function theorem is an important result in multivariable calculus that provides conditions under which a function can be inverted locally and describes the relationship between the derivatives of a function and its inverse.

Let $\Omega \subseteq \mathbb{R}^n$ be an open set, and let $\vec{\boldsymbol{f}} : \Omega \to \mathbb{R}^n$ be a continuously differentiable function. Suppose that at a point $\vec{\boldsymbol{x}}_0 \in \Omega$, the Jacobian matrix $\boldsymbol{J}_{\vec{f}}(\vec{\boldsymbol{x}}_0) := \frac{\partial \vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}_0)}{\partial \vec{\boldsymbol{x}}}$ is

invertible (i.e., has a non-zero determinant). Then, the *inverse function theorem** states that there exists a neighborhood $U \subset \Omega$ of $\vec{\boldsymbol{x}}_0$ and a neighborhood $V \subset \mathbb{R}^n$ of $\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}_0)$ such that:

- $\vec{\boldsymbol{f}}$ is a bijection from U to V; in other words, $\vec{\boldsymbol{f}}$ is locally bijective where the Jacobian $\boldsymbol{J}_{\vec{\boldsymbol{f}}}(\vec{\boldsymbol{x}}_0)$ is invertible.
- The inverse function $\vec{\boldsymbol{f}}^{-1} : V \to U$ locally exists and is continuously differentiable on V.
- The Jacobian matrix of the inverse function $\vec{\boldsymbol{f}}^{-1}$ at $\vec{\boldsymbol{f}}(\vec{\boldsymbol{x}}_0)$ is the inverse of the Jacobian matrix of $\vec{\boldsymbol{f}}$ at $\vec{\boldsymbol{x}}_0$, that is,

$$\boldsymbol{J}_{\vec{\boldsymbol{f}}^{-1}}\left(f(\vec{\boldsymbol{x}}_0)\right) = \left(\boldsymbol{J}_{\vec{\boldsymbol{f}}}(\vec{\boldsymbol{x}}_0)\right)^{-1}.$$

A consequence of this theorem is that if $\vec{\boldsymbol{x}} = \vec{\boldsymbol{f}}(\vec{\boldsymbol{X}})$ is a continuously differentiable function on Ω with $\vec{\boldsymbol{x}} = [x, y, z]^T$ and $\vec{\boldsymbol{X}} = [X, Y, Z]^T$ whose Jacobian matrix

$$\boldsymbol{J}_{\vec{\boldsymbol{f}}} := \frac{\partial \vec{\boldsymbol{f}}}{\partial \vec{\boldsymbol{X}}} = \begin{pmatrix} \frac{\partial x}{\partial X} & \frac{\partial y}{\partial X} & \frac{\partial z}{\partial X} \\ \frac{\partial x}{\partial Y} & \frac{\partial y}{\partial Y} & \frac{\partial z}{\partial Y} \\ \frac{\partial x}{\partial Z} & \frac{\partial y}{\partial Z} & \frac{\partial z}{\partial Z} \end{pmatrix}$$

is invertible for all $\vec{\boldsymbol{x}} \in \Omega$, then the inverse function $\vec{\boldsymbol{X}} = \vec{\boldsymbol{f}}^{-1}(\vec{\boldsymbol{x}})$ exists for all points $\vec{\boldsymbol{x}} \in \vec{\boldsymbol{f}}(\Omega)$ and its Jacobian

$$\boldsymbol{J}_{\vec{\boldsymbol{f}}^{-1}} := \frac{\partial \vec{\boldsymbol{f}}^{-1}}{\partial \vec{\boldsymbol{x}}} = \begin{pmatrix} \frac{\partial X}{\partial x} & \frac{\partial Y}{\partial x} & \frac{\partial Z}{\partial x} \\ \frac{\partial X}{\partial y} & \frac{\partial Y}{\partial y} & \frac{\partial Z}{\partial y} \\ \frac{\partial X}{\partial z} & \frac{\partial Y}{\partial z} & \frac{\partial Z}{\partial z} \end{pmatrix}$$

is the inverse of the Jacobian matrix of $\vec{\boldsymbol{f}}$, that is, $\frac{\partial \vec{\boldsymbol{f}}^{-1}}{\partial \vec{\boldsymbol{x}}} = \left(\frac{\partial \vec{\boldsymbol{f}}}{\partial \vec{\boldsymbol{X}}}\right)^{-1}$, or more explicitly,

$$\begin{pmatrix} \frac{\partial X}{\partial x} & \frac{\partial X}{\partial y} & \frac{\partial X}{\partial z} \\ \frac{\partial Y}{\partial x} & \frac{\partial Y}{\partial y} & \frac{\partial Y}{\partial z} \\ \frac{\partial Z}{\partial x} & \frac{\partial Z}{\partial y} & \frac{\partial Z}{\partial z} \end{pmatrix} = \begin{pmatrix} \frac{\partial x}{\partial X} & \frac{\partial x}{\partial Y} & \frac{\partial x}{\partial Z} \\ \frac{\partial y}{\partial X} & \frac{\partial y}{\partial Y} & \frac{\partial y}{\partial Z} \\ \frac{\partial z}{\partial X} & \frac{\partial z}{\partial Y} & \frac{\partial z}{\partial Z} \end{pmatrix}^{-1}.$$

Example A.2. *Using the inverse function theorem*

Consider the coordinate transformation from polar to Cartesian coordinates given by:

$$x(r, \theta) = r\cos\theta \qquad\qquad y(r, \theta) = r\sin\theta$$

Using the inverse function theorem, find the partial derivatives $\frac{\partial\theta}{\partial x}$, $\frac{\partial\theta}{\partial y}$, $\frac{\partial r}{\partial x}$, and $\frac{\partial r}{\partial y}$.

Solution:

Let's find those derivatives using a direct method and then the inverse function theorem.

Method 1: direct calculation using known inverse relations

From the standard polar-to-Cartesian transformation, we know that the inverse relations are:

$$r = \sqrt{x^2 + y^2} \qquad\qquad \theta = \arctan2(y, x)$$

Let's first calculate $\frac{\partial r}{\partial x}$. Since $r = \sqrt{x^2 + y^2}$, we apply the chain rule:

$$\frac{\partial r}{\partial x} = \frac{\partial}{\partial x}\left[\sqrt{x^2 + y^2}\right] = \frac{1}{2}\left(x^2 + y^2\right)^{-\frac{1}{2}} 2x = \frac{x}{\left(x^2 + y^2\right)^{\frac{1}{2}}} = \frac{x}{r} = \cos\theta.$$

The proof of $\frac{\partial r}{\partial y} = \sin\theta$ follows analogously and is left as an exercise to the reader.

Now let us find $\frac{\partial\theta}{\partial x}$. Considering that $\tan\theta = \frac{y}{x}$, we implicitly differentiate both sides of this equality with respect to x (recall also that $\frac{\mathrm{d}}{\mathrm{d}\theta}\tan\theta = \sec^2\theta$):

$$\frac{\partial}{\partial x}[\tan\theta] = \frac{\partial}{\partial x}\left[\frac{y}{x}\right]$$

$$\underbrace{\frac{\mathrm{d}}{\mathrm{d}r}[\tan\theta]}_{=0}\frac{\partial r}{\partial x} + \underbrace{\frac{\mathrm{d}}{\mathrm{d}\theta}[\tan\theta]}_{\sec^2\theta}\frac{\partial\theta}{\partial x} = -yx^{-2} \qquad \text{(using the chain rule)}$$

$$\sec^2\theta\frac{\partial\theta}{\partial x} = -\frac{y}{x^2}$$

and, since $\sec\theta = \frac{1}{\cos\theta} = \frac{r}{x}$, we obtain:

$$\frac{r^2}{x^2}\frac{\partial\theta}{\partial x} = -\frac{y}{x^2};$$

solving for $\frac{\partial\theta}{\partial x}$, and considering that $\sin\theta = \frac{y}{r}$, we get:

$$\frac{\partial\theta}{\partial x} = -\frac{y}{r^2} = -\frac{\sin\theta}{r}.$$

The proof of $\frac{\partial\theta}{\partial y} = \frac{\cos\theta}{r}$ is done similarly and is left as an exercise to the reader.

Method 2: The inverse function theorem approach

The inverse function theorem states that if we have a transformation $\vec{f} : \mathbb{R}^2 \to \mathbb{R}^2$ given by:

$$\vec{f}(r,\theta) = \begin{pmatrix} x(r,\theta) \\ y(r,\theta) \end{pmatrix} = \begin{pmatrix} r\cos\theta \\ r\sin\theta \end{pmatrix}$$

then the Jacobian matrix of the inverse transformation is $\boldsymbol{J}_{\vec{f}^{-1}} = \left(\boldsymbol{J}_{\vec{f}}\right)^{-1}$ where $\boldsymbol{J}_{\vec{f}}$ is the Jacobian matrix of $\vec{f}$.

Let's compute the Jacobian matrix $\boldsymbol{J}_{\vec{f}}$:

$$\boldsymbol{J}_{\vec{f}} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix}$$

To find the inverse of this 2×2 matrix, we use the formula:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}^{-1} = \frac{1}{ad-bc}\begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

and using this equation we compute the inverse:

$$\boldsymbol{J}_{\vec{f}}^{-1} = \frac{1}{r}\begin{pmatrix} r\cos\theta & r\sin\theta \\ -\sin\theta & \cos\theta \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\frac{\sin\theta}{r} & \frac{\cos\theta}{r} \end{pmatrix}$$

The inverse function theorem gives us, through the inverse Jacobian matrix, all the partial derivatives we seek:

$$\boldsymbol{J}_{\vec{f}}^{-1} = \begin{pmatrix} \frac{\partial r}{\partial x} & \frac{\partial r}{\partial y} \\ \frac{\partial \theta}{\partial x} & \frac{\partial \theta}{\partial y} \end{pmatrix} = \begin{pmatrix} \cos\theta & \sin\theta \\ -\frac{\sin\theta}{r} & \frac{\cos\theta}{r} \end{pmatrix}$$

These results match exactly what we obtained through direct calculation and implicit differentiation.

Let's verify our analytical results using symbolic computation. The following Python code uses SymPy to compute the Jacobian and its inverse:

```
# Import the necessary libraries
from sympy import symbols, sin, cos, diff, Matrix, simplify

# Define the symbols
r, t = symbols('r t')

# Define the coordinate transformation functions
x = r*cos(t)
y = r*sin(t)
```

```
# Form the Jacobian matrix
J = Matrix([[diff(x, r), diff(x, t)],
            [diff(y, r), diff(y, t)]])

# Calculate the inverse of the Jacobian matrix
inv_J = simplify(J.inv())

# Extract and display the partial derivatives
print('dr_dx =', inv_J[0,0])     # prints dr_dx = cos(t)
print('dr_dy =', inv_J[0,1])     # prints dr_dy = sin(t)
print('dt_dx =', inv_J[1,0])     # prints dt_dx = -sin(t)/r
print('dt_dy =', inv_J[1,1])     # prints dt_dy = cos(t)/r
```

A.9 Orthogonal matrices

An $n \times n$ matrix $\boldsymbol{Q}$ is called an *orthogonal matrix* if its columns (or equivalently, rows) form an orthonormal set of vectors. Mathematically, this means: $\boldsymbol{Q}\boldsymbol{Q}^T = \boldsymbol{Q}^T\boldsymbol{Q} = \boldsymbol{I}$. This property implies that the inverse of an orthogonal matrix is equal to its transpose: $\boldsymbol{Q}^{-1} = \boldsymbol{Q}^T$.

Orthogonal matrices have several important properties:

- **Preservation of norms:** If $\boldsymbol{Q}$ is an orthogonal matrix, then for any vector $\vec{\boldsymbol{x}} \in \mathbb{R}^n$, the Euclidean norm of $\vec{\boldsymbol{x}}$ is preserved under multiplication by $\boldsymbol{Q}$. That is: $\|\boldsymbol{Q}\vec{\boldsymbol{x}}\| = \|\vec{\boldsymbol{x}}\|$. This property ensures that the transformation represented by $\boldsymbol{Q}$ does not stretch or shrink vectors.

- **Preservation of dot products:** For any two vectors $\vec{\boldsymbol{x}}$ and $\vec{\boldsymbol{y}}$ in $\mathbb{R}^n$, the dot product is preserved under multiplication by an orthogonal matrix $\boldsymbol{Q}$. Specifically: $(\boldsymbol{Q}\vec{\boldsymbol{x}}) \cdot (\boldsymbol{Q}\vec{\boldsymbol{y}}) = \vec{\boldsymbol{x}} \cdot \vec{\boldsymbol{y}}$. This property is closely related to the preservation of norms, as the dot product determines angles between vectors, and orthogonal transformations preserve angles.

- **Determinant equals ± 1:** The determinant of an orthogonal matrix $\boldsymbol{Q}$ satisfies: $\det(\boldsymbol{Q}) = \pm 1$. In case that $n = 3$, if $\det(\boldsymbol{Q}) = 1$, the matrix represents a *rotation* around an axis; on the other hand, if $\det(\boldsymbol{Q}) = -1$, the matrix represents a *reflection* across a plane.

- **Eigenvalues lie on the unit circle:** The eigenvalues of an orthogonal matrix $\boldsymbol{Q}$ have magnitude 1. That is, if λ is an eigenvalue of $\boldsymbol{Q}$, then: $|\lambda| = 1$. In the complex plane, the eigenvalues lie on the unit circle. For real orthogonal matrices, the eigenvalues are either ± 1 or come in complex conjugate pairs of the form $\exp(+i\theta) = \cos\theta + i\sin\theta$ and $\exp(-i\theta) = \cos\theta - i\sin\theta$.

- **Orthogonal columns and rows:** The columns (and rows) of an orthogonal matrix $\boldsymbol{Q}$ form an orthonormal basis for $\mathbb{R}^n$. This means that each column vector has unit length: $\|\vec{\boldsymbol{q}}_i\| = 1$ for all $i = 1, 2, \ldots, n$, and that any two distinct column vectors are orthogonal: $\vec{\boldsymbol{q}}_i \cdot \vec{\boldsymbol{q}}_j = 0$ for $i \neq j$. Similarly, the rows of $\boldsymbol{Q}$ also form an orthonormal basis.
- **Inverse equals transpose:** As mentioned earlier, the inverse of an orthogonal matrix is equal to its transpose: $\boldsymbol{Q}^{-1} = \boldsymbol{Q}^T$. This property makes orthogonal matrices computationally efficient, as computing the inverse of a general matrix can be expensive, while transposing a matrix is straightforward.
- **Product of orthogonal matrices is orthogonal:** If $\boldsymbol{Q}_1$ and $\boldsymbol{Q}_2$ are orthogonal matrices, then their product $\boldsymbol{Q}_1\boldsymbol{Q}_2$ is also an orthogonal matrix.

A.10 Chain rule

The *chain rule** is a fundamental tool in calculus used to determine the derivative of a composite function. Think of it as a way to calculate how a "change" affects a number of interconnected dependencies.

The basic case: composition of two functions If z depends on y, and y depends on x (expressed as $z = f(y)$ and $y = g(x)$), then the rate of change of z with respect to x is the product of their individual rates:

$$\frac{\mathrm{d}z}{\mathrm{d}x} = \frac{\mathrm{d}z}{\mathrm{d}y}\frac{\mathrm{d}y}{\mathrm{d}x} = f'(y)g'(x) = f'(g(x))g'(x).$$

The total derivative In many physical systems, a function f depends on time t both directly and indirectly through other variables $x_i(t)$. For a function $f(t, \vec{\boldsymbol{x}}(t))$ where $\vec{\boldsymbol{x}}(t) = [x_1(t), \ldots, x_m(t)]^T$, the *total derivative** accounts for all these paths of influence:

$$\frac{\mathrm{d}f}{\mathrm{d}t} = \frac{\partial f}{\partial t} + \sum_{i=1}^{m} \frac{\partial f}{\partial x_i}\frac{\mathrm{d}x_i}{\mathrm{d}t} = \frac{\partial f}{\partial t} + \frac{\partial f}{\partial \vec{\boldsymbol{x}}} \cdot \frac{\mathrm{d}\vec{\boldsymbol{x}}}{\mathrm{d}t}$$

The first term $\frac{\partial f}{\partial t}$ represents the explicit change over time, while the second term captures the change "channeled" through the moving variables x_i. For example, the derivative of a function $f(x, y)$ where x and y are functions of t is given by:

$$\frac{\mathrm{d}f(t)}{\mathrm{d}t} = \frac{\partial f}{\partial x}\frac{\mathrm{d}x}{\mathrm{d}t} + \frac{\partial f}{\partial y}\frac{\mathrm{d}y}{\mathrm{d}t}.$$

General multi-variable case If we have a function $f(u_1(\vec{\boldsymbol{x}}),\dots,u_m(\vec{\boldsymbol{x}}))$ where each u_i depends on a vector of independent variables $\vec{\boldsymbol{x}} = [x_1,\dots,x_n]^T$, the partial derivative with respect to any specific variable x_j is:

$$\frac{\partial f}{\partial x_j} = \sum_{i=1}^{m} \frac{\partial f}{\partial u_i}\frac{\partial u_i}{\partial x_j} \quad \text{for } j = 1,\dots,n.$$

this total change in f is the "sum of all possible paths" through which x_j can influence f. In vector notation, this is a dot product between the gradient of f and the partial derivative of the vector $\vec{\boldsymbol{u}}$ with respect to x_j:

$$\frac{\partial f}{\partial x_j} = \frac{\partial f}{\partial \vec{\boldsymbol{u}}} \cdot \frac{\partial \vec{\boldsymbol{u}}}{\partial x_j} = \nabla_{\vec{\boldsymbol{u}}} f \cdot \frac{\partial \vec{\boldsymbol{u}}}{\partial x_j}$$

Now, when mapping a vector $\vec{\boldsymbol{x}} \in \mathbb{R}^n$ to a vector $\vec{\boldsymbol{y}} \in \mathbb{R}^m$ via $\vec{\boldsymbol{g}} : \mathbb{R}^n \to \mathbb{R}^m$, i.e. $\vec{\boldsymbol{y}} = \vec{\boldsymbol{g}}(\vec{\boldsymbol{x}})$, and then to $\vec{\boldsymbol{z}} \in \mathbb{R}^p$ via $\vec{\boldsymbol{f}} : \mathbb{R}^m \to \mathbb{R}^p$, that is $\vec{\boldsymbol{z}} = \vec{\boldsymbol{f}}(\vec{\boldsymbol{y}}) = \vec{\boldsymbol{f}}(\vec{\boldsymbol{g}}(\vec{\boldsymbol{x}}))$, the chain rule is expressed through the multiplication of Jacobian matrices:

$$\boldsymbol{J}_{\vec{\boldsymbol{f}}\circ\vec{\boldsymbol{g}}}(\vec{\boldsymbol{x}}) = \underbrace{\frac{\partial \vec{\boldsymbol{f}}}{\partial \vec{\boldsymbol{x}}}}_{p\times n} = \boldsymbol{J}_{\vec{\boldsymbol{f}}}(\vec{\boldsymbol{g}}(\vec{\boldsymbol{x}}))\boldsymbol{J}_{\vec{\boldsymbol{g}}}(\vec{\boldsymbol{x}}) = \underbrace{\frac{\partial \vec{\boldsymbol{f}}}{\partial \vec{\boldsymbol{g}}}}_{p\times m}\underbrace{\frac{\partial \vec{\boldsymbol{g}}}{\partial \vec{\boldsymbol{x}}}}_{m\times n} = \frac{\partial \vec{\boldsymbol{f}}}{\partial \vec{\boldsymbol{y}}}\frac{\partial \vec{\boldsymbol{g}}}{\partial \vec{\boldsymbol{x}}}.$$

A.11 Linear transformation

A *linear transformation*[★] is a function between two vector spaces that preserves the operations of vector addition and scalar multiplication. Let V and W be vector spaces over the real numbers $\mathbb{R}$. A function $T : V \to W$ is called a *linear transformation* if, for all vectors $\vec{\boldsymbol{u}}$, $\vec{\boldsymbol{v}}$ in V and all scalars $c \in \mathbb{R}$, the following two properties hold:

- **Additivity:** $T(\vec{\boldsymbol{u}} + \vec{\boldsymbol{v}}) = T(\vec{\boldsymbol{u}}) + T(\vec{\boldsymbol{v}})$.
- **Homogeneity** (or multiplication by a scalar): $T(c\vec{\boldsymbol{u}}) = cT(\vec{\boldsymbol{u}})$·

Linear transformation have the following properties:

- The composition of two linear transformations is also a linear transformation. In other words, if $T_1 : V \to W$ and $T_2 : W \to Z$ are two linear transformations, then their composition, denoted by $T_2 \circ T_1 : V \to Z$, where $\big(T_2 \circ T_1\big)(\vec{\boldsymbol{u}}) = T_2\big(T_1(\vec{\boldsymbol{u}})\big)$, is also a linear transformation.
- A linear transformation is invertible if and only if its matrix representation is invertible (i.e., has a non-zero determinant for square matrices). In other words, an invertible linear transformation effectively "undoes" the original transformation.

- Linear transformations preserve linear combinations; in other words, for any vectors $\vec{\boldsymbol{u}}_1, \ldots, \vec{\boldsymbol{u}}_n \in V$ and scalars $c_1, \ldots, c_n \in \mathbb{R}$ it holds that

$$f(c_1\vec{\boldsymbol{u}}_1 + \cdots + c_n\vec{\boldsymbol{u}}_n) = c_1 f(\vec{\boldsymbol{u}}_1) + \cdots + c_n f(\vec{\boldsymbol{u}}_n).$$

In finite-dimensional spaces, every linear transformation from $\mathbb{R}^n$ to $\mathbb{R}^m$ can be represented by an $m \times n$ matrix $\boldsymbol{A}$ such that $T(\vec{\boldsymbol{x}}) = \boldsymbol{A}\vec{\boldsymbol{x}}$, where $\vec{\boldsymbol{x}}$ is a column vector in $\mathbb{R}^n$. In this case, linear transformation can be understood as a special "magic powder" that you can sprinkle on any vector drawn as an arrow on the blackboard. When you apply this magic powder, the arrow changes in such a way that it might stretch or shrink the vector and it might rotate the arrow to point in a different direction. However, no matter how the arrow changes, two important things always happen:

1. If you apply the magic powder to the sum of two arrows, the result is the same as applying the powder to each arrow separately and then adding those results together.
2. If you apply the magic powder to an arrow that has been scaled (made longer or shorter by a number), the result is the same as scaling the transformed arrow by that number.

Because of these properties, this magic powder represents a linear transformation, that is a function that changes vectors in a way that preserves their linear structure. It never bends or twists the space V to map a vector in W; it only stretches, shrinks, rotates, or flips vectors, always keeping the origin fixed.

A.12 Positive definite matrix

Let's consider a symmetric matrix $\boldsymbol{A} \in \mathbb{R}^{n\times n}$. $\boldsymbol{A}$ is said to be *positive definite** when at least one of the following conditions is met:

- The product $\vec{\boldsymbol{x}}^T\boldsymbol{A}\vec{\boldsymbol{x}}$ is always positive, that is $\vec{\boldsymbol{x}}^T\boldsymbol{A}\vec{\boldsymbol{x}} > 0$, for any vector $\vec{\boldsymbol{x}} \in \mathbb{R}^n \setminus \{\vec{\boldsymbol{0}}\}$.
- All eigenvalues of $\boldsymbol{A}$ are positive.
- *Sylvester's criterion**: all *principal minors of* $\boldsymbol{A}$ are positive, that is, $\det(\boldsymbol{A}_{1:i,1:i}) > 0$ for all $i = 1, \ldots, n$.

It can be shown that if any of these conditions is satisfied, the other two are automatically fulfilled. Sometimes $\boldsymbol{A} > 0$ is written to denote that $\boldsymbol{A}$ is positive definite. The reader may wonder: what are positive definite matrices used for? The answer lies in the fact that for these types of matrices, there exist super-efficient numerical algorithms, from a computational point of view, for solving systems of linear equations $\boldsymbol{A}\vec{\boldsymbol{x}} = \vec{\boldsymbol{b}}$, such as the Cholesky decomposition or the conjugate gradient method;

similarly, this property is used by many algorithms for minimizing convex functions, both linear and non-linear, in several dimensions. Additionally, positive definite matrices have very particular properties:

- The determinant of a positive definite matrix $\boldsymbol{A}$ is always positive, that is $\det(\boldsymbol{A}) > 0$, so the inverse $\boldsymbol{A}^{-1}$ exists. Moreover, $\boldsymbol{A}^{-1}$ is also positive definite.
- If matrices $\boldsymbol{A}$ and $\boldsymbol{B}$ are positive definite, their sum $\boldsymbol{A}+\boldsymbol{B}$ is also positive definite.
- The diagonal terms A_{ii} of the positive definite matrix $\boldsymbol{A}$ are all positive, so its trace will also be positive, that is $\mathrm{tr}(\boldsymbol{A}) > 0$.
- The inequality $A_{ij} < \sqrt{A_{ii}A_{jj}}$ is always satisfied for all $i, j \in 1, \ldots, n$.
- The term with the largest absolute value of a positive definite matrix $\boldsymbol{A}$ always lies on the diagonal, that is, $\max_{i,j} |A_{ij}| < \max_i A_{ii}$ for all $i, j \in 1, \ldots, n$.
- Every positive definite matrix $\boldsymbol{A}$ has a unique square root matrix $\boldsymbol{R}$ such that $\boldsymbol{R}^T\boldsymbol{R} = \boldsymbol{A}$. The matrix $\boldsymbol{R}$ is also positive definite.
- Remember that, by definition, every positive definite matrix is symmetric.

A.13 Neumann expansion

A.14 Taylor series expansion

Taylor series expansions, named after the British mathematician Brook Taylor (1685–1731), approximate functions by polynomials around a point. There are two common forms: one-dimensional and multidimensional, which are summarized below.

A.14.1 Taylor series expansion in one variable

The *Taylor series** of a real or complex function $f(x)$ that is infinitely differentiable in a neighborhood of a real or complex point x_0 is given by the power series

$$f(x_0 + \Delta x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} \Delta x^n,$$

that is,

$$f(x_0 + \Delta x) = f(x_0) + \frac{f'(x_0)}{1!}\Delta x + \frac{f''(x_0)}{2!}\Delta x^2 + \frac{f^{(3)}(x_0)}{3!}\Delta x^3 + \cdots.$$

Equivalently, by setting $\Delta x = x - x_0$, one may write

$$f(x) = f(x_0) + \frac{f'(x_0)}{1!}(x - x_0) + \frac{f''(x_0)}{2!}(x - x_0)^2 + \frac{f^{(3)}(x_0)}{3!}(x - x_0)^3 + \cdots,$$

where $n!$ denotes the factorial of n and $f^{(n)}(x_0)$ is the n-th derivative of f evaluated at x_0; the zeroth derivative, $f^{(0)}$, is defined as the function f itself, and the terms Δx^0 and 0! are both equal to 1.

When $x_0 = 0$, the Taylor series is also known as the *Maclaurin series*, in honor of the Scottish mathematician Colin Maclaurin (1698–1746).

The main purpose of Taylor series is to approximate complicated but differentiable functions by means of simple polynomials in Δx, which are easier to evaluate, differentiate, and integrate. This approximation is valid only near the expansion point x_0; if the distance Δx is small, the approximation is expected to be highly accurate.

A.14.2 Taylor series expansion in several variables

The Taylor series for multivariate functions around the point $(a_1,\ a_2,\ \ldots, a_d)$ is defined as

$$f(x_1,\ldots,x_d) = \sum_{n_1=0}^{\infty} \cdots \sum_{n_d=0}^{\infty} \frac{(x_1-a_1)^{n_1}\cdots(x_d-a_d)^{n_d}}{n_1!\cdots n_d!} \left(\frac{\partial^{n_1+\cdots+n_d} f}{\partial x_1^{n_1}\cdots\partial x_d^{n_d}}\right)(a_1,\ldots,a_d).$$

For a function of two variables x and y, the Taylor approximation around the point (x_0, y_0) that keeps only second-order terms is

$$\begin{aligned} f(x_0+\Delta x, y_0+\Delta y) \approx f\left(x_0,y_0\right) &+ \Delta x \left.\frac{\partial f}{\partial x}\right|_{(x_0,y_0)} + \Delta y \left.\frac{\partial f}{\partial y}\right|_{(x_0,y_0)} \\ &+ \frac{1}{2!}\left[\Delta x^2 \left.\frac{\partial^2 f}{\partial x^2}\right|_{(x_0,y_0)} + 2\Delta x\Delta y \left.\frac{\partial^2 f}{\partial x\partial y}\right|_{(x_0,y_0)} + \Delta y^2 \left.\frac{\partial^2 f}{\partial y^2}\right|_{(x_0,y_0)}\right], \end{aligned}$$

and is called the *second-order truncated Taylor expansion*.

In the multivariate case, the Taylor expansion around the point $\vec{\boldsymbol{x}}_0$ that keeps only second-order terms is expressed compactly using the gradient ∇f and the Hessian matrix $\boldsymbol{H}$ of f evaluated at the point $\vec{\boldsymbol{x}}_0$:

$$f(\vec{\boldsymbol{x}}_0 + \Delta\vec{\boldsymbol{x}}) \approx f(\vec{\boldsymbol{x}}_0) + \nabla f(\vec{\boldsymbol{x}}_0)^T \Delta\vec{\boldsymbol{x}} + \frac{1}{2}\Delta\vec{\boldsymbol{x}}^T \boldsymbol{H}(\vec{\boldsymbol{x}}_0)\Delta\vec{\boldsymbol{x}},$$

where $\Delta\vec{\boldsymbol{x}} = \vec{\boldsymbol{x}} - \vec{\boldsymbol{x}}_0$ is the vector of increments,

$$\nabla f(\vec{\boldsymbol{x}}_0) = \begin{pmatrix} \frac{\partial f(\vec{\boldsymbol{x}}_0)}{\partial x_1} & \frac{\partial f(\vec{\boldsymbol{x}}_0)}{\partial x_2} & \cdots & \frac{\partial f(\vec{\boldsymbol{x}}_0)}{\partial x_d} \end{pmatrix}^T$$

is the gradient of f at $\vec{\boldsymbol{x}}_0$, and

$$\boldsymbol{H}(\vec{\boldsymbol{x}}_0) = \begin{pmatrix} \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_1^2} & \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_1\partial x_2} & \cdots & \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_1\partial x_d} \\ \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_2\partial x_1} & \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_2^2} & \cdots & \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_2\partial x_d} \\ \vdots & \vdots & \ddots & \vdots \\ \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_d\partial x_1} & \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_d\partial x_2} & \cdots & \frac{\partial^2 f(\vec{\boldsymbol{x}}_0)}{\partial x_d^2} \end{pmatrix}$$

is the Hessian matrix of f at $\vec{x}_0$. The gradient is a column vector that contains the first-order partial derivatives of f with respect to each variable, and the Hessian is a symmetric square matrix that contains the second-order partial derivatives of f with respect to each pair of variables.

A.14.3 Small-angle approximation

The small-angle approximation is widely used in engineering when the angles involved are very small (usually less than 5° in magnitude), because it simplifies derivations considerably. It is valid in the limit as the angle tends to zero and is essentially a first-order approximation (linearization) of trigonometric functions obtained by truncating their Taylor series, so when the angle is measured in radians,

$$\sin x \approx x$$
$$\cos x \approx 1 \text{ or also } 1 - \frac{x^2}{2} \text{ (if a second-order approximation is used)}$$
$$\tan x \approx x.$$

This approximation can be justified by observing the Taylor series of the trigonometric functions

$$\sin x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n+1)!} x^{2n+1} = x - \frac{x^3}{3!} + \frac{x^5}{5!} - \cdots \text{ for all } x$$
$$\cos x = \sum_{n=0}^{\infty} \frac{(-1)^n}{(2n)!} x^{2n} = 1 - \frac{x^2}{2!} + \frac{x^4}{4!} - \cdots \text{ for all } x$$
$$\tan x = \sum_{n=1}^{\infty} \frac{B_{2n}(-4)^n(1-4^n)}{(2n)!} x^{2n-1} = x + \frac{x^3}{3} + \frac{2x^5}{15} + \cdots \text{ for } |x| < \frac{\pi}{2},$$

where the B_{2n} are called *Bernoulli numbers*, which are defined by

$$B_m(n) = \sum_{k=0}^{m} \sum_{v=0}^{k} (-1)^v \binom{k}{v} \frac{(n+v)^m}{k+1}.$$

When the angle x is less than one radian, the powers x^2, x^3, ... decrease rapidly and, therefore, very few of them are needed. The smaller x is, the faster the powers x^2, x^3, ... decrease, which justifies truncating the Taylor series.

Using the following Python code:

```
import numpy as np

# Calculations for generating Table A.1
theta = np.deg2rad(np.arange(0, 11, 1))
s = np.sin(theta);                error_s = 100*np.abs(s - theta)/s
```

Table A.1: *Value of the trigonometric functions for different angles*

θ (Degrees)	θ (Radians)	$\sin(\theta)$ (Error)	$\cos(\theta)$ (Error)	$\tan(\theta)$ (Error)
0	0	0 (0%)	1 (0%)	0 (0%)
1	0.0175	0.0175 (0.0051%)	0.9998 (0.0152%)	0.0175 (0.0102%)
2	0.0349	0.0349 (0.0203%)	0.9994 (0.0610%)	0.0349 (0.0406%)
3	0.0524	0.0523 (0.0457%)	0.9986 (0.1372%)	0.0524 (0.0914%)
4	0.0698	0.0698 (0.0813%)	0.9976 (0.2442%)	0.0699 (0.1625%)
5	0.0873	0.0872 (0.1270%)	0.9962 (0.3820%)	0.0875 (0.2540%)
6	0.1047	0.1045 (0.1830%)	0.9945 (0.5508%)	0.1051 (0.3658%)
7	0.1222	0.1219 (0.2492%)	0.9925 (0.7510%)	0.1228 (0.4980%)
8	0.1396	0.1392 (0.3257%)	0.9903 (0.9828%)	0.1405 (0.6507%)
9	0.1571	0.1564 (0.4124%)	0.9877 (1.2465%)	0.1584 (0.8238%)
10	0.1745	0.1736 (0.5095%)	0.9848 (1.5427%)	0.1763 (1.0175%)

```
c = np.cos(theta);                  error_c = 100*np.abs(c - 1)/c
t = np.tan(theta);                  error_t = 100*np.abs(t - theta)/t

#           angle (deg)  angle (rad)  sine        cosine      tangent
np.c_[np.rad2deg(theta), theta,       s, error_s, c, error_c, t, error_t]
```

we obtain Table A.1. There we can see, for example, that the small-angle approximation errors for 5° for the sine, cosine, and tangent of x are 0.127%, 0.382%, and 0.254%, respectively. Here the functions `deg2rad` and `rad2deg` perform the conversion from degrees to radians and from radians to degrees, respectively.

A.15 Change of variables in multiple integrals

The change of variables in multiple integrals is a technique for evaluating multiple integrals by transforming the variables of integration. It allows us to transform a complicated integral into a simpler one, often by converting to a more convenient coordinate system. The main idea is to simplify the region of integration or the integrand itself by using a suitable transformation.

Let $\vec{\boldsymbol{x}} = \vec{\boldsymbol{x}}(\vec{\boldsymbol{u}})$, be a smooth one-to-one map from a region $U \subset \mathbb{R}^n$ (with coordinates $\vec{\boldsymbol{u}} = (u_1, \ldots, u_n)$) to a region $\Omega \subset \mathbb{R}^n$ (with coordinates $\vec{\boldsymbol{x}} = (x_1, \ldots, x_n)$). Then, for an integrable scalar function f, we have:

$$\int_\Omega f(\vec{\boldsymbol{x}})\, \mathrm{d}\vec{\boldsymbol{x}} = \int_U f\big(\vec{\boldsymbol{x}}(\vec{\boldsymbol{u}})\big)\, |\det \boldsymbol{J}(\vec{\boldsymbol{u}})|\, \mathrm{d}\vec{\boldsymbol{u}}, \tag{A.1}$$

where $\boldsymbol{J}(\vec{\boldsymbol{u}})$ is the Jacobian matrix of the transformation:

$$\boldsymbol{J}(\vec{\boldsymbol{u}}) = \frac{\partial(x_1,\dots,x_n)}{\partial(u_1,\dots,u_n)} = \begin{pmatrix} \frac{\partial x_1}{\partial u_1} & \cdots & \frac{\partial x_1}{\partial u_n} \\ \vdots & \ddots & \vdots \\ \frac{\partial x_n}{\partial u_1} & \cdots & \frac{\partial x_n}{\partial u_n} \end{pmatrix}.$$

The quantity $|\det \boldsymbol{J}|$ is the local factor by which volumes are scaled under the transformation. In 2D it scales areas, and in 3D it scales volumes. The absolute value appears because an orientation reversal (negative determinant) should not produce negative area or volume elements.

Typical workflow

To compute a multiple integral by change of variables:

1. Choose a transformation $\vec{\boldsymbol{x}} = \vec{\boldsymbol{x}}(\vec{\boldsymbol{u}})$ that simplifies the region and/or integrand.
2. Compute the Jacobian matrix $\boldsymbol{J}$ and its determinant.
3. Rewrite the integrand as $f(\vec{\boldsymbol{x}}(\vec{\boldsymbol{u}}))$.
4. Rewrite the integration domain in the new coordinates.
5. Integrate using $|\det \boldsymbol{J}|\, \mathrm{d}\vec{\boldsymbol{u}}$.

Example A.3. *From rectangular to polar coordinates*

Compute the double integral of the function $f(x,y) = 1$ over a disk of radius R centered at the origin, $\Omega = \{(x,y) : x^2 + y^2 \le R^2\}$:

$$\iint_\Omega f(x,y)\,\mathrm{d}A = \iint_{x^2+y^2\le R^2} f(x,y)\,\mathrm{d}A = \int_{-R}^{R}\int_{-\sqrt{R^2-x^2}}^{\sqrt{R^2-x^2}} f(x,y)\,\mathrm{d}y\,\mathrm{d}x,$$

where $\mathrm{d}A$ is the area element in Cartesian coordinates.

Solution:

Computing this integral directly in Cartesian coordinates is cumbersome due to the curved boundary. We use the change of variables to polar coordinates to simplify both the integrand and the domain.

The transformation $\vec{\boldsymbol{x}} = \vec{\boldsymbol{x}}(r,\theta)$ is defined by:

$$x = r\cos\theta, \qquad y = r\sin\theta,$$

with $r \geq 0$ and $\theta \in [0, 2\pi)$. The new domain in polar coordinates is the rectangle $U = [0, R] \times [0, 2\pi)$, which is much simpler to integrate over. Following the workflow, we compute the Jacobian matrix of the transformation:

$$\boldsymbol{J}(r,\theta) = \frac{\partial(x,y)}{\partial(r,\theta)} = \begin{pmatrix} \frac{\partial x}{\partial r} & \frac{\partial x}{\partial \theta} \\ \frac{\partial y}{\partial r} & \frac{\partial y}{\partial \theta} \end{pmatrix} = \begin{pmatrix} \cos\theta & -r\sin\theta \\ \sin\theta & r\cos\theta \end{pmatrix},$$

and its determinant:

$$\det \boldsymbol{J}(r,\theta) = r\cos^2\theta + r\sin^2\theta = r.$$

Since $r \geq 0$, we have $|\det \boldsymbol{J}| = r$, so the area element transforms as $\mathrm{d}A = r\,\mathrm{d}r\,\mathrm{d}\theta$. Applying (A.1), the area of the disk is:

$$\iint_{x^2+y^2\leq R^2} 1\,\mathrm{d}A = \int_0^{2\pi}\int_0^R r\,\mathrm{d}r\,\mathrm{d}\theta = \int_0^{2\pi} \frac{R^2}{2}\,\mathrm{d}\theta = \pi R^2,$$

which is the well-known formula for the area of a disk of radius R.

A.16 Divergence theorem

In vector calculus, the *divergence theorem*★, also called *Gauss's theorem*★ or the *Ostrogradsky theorem*★, relates the flux of a vector field through a closed surface $\partial\Omega$ to the divergence of that field in the enclosed volume Ω. It converts a surface integral into a volume integral, and vice versa.

Let H and Ω be two sets in $\mathbb{R}^3$, i.e. $H, \Omega \subset \mathbb{R}^3$, such that $\Omega \subset H$ is closed and bounded, and let the boundary $\partial\Omega$ be an oriented surface with outward unit normal vector $\hat{\boldsymbol{n}}$. Let $\vec{\boldsymbol{g}} : H \to \mathbb{R}^3$ be a C^1 vector field[12], written as $\vec{\boldsymbol{g}} = g_x\hat{\boldsymbol{e}}_x + g_y\hat{\boldsymbol{e}}_y + g_z\hat{\boldsymbol{e}}_z$. Then:

$$\iiint_\Omega \nabla\cdot\vec{\boldsymbol{g}}\,\mathrm{d}V = \oiint_{\partial\Omega} \vec{\boldsymbol{g}}\cdot\hat{\boldsymbol{n}}\,\mathrm{d}S$$

where $\mathrm{d}V$ is a volume differential element of Ω, $\mathrm{d}S$ is a surface differential element of $\partial\Omega$, and $\nabla\cdot\vec{\boldsymbol{g}}$ is the *divergence*★ of the vector field $\vec{\boldsymbol{g}}$, defined as the scalar function:

$$\nabla\cdot\vec{\boldsymbol{g}} := \operatorname{div}\vec{\boldsymbol{g}} := \frac{\partial g_x}{\partial x} + \frac{\partial g_y}{\partial y} + \frac{\partial g_z}{\partial z}.$$

[12] C^1 is the set of functions whose the components are continuous first-order partial derivatives; in other words, $\vec{\boldsymbol{g}} \in C^1$ means that $\frac{\partial g_i}{\partial x_j}$ exists and is continuous for all i, j.

The divergence theorem, can be generalized to higher dimensions and to more general vector fields, such as tensor fields. For example, if $\boldsymbol{\sigma} : H \to \mathbb{R}^{3\times 3}$ is a C^1 tensor field, the divergence theorem can be expressed as:

$$\iiint_{\Omega} \nabla \cdot \boldsymbol{\sigma} \, \mathrm{d}V = \oiint_{\partial\Omega} \boldsymbol{\sigma}\hat{\boldsymbol{n}} \, \mathrm{d}S$$

where:

$$\boldsymbol{\sigma} = \begin{pmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{pmatrix},$$

and $\nabla \cdot \boldsymbol{\sigma}$ is the vector field defined as:

$$\nabla \cdot \boldsymbol{\sigma} := \operatorname{div} \boldsymbol{\sigma} := \begin{pmatrix} \frac{\partial \sigma_{xx}}{\partial x} + \frac{\partial \sigma_{xy}}{\partial y} + \frac{\partial \sigma_{xz}}{\partial z} \\ \frac{\partial \sigma_{yx}}{\partial x} + \frac{\partial \sigma_{yy}}{\partial y} + \frac{\partial \sigma_{yz}}{\partial z} \\ \frac{\partial \sigma_{zx}}{\partial x} + \frac{\partial \sigma_{zy}}{\partial y} + \frac{\partial \sigma_{zz}}{\partial z} \end{pmatrix}.$$

A.17 Outer product

The *outer product** of two vectors $\vec{\boldsymbol{a}} \in \mathbb{R}^m$ and $\vec{\boldsymbol{b}} \in \mathbb{R}^n$ is defined as the $m \times n$ matrix obtained by multiplying each element of $\vec{\boldsymbol{a}}$ by each element of $\vec{\boldsymbol{b}}$:

$$\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}} = \vec{\boldsymbol{a}}\vec{\boldsymbol{b}}^T = \begin{pmatrix} a_1 b_1 & a_1 b_2 & \cdots & a_1 b_n \\ a_2 b_1 & a_2 b_2 & \cdots & a_2 b_n \\ \vdots & \vdots & \ddots & \vdots \\ a_m b_1 & a_m b_2 & \cdots & a_m b_n \end{pmatrix}$$

The outer product has several important properties:

- The outer product is not commutative, that is $\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}} \neq \vec{\boldsymbol{b}} \otimes \vec{\boldsymbol{a}}$ in general.
- The outer product is bilinear, meaning that it is linear in each argument separately. Specifically, for any vectors $\vec{\boldsymbol{a}}, \vec{\boldsymbol{a}}' \in \mathbb{R}^m$, $\vec{\boldsymbol{b}}, \vec{\boldsymbol{b}}' \in \mathbb{R}^n$, and scalars $c, d \in \mathbb{R}$, we have:

$$(c\vec{\boldsymbol{a}} + d\vec{\boldsymbol{a}}') \otimes \vec{\boldsymbol{b}} = c(\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}) + d(\vec{\boldsymbol{a}}' \otimes \vec{\boldsymbol{b}})$$
$$\vec{\boldsymbol{a}} \otimes (c\vec{\boldsymbol{b}} + d\vec{\boldsymbol{b}}') = c(\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}) + d(\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}')$$

- The outer product can be used to construct rank-1 matrices. Specifically, any matrix $\boldsymbol{A}$ of rank 1 can be expressed as the outer product of two vectors $\vec{\boldsymbol{a}}$ and $\vec{\boldsymbol{b}}$, such that $\boldsymbol{A} = \vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}$.
- The outer product is related to the dot product through the following identity: for any vectors $\vec{\boldsymbol{a}}, \vec{\boldsymbol{b}} \in \mathbb{R}^m$, we have $\vec{\boldsymbol{a}} \cdot \vec{\boldsymbol{b}} = \operatorname{tr}(\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}})$, where tr denotes the trace of a matrix.

Example A.4. *Divergence of the outer product*

Show that the divergence of the outer product $\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}$ for vectors $\vec{\boldsymbol{a}} \in \mathbb{R}^m$ and $\vec{\boldsymbol{b}} \in \mathbb{R}^n$ is given by the vector:

$$\nabla \cdot (\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}) = (\nabla \cdot \vec{\boldsymbol{b}})\vec{\boldsymbol{a}} + \frac{\partial \vec{\boldsymbol{a}}}{\partial \vec{\boldsymbol{x}}}\vec{\boldsymbol{b}}. \tag{A.2}$$

Solution:

This result can be derived using the product rule for differentiation and the properties of the divergence operator. Let's go through the steps to derive this identity. The i-row of the left-hand side can be expressed in index notation, for $i = 1, \ldots, m$, as:

$$\left[\nabla \cdot (\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}})\right]_i = \sum_{j=1}^{n} \frac{\partial}{\partial x_j}(a_i b_j).$$

Applying the product rule for differentiation, we have:

$$\frac{\partial}{\partial x_j}(a_i b_j) = a_i \frac{\partial b_j}{\partial x_j} + b_j \frac{\partial a_i}{\partial x_j}.$$

From these last two equations, we can identify the resulting terms in standard vector notation, for $i = 1, \ldots, m$:

- $\sum_{j=1}^{n} a_i \frac{\partial b_j}{\partial x_j} = a_i(\nabla \cdot \vec{\boldsymbol{b}})$; this corresponds to the vector $(\nabla \cdot \vec{\boldsymbol{b}})\vec{\boldsymbol{a}}$.
- $\sum_{j=1}^{n} b_j \frac{\partial a_i}{\partial x_j} = \left[\frac{\partial \vec{\boldsymbol{a}}}{\partial \vec{\boldsymbol{x}}}\vec{\boldsymbol{b}}\right]_i$; this corresponds to the Jacobian of $\vec{\boldsymbol{a}}$ multiplied by $\vec{\boldsymbol{b}}$, which is the vector $\frac{\partial \vec{\boldsymbol{a}}}{\partial \vec{\boldsymbol{x}}}\vec{\boldsymbol{b}}$.

Summing these results gives the final identity:

$$\nabla \cdot (\vec{\boldsymbol{a}} \otimes \vec{\boldsymbol{b}}) = (\nabla \cdot \vec{\boldsymbol{b}})\vec{\boldsymbol{a}} + \frac{\partial \vec{\boldsymbol{a}}}{\partial \vec{\boldsymbol{x}}}\vec{\boldsymbol{b}}.$$

A.18 Localization theorem

The localization theorem lets us deduce that a function is identically zero using only its continuity and the value of its integral. It can be stated as follows: Let $f(x) : \Omega \to \mathbb{R}$ be a continuous function defined on an open connected set $\Omega \subset \mathbb{R}^n$. If the integral of f over *every* subdomain $D \subset \Omega$ equals zero:

$$\int_D f(\vec{\boldsymbol{x}})\, \mathrm{d}\vec{\boldsymbol{x}} = 0$$

then $f(\vec{\boldsymbol{x}}) = 0$ for all $\vec{\boldsymbol{x}} \in \Omega$.

Proof: Assume that there exists a point $\vec{\boldsymbol{x}}_0 \in \Omega$ such that $f(\vec{\boldsymbol{x}}_0) \neq 0$. Without loss of generality, we can assume that $f(\vec{\boldsymbol{x}}_0) > 0$ (the case where $f(\vec{\boldsymbol{x}}_0) < 0$ can be handled similarly). Since f is continuous, there exists a neighborhood around $\vec{\boldsymbol{x}}_0$ where f remains positive. Specifically, there exists an $\epsilon > 0$ such that for all $\vec{\boldsymbol{x}}$ in the ball $\mathscr{B}(\vec{\boldsymbol{x}}_0, \epsilon)$, we have $f(\vec{\boldsymbol{x}}) > 0$. Now, consider the subdomain $D = \mathscr{B}(\vec{\boldsymbol{x}}_0, \epsilon)$. The integral of f over this subdomain is given by:

$$\int_D f(\vec{\boldsymbol{x}})\, \mathrm{d}\vec{\boldsymbol{x}} > 0$$

This contradicts the assumption that the integral of f over every subdomain is zero. Therefore, our initial assumption must be false, and we conclude that $f(\vec{\boldsymbol{x}}) = 0$ for all $\vec{\boldsymbol{x}} \in \Omega$. This completes the proof.

A.19 Some vector calculus identities

Let $\vec{\boldsymbol{u}}$, $\vec{\boldsymbol{v}}$, and $\vec{\boldsymbol{w}}$ be three arbitrary vectors $\mathbb{R}^3 \to \mathbb{R}^3$; also let α, β, and γ be three arbitrary scalar fields $\mathbb{R}^3 \to \mathbb{R}$. Here we list some useful vector calculus identities that are often used in physics and engineering:

- Gradient of a product of a scalar and a vector:

$$\nabla(\alpha \vec{\boldsymbol{v}}) = \alpha \nabla \vec{\boldsymbol{v}} + \vec{\boldsymbol{v}} \otimes \nabla \alpha \tag{A.3}$$

- Divergence of a product of a scalar and a vector:

$$\nabla \cdot (\alpha \vec{\boldsymbol{v}}) = \alpha \nabla \cdot \vec{\boldsymbol{v}} + \nabla \alpha \cdot \vec{\boldsymbol{v}} \tag{A.4}$$

- Divergence of an outer product of two vectors:

$$\nabla \cdot (\vec{\boldsymbol{u}} \otimes \vec{\boldsymbol{v}}) = (\nabla \cdot \vec{\boldsymbol{v}})\, \vec{\boldsymbol{u}} + \frac{\partial \vec{\boldsymbol{u}}}{\partial \vec{\boldsymbol{x}}}\, \vec{\boldsymbol{v}}$$

- Time derivative of a cross product:

$$\frac{\partial}{\partial t}(\vec{\boldsymbol{r}} \times \vec{\boldsymbol{s}}) = \frac{\partial \vec{\boldsymbol{r}}}{\partial t} \times \vec{\boldsymbol{s}} + \vec{\boldsymbol{r}} \times \frac{\partial \vec{\boldsymbol{s}}}{\partial t}, \tag{A.5}$$

Disclaimer

I am not a native speaker of English. To ensure clarity and stylistic consistency, I used large-language-model (LLM) services (GitHub Copilot, Perplexity, Gemini, Claude, and DeepSeek). These tools acted exclusively as editorial aids: they polished grammar, improved phrasing, and helped maintain a coherent narrative flow.

The intellectual content of this book, however, is my own. After extensive study of primary and secondary sources in continuum mechanics, I produced original drafts for every chapter and section. I then submitted each draft to an LLM with a prompt similar to:

"Adopt the voice of an experienced professor of continuum mechanics. Using the draft below, compose a section on topic XXX that reads as a single, logically ordered narrative. Begin with fundamental principles and progress systematically to more advanced concepts, interweaving physical intuition with mathematical rigor. Enhance pedagogical clarity—through explanatory remarks, illustrative notes, while preserving technical accuracy."

Following each LLM response, I reviewed the output line by line, correcting any inaccuracies, eliminating bias, and retaining only those passages that faithfully reflected my intended exposition. The final manuscript is therefore a synthesis of my own research, analysis, and didactic choices, refined—but not fundamentally altered—by the language models named above.



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